VU MEDICAL ZONE

Admin: Amaan Khan

BT 404 ppt

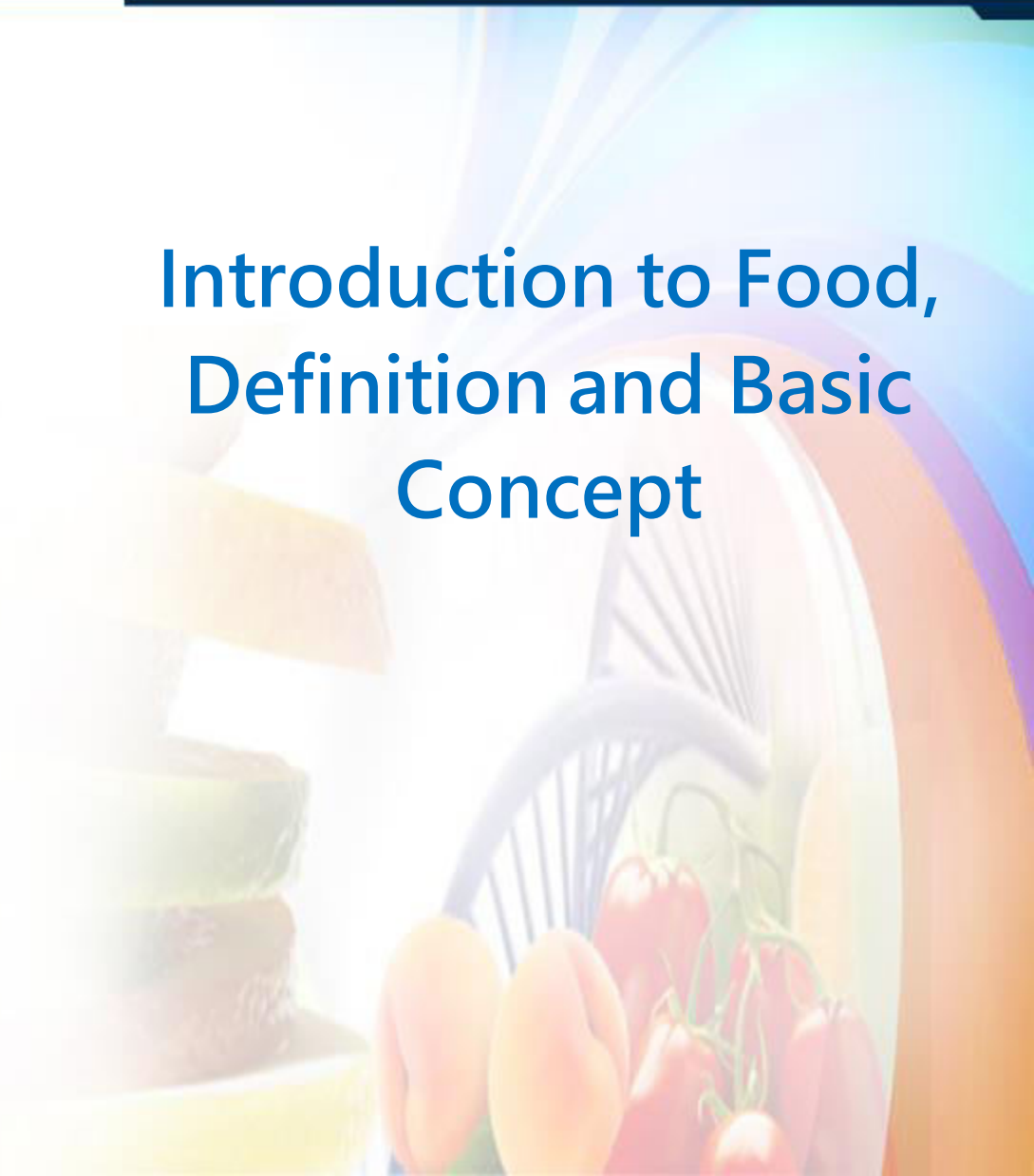
Food Biotechnology

Module #1



Food Biotechnology

Introduction to Food, Definition and Basic Concept



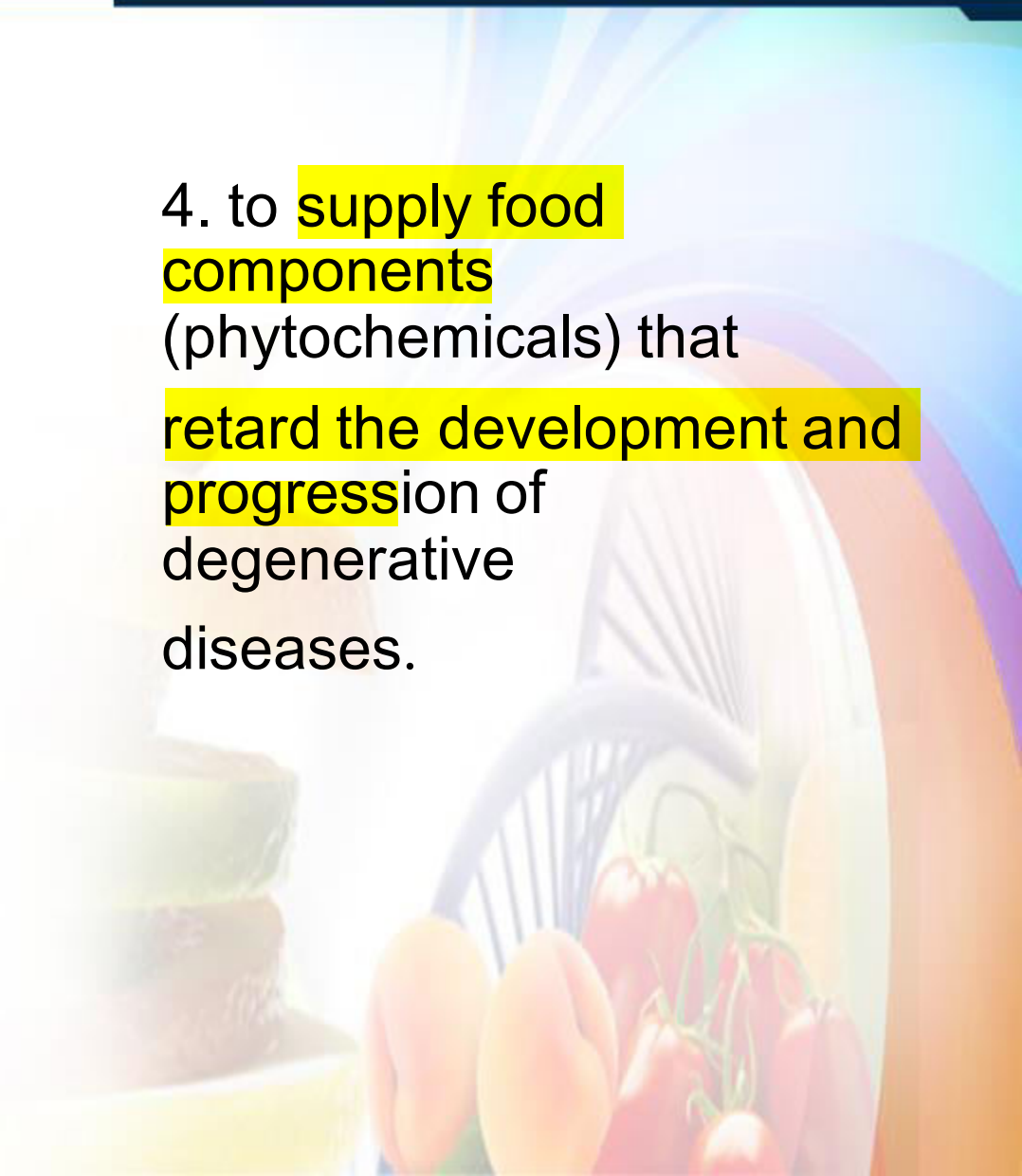
Introduction to Food, Definition and Basic Concept

Foods substances that are ingested by living organisms and can be metabolized to perform following functions:

1. source of energy;
2. source of raw materials for growth and development;
3. to supply minute amount of nutrients to regulate vital metabolic processes;

Introduction to Food, Definition and Basic Concept

4. to supply food components (phytochemicals) that retard the development and progression of degenerative diseases.



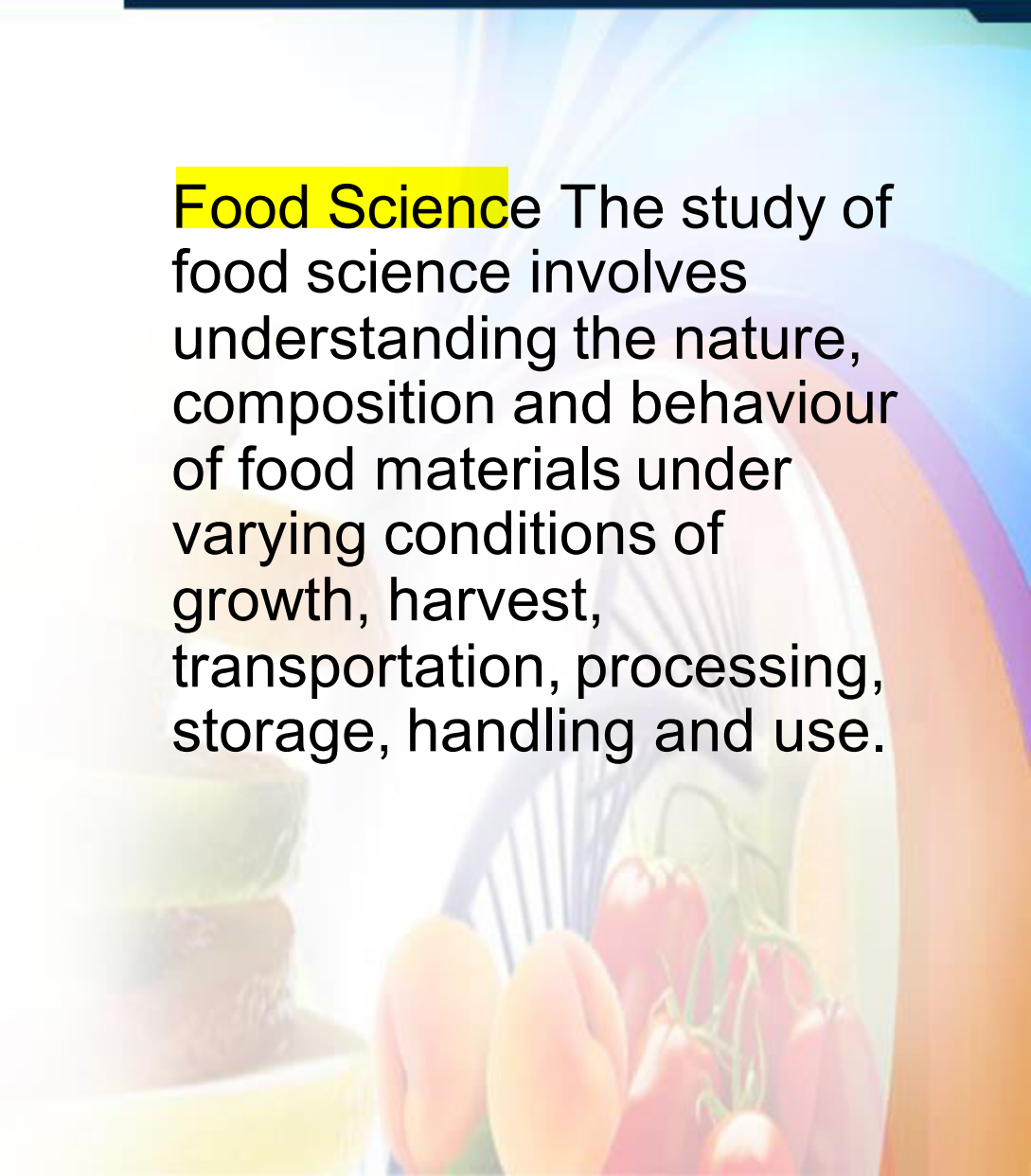
Introduction to Food, Definition and Basic Concept

The chemical components of food, which perform these functions are called *nutrients*.

Nutrition: The study of various nutrients, their functions, food sources and their utilisation by the human body and their effect on human wellbeing.

Introduction to Food, Definition and Basic Concept

Food Science The study of food science involves understanding the nature, composition and behaviour of food materials under varying conditions of growth, harvest, transportation, processing, storage, handling and use.



Introduction to Food, Definition and Basic Concept

Food Technology: is the application of food science & engineering principles to the selection, preservation, processing, packaging, distribution, and use of safe, nutritious, and wholesome food.



Introduction to Food, Definition and Basic Concept

Food Security:

Food security is achieved when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life. (800 million people suffer from hunger).

Introduction to Food, Definition and Basic Concept

Food Industry:

Food is not only our most important need, but the food industry is the largest of all industries, employing tens of millions of people in all areas such as growing, fishing, processing, transportation, storage, preparation, inspection, distribution, sales, and marketing.



Introduction to Food, Definition and Basic Concept

Food Biotechnology:

Application of technology to modify genes of animals, plants, and microorganisms to create new species which have desired production, marketing, or nutrition related properties. Called genetically engineered (GE) or genetically modified (GM) foods.

Food Biotechnology

Module # 2



Food Biotechnology

Food Composition

Food composition

The nutrients

found in foods include carbohydrates, proteins, fats, minerals, vitamins, water and fibre

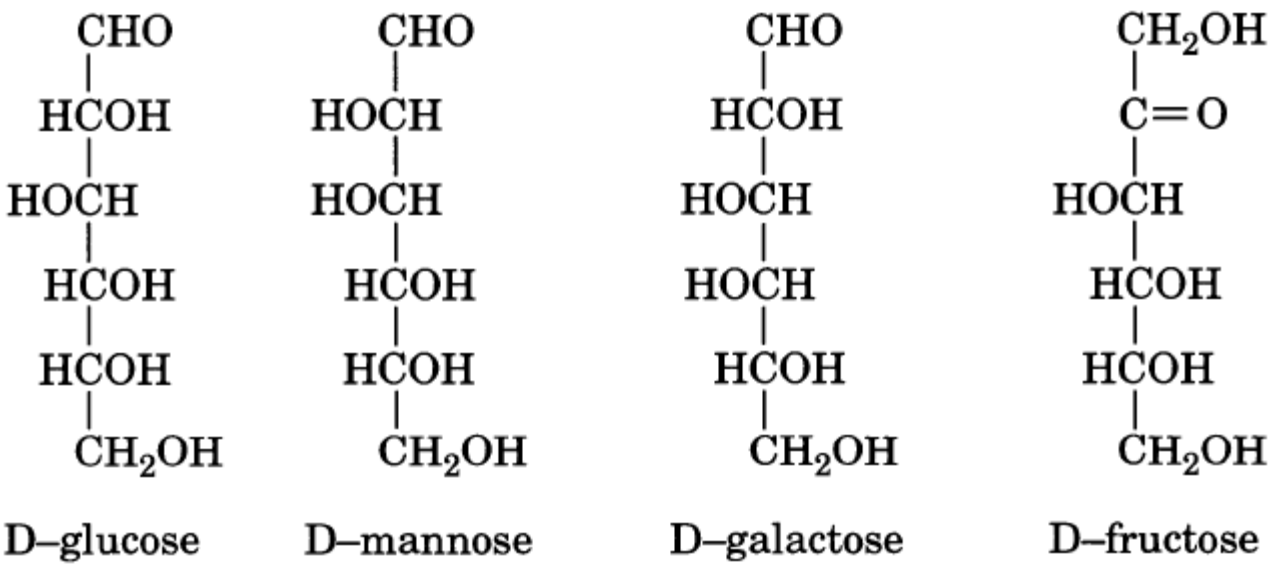
Food composition

Carbohydrates: In our diet, about 65 to 80 per cent of the energy is supplied by carbohydrates.

Carbohydrates contain carbon, hydrogen and oxygen.

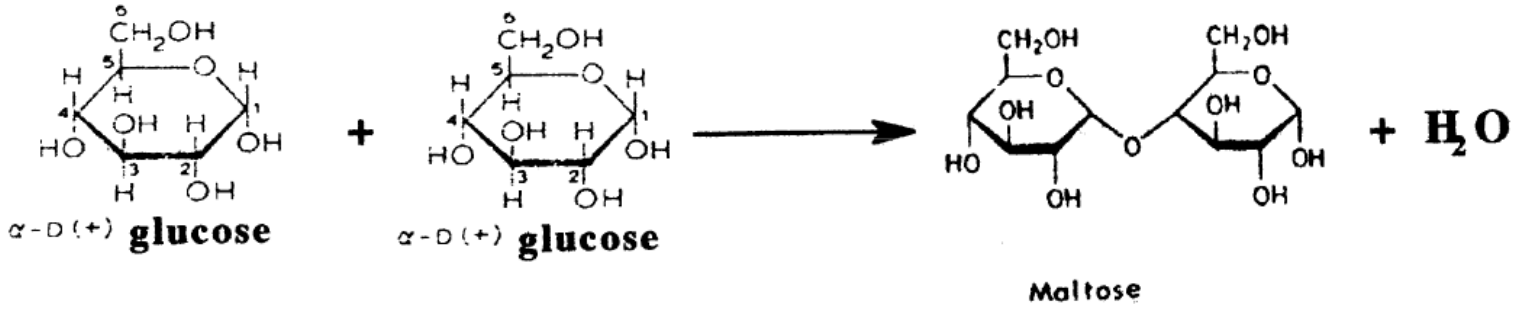
Monosacharides : are the the simplest units of the carbohydrates because these cannot be broken down any further by hydrolysis.

Food composition



More complex sugars may be formed as two or three molecules of monosaccharides join together. When this happens, water molecules are formed:

glucose + glucose → maltose + water



Food composition

Food composition

Polysaccharides

Polysaccharides are polymers (compounds of many smaller molecules) of the ten or more simple monosaccharides joined together.

However most polysaccharides in nature consist of hundreds to several thousand of monosaccharide units..

Food Composition

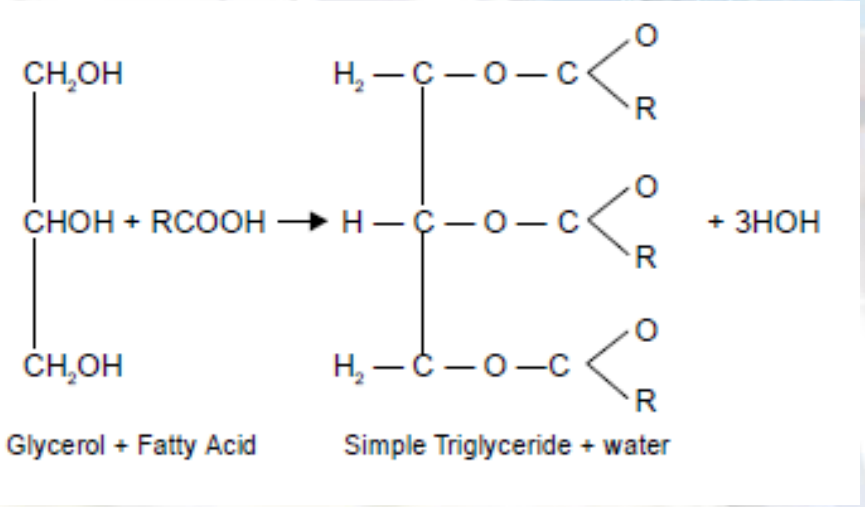
Fibers: Dietary fiber includes the nondigestible carbohydrates. These may be either water soluble or water-insoluble. The water-insoluble group, which includes wheat products and wheat bran. The water-soluble fibers, includes such as some cereal brans and pectin

Food Composition

Oils and fats belong to a naturally occurring substances called **lipids**. The common characteristics of lipids are:

1. They are soluble in organic solvents (ether, acetone etc.)
2. They are insoluble in water
3. Most of them are derivatives of fatty acids

Food Composition



Food Composition

Vegetable oils Groundnut, sesame, mustard, coconut, safflower, coconut, corn, cotton seed, soyabean and palm oil.

Animal fats Butter, ghee and cream from milk, lard from hogs and tallow from cattle.

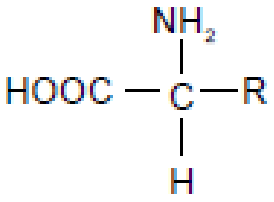
Manufactured fats *Vanaspati* and margarine.

Food composition

Proteins: Proteins are present in all living plant and animal tissues.

Composition Proteins, like fats and carbohydrates, contain carbon, hydrogen and oxygen. In addition they contain about sixteen per cent nitrogen, which distinguishes the proteins from carbohydrates and fats.

Some proteins contain sulphur, phosphorus, iron or other minerals also. Proteins are built from simpler compounds called *amino acids*. The amino acids contain a basic (amino — NH2) and an acidic (carboxyl — COOH) group in their molecules.



Food Composition

Food Composition

Water: Cellular material contains an abundance of water as it essential for survival of organisms due to its critical role in maintenance of biological functions.

Food Composition

Minerals:

Human beings require 14 different elements for health and growth. These include calcium, phosphorus, sodium, chlorine, potassium, magnesium and sulphur, which are present in appreciable amounts. the other iron, iodine, manganese, copper, zinc, cobalt and fluorine are found in traces.

Food Composition

Vitamins: are essential for the maintenance of good health as they catalyse various body processes, which help to utilize all the nutrients supplied and regulate a number of functions in the body. Vitamins are either fat-soluble and water-soluble. Fat-soluble vitamins include A, D, E and K, and water-soluble vitamins include the B-group and vitamin C.

Food Biotechnology

Module # 3



Food Biotechnology

Water Contents in Food

Food composition

Water Contents in Food

Moisture content is quantity of water present in various food

products and ingredients

Food composition

Water Contents in Food

It exists in three forms in foods

1. Free Water
2. Chemically adsorbed water
3. Chemically bound water

Water Contents in Food

Composition

Importance of Water in Foods

1. Water as Quality Factor in Preservation :

such as in products like dried milk, dehydrated vegetables, spices and herbs.

2. Quality factor of products

Such as jams, jellies, fruit syrups, prepared cereals etc.

Food composition

3. Convenience in Transportation :
Reduced moisture food is helpful in packaging and shipping such as concentrated milk, liquid cane sugar, concentrated fruits juices.

4. Moisture as compositional standards
Such as in foods cheese, flour, pineapple, meat etc.

Water Contents in Food

5. Moisture as nutritional Standards

6. Moisture data as expressed in analytical results of many foods such as food contents on wet basis and dry basis

Water Contents in Food

Water Activity is a concept different than water contents and it has important implications such as non-enzymic browning, lipid oxidization, degradation of vitamins, enzymatic hydrolysis, protein denaturation, and starch gelatinization and retrogradation. Shelf life, Microbial growth

Food Biotechnology

Module # 4



Food Biotechnology

Carbohydrates

Food composition

Carbohydrates

Monosaccharides:

are simple sugars ofter with suffix “ose” such as ketoses or aldoses.

Food composition

Carbohydrates

Oligosaccharides:

Consistes of 2-10 units of sugars, mostly disaccharidese are very important such as

1. Sucrose,
2. Fructoses
3. lactoses.
4. Maltoses

Food composition

Carbohydrates

Polisaccharides:

Polysaccharides are built of repeat units of monosaccharides

systematically named with the suffix *-an*. The generic name for polysaccharides is *glycan* and these can be *homoglycans* consisting of the one type of monosaccharide or *heteroglycans* consisting of

two or more types of monosaccharide.

Food composition

Carbohydrates

Starch:

Starch occurs in the form of semi-crystalline granules ranging in size from 2 to 100 µm, and consists of two types of glucan: *amylose* and *amylopectin*.

Food composition

Carbohydrates

Glycogen:

The polysaccharide that animals use for the shortterm storage of food energy in the liver and muscles is known as *glycogen*.Branching aids the rapid release of glucose since the enzymes that release glucose attack on the non-reducing ends, cleaving one glucose molecule at a time.

Food composition

Carbohydrates

Cellulose:

The most abundant molecule in nature. Makes up dietary fibre.

Food composition

Carbohydrates

Pectin:

Form gels in food products such as jams and preserves.

Carbohydrates

Gums:

High affinity with water and instead of making gels they form high viscosity liquids.

Food Biotechnology

Module # 4



Food Biotechnology

Carbohydrates

Food composition

Carbohydrates

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Food composition

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Food composition

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Food composition

Carbohydrates

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The most abundant molecule in nature. Makes up dietary fibre.

Food composition

Carbohydrates

Pectin:

Form gels in food products such as jams and preserves.

Carbohydrates

Gums:

High affinity with water and instead of making gels they form high viscosity liquids.

Food Biotechnology

Module # 6



Food Biotechnology

Fat contents in foods

Fat contents in foods

Lipids provide structure of cells and are source of energy, They are also carrier of fat soluble vitamins.

They are either fats or oils.

They are all fatty acid esters and may exist in form of triglycerides, phospholipids, sterols and tocopherols.

Food composition

Fat contents in foods

They are classified as

1. Simple lipids
2. Complex lipids
3. Derived lipids

Food composition

Fat contents in foods

Fatty acids:

Saturate and non saturated
Cis and trans

Food composition

Fat contents in foods

Systematic name	Trivial name	Lipid number
Butanoic acid	Butyric acid	4:0
Hexanoic acid	Caproic acid	6:0
Octanoic acid	Caprylic acid	8:0
Decanoic acid	Capric acid	10:0
Dodecanoic acid	Lauric acid	12:0
Tetradecanoic acid	Myristic acid	14:0
Hexadecanoic acid	Palmitic acid	16:0
cis-9-Hexadecenoic acid	Palmitoleic acid	16:1, *cis*-Δ^{9}
Octadecanoic acid	Stearic acid	18:0
cis-9-Octadecenoic acid	Oleic acid	18:1, *cis*-Δ^{9}
cis, cis-9,12-Octadecadienoic acid	Linoleic acid	18:2, *cis,cis*-$\Delta^{9,12}$
all-*cis*-9,12,15-Octadecatrienoic acid	Linolenic acid	18:3, *cis,cis,cis*-$\Delta^{9,12,15}$
Eicosanoic acid	Arachidic acid	20:0
all-*cis*-5,8,11,14-Eicosatetraenoic acid	Arachidonic acid	20:4, *cis,cis,cis,cis*-$\Delta^{5,8,11,14}$
cis-13-Docosenoic acid	Erucic acid	22:1, *cis*-Δ^{13}

Food composition

Fat contents in foods

Structure of Triglycerides:

MUFA

PUFA

Polymorphism

α, β, and β'.

Food composition

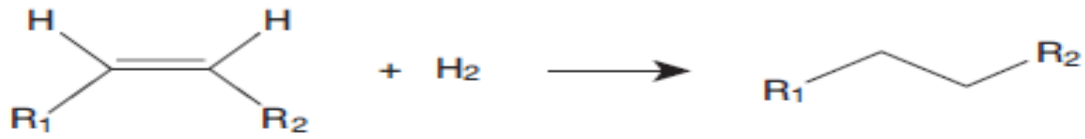
Processing of Fats and oils:

1. Hydrogenation oils are exposed to hydrogen gas under conditions of high temperature (150–180◦C) and pressure (2–10 atm) in the presence of a nickel catalyst.

Food composition

Fat contents in foods

Hydrogenation (main reaction):

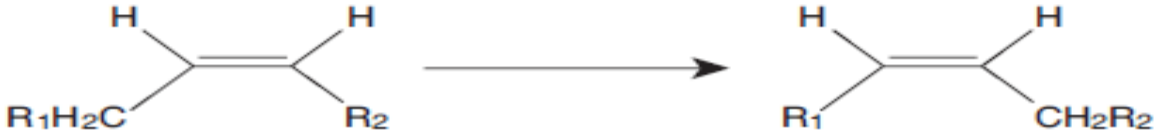


Isomerization (side reaction 1):



cis *trans*

Double bond migration (side reaction 2):



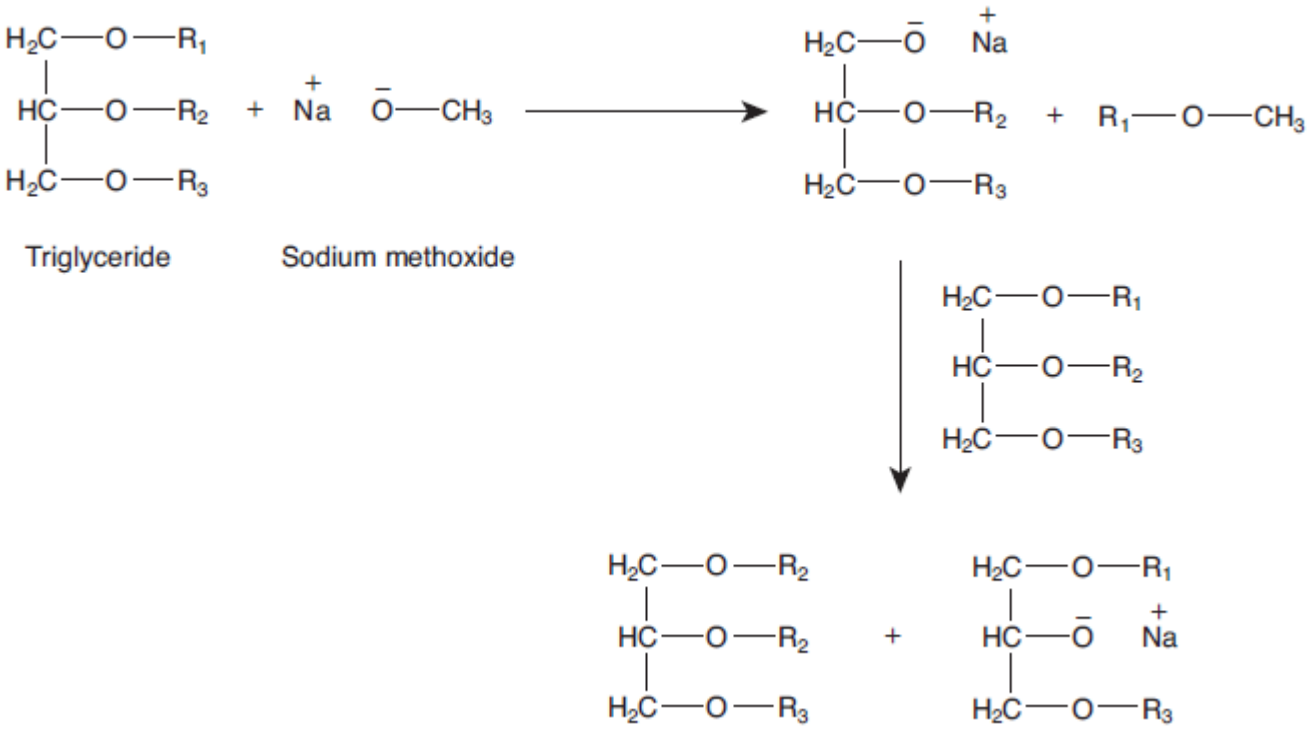
Fat contents in foods

Interesterification:

achieved through heating fats at high temperature (<200◦C) for long periods, but it is more efficient to employ catalysts that speed up the process (30 min) and lower the temperature required (50◦C). Sodium methoxide is the most popular catalyst

Food composition

Fat contents in foods



Food Biotechnology

Module # 7



Food Biotechnology

Minerals

d composition

Minerals

After remove of all water organic matter from the food products what remains is the inorganic material or ash. The human body consists of about 3% minerals, most of which is in the skeletal system.

Minerals are generally categorized as "major“ and "trace" based on the amounts in the body.

Food Composition

Minerals

1. Iron:

is a trace element, Iron is an essential part of both the blood pigment hemoglobin and the muscle pigment myoglobin. It is included in some enzymes. Deficiencies of iron cause anemia.

iron phosphates and iron phytates found in Plants are poorly absorbed.

Toxicity is also possible.

Food Composition

Minerals

Calcium:

It is required for bone and tooth structure and is necessary for the function of nerves and musc1es. Calcium is also needed in the blood clotting mechanism. Deficiencies can lead to osteoporosis. Vitamin D and Lactose help absorption.

Food Composition

Minerals

Sodium:

In humans, sodium is required in the extracellular fluids to maintain osmotic equilibrium and body-fluid volume. Sodium has been shown to aggravate hypertension (high blood pressure) and it is recommended that daily consumption levels be kept between 1100 and 3300 mg.

Food Composition

Minerals

Chlorine

Chlorine is also involved in extracellular fluids as the major negatively charged ion.

It has a role in controlling blood pH and is necessary in the production of hydrochloric acid of gastric juice. If great losses of body fluids occur, as in vigorous exercise, the amount of chlorine may have to be replenished.

F coMnpositiMinerals

Potassium

is associated with the function of musc1es and nerves and with the metabolism of carbohydrates.

It is important in maintaining the fluid volume inside cells, and the acid-base balance. Good sources of potassium are meats, eggs, oranges, bananas, and fresh milk. Cell membranes are quite permeable to potassium, but as it leaks out, a highly active membrane pump returns it to the cell in exchange for sodium. If 6% of the potassium contained in the cells were to escape into the blood,
the heart would stop.

Fod composition Minerals

Phosphorus

About 85% is found combined with calcium. It is also part of the body's major buffers (phosphoric acid and its salts) and it is part of both DNA and RNA. It is part of some lipids .

It plays a key role in energy transfer as it is part ofthe energy carrier of the cells, ATP. Some sources of phosphorus are meats, fish, eggs, and nuts..

Food Composition

Minerals

Magnesium:

Magnesium is a minor component of bones and is present in soft tissue cells, where it is involved in protein synthesis.

Sulfur:

Sulfur is present in virtually all proteins Deficiencies of sulfur are associated with protein deficiency and if foods containing sufficient protein are eaten then sulfur amounts will be adequate.

Food Composition

Minerals

Trace elements:

Iodine , Flourine
Copper, Cobalt
Zinc, Mangnese
Selenium, Vanadium
Silicon, Tin,
Chromium
Aluminium boron cadmium.

Food Biotechnology

Module # 8



Food Biotechnology

Vitamins

Food Composition

Vitamins

Fat Soluble vitamins:

Vitamin A:

Carotene conents in plant foods converts to Vit A. Important for vision and resistance to infection. Bone and teeth formation.

1 IU= 0.3 RE

1 RE= 6 microgram b-carotene

Food Composition

Vitamins

Vitamin D:

Vitamin D (calciferol or activated ergosterol) is necessary for normal tooth and bone formation. It is now expressed in micrograms (Ilg) as cholecalciferol and 10 Ilg = 400 IU. Vitamin D, as vitamin A is also routinely added to milk or vegetable oil.

Food Composition

Vitamins

VitaminE

Vitamin E has four different forms (the tocopherols) that have the same name except with the prefixes alpha-, beta-, gamma-, and delta-. It serves as an antioxidant that serves to prevent the oxidation of body components.

Good sources of Vitamin E are corn oil, cottonseed oil, and peanut oil. Deficiency effects reproduction, injury to the central nervous system, growth retardation, muscular dystrophy, and interference with normal heart action. The RDA for vitamin E is now given in milligrams of alpha-tocopherol equivalents (a-TE).

Vitamins

VitaminK

Vitamin K is essential for clotting of blood. Cabbage, spinach, cauliflower, and liver are especially good sources and it is also synthesized by bacteria in the human intestine. Antibiotic therapy that destroys intestinal organisms can produce deficiencies of vitamin K.

Food Composition

Vitamins

Vitamin BI (Thiamin)

It is also necessary for nerve function, appetite, and normal digestion as well as growth, fertility, and lactation. The symptoms ofthiamin deficiency are retardation of growth, palpitation and enlargement of the heart, hypertension, and beri-beri.

Food Composition

Vitamins

Vitamin B2 (Riboflavin)

Vitamin B2 makes up apart of enzyme systems involved in the oxidation and reduction of different materials in the body. Deficiency of riboflavin generally results in growth retardation and may result in vision impairment, scaling of the skin, and lesions on mucous tissues. Neuritis is another effect of deficiency.

Food Composition

Vitamins

Niacin (Nicotinic Acid)

Niacin is part of enzyme systems regulating reduction reactions in the body. It is also a compound that dilates blood vessels. It is part of the coenzyme nicotinamide adenine dinucleotide (NAD) which is involved in glucose breakdown. Deficiency of niacin causes pellagra (a disease that causes diarrhea, dermatitis, nervous disorders, and sometimes death). Good sources are yeast, meat, fish, poultry, peanuts, legumes, and whole grain.

Vitamins

Vitamin B6 *(Pyridoxine)*

Other substances closely related to this vitamin are pyridoxal and pyridoxamine.

It removes CO2 from the acid group (COOH) of certain amino acids and transfers amine groups (NH2) from one compound to another in the body.

Food Composition

Vitamins

Biotin:

Biotin is reported to be a coenzyme in the synthesis of aspartic acid, involves fixation of carbon dioxide. Deficiency of this compound is unusual, although it can be tied up by a substance in raw egg whites called aviden and the deficiency has been demonstrated in feeding studies with mice. Because ofthe production ofbiotin by microbial flora of the intestines, the dietary requirement for this compound is unknown. Liver is an excellent source of biotin and peanuts, peas, beans, and whole cooked eggs are good sources.

Food Composition

Vitamins

Pantothenic Acid

Pantothenic acid, a vitamin required for normal growth, nerve development, and normal skin. Organs of animals (liver, heart, kidney), eggs, whole wheat products, and peanuts are excellent sources.

Food Composition

Vitamins

Choline

It is a component of ceIl membranes and brain tissue and it functions as part ofthe substance acetylcholine, which is one ofthe brain's principal neurotransmitters.

Food Composition

Vitamins

***Vitamin** B 12*
(Cyanocobalamin)

It is required for the normal development of red blood cells, and a deficiency in vitamin B12 causes acute pernicious anemia and a variety of other disorders.

Food Composition

Vitamins

Vitamin C:

Ascorbic acid is required for the formation of intercellular substances in the body including dentine, cartilage, and the protein network of bone. Hence, it is important in tooth formation, the healing ofbroken bones, and the healing ofwounds.

Food Biotechnology

Module # 9



Food Biotechnology

Fibers

Food composition

Fiber

Dietary fiber includes the nondigestible carbohydrates. It is either

1. Water soluble
2. Water insoluble

Food composition

Fiber

Cellulose:
main component of plants and industrially important material such as wood , paper and cotton.

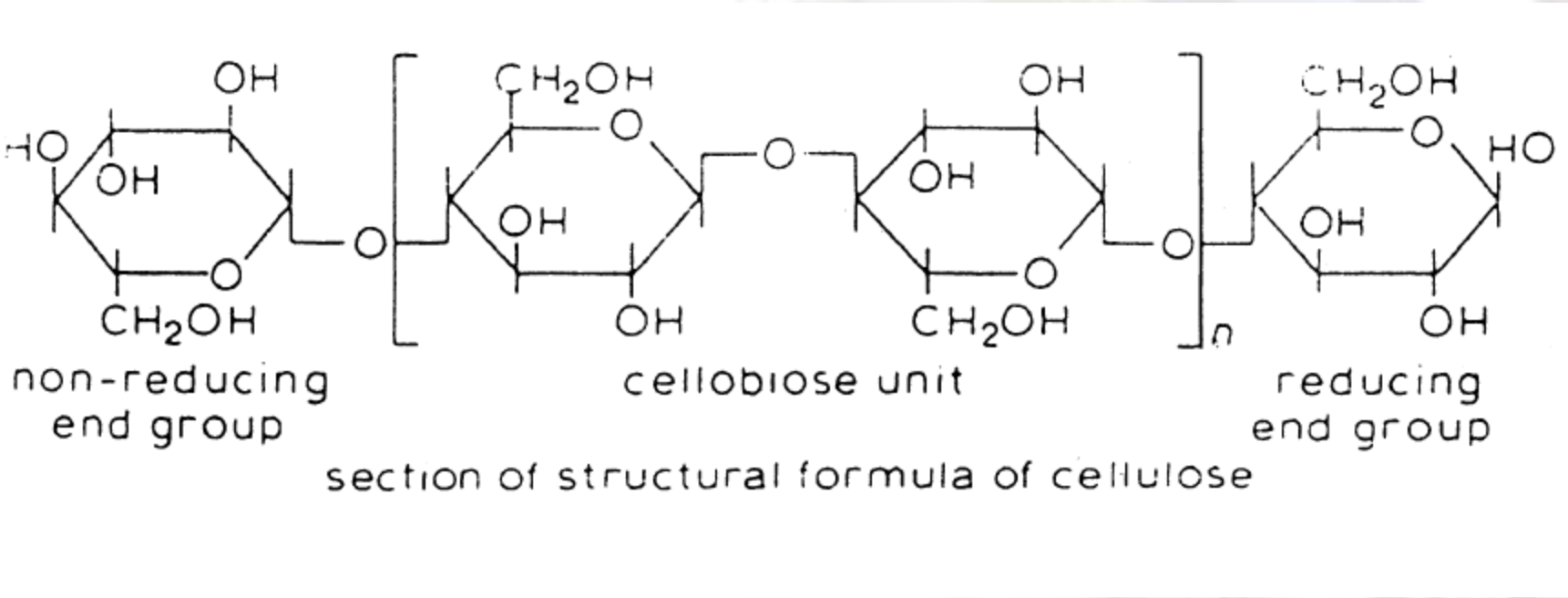
It is water insoluble and gives mechanical strength. The molecule is elongated and rigid. It needs enzyme cellulases to digest cellulose.

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Food composition

Fiber

Chemically it is linear polymer of D-glucose with beta 1,4 bonds



Food composition

Fiber

Treating cellulose with dilute acids yields microcrystalline celluloses for food applications.

Food composition

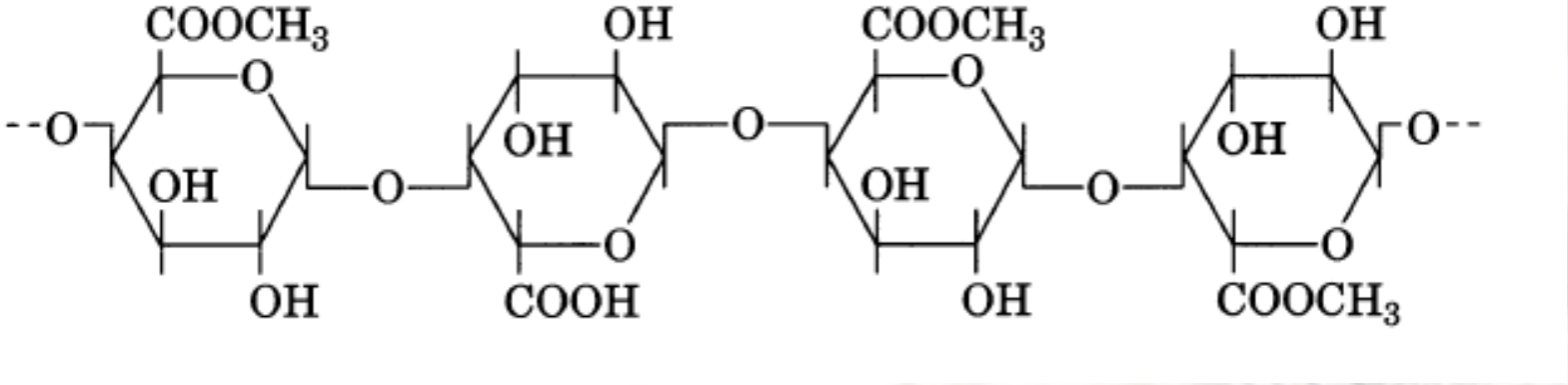
Fiber

Pectin:
Is water soluble fiber existing in intracellular spaces of plant tissues.
Structurally they are heteropolysaccharides

Food composition

Fiber

The unbranched chains consist of polygalacturonic acid molecules joined with a 1,4 linkage. The galacturonic acid units may be esterified (an acid is combined with an alcohol with methyl alcohol:



Food composition

Fiber

Slow set pectin have 50 to 70 % esterification and it sets slowly with addition of more sugar and acids.

Rapid set pectin has above 70% esterification and is used in products such as jams and preserves.

Low methoxy pectin is used as low calorie products because it needs less sugar and divalent ions such as calcium.

Food composition

Fiber

Gums are water soluble fibers. They are hydrophilic compounds that act as gelling , stabilizing and suspending agents with less concentration.

They can be

Exudate gums

Bean gums

Seed gums

Seaweed gums

They are also heteropolysaccharides with multiple applicatoins in variety of food products

Food Biotechnology

Module # 10



Food Biotechnology

Proximate Analysis

Food Composition

Proximate Analysis

Food Matrix is complex and consists of **carbohydrates, proteins, fats, minerals, vitamins, water and fibre**.

Broader classes of these closely related groups are termed as **proximate constituents.** Proximate analysis puts together all these closely related groups together.

Proximate Analysis
Ostion

Crude Fiber:

Refers to organic residues of vegetable origin consisting of cellulose, lignins and pentosans.

Carbohydrates and proteins are removed by boilling acid and alkali treatment.

Quality factor in many foods and it is inversely proportional to food value of products.

Food composition

Proximate Analysis

Moisture contents:
60-95 % water contents in foods.
Fruits and vegetalbes and pulses all have varying degree of water.
Its measure of stability and quality.

It exists in various forms such as free, bound and hence removal of water needs varying treatment to measure moisture contents.
there are many variables to consider for accurate measurement during drying process.

Food composition

Proximate Analysis

Crude Fat:

Also called ether extract, it includes phospholipids, lecithing, sterol, fatty acids, waxes, carotenoids, chlorophyl and other pigments.

Soxlet method of extraction for fats.

Food composition

Proximate Analysis

Ash contents:

Inorganic material after burning organic matter in food. Original form of minerals may change to carbonates or oxide, or phosphates, chloride, sulphate form. Change in chemical nature may lead to certain volatiles such as sulphides and halides.

Procedures must be followed thoroughly. It is important for identification, quality, adulteration etc.

Food composition

Proximate Analysis

Crude Proteins:

Nitrogen is a central element of proteins but it is also found in other compounds such as nucleic acids, alkaloids, nitrogen lipids and vitamins etc.

Kjeldahl Method to determine proteins involves these steps

1. Sample digestion or wet oxidation
2. Distillation of ammonia
3. Titration of liberated ammonia

Food Biotechnology

Module # 11



Food Biotechnology

Estimation of moisture

Estimation of Moisture

There are different methods for moisture analysis depending upon type of food and requirement of analysis.

1. Drying methods
2. Distillation methods

Food Composition

Estimation of Moisture

Oven methods:

Many different types of oven can be used.

Basic principle involves drying of food and calculating the weight loss referred to as moisture contents.

Need to consider type of foods, such as oily foods, spices or sugars they all need different treatments.

Estimation of Moisture

Air Oven Method:

1. Dry the flat bottom dish with tightly fitted lid at 100ºCwith heat first and weigh and tare the balance.
2. Add 2-10 g of food, loose the cover and heat for 2-3 hrs at 100ºC.
3. Remove and cool in dessicator, repeat the process till 2 mg different persists.
4. Calculate moisture loss

Food Composition

Estimation of Moisture

$$\% \text{ Moisture (wt/wt)} = \frac{\text{wt } H_2O \text{ in sample}}{\text{wt of wet sample}} \times 100$$

$$\% \text{ Moisture (wt/wt)} = \frac{\text{wt of wet sample} - \text{wt of dry sample}}{\text{wt of wet sample}} \times 100$$

Food composition

Estimation of Moisture

Vacuum Oven:

Suitable for spices, meats or sugar samples.

Pressure of 100 mg or less

4-6 hrs heating and 60-70ºC.

Remove dish and cool in dessicator.

Again heat for 1 hour and cool in dessicator till 2-3mg difference exists.

Calculate moisture loss.

Estimation of Moisture

Infra red drying:

Infra red lamps with a filament temperature of 2000-2500 K can penetrate heat to food and shorten drying time to 20-25 minutes only.

Food Composition

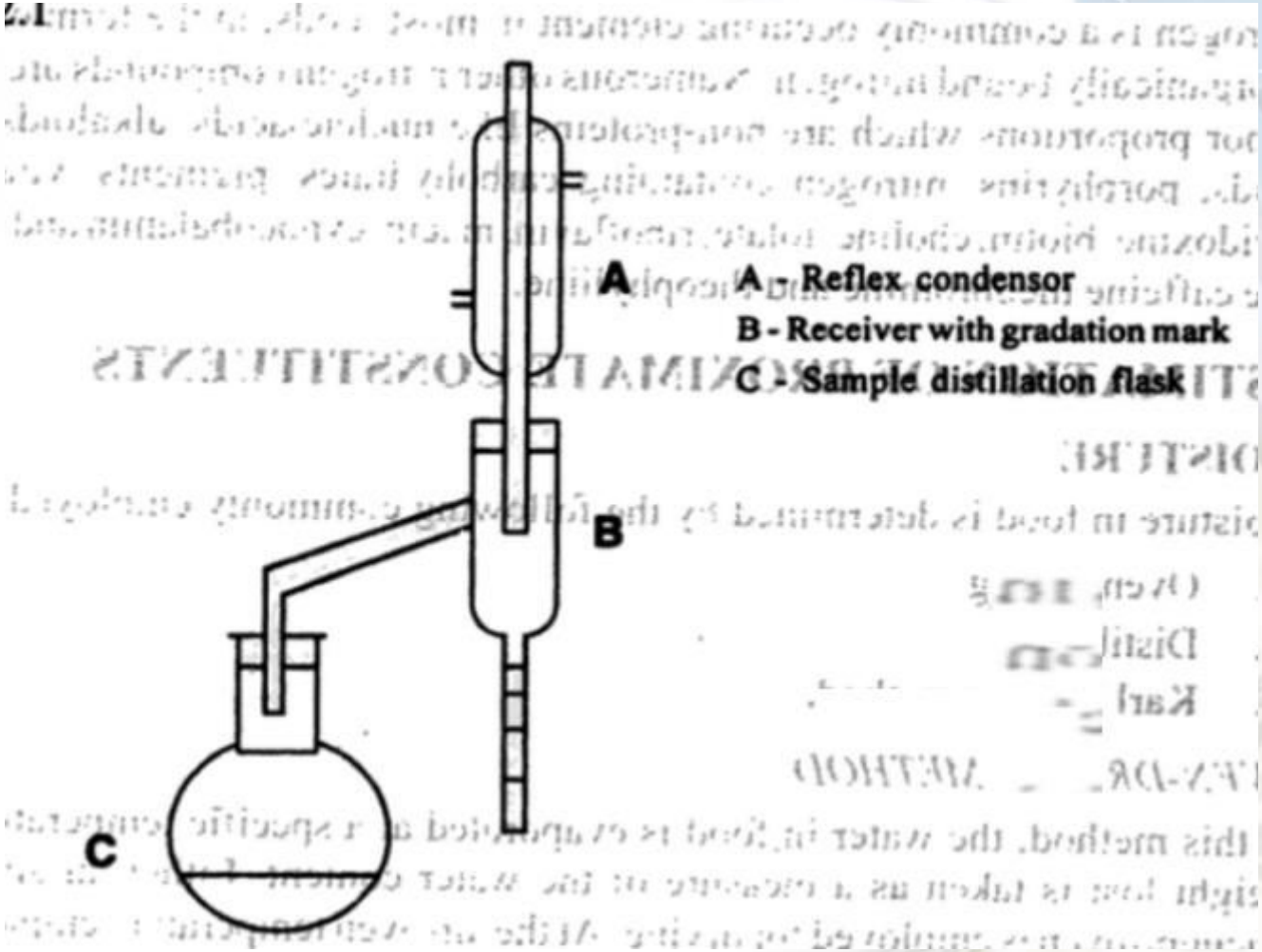
Estimation of Moisture

Distillation Method:

Food sample is heated with a solvent with higher boiling point and lower specific gravity than water. Solvent include xylene, toluene and carbon tetrachloride etc.
Water is received in a graduate cylinder and is measure directly from the graduated cylinder.

Food composition

Estimation of Moisture



Estimation of Moisture Content by Distillation

Method:

1. Add 10-40 g sample into flask and 75-100 ml solvent to cover food product.
2. Distill @ 1-2 drop per minute unless all clear water is achieved.
3. Wash the condenser and continue condensing for 5 minutes.
4. Cool and read the volume of water.

Food Biotechnology

Module # 12



Estimation of Ash in Foods

Food composition

Ash content

Total Ash:

Food sample is ignited in specific conditions and residue is weighed as total ash.

1. In silica dish weigh 2-3 g of food, pre dry if necessary.
2. Ignite over burner for 4-6 hrs /or use muffle furnace at 500ºC overnight ill white greyish ash achieved.

Food composition

Ash content

3. Cool the dish in a dessicator and weigh the dish

4. Calculate the residue as total ash.

Food composition

Ash content

$$\%\ \text{ash (dry basis)} = \frac{\text{wt after ashing} - \text{tare wt of crucible}}{\text{original sample wt} \times \text{dry matter coefficient}} \times 100$$

Food Composition

Ash content

Wet Ashing:

We use oxidizing agents or acids or their combinations to solubilize minerals without volatolization.

1. Weigh 1 g previously dried sample in Erlenmeyer flask (previously dried and acid washed).
2. Add 3 ml H_2SO_4 and 5 ml HNO_3. (with Blank)

Food Composition

Ash content

3. Heat to 200ºC till fume cease.

4. Add HNO3 and then boil and cool at room temperature

5 . Add ultrapure water in flastk and make up volume for further mineral analysis.

Food Composition

Ash content:

Water Soluble Ash:

Ash soluble in water gives information about fruit contents in jams and preserves.

Solution of water is heated and filtered to know soluble and insoluble fractions.

F Ash protein composition

Acid Soluble Ash:

Acid insoluble ash is measure of added minerals to foods such as dirt, sand or talc.

Food Biotechnology

Module #13



Estimation of Carbohydrates

Sample Preparation

It depends on the nature of the food being analyzed.

Aqueous solutions require very little preparation prior to analysis.

nuts, cereals, fruit, breads and vegetables.

Estimation of Carbohydrates

Sample Preparation

In these foods it is usually necessary to isolate the carbohydrate from the rest of the food before it can be analyzed. by solvent extraction.

Estimation of Carbohydrates

Isolation method depends on type of carbohydrate food matrix the purpose of analysis, however, there are some procedures that are common to many isolation techniques. For example, foods are usually dried under vacuum (to prevent thermal degradation), ground to a fine powder (to enhance solvent extraction) and then defatted

Estimation of Carbohydrates

low molecular weight carbohydrates are extracted from foods by boiling a defatted sample with an 80% alcohol solution. Monosaccharides and oligosaccharides are soluble in alcoholic solutions.

Estimation of Carbohydrates

The soluble components can be separated from the insoluble components by filtering. These two fractions can then be dried and weighed to determine their concentrations.

Estimation of Carbohydrates

Clarification:

Done to remove interfering substances for spectroscopic analysis or endpoint determinations. Heavy metal salts (such as lead acetate) which form insoluble complexes with interfering substances that can be removed by filtration or centrifugation.

Estimation of Carbohydrates

Ion-exchange.

Many monosaccharides and oligosaccharides are polar non-charged molecules and can therefore be separated from charged molecules by passing samples through ion-exchange columns.

Estimation of Carbohydrates

Ion-exchange.

By using a combination of a positively and a negatively charged column it is possible to remove most charged contaminants.

Estimation of Carbohydrates

Chromatographic and Electrophoretic methods

TLC, GC and HPLC are commonly used to separate and identify carbohydrates. Carbohydrates are separated on the basis of their differential adsorption on the basis of their partition coefficients, polarities or sizes, depending on the type of column used.

Estimation of Carbohydrates

HPLC is capable of rapid, specific, sensitive and precise measurements. In addition, GC requires that the samples be volatile, which usually requires that they be derivitized, whereas in HPLC samples can often be analyzed directly.

Estimation of Carbohydrates

HPLC and GC are commonly used in conjunction with NMR or mass spectrometry so that the chemical structure of the molecules that make up the peaks can also be identified.

Estimation of Carbohydrates

Carbohydrates can also be separated by electrophoresis after they have been derivitized to make them electrically charged, *e.g.,* by reaction with borates. A solution of the derivitized carbohydrates is applied to a gel and then a voltage is applied across it.

Estimation of Carbohydrates

The carbohydrates are then separated on the basis of their size: the smaller the size of a carbohydrate molecule, the faster it moves in an electrical field.

Estimation of Carbohydrates

Chemical methods

reducing nature of mono and oligosaccharides reacts with other components to yield precipitates or colored complexes which can be quantified. The concentration of carbohydrate can be determined gravimetrically, spectrophotometrically or by titration.

Methods For Estimation of Carbohydrates

Food composition

Non-reducing carbohydrates are first hydrolyzed to make them reducing. It is possible to determine the concentration of both non-reducing and reducing sugars by carrying out an analysis for reducing sugars before and after hydrolyzation.

Methods For Estimation of Carbohydrates

Food composition

Many different chemical methods are available for quantifying carbohydrates. Most of these can be divided into three catagories: titration, gravimetric and colorimetric.

Estimation of Carbohydrates

Titration Methods

The Lane-Eynon method is an example of a tritration method of determining the concentration of reducing sugars in a sample. A known amount of boiling copper sulfate solution and a methylene blue indicator. The reducing sugars in the carbohydrate solution react with the copper sulfate present in the flask.

Estimation of Carbohydrates

Once all the copper sulfate in solution has reacted, any further addition of reducing sugars causes the indicator to change from blue to white. The volume of sugar solution required to reach the end point is recorded.

A calibration curve by carrying out the experiment with a series of standard solutions of known carbohydrate concentration.

Estimation of Carbohydrates

Colorimetric Methods

Sugars react with the anthrone reagent under acidic conditions to yield a blue-green color.

Enzymatic Methods

Analytical methods based on enzymes rely on their ability to catalyze specific reactions.

Estimation of Carbohydrates

These methods are rapid, highly specific and sensitive to low concentrations and are therefore ideal for determination of carbohydrates in foods

Food Biotechnology

Module #14



Estimation of Proteins

Protein analysis is important for:

1. Blological activity

Some proteins, including enzymes or enzyme inhibitors, are relevant to food science and nutrition: for instance. the proteolytic enzymes in the tenderization of meats, pectinases in the ripening of fruits, and trypsin inhibitors in legume seeds are proteins.

Estimation of Proteins

2. Functional property investigation.

Proteins in various types of food have unique food functional properties: for example, gliadin and glutenins in wheat flour for breadrnaking, casein in milk for coagulation into cheese products, and egg albumen for foaming.

3. Nutrition labelling.

Estimation of Proteins

Protein analysis is required when you want to know:

1. Total protein content
2. Amino add composition
3. Content of a particular protein in a mixture
4. Protein content during isolation and purification of a protein

Estimation of Proteins

Protein analysis is required when you want to know:

5. Nonprotein nitrogen

6. Nutritive value (digestibility, protein efficiency ratio, or nitrogen balance) of a protein

Estimation of Proteins

Kjeldahl method :

About 1000 years old method , Developed by Kjeldahl in 1883 .

Applied on variety of substances e.g. meat ,grains ,waste water soil etc. With the passage of time technique and apparatus modified

The basic principle is same still now

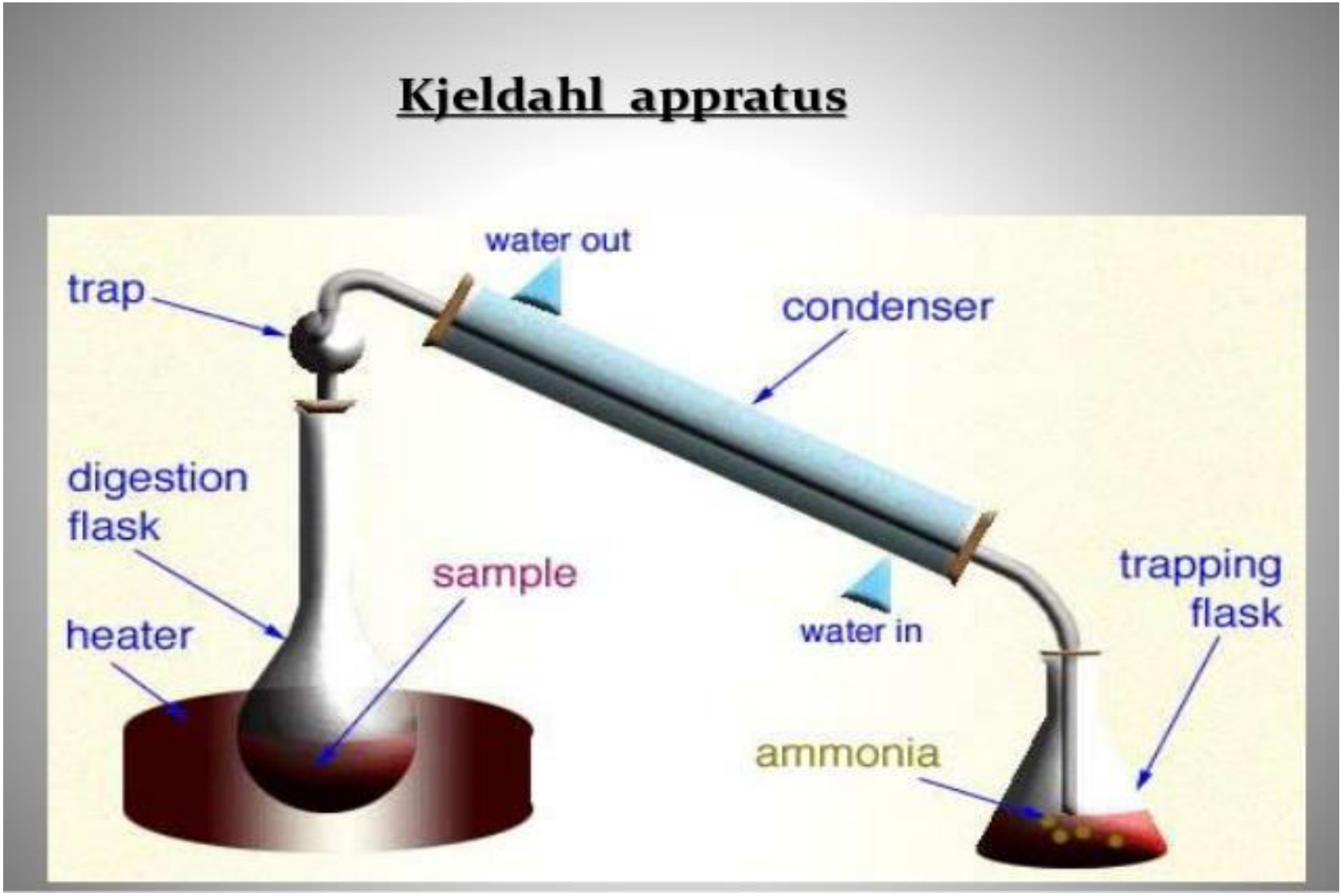
Estimation of Proteins

Estimation of nitrogen by digesting food in strong acid

Steps: The process is divided into 3 main steps.

1. Digestion
2. Distillation
3. Titration

Estimation of Proteins



Estimation of Proteins

Digestion

- Proteins are first digested
- Digestion is carried out with concentrated H_2SO_4
- In presence of an inorganic catalyst
- Digestion converts any nitrogen in food to ammonia and other organic matter to CO_2 and H_2O

Estimation of Proteins

Distillation

- Digestion flask is connected with receiving flask
- Ammonium is formed using distillation
- capture of ammonium occurs with a weak acid (boric acid)

Estimation of Proteins

Titration

- The third step is quantification of the ammonium done by titration
- Titration with a strong acid (sulfuric acid) & suitable indicator to determine end point of the reaction

Estimation of Proteins

Dumas method:

This is an automated instrumental technique

use to measure protein concentration of food

described by a scientist Dumas in 1831

Is older than the Kjeldahl

Estimation of Proteins

Sample of known mass is combusted at high temperature (900^0C) chamber in the presence of oxygen

This lead to the release of Carbon dioxide and water.

Estimation of Proteins

The method has three steps:

- Combustion
- Reduction
- Separation
- Detection

END

Module #15



Estimation of Fat

Some of the most important properties of concern to the food analyst are:

Total lipid concentration

Type of lipids present

Physicochemical properties of lipids

Structural organization of lipids within a food

Estimation of Fat

Importance of lipid analysis

1. Nutritional importance (omega 3, saturated fat, cholesterol, fat sol vitamins).

2. Affect oxidative stability of foods (stability of many foods is affected by fatty acid composition or presence of enzymes that act on lipids such as lipoxygenase, lipase etc).

Estimation of Fat

3.Affect physical properties of foods (melting behavior in chocolate, margarine, etc.)

Estimation of Fat

Sample Selection and Preservations Depends on

Type of food – meat, milk, margarine.

The nature of the lipid component – volatility, physical state

Type of analytical procedure used – solvent extraction, instrumental technique

Estimation of Fat

Determination of Total Lipid Concentration

Reasons for analysis

- Economic
- Legal
- Health
- Quality
- Processing

Estimation of Fat

Solvent Extraction

Drying sample – solvent cannot easily penetrate foods containing water

Particle size reduction – finely ground

Acid hydrolysis – to release bound lipids into easily extractable forms

Solvent selection – choose the best solvent for the extraction

Estimation of Fat

Ideal Solvent For Fat Extraction

- High solvent power for lipids
- Low solvent power for nonlipids
- No residue
- Evaporate easily
- Low boiling point
- Non flammable / not explosive

Estimation of Fat

Ideal Solvent For Fat Extraction

- Nontoxic
- Low surface tension with food
- Single compound
- Non-hygroscopic

Estimation of Fat

Ethyl ether is used a lot but is

- Very flammable,
- Explosion hazard
- Forms peroxides
- Expensive.
- Petroleum ether is an excellent solvent for lipids
- More selective for more hydrophobic lipids
- Non hygroscopic
- Less flammable

Estimation of Fat

Batch Solvent Extraction

Mixing sample with organic solvent in separating funnel

Shake vigorously and allow the separation either by gravity or centrifugation.

Estimation of Fat

Batch Solvent Extraction

Aqueous phase is decanted off and the concentration of lipid in the solvent is determined by evaporating off the solvent and measure the mass lipid remaining

Have to be repeated a few times

Estimation of Fat

Semi-Continuous Solvent Extraction

Used to increase efficiency of lipid extraction from foods

Common method: Soxhlet extraction

Solvent extracts the lipids and carries them into the flask

The lipids still remain in the solvent due to low volatility

Estimation of Fat

Continuous Solvent Extraction

Commonly used is Goldfisch method

Similar to soxhlet method except the extraction chamber is designed.

Solvent trickles through the sample rather than building up around it

Estimation of Fat

Continuous Solvent Extraction

Disadvantage: channeling of the solvent can occur

i.e. solvent may take certain routes through the sample

Estimation of Fat

Accelerated Solvent Extraction

By increase the temperature and pressure normally used

The effectiveness of the lipid extraction increases as its temperature increases, but the pressure must also be increased to keep the solvent in the liquid state.

Advantage: reduce the amount of solvent

Estimation of Fat

Supercritical Fluid Extraction

Pressurized CO_2 is heated above a certain critical temperature to become supercritical fluid

This fluid behaves like a gas to easily penetrate into a sample and extract lipid while it also behaves like a liquid to dissolve a large quantity of lipids.

Estimation of Fat

Supercritical Fluid Extraction

The CO_2 extracts the lipid, and forms a separate solvent layer, which is separated from the aqueous components

Estimation of Fat

Non-solvent Liquid Extraction Methods

Babcock Method

A specified amount of milk is accurately pipetted into Babcock bottle. Sulfuric acid: breaks down the fat globule membrane than surrounds the droplet and thereby release the fat. Lipid is removed from the aqueous phase by centrifuging while it is hot (55-60°C).

Estimation of Fat

- Non-solvent Liquid Extraction Method
- Babcock method
- Surfactant Method
- Gerber Method

Food Biotechnology

Module #16



Estimation of Crude Fiber In Food

Definitions of Dietary Fiber

The term "dietary fiber" was originally synonymous With "roughage"—the nondigestible plant matter, principally of cell wall origin, that is responsible for fecal bulking.

Estimation of Crude Fiber In Food

Dimethyl Sulfoxide (DMSO)–Enzymatic–Chemical for NSP

The enzymatic–chemical methods were devised for specifically measuring NSP. They use enzymatic digestion to remove starch and protein.

Estimation of Crude Fiber In Food

Dimethyl Sulfoxide (DMSO)–Enzymatic–Chemical for NSP

And then measure remaining nondigestible polysaccharide specifically, as monosaccharide released by acid hydrolysis.

Estimation of Crude Fiber In Food

Enzymatic–Chemical/Gravimetric

The enzymatic–chemical/gravimetric method removes digestible food components enzymatically.

Estimation of Crude Fiber In Food

Enzymatic–Chemical/Gravimetric

And measures nondigested polysaccharides including undigested starch, chemically, but includes a lignin residue determined gravimetrically, with allowance for protein and ash content.

Estimation of Crude Fiber In Food

Detergent–gravimetric/Enzymatic–Gravimetric

This approach attempts to overcome the loss of soluble dietary fiber in the detergent extract of detergent methods. Duplicate samples are extracted, one with hot neutral detergent to measure detergent fiber, and the other with buffer and amyloglucosidase to measure soluble fiber.

Estimation of Crude Fiber In Food

Detergent–gravimetric/Enzymatic–Gravimetric

The detergent fiber and soluble fiber are added to obtain a measure of total fiber.

Estimation of Crude Fiber In Food

Enzymatic–Gravimetric

Method for measuring polysaccharides, lignin, and associated substances that are resistant to digestion by the alimentary enzymes of humans. Enzymatic digestion is used to remove most starch and protein.

Estimation of Crude Fiber In Food

Enzymatic–Gravimetric

Residual protein and ash content are deducted from the residue weight with the aim of obtaining a measure of NSP plus lignin by difference.

Estimation of Crude Fiber In Food

Nonenzymatic Gravimetric

The nonenzymatic gravimetric procedure was developed to give a rapid measurement of dietary fiber in samples, such as ripe fruit, that contained little starch and protein, and enough fiber relative to starch to enable starch digestion to be dispensed with.

Estimation of Crude Fiber In Food

Introduction

Some of the most important properties of concern to the food analyst are:

- Total lipid concentration
- Type of lipids present
- Physicochemical properties of lipids
- Structural organization of lipids within a food

Food Biotechnology

Module #17



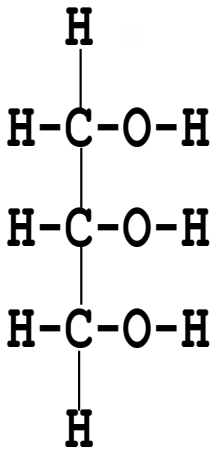
Fats and Oils

Oils and fats – important for storing chemical energy in living things.

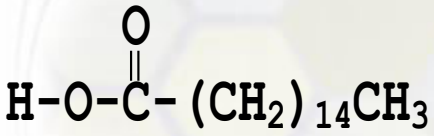
Oils are liquids – fats are solids.

Made from esters of propane-1,2,3,triol (glycerol) + long chain carboxylic acids RCOOH

Fats and Oils



Propane-1,2,3-triol

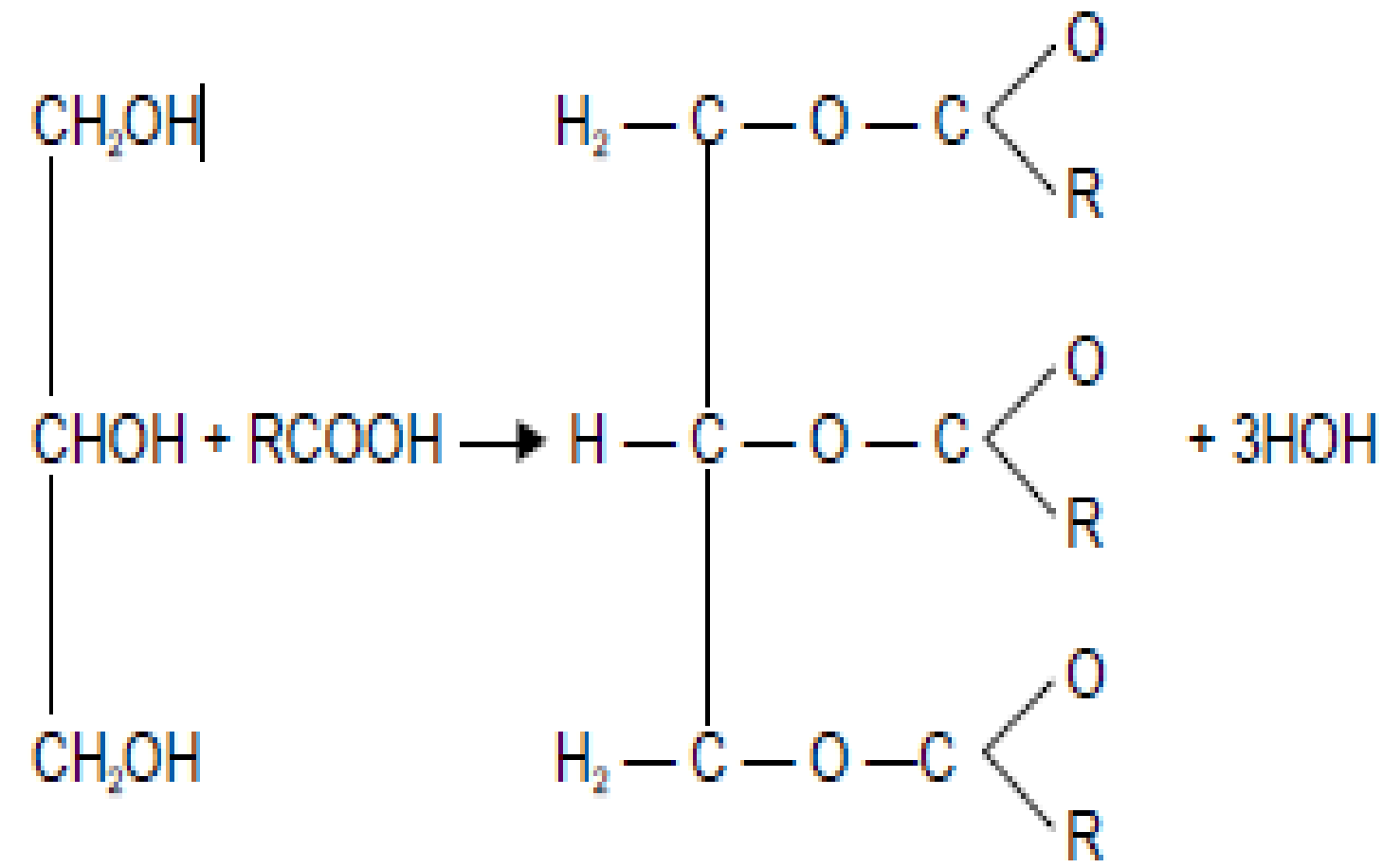


palmitic acid

Fats and Oils

S. no.	Food	Total fat (%)
1.	Ghee	99
2.	Butter	81
3.	Coconut	62
4.	Sunflower	52
5.	Gingelly	43
6.	Groundnut, Mustard	40
7.	Safflower	26
8.	Soyabean	20
9.	Fatty fish	5 to 21
10.	Egg	13
11.	Meat, Poultry	1 to 13
12.	Milk	1.5 to 7
13.	Cereals and Pulses	0.5 to 6
14.	Corn	4
15.	Vegetables	0.1 to 1

Fats and Oils



Fats and Oils

Composition and Classification

Oils and fats are composed of the elements carbon, hydrogen and oxygen. Fats are built up by linking together a number of individual fatty acids with glycerol.

Fats and Oils

Common Fatty Acids

Structure	Traditional name	Origin of name
$CH_3(CH_2)_{14}COOH$	palmitic acid	palm oil
$CH_3(CH_2)_{16}COOH$	stearic acid	suet (Greek: *stear*)
$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$	oleic acid	olive oil
$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	linoleic acid	oil of flax (Latin: *linum*)

Fats and Oils

Selection and Buying of Oils and Fats

Criteria for selection of oil are

presence of the natural characteristic aroma, natural colour, clarity, freedom from admixture with other kinds of oils, freedom from solid particles and flat or rancid odour

Fats and Oils

Nutritive Value and Digestibility

•Fats and oils are concentrated sources of energy

- Each gram of pure fat or oil supplies nine kilocalories
- Fats impart special flavor and texture to our foods

Fats and Oils

Physical and Chemical Properties

- **Melting Point** All food fats are mixtures of triglycerides, and therefore, do not have a sharp melting point, but melt over a range of temperatures.

Fats and Oils

Physical and Chemical Properties

- **Creaming** of Fats Solid fats like butter and margarine can be creamed or made soft and fluffy by the incorporation of air

Fats and Oils

Plasticity of Fats:

Fats are mouldable and can be creamed to exhibit plasticity. Such fats do not flow at room temperature and are thus solid fats. The spreading quality of butter is the result of its plastic nature. Plastic fats are composed of a mixture of triglycerides and not of one kind of a molecule.

Fats and Oils

Physical and Chemical Properties

- **Emulsification** The specific gravity of oils and fats is about 0.9, which indicates that they are lighter than water.

Fats and Oils

Physical and Chemical Properties

- **Emulsification**

They can form an emulsion with water when beaten up with it to form tiny globules in the presence of suitable emulsifying agent.

- E.g. butter

Fats and Oils

Fats are used in food preparation

(i) as a medium of cooking.

(ii) as a shortening biscuits, pastry and cakes.

(iii) to add richness and flavour

(iv) seasoning of vegetables and salads.

Fats and Oils

Effect of Prolonged Heating on Nutritive Value of Fats and Oils

- Thermal oxidation which is accelerated by the addition of soda and water. Consumption of thermally damaged fat is harmful.

Fats and Oils

Changes in Fats During Storage

•Fats and oils undergo certain undesirable changes during storage, which result in spoilage.

- Flavour Reversion
- Rancidity Spoilage

Food Biotechnology

Module #18



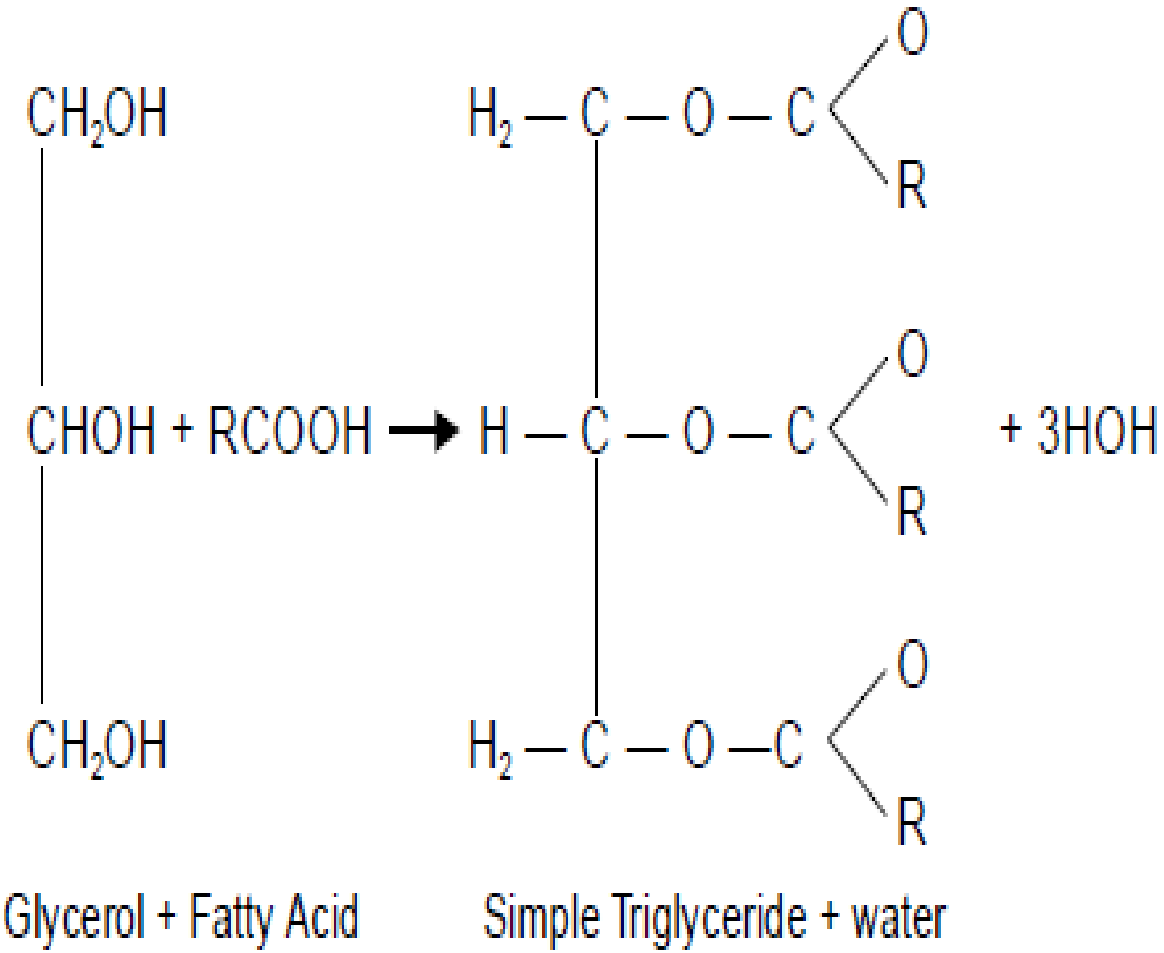
Food Biotechnology

Composition and Classification of fats & Oils

Composition and Classification of fats & Oils

Fats and oils are mixture of different triglycerides. Triglycerides are formed by mixing of glycerol and fatty acids with elimination of a water molecule.

Composition and Classification of fats & Oils



Composition and Classification of fats & Oils

Monoglycerides

Diglycerides

Simple triglycerides

Mixed triglycerides

Natural fats consists of mixed triglycerides and natural substances responsible for unique flavour or taste.

Composition and Classification of fats & Oils

Solid fats such as butter and ghee more saturated fatty acids.

Liquid fats such as soybean oil, canola oil etc. have less saturated fatty acids.

Composition and Classification of fats & Oils

No. of carbons	Fatty acid		Coconut oil (%)	Groundnut oil (%)	Corn (%)	Cotton seed (%)	Lard (%)	Mutton tallow (%)	Butter (%)
	Satura-ted	Unsatu-rated							
4	Butyric		–	–	–	–	–	–	3
6	Caproic		0.5	–	–	–	–	–	2
14	Myristic		18	–	–	1	1	2	10
16	Palmitic		9	6	6	21	28	34	30
18	Stearic		2	5	2	2	8	19	11
18		Oleic	7	61	37	25	56	43	30
18		Linoleic[2]	2	22	54	50	5	2	3
	Other acids		61.5	6	1	1	2	–	11
	Total		100	100	100	100	100	100	100

Composition and Classification of fats & Oils

Fatty acids are described as either short or long chain or saturated and unsaturated fatty acids.

small fatty acids such as acetic acid in vinegar, caproic and butyric acids in butter.

12-18 carbon chain length such as palmitic and stearic acids in lard and tallow.

Composition and Classification of fats & Oils

Fatty acids are mono and polyunsaturated fatty acids.

18 carbon oleic and linoleic found in olive and corn respectively.

Vegetable oils are more unsaturated and considered more healthy.

Composition and Classification of fats & Oils

END

Animal source of fats are more saturated and considered less healthy and the amount consumed should be restricted.

Trans fatty acid contents in hydrogenated fats are another factor making them more harmful for health.

Food Biotechnology

Module #19



Nutritive value of Fats and Oils

Nutritive value of Fats and Oils

Fats and Oils are source of energy. One gram of fats and oils produces 9 kilocalories of energy so they are concentrated form of energy.

There are many other functions such as they act as carrier for fat soluble vitamins A, D, E and K.

Nutritive value of Fats and Oils

Fats and Oils have more palatable satiety value, means they take more time leaving the stomach.

Digestion of fats starts in small intestine call duodenum. Gall bladder produces bile and pancreas produces lipase through a common duct that goes to duodenum.

Nutritive value of Fats and Oils

Bile neutralizes acidic pH of food in duodenum so that the work of pancreas enzymes starts.

Lipids are emulsified and surface area is enlarged to facilitate work of pancreas enzyme to split triglycerides into glycerol and fatty acids and mono and diglycerides.

Nutritive value of Fats and Oils

Bile salts disperse glycerol and fatty acids into micelles.

Rest of mono and diglycerides are also broken down to glycerol and fatty acids by intestinal lipases, made into micelles and absorbed by intestine.

Essential and non essential fatty acids.

Nutritive value of Fats and Oils

Essential fatty acids are linoleic acids

(omega-6) and a-linoleic acids (omega-3).

Deficiency of EFAs results in impaired brain growth, mental retardation and learning difficulties, dermatitis (dryness of skin), hair loss, and poor wound healing.

Nutritive value of Fats and Oils

Saturated fatty acids can raise LDL (bad Cholestrol), that may lead to heart disease.

Similarly trans fats can build up plaque in arteries and lead to heart diseases.

Food Biotechnology

Module #20



Physical properties of fats and oils



Physical properties of fats and oils

Melting Point

All food fats are mixtures of triglycerides, and therefore, do not have a sharp

melting point, but melt over a range of temperatures.

Physical properties of fats and oils

Creaming of Fats:

Solid fats like butter and margarine can be creamed or made soft and fluffy

by the incorporation of air. Fat and sugar are usually creamed together in the preparation of cakes

Physical properties of fats and oils

Plasticity of Fats

Fats are mouldable and can be creamed to exhibit plasticity. Such fats do not flow at room temperature and are thus solid fats. The spreading quality of butter is the result of its plastic nature. Plastic fats are composed of a mixture of triglycerides and not of one kind of a molecule. They, therefore, do not have a sharp melting

Physical properties of fats and oils

Emulsification

Lipids can form an emulsion with water when beaten up with it to form tiny globules in the presence of suitable emulsifying agent. Butter is an emulsion, so also is cream. The presence of minute amounts of milk protein helps to stabilise these emulsions. Lecithin, a phospholipid from egg yolk helps to stabilise mayonnaise.

Physical properties of fats and oils

Smoke Point:

The smoke point is the temperature at which a fat or oil gives off a thin bluish smoke. Foods that are fried are added to the hot oil just before it reaches its smoke point. Fats and oils with low molecular weight fatty acids (those with a short chain length) have low smoke point.

Physical properties of fats and oils

Normally, oils that are selected for deep fat frying are those, which have a high smoke point. If oils with low smoke points are used for deep fat frying, then the foodstuff is fried at a lower temperature. Repeated use of the same sample of oil for frying results in a decrease in its smoke point and ultimately in its decomposition.

Food Biotechnology

Module #21



Refining of Crude Oils

Refining of Crude Oils

Most of oil used now a days is refined.

In the refining process the impurities in oils are removed.

The impurities in the oils include moisture, free fatty acids, colouring pigments, resins, gums and sometimes vitamins. The colour, flavour, odour and clarity as also the shelf-life of the oil are affected by impurities.

Refining of Crude Oils

Refining is a 5 step process

1. **Degumming :**

Some impurities in the crude oil form gums in the presence of water. Such impurities are removed by adding water to the warm oil and centrifuging it to remove the denser gum particles. Thus, a clarified oil is obtained as an upper layer, which is siphoned off.

Refining of Crude Oils

2. Neutralising

All crude oils contain some free fatty acids formed due to spoilage. The oil is

neutralized with caustic soda solution and the insoluble soaps formed from the free fatty

acids, which settle at the bottom are removed..

Refining of Crude Oils

3. Washing and Drying:

The last traces of soap from the oil are removed by washing. The lower aqueous soap layer is run off and the oil layer is dried under vacuum. The oil thus obtained has a yellowish colour and a distinct odour. To remove these, the oil is bleached and deodourised..

Refining of Crude Oils

4. Bleaching

To absorb the colouring matter from the oil, adsorbing substances such as activated carbon and fuller's earth are added to the warmed oil. The mixture is stirred with maintenance of partial vacuum. After, all the coloured matter is adsorbed, the mixture is put through filter presses to get a clear colourless oil.

Refining of Crude Oils

5. Deodourising

Finally the oil is deodourised by injecting steam with agitation to remove all odourous material as vapour. The deodourised oil is packed as such or blended with other oils and packed. The refined oil has to be stored under an inert gas such as nitrogen or vacuum packed to prevent oxidation.

Food Biotechnology

Module #22



Food Biotechnology

Different Uses/Products of Fats and Oils

Different Uses/Products of Fats and Oils

Role in Cookery:

The role of different types of fats and oils in cookery is largely based on their composition and properties. Thus, liquid fats or oils with a high smoke point are used for deep-fat frying purposes and likewise, solid fats like butter and margarine are used as shortening and tenderizing agents in foods.

Different Uses/Products of Fats and Oils

Fats are used in food preparation

(i) as a medium of cooking.

(ii) as a shortening as in biscuits, pastry and cakes.

(iii) to add richness and flavour

(iv)seasoning of vegetables and salads.

Different Uses/Products of Fats and Oils

As a Medium of Cooking:

Foods get cooked in fat in shorter time than when cooked in water. Fried foods, *have a crisp* texture and a delectable flavour. The high temperature used in frying destroys harmful bacteria, thus making the food safe for consumption.

Different Uses/Products of Fats and Oils

As a Medium of Cooking:

Some fat is absorbed by the food and the calorific value of the food is increased when it is cooked in fat or oil

Different Uses/Products of Fats and Oils

As a Shortening:

In many preparations, such as cakes, biscuits *fats or oils are added to improve* the texture. The fat covers the surface of the flour particles and prevents the sticking of particles together.

Different Uses/Products of Fats and Oils

As a Shortening:

Many factors such as the nature of the fat or oil, the amount added, the temperature, presence of other ingredients, manipulation and the extent of mixing, affect the shortening power.

Different Uses/Products of Fats and Oils

As a Seasoning:

Fats and oils are used to season most food preparations. In sweet preparations, fats, such as butter, ghee, vanaspati are used, as they have mild flavour, which blends with the sweet preparation.

Food Biotechnology

Module #23



Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- Fats and oils undergo certain undesirable changes during storage, which result in spoilage.
- Flavour Reversion
- Rancidity Spoilage

Storage and Shelf life of Fats and Oils

- **Flavour Reversion**

The fats undergo a peculiar change before the onset of rancidity. The characteristic flavour is lost and the fat or oil has a flat taste and a greasy feel on the tongue. This is known as flavour reversion and precedes rancidity changes.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- *Rancidity*

Spoilage of fats results in off flavour and renders the fat inedible. These changes are known as rancidity of fats. Fats and oils can get rancid by the action of

(a) air (oxidation),

(b) water (hydrolysis) and

(c) Enzymes (enzymatic breakdown).

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

•**Hydrolysis** is the decomposition of fats (triglyceride molecules) to glycerol and free fatty acids. Presence of moisture, microorganisms and the enzyme lipase hastens the hydrolytic breakdown. This kind of spoilage is known as hydrolytic rancidity.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- **Oxidation** of unsaturated fats leads to oxidative rancidity. Thus, oils or fats containing more double bonds (unsaturation) are more likely to develop oxidative rancidity than those with few double bonds.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- **Oxidation**

This oxidation is a chain reaction initiated by the production of free radicals.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

Enzymatic breakdown normally accompanies hydrolysis

Storage and Shelf life of Fats and Oils

Prevention of Fat Spoilage

- Keep fats and oils in dry, tightly covered containers to ensure exclusion of air and moisture.

Storage and Shelf life of Fats and Oils

- Keep the container sealed until needed.
- Keep fat in a container having a narrow opening to prevent undue exposure.
- Store in a cool, dry place away from cooking area, where the temperature and humidity fluctuations are not great.

Storage and Shelf life of Fats and Oils

Prevention of Fat Spoilage

Addition of antioxidants, such as tocopherols, and other phenolic compounds such as BHA1, BHT2, or propyl gallate, are used to retard rancidity in commercial fatty products

Food Biotechnology

Module #24



Introduction To Dairy Products

TYPES OF DAIRY PRODUCTS

- Milk
- Cream
- Cultured Dairy Products
- Frozen Dairy Products
- Concentrated Dairy Products
- Non-Dairy Products
- Butter
- Cheese

MAIN DAIRY FOODS



CHEESE







YOGHURT



FROMAGE FRAIS



MILK



CREAM



ICE CREAM

Introduction to Dairy Products

Milk

Can be plain or flavored (chocolate, strawberry, etc.)

Usually fortified with VITAMIN D

It is generally then processed in the following ways before it is sent to stores:

Pasteurized: Process of heating to destroy harmful bacteria

Introduction to Dairy Products

Milk

Homogenized: Process of agitating milk to help distribute the fat throughout so it's uniform in texture (not clumpy

Introduction to Dairy Products

Milk

Other types of milk

Flavoured Milk – usually UHT MILK, either semi-skimmed or skimmed.

Dried Skimmed Milk, Evaporated Milk, Condensed Milk full cream or skimmed

Introduction to Dairy Products

Milk

UHT Milk – this milk owes its long life to ultra high temperature

132*c for one second followed by packaging

Introduction to Dairy Products

Cream

Cream is a more concentrated form of milk

Once a cow is milked, the solids float to top (milkfat) and they are skimmed off and this becomes cream!

It comes in the following varieties:

Heavy (whipping) Cream

Higher percentage of fat (85% cream, 15% milk)

Introduction to Dairy Products

Cream

Light (whipping) Cream

Lower percentage of fat (70% cream, 30% milk)

Half & Half

Even less fat (half 50% cream, half 50% milk)

To Make Whipped Cream:

Use cold bowl and whip cream until frothy… to sweeten, gradually add sugar little by little

Introduction to Dairy Products

Cream

DO NOT OVERBEAT, or it will deflate and turn into butter

Introduction to Dairy Products

Cultured Products

Made from cultured, or specially grown bacteria

Usually thick in texture & tangy in flavor

Examples of cultured dairy products include:

Yogurt

Introduction to Dairy Products

Cultured Products

Yogurt

This is the dairy product with the lowest amt of fat

Can be substituted for sour cream to reduce the fat in a recipe

Sour Cream

Buttermilk

Introduction to Dairy Products

Frozan Products

Dairy products that have been prepared and stored at very low temperatures

ICE CREAM

Made from milk, cream, sugar and flavoring ,Generally has about 6-8 grams of fat

REDUCED FAT has about 4-5 grams of fat, LOWFAT has less than 3 grams of fat

NONFAT has less than 0.5 grams of fat

Introduction to Dairy Products

Frozan Products

SHERBERT

Made from milk, sugar and fruit juice

FROZEN YOGURT

Made from cultured dairy product, sugar & flavoring

Introduction to Dairy Products

Concentrated Products

Dairy products that have had the water or liquids removed to increase the density

Examples include:

EVAPORATED MILK

Has some water removed, Can be reconstituted and used as fresh milk.

Available in cans

SWEETENED CONDENSED MILK

Has water removed and sweetener added

Used most commonly in baking

Cannot replace fresh milk or evaporated milk

Available in cans

Introduction to Dairy Products

Concentrated Products

NONFAT DRY MILK POWER

Used by chefs because it does not spoil and it costs less than fresh milk

Can be reconstituted and used as fresh milk

Comes powdered in packets (boxes)… think hot cocoa

Introduction to Dairy Products

Butter & Products

Made by churning cream (either sweetened cream or sour cream) into butter

It is usually then colored artificially and either salted or left unsalted and packaged then sold

Whipped butter just incorporates more air into the churning process resulting in a less dense end product

Introduction to Dairy Products

Butter & Products

It can be frozen for a longer shelf-life but should ideally be refrigerated

Can spoil if left out (resulting in BITTER BUTTER… remember Betty?!)

Offers saturated fat (animal product)

Introduction to Dairy Products

Cheese

TYPES:

UNRIPENED sold immediately, not allowed to age

Ex. Cottage cheese, cream cheese, ricotta cheese

Better for cooking because they're more blendable

RIPENED curds are packaged and aged (sometimes for years)

Ex. Cheddar, Muenster, Provolone, Swiss…

The softer the cheese, the better it is for you… while all cheeses have saturated fats, harder cheeses have higher levels

Introduction to Dairy Products

Cheese

PROCESSED chemically made or altered

Ex. Velveeta, cheese sauces, imitation cheese

These tend to create really smooth, creamy cheeses & cheese sauces

Food Biotechnology

Module #25



Microbiology of Milk

- Milk is a highly nutritious media for bacteria
- Can spoil easily
- bacteria can enter milk during milking process and handling
- Refrigeration improved milk industry, but still other improvements needed to be done to reduce No's of non pathogenic bacteria

Microbiology of Milk

- **Non Pathogens in milk**
- Air borne contaminants : Pseudomonas, Flavobacterium, Alcaligenes, coliforms, micrococcus, fungi
- Psychrotrophs :grow at 5°c and less and make up a large number of the microbes in milk, they are natural soil inhabitants and wide spread.

Microbiology of Milk

- **Non Pathogens in milk**

Spoilage of refrigerated milk consists usually of bitter, rancid, fruity flavours

Microbiology of Milk

- **Sources of Pathogens in milk**
- Animal itself : Mycobacterium bovis, Brucella species, Coxiella burneti ,Campylobacter sp.
- Humans : Salmonella, Shigella
- Environment : Bacillus anthracis, Clostridium perfringens, Clostridium botulinum

Microbiology of Milk

Pasteurization of Milk

Pasteurization:

Kills and removes all pathogens(100%)

Kills most non pathogens (99%) Surviving organisms are spore formers, thermoduric and thermophilic organisms

Microbiology of Milk

- **Pasteurization of Milk**

Common thermoduric organisms:

Bacillus, Micrococci, Lactococci, Arthrobacter Microbacteria, Corynebacteria

Pasteurization provide a product with average shelf life 7-10 days(LTLT-HTST)

Microbiology of Milk

- **Pasteurization of Milk**

Types and numbers of bacteria in milk depend on the microbial load before pasteurization

Microbiology of Milk

- **Pasteurization of Milk**

No of non pathogens in raw milk is 500,000 to 3,000,000 organisms/ml of milk

While No of non pathogens in pasteurized milk is 2,000 to 20,000 organisms /ml

Food Biotechnology

Module #26



Microbiology of Milk

- Milk is a highly nutritious media for bacteria
- Can spoil easily
- Initially milk may have few organisms , but bacteria can enter milk during milking process and handling
- Refrigeration improved milk industry, but still other improvements needed to be done to reduce No's of non pathogenic bacteria

Microbiology of Milk

- **Non Pathogens in milk**
- Air borne contaminants : Pseudomonas, Flavobacterium, Alcaligenes, coliforms, micrococcus, fungi
- Psychrotrophs :grow at 5°c and less and make up a large number of the microbes in milk, they are natural soil inhabitants and wide spread.

Microbiology of Milk

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Microbiology of Milk

- **Sources of Pathogens in milk**
- Animal itself : Mycobacterium bovis, Brucella species, Coxiella burneti ,Campylobacter sp.
- Humans : Salmonella, Shigella
- Environment : Bacillus anthracis, Clostridium perfringens, Clostridium botulinum

Microbiology of Milk

Pasteurization of Milk

LTLT : Low temp. Long time 63°C 30 minutes

HTST: High temp. Short time72°C 15 seconds

UHT : Ultra high temp.,130-135 °C 1 second

Microbiology of Milk

Pasteurization of Milk

Pasteurization:

Kills and removes all pathogens(100%)

Kills most non pathogens (99%) Surviving organisms are spore formers, thermoduric and thermophilic organisms

Microbiology of Milk

- **Pasteurization of Milk**

Common thermoduric organisms:

Bacillus, Micrococci, Lactococci, Arthrobacter Microbacteria, Corynebacteria

Pasteurization provide a product with average shelf life 7-10 days(LTLT-HTST)

Microbiology of Milk

- **Pasteurization of Milk**

Types and numbers of bacteria in milk depend on the microbial load before pasteurization

Microbiology of Milk

- **Pasteurization of Milk**

No of non pathogens in raw milk is 500,000 to 3,000,000 organisms/ml of milk

While No of non pathogens in pasteurized milk is 2,000 to 20,000 organisms /ml

Microbiology of Milk

- **Pasteurization of Milk**

Different methods to count non pathogens in milk such as:

Viable count(pour plate method).

Dye reduction test Direct (clump) microscopic count

Microbiology of Milk

Viable count method

Pour plate method

- Deliver 1ml of milk to 9ml of sterile saline tube(10-1), mix and transfer 1ml diluted milk from tube 1 to tube 2 (10-2).
- Deliver 1ml of milk to an empty Petri dish ,1ml of diluted milk 10-1,10-2 to empty Petri dishes .
- Pour 9ml of molten nutrient agar to each Petri dish, mix well

Microbiology of Milk

- Leave to solidify, then incubate at 37 c 48 hrs.
- Count No of colonies and multiply by dilution factor to determine No of viable non pathogens on milk
- A plate containing more than 300 colonies should not counted, plate with more dilution is counted instead.

Microbiology of Milk

Dye reduction test

A quick test for determination of microbial load of milk

Use Methylene blue or resazurin dyes

Add 1ml of dye to 10ml of milk, incubate 1 hr at 36c

If there is change in color , it means high bact. Load .It is not good quality milk, shorten expiration date.

Microbiology of Milk

- If no change in color (still blue), low bact. Load , good quality milk , expand expiration period
- A long history of use in the dairy industry, especially to measure microbial quality from raw milk
  - Outbreaks have been associated with cheeses

Microbiology of Milk

- Salmonella serovar Zanzibar in goat milk cheese
- L. monocytogenes in Mexican-style cheese

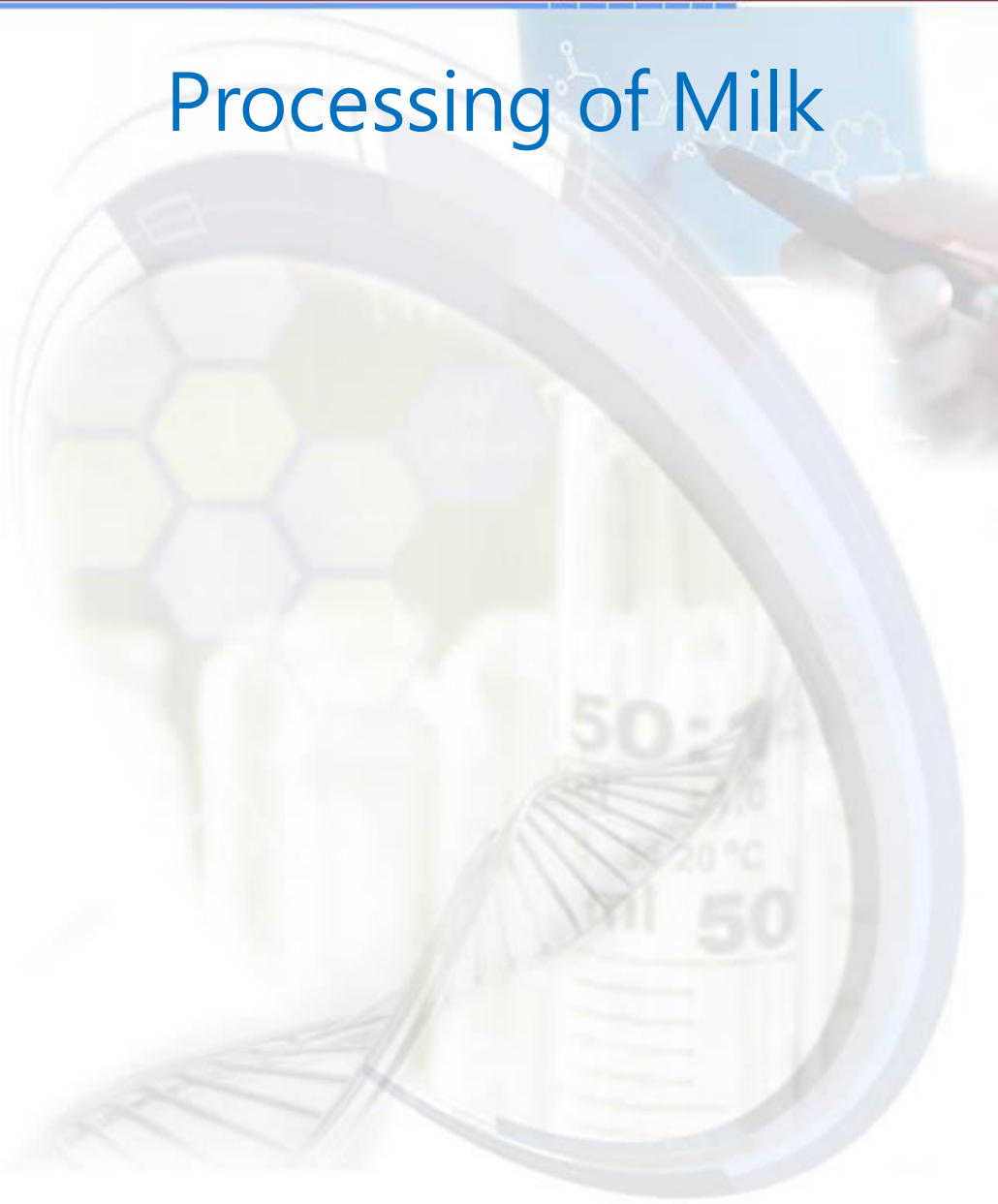
Food Biotechnology

Module #27



Food Biotechnology

Processing of Milk



Processing of Milk

Introduction:

Raw milk is associated with pathogenic diseases such as tuberculosis, brucellosis or typhoid fever, among others.

Hygienic milk production, proper handling and storage of milk, and appropriate heat treatment can reduce or eliminate pathogens in milk.

In many pasteurization of milk is mandatory.

•

Processing of Milk

- Milk may be given various treatments like heating, concentrating, drying or altering its pH to obtain a number of different products to preserve milk and add variety to our meals.

This is done by:

1. exposing milk to high temperatures at which microorganisms are killed,

Processing of Milk

2. binding or reducing the water present in milk and thus making moisture unavailable to micro-organisms.

3. by increasing the acidity of milk to a level that does not permit growth of spoilage microorganisms, and yet imparts a desirable sour flavour to the product.

Processing of Milk

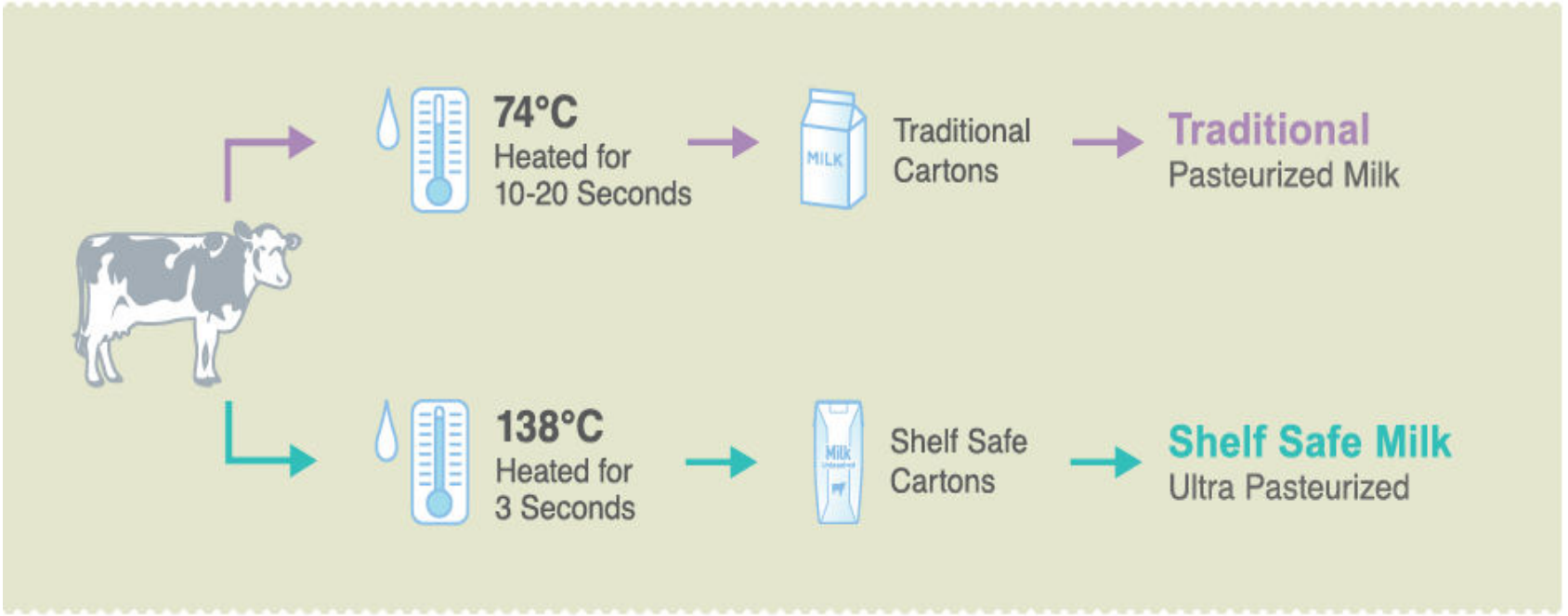
Types of Processing Methods:

Pasteurization

- ***Holding Method*** - 62.7°C for 30 minutes.
- ***Higher Temperature Short Time Method*** **(HTST)**- 71.6°C at least for 15 seconds.
- ***Ultra High Temperature Process***- Milk is heated to a temperature of 89–90°C or more for 1 second or less.

All followed by cooling at 10°C

Processing of Milk



Processing of Milk

Concentrated Milks:

- Such as evaporated milk, sweetened condensed milk and dried milk made from whole or skimmed milk.
- Varying amounts of water are removed to make the concentrated milk products.
- These products have a longer shelf-life and some can be reconstituted to their original form.

Processing of Milk

Evaporated Milk:

- more than half the water from milk under vacuum is removed.
- It is necessary to fore-warm milk 10 minutes to prevent coagulation of casein during sterilization after it is canned.

Processing of Milk

Condensed Milk:

- Is concentrated to about 1/3 of its original volume and has about 15 percent sugar added to the milk.
- Since the finished product contains at least 40 per cent of sugar, which acts as a preservative, it is not necessary to sterilize it before canning.

Processing of Milk

Dehydrated Milk Products:

- Includes whole milk powder, skimmed milk powder, infant milk foods and malted milk.

Processing of Milk

Skimmed Milk Products:

- In production of skimmed milk powder, the milk fat is removed or skimmed from the milk, before the milk is dehydrated to a moisture content less than 4 per cent.

Food Biotechnology

Module #28



Other Dairy Products



Other Dairy Products

Curd (Yogurt):

- Curds is a very popular milk product in India.
- It ranks second to milk in the extent of consumption.
- Milk is boiled and cooled to about 50°C and a teaspoon of curd from an earlier batch of curd is added and mixed thoroughly.
- The lactic acid bacteria present in this sample of curd curdle the milk.

Other Dairy Products

- They utilize the lactose in milk and break it down to lactic acid.
- The formation of lactic acid increases the acidity of milk. When the pH reaches 4.6, the milk protein, casein, coagulates as curd.
- Since the process is gradual, the milk serum is bound in the mass of coagulated proteins.

Other Dairy Products

- The optimum temperature for the formation of curd is 35-40°C.
- The time needed for curd formation varies from 8-12 hours depending on atmospheric temperature.

Other Dairy Products

- Once made, curd keeps well at refrigerator temperature of 2–3 days. It is used as dressing on salads made from fresh vegetables.
- It combines well with plain cooked rice.
- Preparation of curd is one method of prolonging the shelf-life of milk by several days.

Other Dairy Products

- Soured milk does not readily undergo proteolysis and other undesirable changes.
- Curds can be held at refrigerator temperature for several days without loss of acceptability.
- It is reported that the riboflavin and thiamin content increases during curd formation.

Other Dairy Products

- It is also reported that fermented milk inhibits the growth of *Bacillus typhosus, Bacillus dysenteriae*, and *Vibrio cholerae* to a certain extent.

Other Dairy Products

Paneer:

- Is prepared by addition of lemon juice or citric acid to hot milk and precipitating the casein.
- The liquid released in this process is known as *whey*, which contains most of the soluble nutrients from milk.
- Paneer contains about 18 per cent protein and is a good source of it.

Other Dairy Products

Cheese:

- The milk is subjected to several process steps to produce cheese.
- The milk it held at about 27°C in vats and a lactic acid culture is added.
- When the milk gets acidic, rennet is added to it and the milk is allowed to coagulate.

Other Dairy Products

- The curd formed is cut into strips and heated to about 37°C with stirring to separate the whey.
- The whey is drained.
- Salt is mixed with the curd and it is pressed to remove further amounts of whey.

Other Dairy Products

- The cheese formed is coated with paraffin to prevent loss of moisture and allowed to ripen at low temperature.
- Cheese contain about 24 per cent protein and is thus a concentrated source of protein.

Other Dairy Products

Buttermilk

- is made by adding water to curd and churning to remove fat in the form of butter.
- The energy value is thus reduced due to fat removal.
- The concentration of the other nutrients is reduced in proportion to the dilution.
- It is known to re-establish intestinal flora after an attack of diarrhoea.

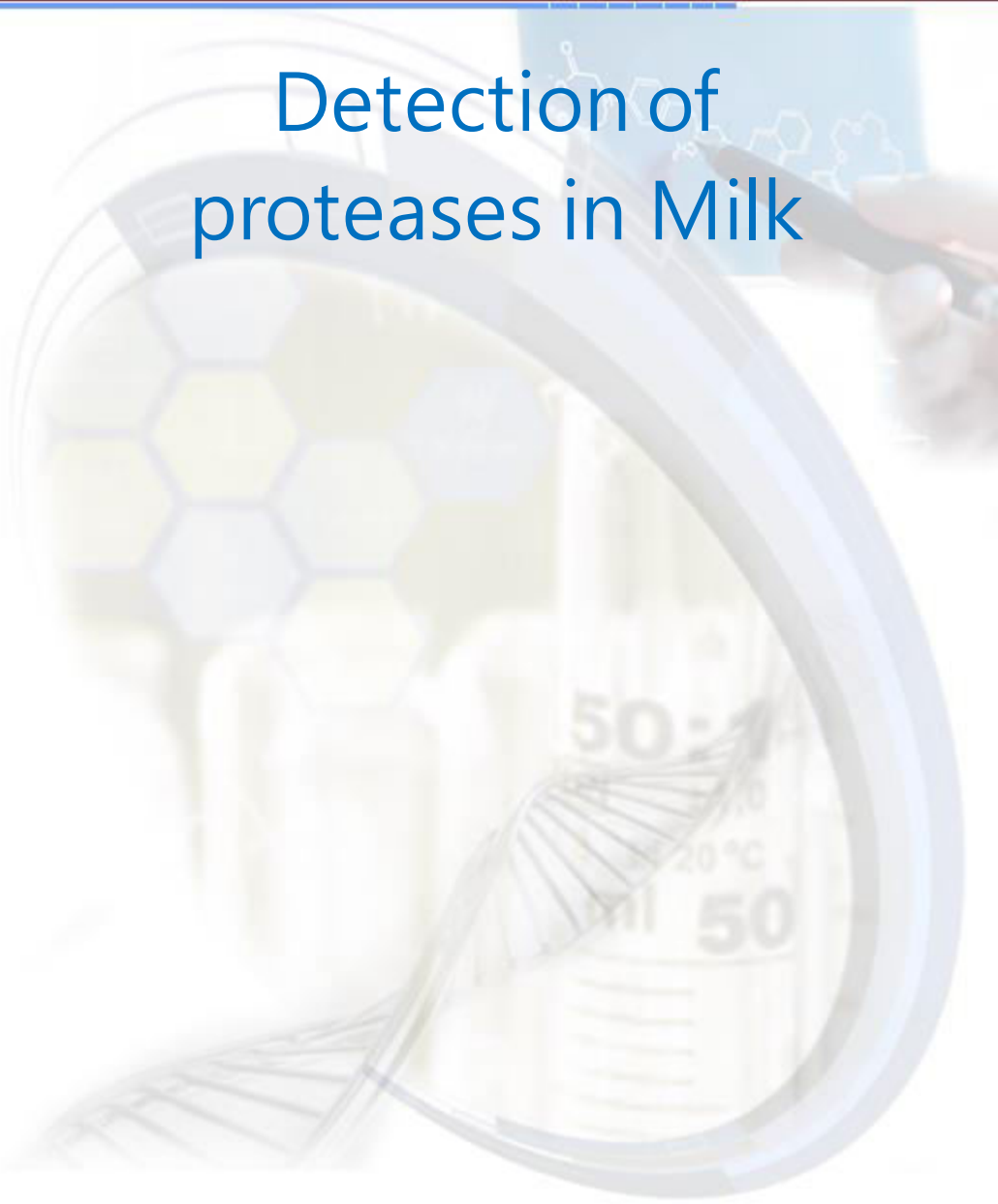
Food Biotechnology

Module #29



Food Biotechnology

Detection of proteases in Milk



Detection of proteases in Milk

Introduction:

Proteases are enzymes that degrade proteins. Raw milk may have proteolytic activity from endogenous or indigenous origin. Wherever they come from, proteases can cause unpleasant flavours and odours in milk.

Detection of proteases in Milk

- Several bacteria present in raw milk can produce proteases. Among them, those from psychrotrophic microorganisms (*Bacillus subtilis*), which produce enzymes under refrigeration, cause the most serious problems.
- It secretes a thermo-resistant protease which may cause proteolysis in pasteurized or sterilized milk.

Detection of proteases in Milk

Sample preparation

Milk sample need to be chosen that is required for analysis.

The standard protease from Bacillus subtilis is needed and required concentrations are made for calibration curve.

Detection of proteases in Milk

Enzyme determination

Reagents

- Disodium hydrogenphosphate,
- Sodium dihydrogenphosphate,
- Trichloracetic acid.

•Solutions

•Prepare a pH 7.2 buffer solution by mixing 36 ml of sodium hydrogenphosphate (5 mM) and 14 ml of sodium dihydrogenphosphate (5 mM) and diluting with distilled water to 1000 ml.

•This buffer is used to prepare a 1% azocasein solution.

•A 5% aqueous solution of trichloracetic acid was also required.

Detection of proteases in Milk

- **Determination of protease activity**
- The assay for proteolytic activity involves combining 1 ml of the azocasein solution with 100 ml of the sample which contains the enzyme.
- The contents of the tubes are mixed and incubated at 35.5 degree C for 15 min. The reaction is stopped by the addition of 2 ml of 5% TCA.

Detection of proteases in Milk

The absorbance of the supernatant is recorded at 345 nm in a UV/Visible spectrophotometer.

Activity calculated against standard curve.

END

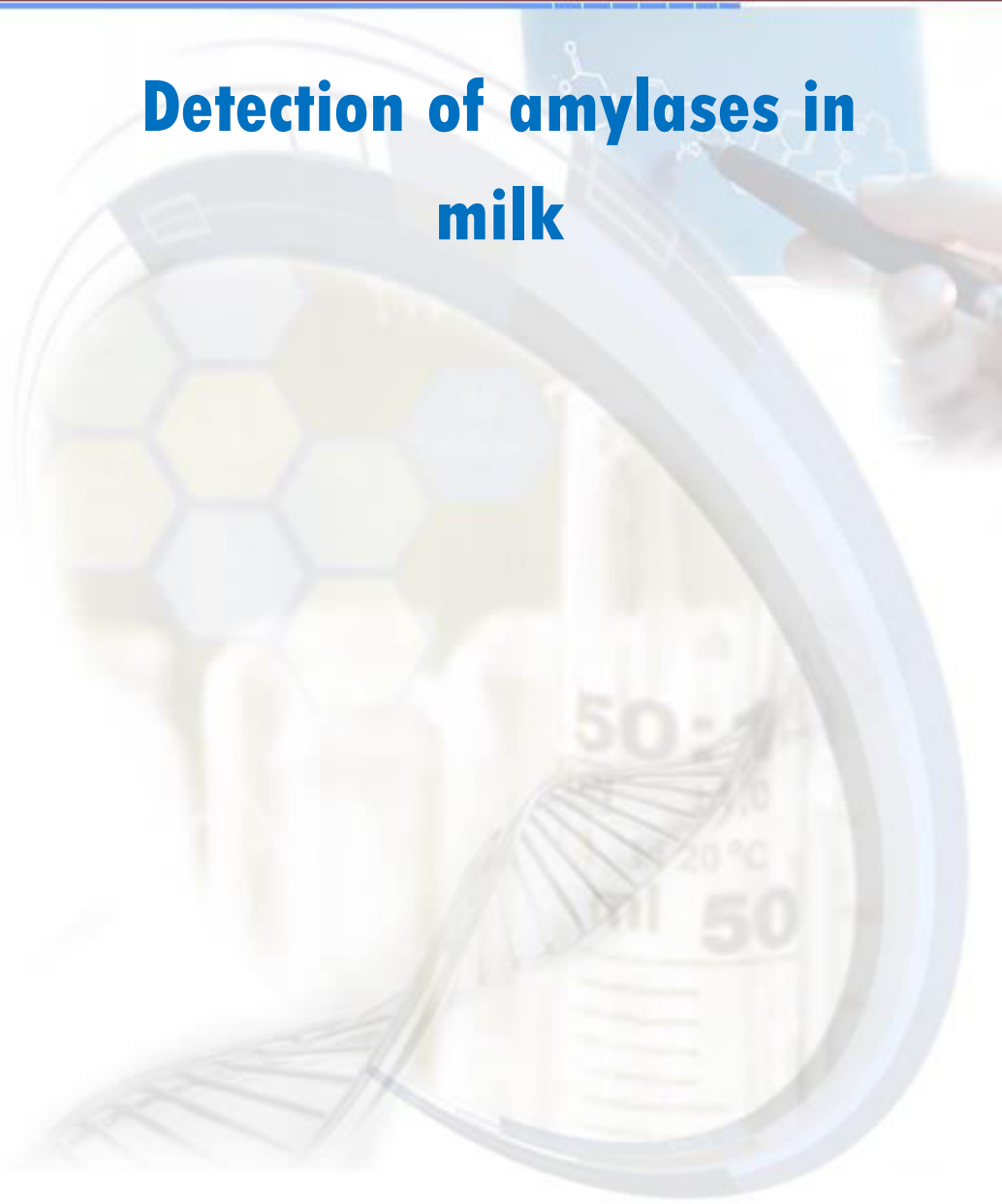
Food Biotechnology

Module #30



Food Biotechnology

Detection of amylases in milk



Detection of amylases in milk

Procedure:

- Amylase activity in milk is a critical quality parameter.
- The amylase activity in milk is often referred as diastatic power and refers to the production of reducing substances by the action of alpha & beta amylases.

Detection of amylases in milk

- The measurement involves digestion of the soluble matter with a malt infusion (extract) and following increase in reducing substances by measuring reduction of Fehling's Solution.
- Specifically measuring alpha amylase activity (dextrinizing activity) is more complicated and is based on using a limit dextrin as substrate.

Detection of amylases in milk

- Limit dextrin is prepared by action of beta amylase (free of alpha amylase activity).
- The beta amylase clips maltose units off the nonreducing end of the starch molecule until an alpha-1,6-brach point is encountered.
- The resulting product is a beta limit dextrin that serves as the substrate for the endo cleaving alpha amylase.

Detection of amylases in milk

- A malt infusion is added to the previously prepared limit dextrin substrate and aliquots removed periodically to a solution of dilute iodine.
- The alpha amylase activity is measured by changed color of the starch iodine complex in the presence of excess beta amylase used to prepare the limit dextrin.
- The color is compared to a colored disc on a comparator.

Detection of amylases in milk

- This is continued until the color is matched to a color on a comparator.
- The time to reach that color is dextrinizing time and is a measure of alpha amylase activity, a shorter time representing a more active preparation.

END

Food Biotechnology

Module #31



Food Biotechnology

Determination of iodine number of fats

Determination of iodine number of fats

Principle:

- The iodine number is a measure of degree of unsaturation, that is, number of carbon double bonds in relation to the amount of fat or oil.
- Iodine value is defined as the grams of iodine absorbed per 100 gram sample.

Determination of iodine number of fats

- The higher the amount of unsaturation the more iodine is absorbed.
- A quantity of fat or oil dissolved in solvent is reacted with a measured amount of iodine or some other halogen. Halogen addition to double bonds takes place.

Determination of iodine number of fats

Procedure:

- A solution of potassium iodide is added to reduce excess ICl to free iodine.
- The liberated iodine is then titrated with a standardized solution of sodium thiosulfate using a starch indicator, and the iodine value is calculated.

Determination of iodine number of fats

$$\underset{\text{(excess)}}{ICl} + R{-}CH = CH{-}R \rightarrow R{-}CHI{-}CHCl{-}R + \underset{\text{(remaining)}}{ICl}$$

$$ICl + 2KI \rightarrow KCl + KI + I_2$$

$$I_2 + \underset{\text{(blue)}}{\text{starch}} + 2Na_2S_2O_3 \longrightarrow 2NaI + \underset{\text{(colorless)}}{\text{starch}} + Na_2S_4O_6$$

$$\text{Iodine value} = \frac{(B - S) \times N \times 126.9}{W} \times 100$$

where:

Iodine value = g iodine absorbed per 100 g of sample
B = volume of titrant (ml) for blank
S = volume of titrant (ml) for sample
N = normality of $Na_2S_2O_3$ (mol/1000 ml)
126.9 = MW of iodine (g/mol)
W = sample mass (g)

Determination of iodine number of fats

- Calculated iodine value is obtained from fatty acid composition using below equation for triacylglycerol.
- A similar equation allows calculation of iodine value of FFAs.

Iodine value (triglycerides)

= (% hexadecenoic acid × 0.950)

+ (% octadecenoic acid × 0.860)

+ (% octadecadienoic acid × 1.732)

+ (% octadecatrienoic acid × 2.616)

+ (% eicosenoic acid × 0.785)

+ (% docosenoic acid × 0.723) [6]

Determination of iodine number of fats

Applications:

Is used to characterize oils for hydrogenation process in refining, and as an indication of lipid oxidation, since there is a deadline in unsaturation during oxidation.

The calculated value tends to be low for material with a low iodine value and for oils with greater than 0.5% unspecifiable material (e.g, fish oil).

Food Biotechnology

Module #32



Food Biotechnology

Introduction to meat

Meat: Introduction

Meat is defined as the flesh of animals used as food.

meat is a very good source of essential amino acids, and, to a lesser extent, of certain minerals. Although vitamins and essential fatty acids are also present, meat is not usually relied upon for these components in a wellbalanced diet

Meat: Introduction

Meat is one of the animal protein foods used in diet

Meat protein has high biological value.

The proportion of nutrients in meat depend upon the kind of animal, the species and the type of cut.

Meat: Introduction

The average protein content of meat ranges from 16 to 23 per cent and the average fat content ranges from 10 to 40 per cent. Unless the cooking water is discarded, minerals and water soluble vitamins are not lost to a great extent.

Meat provides us vitamin B-complex and some vitamin A, depending on the cut.

Meat: Introduction

Meat is a good source of phosphorus, iron and some trace elements.

The meat of sheep, under 12 months of age, is sold as *lamb*.

After 12 months, it is called *mutton*.

Meat: Introduction

Veal is the meat from cattle 3 to 14.

Between 14 and 52 weeks, the meat is called *calf*.

Meat after one year after birth is called *beef*.

Meat: Introduction

Variety of animal organs are served in food preparations like;

Liver

Kidney

Tongue

Brain etc.

These are less costly than muscle meat.

Meat: Introduction

Meat is also turned into sausages;

Ground beef

Chicken and turkey meat

Theses sausages are of two types;

Ready to eat

Ones which require cooking

Food Biotechnology

Module #33



Structure of Muscles

- ***Muscle Tissue;***
  - Smallest unit of a muscle is a muscle fiber.
  - Fibers join into bundles called *Fasciculi* by connective tissues.
  - The *Fasciculi* then combine with fat deposits are covered with membrane and attached to the bone.

Structure of Muscles

Connective Tissues;

- Bind the meat fibers together.
- Join meat muscle to the bone.
- Are of two types; 1)Collagen 2) Elastin
- Collagen is converted into gelatin during cooking.

Structure of Muscles

Connective Tissues;

- Amount of connective tissues depend on the part of the animal e.g in leg it is in greater amount.

Structure of Muscles

Fatty Tissues;

- These are made up of connective tissues with embedded fat cells.
- Fat is deposited under the skin, around the glands, organs and between and within the muscle fibers.

Structure of Muscles

Fatty Tissues;

- Fat distribution in lean part of meat is called ***marbling*** and well marbled is desired by consumers.

Structure of Muscles

Bones;

- Bone is an indicator of the part of the animal from which the cut of meat is taken.
- Bones are either long or short.
- Long bones are hollow and contain yellow marrow.
- Other bones, which contain red marrow, are spongy inside.

Structure of Muscles

Pigments;

- *Myoglobin* is the pigment in meat that gives the red colour to meat.
- The greater the amount of myoglobin, the darker the colour of meat.
- As the animal ages the amount of myoglobin increases.

Structure of Muscles

Pigments;

- Raw meat changes its colour from red to brownish red upon conditions.

Structure of Muscles

Pigments;

Myoglobin (red pigment)

- Upon refrigeration changes to brown red
- Upon heating changes grayish brown

END

Food Biotechnology

Module #34



Postmortem changes in Meat

When the animal is slaughtered certain type of changes start to take place:

- Stiffening of meat called *Rigor mortis*
- Formation of lactic acid lowering pH
- Swelling of muscle fibers with fluid causing the meat to become hard.

Postmortem changes in Meat

When the animal is slaughtered certain type of changes start to take place:

- The circulatory system stops functioning but the enzymes continue to remain active.
- Oxidation of glycogen brings about production of lactic acid.

Postmortem changes in Meat

- The amount of glycogen stored in a particular animal decides the amount of lactic acid produced after slaughter.
- A lower glycogen storage produces sticky and gummy meat which is undesirable.

Postmortem changes in Meat

- Stiff muscles of the animal start softening after holding for some time.
- This process is called *aging*.
- Tenderness in the meat increases with holding time. Holding conditions should be carefully controlled.

Postmortem changes in Meat

Uncontrolled conditions may allow putrefactive bacteria to grow.
Controlled conditions of holding are:

I. Temperature of holding should have a range of (34–36°F)
II. Carbon dioxide in the atmosphere
III. 70 per cent humidity

Postmortem changes in Meat

This process keeps the meat:

- In a good condition for 3 to 6 weeks.
- After this *aging process*, the meat is ready for sale.
- This process of aging is also called *ripening of meat.*

end

Food Biotechnology

Module #35



Preservation of Meats

Meat is a highly perishable products and soon becomes unfit to eat and possibly dangerous to health through microbial growth, chemical change and breakdown by endogenous enzymes.

Preservation of Meats

Chilling/refrigeration slows or limit the spoilage rate at temperature below the optimal range can inhibit the microbial growth, enzymatic as well as chemical reactions. Storage of fresh meat is done at a refrigeration temperature of 2 to 5°C. Chilling is critical for hygiene, safety, shelf life, appearance and nutritional quality

Preservation of Meats

Curing

- Meat is treated with salt, sugar, sodium nitrate and spices to give different flavors.
- Smoke is used some times to give a desirable flavor and to improve the keeping quality of meat.

Preservation of Meats

Freezing:

- This is a common method of preserving meat.
- Nutrient loss is least during freezing.
- The one undesirable effect, freezing has on meat is that, the meat when cooked, is less juicy.
- This method is commonly used for storing cooked meat.

Preservation of Meats

Canning

- Meat can also be preserved by this method.
- Higher temperatures which are used during canning bring about some undesirable changes in flavor and texture.

Preservation of Meats

Radiation and Sterilization

- This is a new method of preservation of meats.
- The meat is sterilized and bacteria present killed.
- According to some people, this meat is not palatable.

end

Food Biotechnology

Module #36



Food Biotechnology

Cooking and Effect of Heat on Meats

Cooking and Effect of Heat on Meats

- Cooking of meat or application of heat brings majorchanges into meat.
- These include colour and texture changes
- Food safety is achieved by reducing bacterial load.

Cooking and Effect of Heat on Meats

- The method of cooking depends on the type of meat.
- Dry heat methods like roasting, broiling, pan frying are used for tender cuts or young animals.

Cooking and Effect of Heat on Meats

- Moist heat methods like pressure cooking, closed pan cooking, steaming are used for less tender cuts.

Cooking and Effect of Heat on Meats

- Most common cooking methods appear to be roasting, broiling and pan-frying.

Cooking and Effect of Heat on Meats

Effect of Heat on Meat

- Thiamin in meat is partially lost during cooking.
- If the food is heated for a long time, the loss is extensive.
- If water or juices, in which the meat is cooked, are discarded, then the loss of the vitamin increases.

Cooking and Effect of Heat on Meats

- Low temperatures are preferred for meat cookery.

Food Biotechnology

Module # 37



Introduction to Cereal Foods

Introduction to Cereal Foods

- Seeds of the grass family
- Major staple food in many countries
- Approx. 65-75 % carbohydrates
- Rice contains about 78% carbohydrates

Introduction to Cereal Foods

- Easy to package and transport
- Used to produce a variety of highly desirable foods
- For humans and animals
- Eaten in many ways

Introduction to Cereal Foods

- Most common cereal foods include:

- Rice
- Wheat
- Maize (corn)
- Millets such as:
  - Jowar
  - Bajra
  - Ragi

Introduction to Cereal Foods

- Can be processed further to make:

- Flour
- Starch
- Oil
- Bran
- Syrup
- Sugar

Introduction to Cereal Foods

- Advantages:
- Inexpensive
- Rich source of carbohydrates
- Gives high yields per acre
- Once harvested; excellent storage capabilities

Introduction to Cereal Foods

- Advantages:
- Can be grown in a variety of areas
- Some even in adverse soil and climatic conditions

End

Food Biotechnology

Module # 38



Cereal Foods

Composition and Structure

Cereal Foods

- Cereal grains are composed of:

- Bran
- Germ
- Endosperm

Cereal Foods

- Bran is outer chaffy coat that covers the kernel
- Endosperm is the large central portion
- Germ is a small structure at the lower end of the endosperm

Cereal Foods

- Bran

- Eliminated when grains are harvested
- Constitutes 5% of the kernel
- Discarded during milling
- High fiber and mineral content
- Good source of thiamin and riboflavin

Cereal Foods

- Aleurone layer

- Located just underneath the bran
- Rich in protein, phosphorus and thiamin
- Some fat
- Makes up 8% of the whole kernel
- Lost in the milling process

Cereal Foods

- Endosperm

- Constitutes about 84-85% of the kernel
- Contains most of the starch and protein
- Little fiber and mineral matter
- Traces of fat
- Low vitamin content

Cereal Foods

- Germ

- Makes up 2-3% of whole kernel
- Rich in fat, protein, minerals and vitamins
- Serves as a store of nutrients for the seed
- Some of it is lost during milling

Cereal Foods

- Bran is indigestible to humans
- Bran goes mainly to animal feeds
- Germ is removed for its high oil content

- Dietary guidelines recommend more fiber in foods: use of bran

Cereal Foods

- Composite Proximate Analysis:

- Protein: 11%
- Fat: 3%
- Carbs: 68%
- Moisture: 12%
- Fiber: 6%

Cereal Foods

- Nutritional Value:
- Varies with each type of cereal grain
- Starch: supplies energy
- Lacks amino acid : Lysine
- Whole cereal vs. Refined cereal

Food Biotechnology

Module # 39



Nutritional Importance of Wheat Crop

Nutritional Importance of Wheat Crop

- Wheat Crop

- Widely cultivated

- Two crops in a year:
- Winter wheat
- Summer wheat

- Several thousand different varieties of wheat

Nutritional Importance of Wheat Crop

- Commonly milled into flour
- Can be grounded to obtain whole wheat flour
- Refined flour (*maida*); used in bread and other baked products
- Whole wheat flour (*atta*); used in preparation of *chapatis, puris, parathas*

Nutritional Importance of Wheat Crop

- Compared to refined wheat flour;
- Whole wheat flour contains more:

- Protein
- Iron
- Thiamin

Nutritional Importance of Wheat Crop

- There are two types of wheat:

- Hard wheat
- Soft wheat

- Hard wheat is higher in protein
- Produce more elastic doughs
- Better suited for making bread

Nutritional Importance of Wheat Crop

- Common used products of wheat:

- Vermicelli
- Broken wheat (*dalia*)
- Noodles
- *Suji*
- Cereals
- Canned soups
- Pasta

Nutritional Importance of Wheat Crop

- Wheat flour may be enriched with:

- Iron
- Calcium

- Can also be enriched by adding small amounts of vitamins

Nutritional Importance of Wheat Crop

- Flour may be used as thickener in canned stews, soups and gravies
- Breakfast cereal products are made from wheat

Food Biotechnology

Module # 40



Food Biotechnology

Nutritional importance of Maize

Nutritional importance of Maize

- Maize:
- Also called Corn
- Mostly used as:
- Animal feed
- Corn starch
- Corn syrup
- Bio-fuel

Nutritional importance of Maize

- Many types of corn are grown:

- Sweet corn
- Popcorn
- Dent corn
- Flour corn

Nutritional importance of Maize

- Number of varieties of corn are present
- Depending on the characteristics of starch present;
- Usually classified as starchy or waxy corn
- Field corn

Nutritional importance of Maize

- Field corn is:
- Lower in protein as compared to wheat
- Deficient in amino acids e.g Lysine
- Varieties have been developed to produce high lysine corn

Nutritional importance of Maize

- Most of corn crop is kept as animal feed

- Considerable amounts are used to produce cornstarch, corn syrup, high-fructose corn syrup an other varies sugar derivatives

Nutritional importance of Maize

- Popcorn
- Puffed cereal

- Specific variety of corn

- Internal moisture creates pressure owing to increased temperature

- Outer shell burst, expanding the grain

Food Biotechnology

Module # 41



Nutritional Importance of Rice

Nutritional Importance of Rice

- Rice:
- Seed of the grass species:
- *Oryza sativa* (Asian rice)
- *Oryza glaberrima* (African rice).

Nutritional Importance of Rice

- Rice cultivation

- Flooding of fields
- Requires ample water
- Labor-intensive

Nutritional Importance of Rice

- Rice is available as

- Milled
- Unpolished
- Polished
- Parboiled
- Flaked
- Puffed

Nutritional Importance of Rice

- Degree of milling and polishing affects the nutrient content
- Highly polished milled rice has very little thiamin
- Parboiled and hand pounded rice contains significant amounts of thiamin

Nutritional Importance of Rice

- Grain types
- This long varieties preferred for table use
- Medium and short varieties used for preparation of rice made after grinding

Nutritional Importance of Rice

- Milling removes the bran, aluerone layer and some germ
- Polishing further removes more than half the mineral matter and most of the vitamins
- Parboiling before milling

Nutritional Importance of Rice

- Rice kernels may be enriched
- Mixed with powder containing vitamins and mineral
- Coated with waterproof edible film to prevent washing off

Nutritional Importance of Rice

- Rice is lower in total protein as compared to wheat
- Does not contain sufficient amounts of certain amino acids
- Usually cooked with twice its own volume of water

Food Biotechnology

Module # 42



Nutritional Importance of Oats

Nutritional Importance of Oats

- Oats
- *Avena sativa*
- Used as livestock feed
- Oatmeal suitable for human consumption

Nutritional Importance of Oats

- Oats most commonly
- Rolled and crushed to make oatmeal
- Grounded into flour
- Used in variety of baked goods such as oatmeal cookies

Nutritional Importance of Oats

- Can grow in colder and wetter climates as compared to wheat
- Harvested in the same manner as wheat
- Moisture content at time of harvest should be no more than 13%

Nutritional Importance of Oats

- Oats can be milled
- May be produced for use as an ingredient of bread
- Or thickener of soups

Nutritional Importance of Oats

- Oat bran has water soluble fiber
- Linked with lower cholesterol levels

Food biotechnology

Module # 43



Food biotechnology

Nutritional Importance of Barley

Nutritional Importance of Barley

- Barley
- *Hordeum vulgare*
- Do not bake as well as wheat products
- Contains little or no gluten
- Not as popular as wheat

Nutritional Importance of Barley

- Hardy grain
- Grows in climate too cold for wheat
- Grows in soils too poor to grow wheat
- Requires a shorter growing time than wheat

Nutritional Importance of Barley

- Spring and winter varieties
- Used as feed for cattle and poultry
- Small amounts of barley flour are produced
- Used for production of malt for brewing

Nutritional Importance of Barley

- Malt
- Grain is soaked in water until moisture content reaches 50%
- Allowed to sprout
- Sprouted barley kiln dried for 24 hours

Nutritional Importance of Barley

- Enzymes that hydrolyze starch to maltose
- Maltose utilized by yeast to produce
- Ethyl alcohol
- Carbon dioxide

Nutritional Importance of Barley

- Malt
- used in the brewing industry
- convert the starches present in barley, rye, rice, corn, or other grains to maltose
- which can then be utilized by yeasts.

Nutritional Importance of Barley

- Malt may also be used in bread baking
- the yeast produce carbon dioxide for leavening
- alcohol produced is largely dissipated during the heating involved in baking.

Food Biotechnology

Module # 44



Other Cereal Grains

Other Cereal Grains

- Other cereal grains are:

- Sorghum
- Buckwheat
- Millet
- Triticale

Other Cereal Grains

- Sorghum

- Four general classes

- 1. sweet sorghum
- 2. broom corn
- 3. grass sorghum
- 4. grain sorghum

Other Cereal Grains

- Some varieties yield glutinous starch
- Used as an alternate to tapioca in WW II
- Deterrent to use sorghum grain for production of starch is its pigmentation of the grains pericarp

Other Cereal Grains

- Buckwheat
- Not a true cereal grain
- Buckwheat belongs to the botanical family *Polygonaceae*
- All cereal grains belong to the botanical family *Gramineae*

Other Cereal Grains

- Dried to about 12% moisture
- Cleaned
- Graded by size
- Milled similarly to wheat

Other Cereal Grains

- Only a few varieties used
- Silverhull used for producing flour because of its higher yield of endosperm
- Most of the flour used for making pancakes

Other Cereal Grains

- Millet
- Used for food in Asia and some of Europe
- Used as hay in Europe and USA
- Some varieties used as food seed for birds and poultry

Other Cereal Grains

- Triticale
- Hybrid of wheat and rye
- Combines high total content of protein in wheat
- With high lysine content of rye
- More adaptable to unfavorable conditions

end

Cereal Foods

Module # 45



Cereal Foods

Processed Cereal Products

Cereal Foods

- A number of processed cereal products are made from grains
- Include products like:

- Cracked wheat
- Semolina
- *Atta*
- *Maida*
- Rice flakes
- Puffed rice

Cereal Foods

- Processes involved:
  - Grinding
  - Roasting
  - Pounding
- Processes increase surface area of product exposed to atmosphere

Cereal Foods

- Processes:
- Decrease preparation time
- Reduce shelf life
- Whole grain shelf life is almost a year or more
- Processed cereal foods shelf life varies from two weeks to a few months

Cereal Foods

- Cracked Wheat

- Also called *dalia*
- Whole wheat coarsely ground
- Losses during milling are very little
- Very nutritious food

Cereal Foods

- Good quality indicated by sweet taste and absence of sour, moldy odor and flavor
- Spoils quickly
- Can be cooked to make and made into *upma, shira* or porridge

Cereal Foods

- Semolina
- *Suji* or *Rawa*
- Fine grain varieties used for preparation of *halwas*
- Large grain varieties used for preparation of *upma*

Cereal Foods

- Selected on basis of:

- Uniformity of size
- Freedom from insect infestation
- Freedom from oxidized or moldy odor
- Grit and bran

Cereal Foods

- *Maida*
- Finely extracted white wheat flour
- Free from bran
- Lower content of protein, iron and B-vitamins

Cereal Foods

- Lower shelf life than semolina
- Large surface area permits faster rate of food spoilage
- Good quality should be free from insect infestation, bad odors and lump formation

Cereal Foods

- Processed rice products
- Rice flakes
- Rice puffs
- Ready-to-eat rice products

Cereal Foods

- Rice flakes are free from bran, broken particles, fragments of the seed coat, insect, stones and bad odor
- Rice flakes retain large parts of iron and B-vitamins
- Rice Puffs are selected for crispiness, freedom from stone, seed coats, sand and dirt

Food Biotechnology

Module #46



Food Biotechnology

Spoilage is defined as ''undesirable changes in the properties of a product i.e. Food (Food Spoilage) , Wood, Marble etc.''

Food spoilage may be

→Physical

→Chemical

→Biological

EFFECT OF SPOILAGE ON FOOD

Food spoilage can effect food by changing its

→ Nutrition Value

→ Organoleptic properties

→ Color

but main drawback or hazard of food spoilage is that safety is compromised

FOOD SPOILAGE AND BIODEGRADATION

Main difference between Food Spoilage and Biodegradation is that biodegradation occurs when

→ Complex molecules and broken down to simpler one and it is crucial for maintenance of ecosystem whereas,

→Food spoilage make food unsafe for consumption

TYPES OF FOOD SPOILAGE

Food spoilage types are

→ Chemical Food Spoilage

1) Biochemical assimilatory Food Spoilage

2) Biochemical Dissimilatory Food Spoilage

→Physical Food Spoilage

1) Mechanical Food Spoilage

2) Fouling /Smelling

TYPES OF FOOD SPOILAGE

Food spoilage types are

→ Chemical Food Spoilage

1)Biochemical Assimilatory Food Spoilage :-

The organism uses the material as food i.e. an energy source.

2)Biochemical Dissimilatory Food Spoilage :-

The chemical change in the food is as a result of waste products from the organisms in question.

TYPES OF FOOD SPOILAGE

END

→ Physical Food Spoilage

1)Mechanical Food Spoilage–

Material is physically disrupted/damaged by the growth or activities of the organisms. ●

2)Soiling/fouling –

Material or product is not necessarily unsafe, but as its appearance has been compromised, it is rendered unacceptable.

Food Biotechnology

Module #47



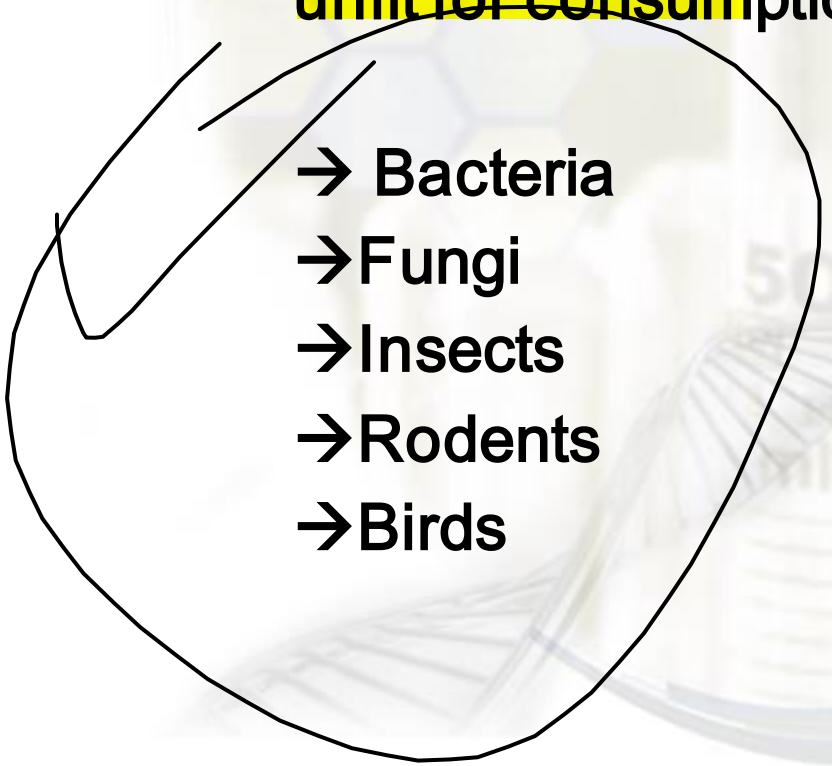
Food Biotechnology

TYPES OF SPOILAGE CAUSING ORGANISMS

TYPES OF SPOILAGE CAUSING ORGANISMS

Introduction :-

Many living organisms (Micro, Macro) cause food spoilage and make it unfit for consumption e.g



→ Bacteria

→Fungi

→Insects

→Rodents

→Birds

1) Bacteria

→ Bacteria greatly spoils the food and cause huge food losses

Example :-

→ Clostridium botulinum (Canned foods i.e. Beverages)

→ Micrococcus (Rancidity)

→ Lactobacillus (Gas formation)

→ Flavobacterium (Decoloration)

FUNGI / MOULD

Fungi and mold are so dangerous as they attack on Agriculture Food crops i.e. Wheat , Rice , Pulses etc.

Examples:-

→Black rot of wheat

→ Black spots on rice kernels

Mould also spoils bakery products i.e. Bread

INSECTS

→Insects can use food (Cereals,pulses,fruits,vegetables etc.) for their nourishment and in return spoil our food and cause huge food losses

→ They deform the food

→ Soften the tissues

RODENTS AND BIRDS

Rodents i.e. Rat/Mouse , Rabbit And Birds greatly spoil and destroy our food (Cereals, Fruits , Vegetables,Meat , cheese etc.)and cause huge food losses

→ Deform food

→ Cause Food Borne illness

→ Make Food Rotten

Food Biotechnology

MODULE 48



Bacterial Food Spoilage

BACTERIAL FOOD SPOILAGE

Bacteria :-

Bacteria are a large diverse group of microscopic, prokaryotic, unicellular organisms. They can be of various shapes (spherical, rod-like or spiral) and may be motile or non-motile. They include both autotrophic and heterotrophic species, and can be aerobic or anaerobic, and many species can thrive under either condition

TYPES OF FOOD SPOILER BACTERIA

Food Spoiler bacteria classified on the basis of their activity or the end product / effect they produce after spoilage .

→ Lipolysis bacteria

→ Pigment former bacteria

→ Gas former Bacteria

LIPOLYSIS BACTERIA

→ Lipolysis is the breakdown of fats and other lipids by hydrolysis to release Fatty Acids

AND

→ Bacteria which cause lipolysis is called lipolysis bacteria e.g. Lactobacillus,leuconostic

→ This bacteria make food containing fats rancid and food give off odour and smell and become spoil

PIGMENT FORMER BACTERIA

→ Sometimes discoloration of food occurs which lowers it aesthetic look and make it unpleasant to consume this is done by Pigment former bacteria

Examples :-

1) Flavobacterium
2) Micrococcus
3) Serratia

GAS FORMING BACTERIA

→Sometimes a canned product swells due to production of gas in it by gas former bacteria

e.g. Beverage can swells due to the activity of Clostridium Botulinum

Examples :-

1)Lactobacillus

2) Leuconostoc

3) Proteus

BENEFICIAL BACTERIAS

→ Despite of spoiling activities some bacteria have beneficial effects for use

Example :-

LAB (Lactic Acid Bacteria) produce lactic acid which is used as starter culture for the production of yoghurt from Milk

→ Proper care should be taken to prevent food spoiling bacteria to spoil food

Food Biotechnology

Module 50



Food Biotechnology

Factors Affecting Microbial Growth

FACTORS AFFECTING MICROBIAL GROWTH

There are many complex reactions and conditions that either inhibit or encourage microbial growth. The availability of oxygen, the temperature (hot or cold), light and other radiation, moisture and dryness, the activity of natural enzymes and the amount of spoilage micro-organisms that are present will all affect the growth of spoilage organisms

FACTORS AFFECTING MICROBIAL GROWTH

Factor affecting Microbial growth are

→ pH

→ Moisture

→Humidity of environment

→Temperature

→Availability of Oxygen

FACTORS AFFECTING MICROBIAL GROWTH

1) pH

→ Micro-organisms are sensitive of pH

→ Best growth at neutral pH 7

→Below pH 4 only few micro-organism can grow

→ pH < 4.6 (no spore forming bacteria can grow)

→ Bacteria are more sensitive to pH as compared to Mould and yeast

FACTORS AFFECTING MICROBIAL GROWTH

2) Water Activity :-

Water activity (aw) is a measure of the water that is available to micro-organisms

→ Pure water aw (1.0)

→Fresh fruits aw (0.97)

→ Bacteria cannot grow at aw (<0.91)

→ Yeast and moulds need less aw to grow

→ Lowest mould growth reported at aw at (0.61)

FACTORS AFFECTING MICROBIAL GROWTH

3) Humidity of the environment :-

The humidity of the environment is important as it affects the a_w of the food as well as the moisture on its surface. Food can pick up moisture from the atmosphere. Under conditions of high relative humidity storage (e.g. in a refrigerator), surface spoilage can take place, unless food is adequately protected by packaging.

FACTORS AFFECTING MICROBIAL GROWTH

4) Temperature :-

Most crucial factor for microbial growth

→ Lowest microbial growth temperature reported is -34 °C and highest is 90 ° C

→ Psychrotrophs optimum temperature(20-30 ° C)

→ Thermophiles optimum temperature (30-40 ° C)

→ Mesophiles optimum temperature is (20-30 ° C)

So if we lowers of increases temperature below or above the range of microbial growth microbial growth can be controlled

FACTORS AFFECTING MICROBIAL GROWTH

5) Oxygen availability :-

Bacteria are classified on the basis of their oxygen demand for growth e.g.

→ Aerobic bacteria (Need oxygen)

→ Anaerobic bacteria (No need of oxygen)

→ Facultative Anaerobic bacteria (survive with or without Oxgyen)

So if O_2 level controlled then microbial growth can be stopped

Food Biotechnology

Module 51



EFFECT OF pH ON MICROBIAL GROWTH

pH :-

pH refers to the hydrogen ion concentration of a solution, a measure of the solution's acidity. It is defined as the negative logarithm of the concentration of H+ ions.

$pH = -\log_{10}[H^+]$

where [H+] is the concentration of H+ ions in moles per litre.

EFFECT OF pH ON MICROBIAL GROWTH

→Different micro-organisms grows at different pH range

→so if we change the (Lower or higher) the pH of a food product below or above the growth range of micro-organisms then we can control their growth and hence preserve food

→so firstly we have to know the pH range at which micro-organisms grows easily at pH range at which their growth stops

EFFECT OF pH ON MICROBIAL GROWTH

pH and microbial growth :-

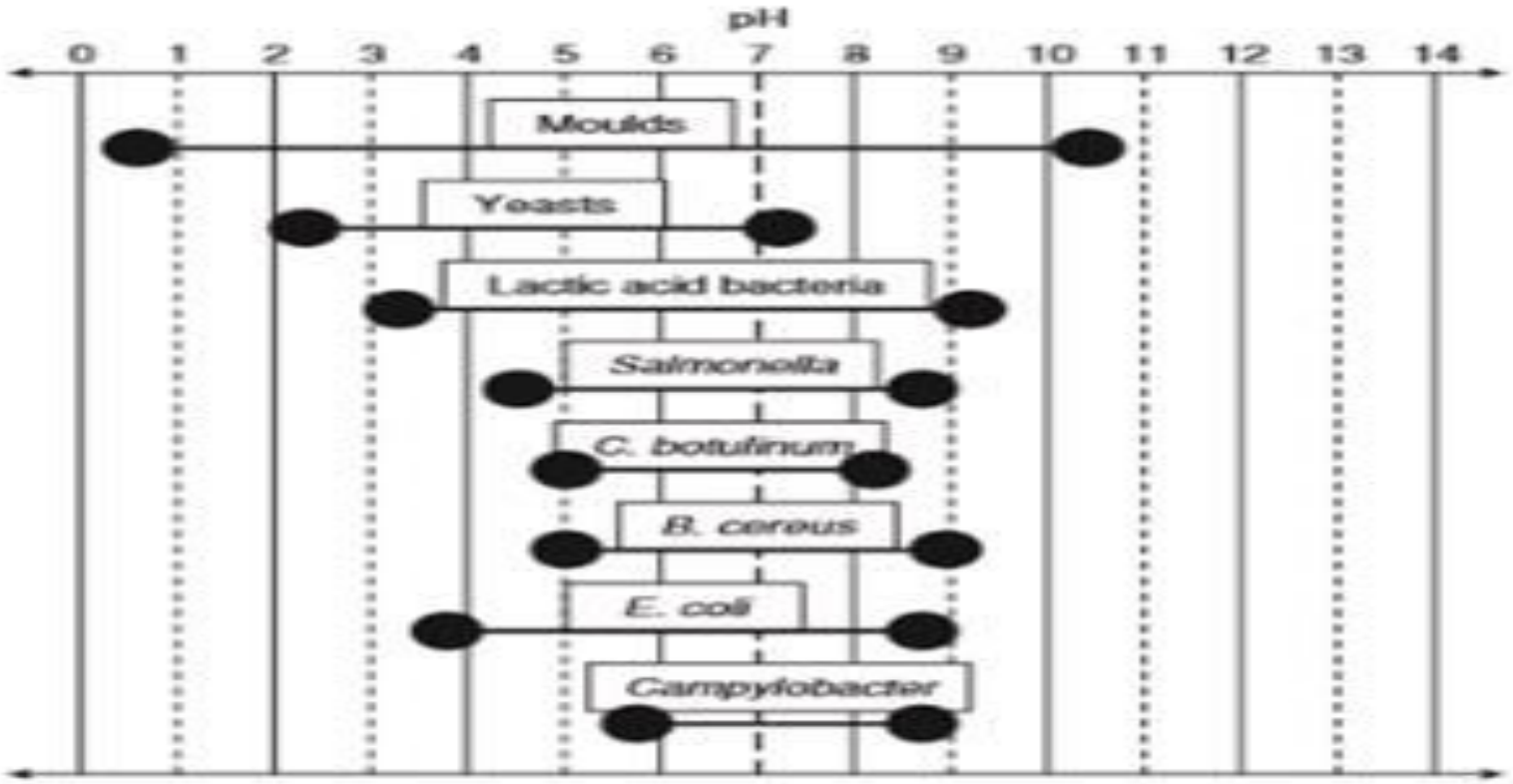
→Micro-organisms best grows at neutral pH of 7

→Below pH 4 only few micro-organism can grow

→ pH < 4.6 (no spore forming bacteria cannot grow)

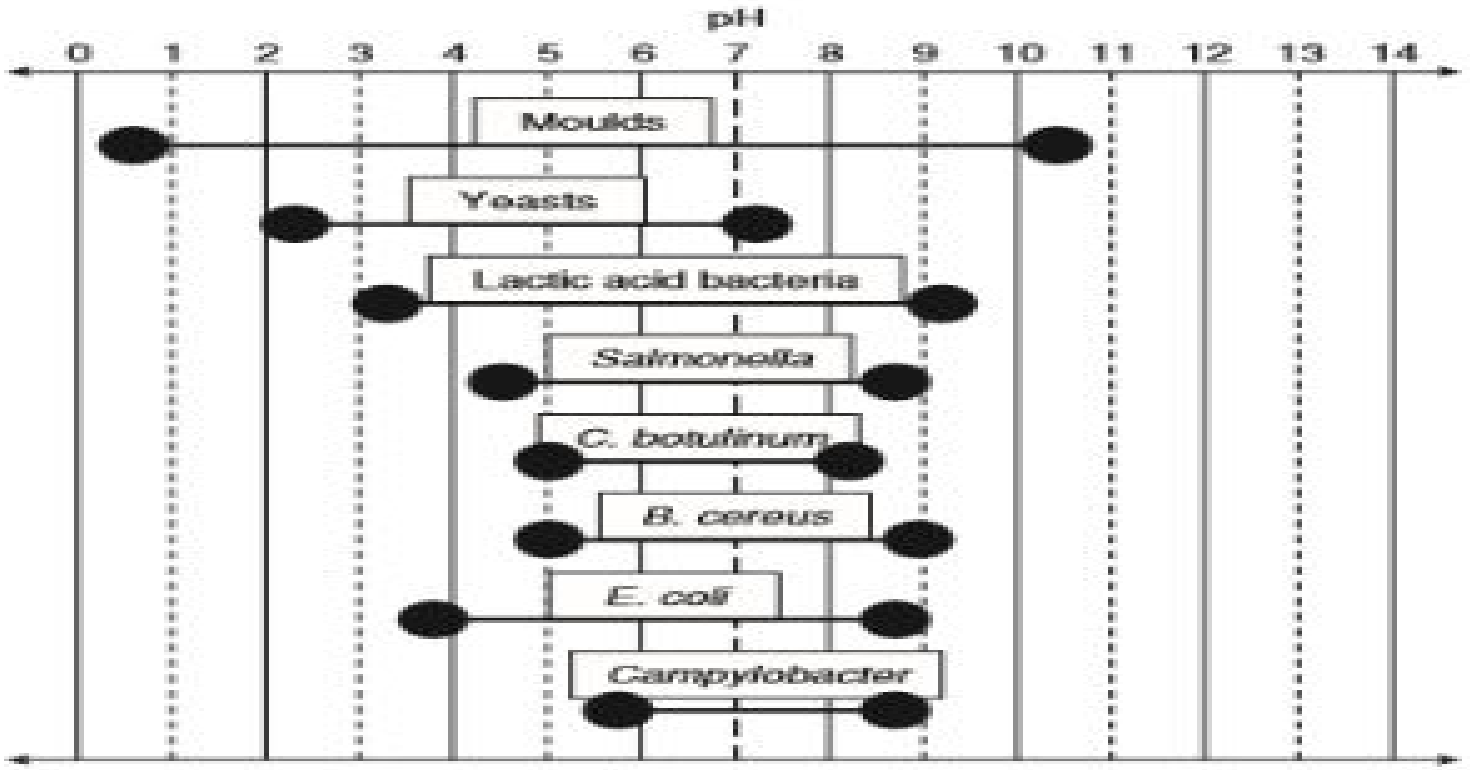
→ Bacteria are more sensitive to pH as compared to Mould and yeast

EFFECT OF pH ON MICROBIAL GROWTH



EFFECT OF pH ON MICROBIAL GROWTH

→pH growth range of selected micro-organisms



ig. 1.9 pH growth ranges for selected micro-organisms.

EFFECT OF pH ON MICROBIAL GROWTH

→ Microbial growth varies as pH varies :-

Figure 1.9 tell us that

→ Moulds can grow at (1-11 pH range)

→Yeast can grow at (2-7 pH range)

→ Bacteria can grow at (3-9 pH range)

→ So if for mould control pH above 11 , for yeast pH below 2 and above 7 , and for bacteria control pH below 3 and above 9 is set

EFFECT OF pH ON MICROBIAL GROWTH

→ Acetic Acid Fermentation is a preservative process mostly used for pickling

→ This process lowers the pH of the mixture so inhibits microbial growth

Food Biotechnology

Module 52



Food Biotechnology

Effect of Moisture for Microbial Growth

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→Micro-organisms growth greatly influence by the availability of moisture

→ If there is high moisture in a food then there are great chances of microbial growth so proper knowledge of relationship between moisture is needed in order to control microbial growth

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

Water Activity (a_w) :-

Water activity is a measure of the water that is available to micro-organisms.

→ High moisture content higher will be the microbial growth hence spoilage

→ If we have to prevent microbial growth and preserve food then we have to lowers the water activity (a_w)

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→ Classification of food on the basis of water content :-

1) Perishable Foods :-

water content (80-95 %)

e.g. Milk, Meat, Tomato

2) Semi Perishable Foods :-

water content (60-80 %)

e.g. Onion , Potato, Apple

3) Stable Foods :-

water content (10-15%)

e.g. Cereals

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→ Perishable foods spoil immediately (few hours) because aw is high and microbial growth is easy

→Semi perishable foods also spoil easily(few days) because of high moisture content

→ Stable foods are less prone to spoilage due to low water content and low microbial growth (months – year)

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→ Water activity and microbial growth :-

Lower the aw lower will be growth of micro-organisms

→ Pure water aw (1.0)

→ Fresh fruits aw (0.97)

→ Bacteria cannot grow at aw (<0.91)

→ Yeast and moulds need less aw to grow

→ Lowest mould growth reported at aw at (0.61)

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

Methods to lowers moisture level in foods:-

→Dehydration

→Sunlight drying

→Water binding agents

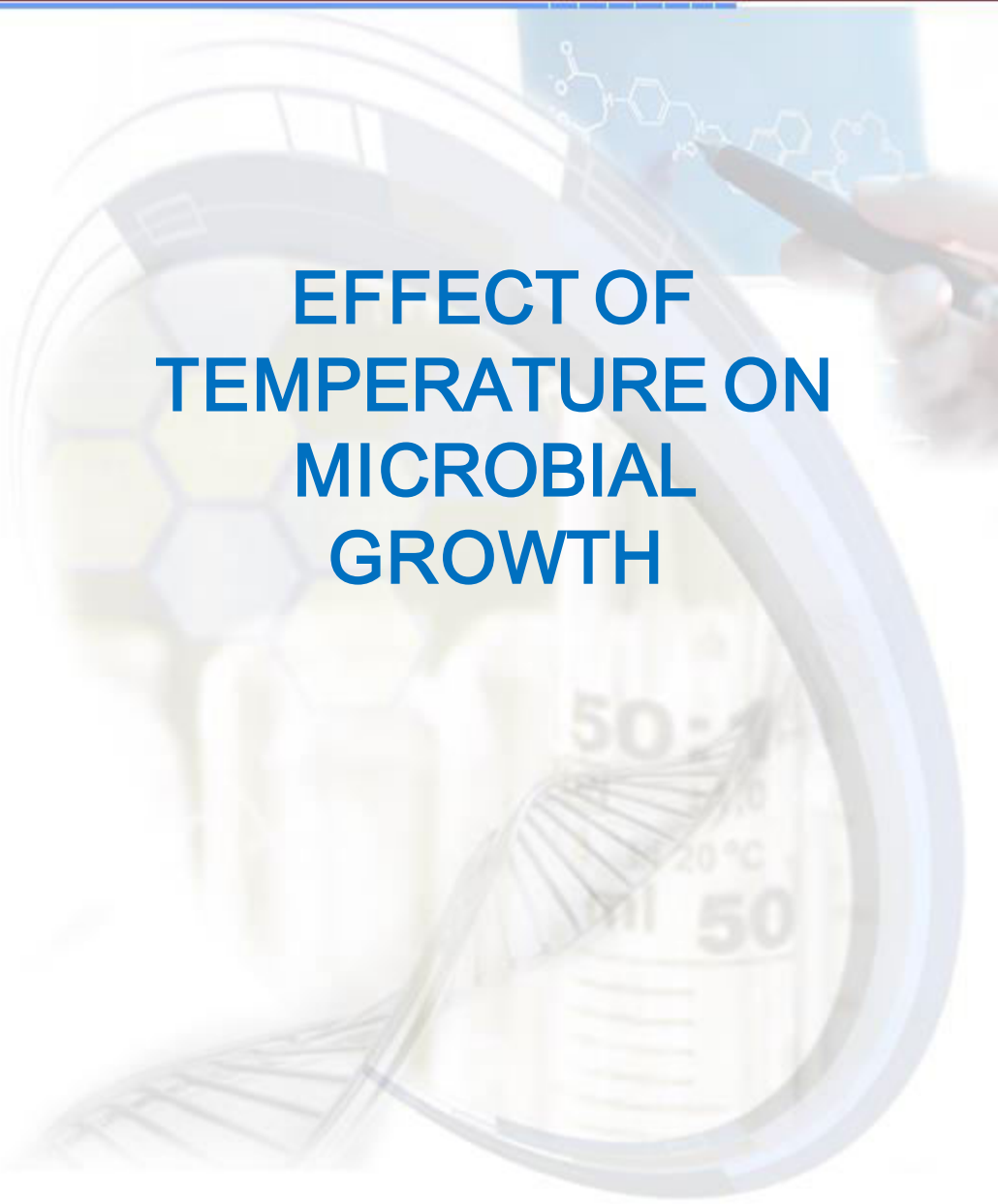
Food Biotechnology

Module 53



Food Biotechnology

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH



EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→Temperature can be considered the most important factor that affects Microbial Growth

→ Micro-organisms have been reported to grow over a wide temperature range; the lowest reported is −34° C and the highest is 90° C

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→ Types of bacteria according to temperature range

1) Thermophiles (40-55 C)

Eubacteria

2) Mesophiles (20-30 C)

Listeria

3) Psychrophiles (7-10 C)

Psychrobacter

So in order to control bacterial growth lowering and increasing temperature technique in order to stop growth of every kind of micro organism

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→Moulds are able to grow over a wide range of pH values and moisture conditions

→They can also tolerate a wider temperature range than bacteria.

→ Many moulds can grow in the refrigerator.

→ Yeasts are not usually found growing in the thermophilic temperature range, but prefer psychrotrophic and mesophilic temperature

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→For growth control temperature should be higher than their optimum growth range temperature or extremely lowers than their optimum growth range temperature

→ But it is not suitable to store all foods in the freezer (−18° C) or even the refrigerator (<8° C), as they loose texture and other desirable features

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

Heat control methods to control Microbial growth :-

→High Temperature :-

1) Pasteurization

2) Sterilization

3) UHT

etc

→ Low Temperature :-

1) Blast freezing

2) Cryogenic freezing

etc.

Food Biotechnology

Module # 49



Food Biotechnology

Fungal Food Spoilage

Fungal Food Spoilage

Fungi are a large group of small chemoheterotrophic organisms.

They do not contain chlorophyll and therefore cannot make their own food by using sunlight.

They are, however extremely adaptable and can utilize almost any organic material.

They are yeast and molds

Fungal Food Spoilage

The conditions affecting the growth of fungi are similar to those affecting bacterial Both yeasts and molds are able to grow in an acidic environment (pH less than 7). The pH range for yeast growth is 3.5 to 4.5 and for molds is 3.5 to 8.0. The low pH of fruits is unfavourable for the growth of bacteria, but yeasts and molds can grow and cause spoilage in fruits.

Fungal Food Spoilage

Fungal growth is characterized by unicellular or multicellular filamentous hyphae, which can often be the cause of physical biodeterioration.

Fungal Food Spoilage

Moulds are responsible for significant spoilage and economic losses in the food chain.

In addition, some species belonging to the genera *Aspergillus*, *Fusarium* and *Penicillium* are mycotoxigenic

Fungal Food Spoilage

Moulds multiply by means of ascospores, zygospores or conidia. The ascospores of some moulds are particularly significant in food spoilage as they are heat resistant.

Botrytis or grey molds

Rhizopus stolonifer (soft rots).

Aspergillus molds.

Fungal Food Spoilage

The presence of this diverse group of fungal secondary metabolites on feed or food presents wide fluctuation from year to year mainly because of the large number of factors affecting fungal invasion and growth.

Fungal Food Spoilage

Examples of common mycotoxins include Aflatoxin,

Ocratoxin A,

Patulin,

Fusarin,

Fumonisin,

deoxynivalenol (DON) and

Zearalenone

END

Fungal Food Spoilage

The presence of this diverse group of fungal secondary metabolites on feed or food presents wide fluctuation from year to year mainly because of the large number of factors affecting fungal invasion and growth.

Fungal Food Spoilage

These contaminants may occur in a wide range of agricultural products such as cereals, fresh and dried fruits, coffee beans, cocoa, coffee and beverages such as beer and wine, while through the consumption of contaminated feed, mycotoxins can be found in meat and dairy products.

Fungal Food Spoilage

Ochratoxin A (OTA) is a secondary metabolite of several species of filamentous fungi (*Penicillium* and *Aspergillus* spp.) with nephrotoxic, immunosuppressive, teratogenic and carcinogenic effects to animals and humans.

Aflatoxin are serious safety concern.

Food Biotechnology

Module # 54



Food Biotechnology

Food Borne Disease: Introduction

Food Borne Disease: Introduction

Food Borne Diseases are on the rise globally.

76 millions case, 325,000 hospitalization and 5000 deaths per annum.

different patterns of food consumption

changing consumer demand

changes in food manufacturing, retail, distribution and storage

Food Borne Disease: Introduction

emergence of new food pathogens

increasing numbers of more susceptible individuals.

Three causes

1. Biological
2. Chemical
3. Physical

Food Borne Disease: Introduction

(1). Pathogenic Bacteria:

More than forty different pathogenic bacteria are known; but Major ones incude

- *Salmonella* spp.,
- *Eschericha coli 0157:H7,*
- *L. monocytogenes,*
- *C. perfringens,*
- *C.botulinum,*
- *S. aureus,* and
- *Campylobacter jejeuni*

Food Borne Disease: Introduction

Symptoms that include headache, muscle pain, nausea, fatigue, chills or fever, stomach or abdominal pain, vomiting, and diarrhoea.

Higher risk population includes pregnant women, young children, older adults, and people with weakened immune systems.

Food Borne Disease: Introduction

The foods that are commonly involved in these food poisoning incidents include meat and poultry and their products, seafood and seafood products, egg and egg products, milk and dairy products, fruits and vegetables.

Food Borne Disease: Introduction

(2). *Viruses*
Foods can be the medium for transmission of certain viruses. Examples of viruses that are known to be food safety hazards are the hepatitis A and E viruses, the Norwalk group of viruses, and rotavirus.

Food Borne Disease: Introduction

(3). ***Parasites***

Several human parasites can be transmitted by foods. The most common human parasites include parasitic protozoan species (e.g., *Entamoeba histolytica, Giardia lambia, Cryptosporidium parvum*), and parasitic worms (*Ascaris lumbricoides, Taenia solium, Trichinella spiralis*).

Food Borne Disease: Introduction

- Other Considerations:

1. Natuarlly occuring substances
2. Unavoidable contaminants
3. Agri residues
4. Industrial Chemicals
5. Heavy metals
6. Food Allergens.

END

Food Biotechnology

Module #55



Plant Toxins



Plant Toxins

- A poisonous substance produced within plants
- Occurs naturally
- Secondary metabolites

Plant Toxins

- Solanine in potatoes
- Inhibits acetyl-choline esterase (a key component in the central nervous system).

•Potatoes contain about 2mg – 15mg / 100g fresh weight

•Amount higher in potatoes exposed to light (as much as 100mg/100g)

Plant Toxins

- Heat stable
- Insoluble in water
- Can cause abdominal pain and diarrhea if consumed in high amounts
- May develop signs on neurological damage
- General awareness of green potatoes is high.

Plant Toxins

- Caffeine
- Theobromine

- Pure alkaloids
- Tea, coffee, and cocoa
- Regarded as stimulants.
- Addictive in nature
- Physiological effects depend on amount consumed

Plant Toxins

Phenylethylamine :

- Found in chocolate
- Vasopressor amine
- Constrict blood vessels
- Can cause migraine in susceptible person

Plant Toxins

Hydrogen Cyanide (HCN):

- Present in tapioca
- Released from glycosides in food such as lima beans.
- Formed from amygdalin

- Varieties rich in cyanogenic glucosides need to be fermented.
- Neurological degeneration and a form of blindness

Plant Toxins

- Legumes contain inhibitors of trypsin and chymotrypsin
- Mostly inactivated by cooking
- Soy beans need to be autoclaved

Plant Toxins

- Some mushrooms can be severely toxic
- Can be fatal
- Snakeroot poisoning

Plant Toxins

- Lectins
- Haemoglutinins

- Present in kidney beans
- Bind to surface of RBC
- Causes clumping

- Toxic: 0.5mg/kg

Plant Toxins

Myristicin :

- Significant in nutmeg
- Lesser amounts in black pepper, carrot and celery
- 10 mg nutmeg can cause euphoria, hallucinations and narcosis
- Nutmeg used in very small amount for flavoring: induces sleep

Plant Toxins

- Most plant toxins are inhibited during cooking
- Some exceptions remain
- Consuming raw food increases chances of toxicity
- Must ensure raw food safety before consuming

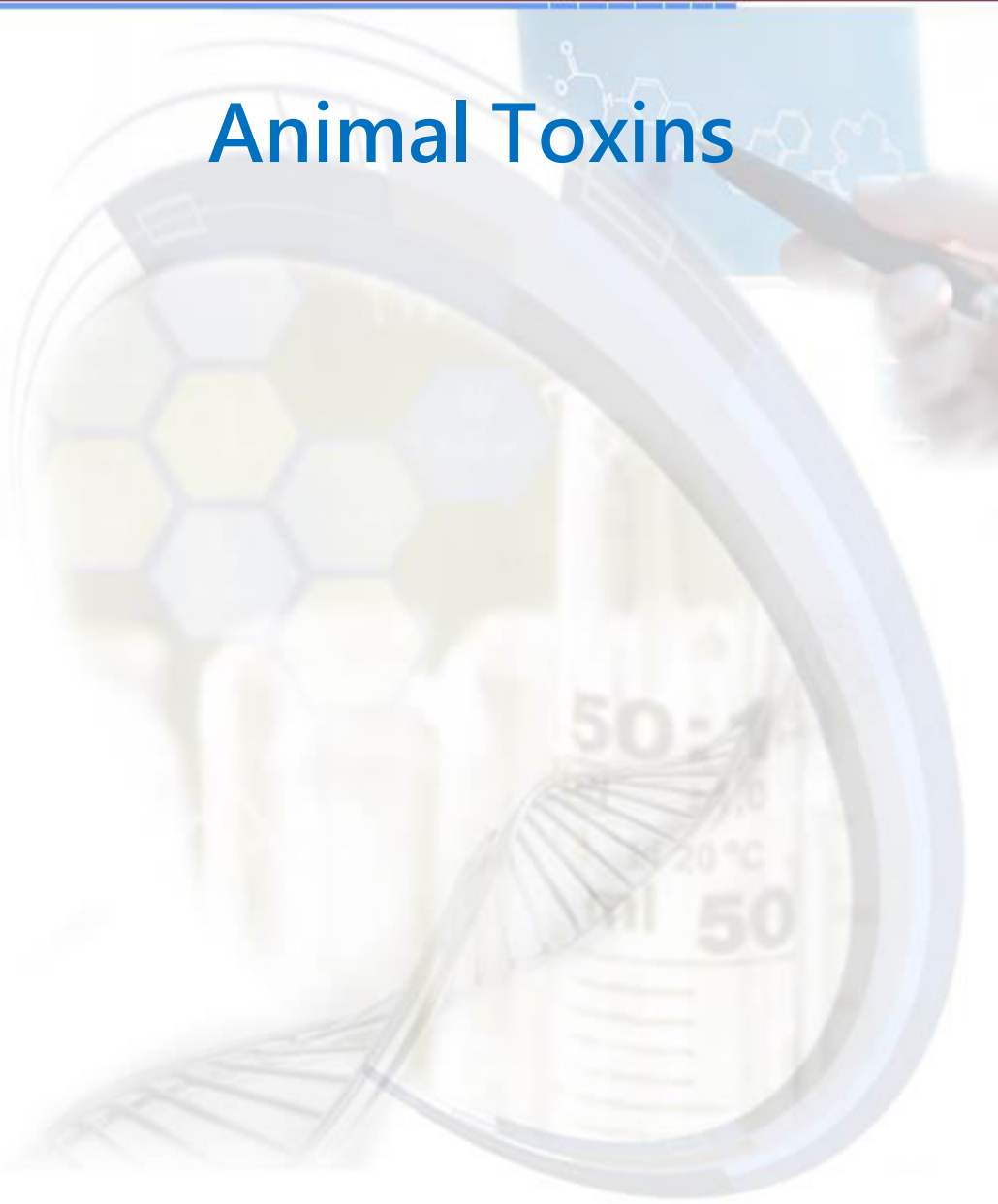
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Food Biotechnology

Module #56



Animal Toxins



Animal Toxins

- A poisonous substance produced within animals
- Produced during metabolism and growth
- Occur naturally

Animal Toxins

Tetrodotoxin :

- Minimum lethal dose: 1.5 mg - 4.0 mg
- Occurs in puffer fish(in liver and ovaries)
- Muscle and testes of puffer fish delicacy in Japan
- Highly skilled food handlers

Animal Toxins

- Block movement of Na+ across membranes of nerve fiber
- Disrupts transmission of electrical impulses
- Can lead to total paralysis and respiratory failure
- Death within 6-24 hours
- No effective treatment

Animal Toxins

- Other animal toxins

- Present in shell fish
- Such as clams, scallops
- Toxin produced due to some planktons

Animal Toxins

- Mussels and Clams
- Poisonous alkaloid
- Can cause food poisoning symptoms

Animal Toxins

- Scrombridae
- Tuna, Sardines, Mackerel
- Held above 10° C for some hours
- Formation of high levels of histamine

Animal Toxins

- Tryptophan converted to histamine in muscles
- Catalyzed by decarboxylase produced by bacteria in fish
- High levels before putrefaction

Animal Toxins

- Headaches, palpitations, gastrointestinal upset, skin flushes and erethrema
- Anti-histamine drugs used to treat

Animal Toxin

- Awareness of symptoms
- Residents near areas with such poisonous plant/animal usually guard against it

END

Food Biotechnology

Module #57



Agricultural Residues

Agricultural Residues

- Residues of pesticides
- May be toxic
- Can make their way into food

Agricultural Residues

- Dichlorodiphenyltrichloroethane (DDT)
- Pesticide
- Used for killing mosquitoes
- Found in food in very high proportions

Agricultural Residues

- Other pesticide residue found in food:

- Dieldrin
- Lindane (BHC)
- Malathion

- Used as spray on fruits, vegetables

Agricultural Residues

- Many of these pesticides have been banned
- Used for many years
- Still present in soil
- Finds its way into human food through crops

Agricultural Residues

Sterols :

- Source of residue
- Used for increased growth in animals
- Mostly for meat operations

Agricultural Residues

- Some sterols have been shown to be carcinogenic
- Residue of sterols may be present in meat

Agricultural Residues

Antibiotics :

- Can be a source of residue
- Used as growth promoters
- Used to combat diseased in animals

Agricultural Residues

- Mastitis in cows treated with antibiotics
- Milk from these cows should be discarded
- Not always the case
- Penicillin in milk can cause allergic reactions to susceptible persons

END

Food Biotechnology

Module #58



Poisoning by Chemicals

Poisoning by Chemicals

- Toxic metals are a major source of chemical poisoning
- Use of contaminated water
- Soil on vegetables poorly washed
- Water pipes (lead)
- Equipment/Containers

Poisoning by Chemicals

Lead :

- Pipes and storage tank
- Soft water more susceptible
- Lead based solder in canning

Poisoning by Chemicals

- Most food subject to regulations of lead content
- Lead poisoning uncommon
- Absorption of lead in children is higher than of adults

Poisoning by Chemicals

Chronic Lead Poisoning

- Symptoms include:

- Anemia
- Poor I.Q performance
- Poor attention span

Poisoning by Chemicals

- Food and water intake contribute to daily lead intake
- Exhaust fumes can be an additional source
- Children from congested cities more susceptible

Poisoning by Chemicals

Mercury :

- Cereal grains treated with antifungal mercury compounds
- Industrial pollution of coast water
- Seafood contaminated with alkyl mercury compounds

Poisoning by Chemicals

- Symptoms of mercury poisoning are variable
- All point to damage to CNS
- Children more sensitive to exposure

Poisoning by Chemicals

Arsenic :

- Widely distributed in nature
- May contaminate chemicals used in food processing
- Important to check for arsenic contamination in food additives

Poisoning by Chemicals

Sodium phosphate :

- Used as stabilizer in milk powder
- Contaminated with arsenic trioxide
- Poisoning of 12,000 infants in Japan

Poisoning by Chemicals

Cadmium :

- Cadmium poisoning come from three sources

- Contaminated water
- Cadmium plated components in food processing machinery
- Zinc plating of containers used to store acidic food

Poisoning by Chemicals

- Lead, Mercury, Arsenic, Cadmium
- Bivalent in nature

- If diet is low in bivalent nutrient metals (Ca, Fe), these contaminants might enter vulnerable systems such as
- Nervous system, kidneys, bone marrow

END

Food Biotechnology

Module #59



Food Biotechnology

Poisoning by microorganisms

Poisoning by microorganisms

- Poisoning is caused by eating contaminated food by the the microorganisms.
- There are two chief kinds of food poisoning caused by bacteria:

1. Botulism
2. Staphylococcus poisoning

Poisoning by microorganisms

Botulism

Poisoning by microorganisms

- It is caused by the presence of toxin in the food
- Foods such as corn, beans, peas, meat, fish are likely to be contaminated.
- Some strains of this organism produce a toxin in the food, which is a potent poison.

Poisoning by microorganisms

- It can cause paralysis of involuntary muscles and may ultimately lead to death due to respiratory failure.
- Toxin can be destroyed by heating.
- Likely to occur if neutral foods are inadequately processed.

Poisoning by microorganisms

Staphylococcus poisoning

Poisoning by microorganisms

- It is caused by staphylococcus aureus.
- Staphylococcus contamination of the food may either be from human or animal sources.
- Nasal passages of many humans, boils and infected wounds are potential sources.

Poisoning by microorganisms

- Strains of *Staphylococcus aureus* produce an enterotoxin.
- It can cause:

1. Salivation
2. Nausea
3. Vomiting
4. Abdominal cramps
5. Diarrhoea.

Poisoning by microorganisms

- The time between the consumption of the food and the appearance of symptoms can range from 1–6 hours.
- Recovery takes about a day or two and mortality is very low.

Poisoning by microorganisms

END

- Employees suffering from *staphylococcal* infections such as colds, boils should not be allowed to handle foods.
- Prevention by refrigeration of foods immediately after making.
- Pasteurization kills all the *staphylococci* that may be present in foods.

Food Biotechnology

Module #60



Food Biotechnology

Food Infections

Food Infections

- Foodborne infection occurs by the consumption of living organisms.
- Symptoms do not appear for at least one day after ingestion.
- There is a vast number of different bacteria involved.

Food Infections

Identification of different pathogens related to food

Food Infections

- *Clostridium botulinum* is a Gram-positive, obligate anaerobe, sporeforming bacterium.
- The botulinum toxin is one of the most potent neurotoxins known which can even cause death.
- Food contaminated with only a few nano grams (10^{-9} ng) can be toxic.

Food Infections

There are five types of botulism:

1. Food-borne botulism
2. Wound botulism
3. Infant botulism
4. Adult infectious botulism
5. Inadvertent botulism

Food Infections

- **Salmonella** is a Gram-negative, rod shaped, motile bacterium, that does not form spores.
- Its is found in fresh and salt water, soil and animal faeces.
- It causes salmonellosis infection and even extremely low doses can cause death.

Food Infections

- *Listeria monocytogens:* are Gram-positive, motile bacteria. They do not form spores, but are quite resistant to desiccation and heat.
- Found in the soil and associated with birds and animals
- Cause disease called listeriosis, which can result in septicaemia, meningitis encephalitis and spontaneous abortions in pregnant females.

Food Infections

- *S. aureus* is a Gram-positive, small round bacterium (coccus), that can produce a heat-stable toxin.
- *S. aureus* is found all over most commonly via humans.
- It is found in the nasal passages, throat, on the skin and in the hair of more than half of healthy people.

Food Infections

- *Escherichia coli* are Gram-negative bacteria found in the intestines of all mammals.
- cause four different classes of illnesses.
- Outbreaks are associated with fecal contamination of water by food handlers, or are due to unprocessed or under-processed foods.

Food Infections

END

- *Campylobacter jejuni* are Gram-negative, microaerophilic, motile, rod- shaped bacteria.
- They are found on raw chickens, in raw milk, and are often carried in the intestines of healthy cattle.
- They can contaminate food via flies on farms and are also found in non chlorinated water.

Food Biotechnology

Module # 61



Sequence of Events in Food Borne Diseases

Sequence of Events in Food Borne Diseases

A sequence of events happens in all food borne diseases.

An understanding of these sequences is helpful in investigating the cause (the source and means of transmission) of a food borne disease. It also helps in recognizing how the sequence can be broken.

Sequence of Events in Food Borne Diseases

1. Initially, there has to be a source of a pathogen.

2. the pathogen has to contaminate a food.

3. Consumption of the food contaminated with a pathogenic virus or parasites

4.It lead to viral or parasitic infection.

Sequence of Events in Food Borne Diseases

For bacterial pathogens (and toxicogenic molds) the contaminated food has to support growth and be exposed for a certain period of time at a suitable temperature to enable the pathogens to grow to produce toxins.

Sequence of Events in Food Borne Diseases

However for some potent pathogens (such as *Esc. coli* 0157:H7) growth may not be necessary to cause a food borne infection.

For intoxication the growth should reach a sufficient level to produce enough toxins so that when the food is consumed, the individual develops the symptoms

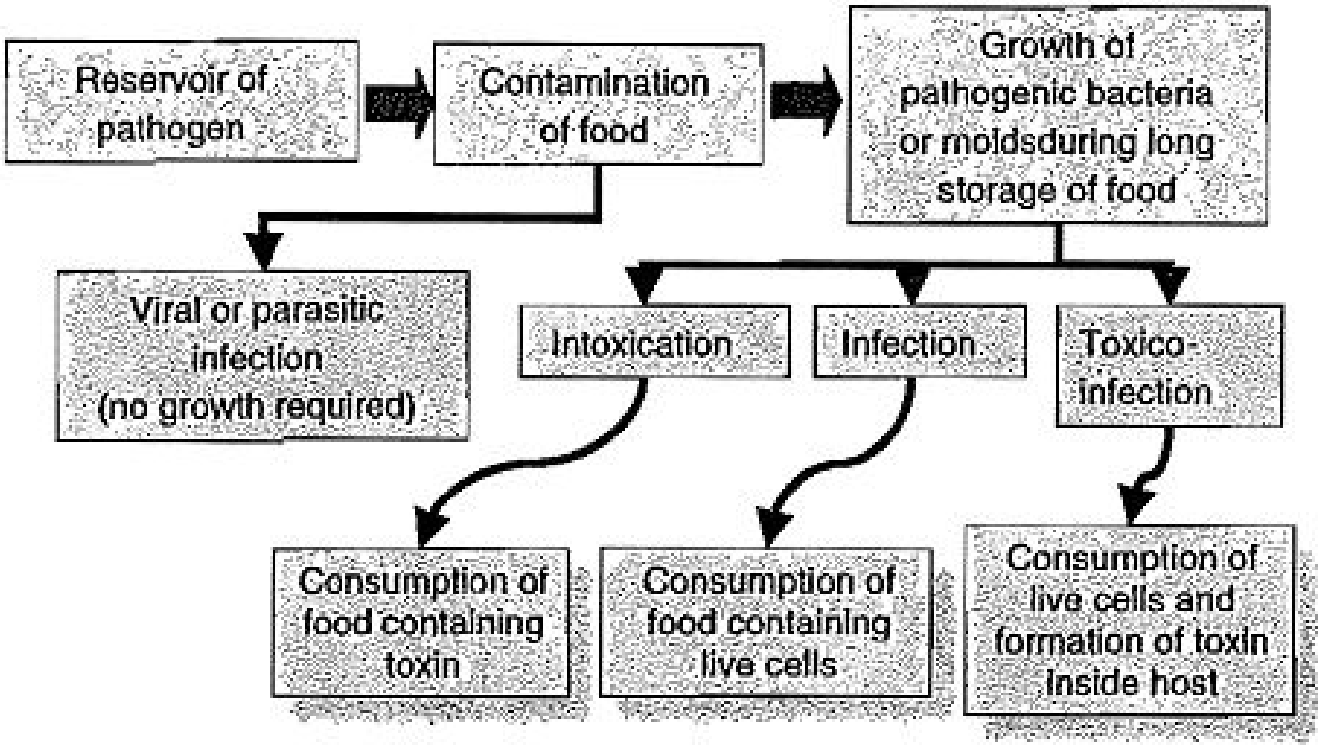
Sequence of Events in Food Borne Diseases

For bacterial infection, viable cells of a pathogen need to be consumed in sufficient numbers, which vary greatly with pathogens, to survive stomach acidity, establish in the digestive tract, and cause illness.

Sequence of Events in Food Borne Diseases

In case of toxicoinfection, viable cells should be consumed either in very high numbers (for those that cannot multiply in the digestive tract, such as *Clo. peifringens*) or in reasonable numbers (for those that can multiply in the digestive tract, such as *Vibrio* cholerae) so that toxins released by them in the digestive tract can produce the symptoms.

Sequence of Events in Food Borne Diseases



Food Biotechnology

Module # 62



Food Biotechnology

Detection of *E.Coli* in Drinking Water

Detection of *E.Coli* in Drinking Water

The Most Probable Number (MPN) method is applicable to the enumeration of coliforms, faecal coliforms and aerogenic *Escherichia coli* in foods and water, including contact water from food manufacturing plants.

Detection of *E.Coli* in Drinking Water

MPN is not intended to be used to isolate and enumerate *E. coli* serotypes associated with human illness, particularly the enterohemorrhagic serotype O157:H7. Many of the pathogenic serotypes do not give a positive faecal coliform reaction.

Detection of *E.Coli* in Drinking Water

Description The MPN procedure involves a multiple tube fermentation technique where three or more decimal dilutions of the sample are inoculated into tubes of broth medium and incubated at a specific temperature and for a specific time.

Detection of *E.Coli* in Drinking Water

The method is progressive; i.e., first determining the presence of coliforms in the tubes, then determining if these tubes also contain faecal coliforms, and then confirming whether *E. coli* is present. Based on the number of tubes indicating the presence / absence of the three groups of organisms, MPN is established

Detection of *E.Coli* in Drinking Water

3. Principle

The terms "coliform" and "faecal coliform" have no taxonomic validity and, therefore, are only meaningful when expressed in terms of the analytical test parameters of medium, time and temperature of incubation. Coliforms, faecal coliforms, and *E. coli* are considered "indicator organisms."

Detection of *E.Coli* in Drinking Water

Which indicate one of the following possibilities:

1. inadequate processing and/or post-processing contamination; and/or
2. microbial growth. The presence of faecal coliforms and *E. coli* may indicate faecal contamination.

Detection of *E.Coli* in Drinking Water

Briefly, this method involves serially diluting out the target organisms in the sample, in 5-replicate aliquots, to extinction. The probable level of the target organisms is then statistically estimated from an MPN table.

Detection of *E.Coli* in Drinking Water

Gas production is used as an indication of ability to ferment lactose from Lauryl Sulfate Tryptose (LST) broth (presumptive coliform test); gas production from Brilliant Green Lactose Bile (BGLB) broth is considered confirmation of coliform presence.

Detection of *E.Coli* in Drinking Water

gas production at 44.5 or 45° C from EC broth is used as confirmation of faecal coliform presence; and appearance of typical nucleated, dark-centred colonies with or without metallic sheen when positive EC broths are streaked onto Levine's Eosin Methylene Blue (L-EMB) agar are indicative of *E. coli.* It must be confirmed by further biochemical tests

Food Biotechnology

Module # 63



Food Preservation Methods

Food Preservation Methods

Inactivation, inhibition, and methods of avoiding recontamination

Food Preservation Methods

Early Methods of Preservation may include:
Using chemicals, e.g. acid and salt:

- pickling in vinegar;
- salting meat;
- sugar to make jam;
- storing in alcohol.

Temperature control:

- sun drying fruit;
- using ice blocks.

Food Preservation Methods

Food technology advanced in the early 19th century with the invention of the canning process.

Since then the range of technologies used to process and preserve foods has expanded and uses a range of physical and chemical techniques

Food Preservation Methods

There have also been many technological changes in the home.

- refrigerators became widely used from the mid-1950s;
- domestic freezers in the 1970s;
- microwave ovens in the 1980s.
- Frozen and ready meals followed these.

Food Preservation Methods

Advances in Technology:

New technologies have given us a greater range of methods to package and store foods. This enables the preservation of nutrients and the extension of food product shelf-life.
Demand has increased for minimally processed foods which retain their freshness.

Food Preservation Methods

MAP Technologies:

Modified atmosphere packaging is a way of extending the shelf life of fresh food products. The technology substitutes the air inside a package with a protective gas mix.

New Packaging technologies are reducing food wastes.

Food Preservation Methods

Nano particles of silver have anti-bacterial properties and can be used on:

- kitchen tools and tableware;
- preparation surfaces;
- plastic food storage containers;
- plastic wrap.

Food Preservation Methods

Encapsulation Technologies:

Encapsulation is "*a process where a continuous thin coating is formed around solid particles, liquid droplets or gas cells that are fully contained within the capsule wall*.

Food Preservation Methods

High Pressure Processing is a cold pasteurisation technique. Food, previously sealed in flexible and water-resistant packaging, is subjected to a high level of hydrostatic pressure (pressure transmitted by water) for a few seconds to a few minutes

Food Biotechnology

Module # 64



Food Biotechnology

Need for Food Preservation

Need for Food Preservation

The main reasons for food preservation are :

To overcome inappropriate planning in agriculture.

To overcome spoilage factors.

Need for Food Preservation

produce value-added products,

and provide variation in diet.

Food Security and food wastage

Need for Food Preservation

In food preservation, the important points that need to be considered are

- The desired level of quality
- The preservation length
- The group for whom the products are preserved

Need for Food Preservation

Detection of

E.Coli

Food Biotechnology

Module # 65



Food Biotechnology

Blanching

Blanching

Blanching is steam treatment of vegetables and some fruits to fix colour and stop enzyme activities.
Its only pre treatment.
It can be followed by heat sterilisation, dehydration and freezing.
also combined with peeling and/or cleaning of food (Chapter 3), to achieve savings in energy consumption, space and equipment costs.

Blanching

Onions and green chillies are exceptions but others suffer quality greatly if blanching is skipped. The factors which influence blanching time are:

- type of fruit or vegetable
- size of the pieces of food
- blanching temperature
- method of heating.

Blanching

Freezing and dehydration temperatures may not inactivate enzymes properly.

In case of canning sterilization temperature may also not be sufficient to warrant blanching.

Blanching

Under blanching is equally bad because of accelerated enzyme activity.
lipoxygenase, polyphenoloxidase, polygalacturonase and chlorophyllase reduce eating and nutritional quality.

Blanching

catalase and peroxidase are found in most vegetable and heat resistant. These are used as marker enzymes particularly peroxidase.

Blanching reduces contaminating microbes helping further processing operations such as sterilization and freezing.

Blanching

Hot water bath and saturated steam chambers are routinely used.

Microwaves are also a possibility.

Need to consider effluent impact by reducing soluble components from food.

Cooling is critical for yield of final product and is efficient for steaming.

END

Food Biotechnology

Module #66



Pasteurization



- Pasteurisation is a relatively mild heat treatment, in which food is heated to below 100°C.
- In low acid foods, for example milk) it is used to minimise pathogenic micro-organisms and to extend the shelf for days.
- In acidic foods is used to minimise pathogens, and to extend the shelf life for several months.

Pasteurization

- Two types of processes are used
- 1. LTLT: 63ºCfor 30 m
- 2. HTST: 71.8ºC for 15 seconds or

88ºC for 1 s, 94ºC for 0.1 s or 100 ºC for 0.01 s for milk.

Other foods are Beer, Eggs, Fruit juices, milk, ice cream etc.

Pasteurization



Pasteurization of packaged foods:

- Some liquids like beer and fruit juices are pasteurized after filling into containers.
- Hot water is normally used if the food is packaged in glass, to reduce risk of thermal shock to the container.

Pasteurization

Metal or plastic containers are processed using steam–air mixtures or hot water as there is little risk of thermal shock.

Pasteurization

Pasteurization of unpackaged foods:

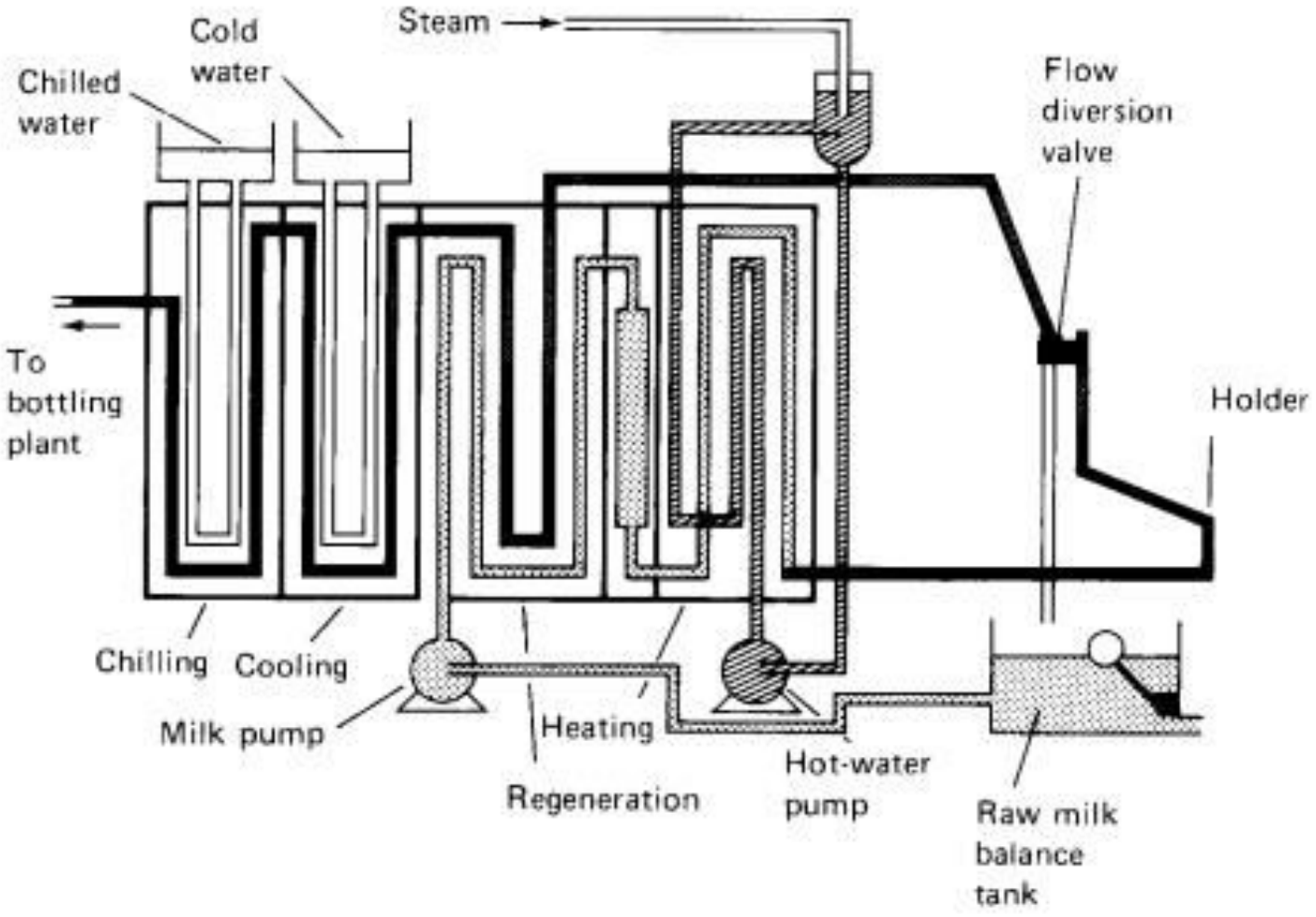
- Swept surface heat exchangers or open boiling pans for small scale batch pasteurization.
- Low viscosity liquids (for example milk, milk products, fruit juices, liquid egg, beers and wines) usually employs plate heat exchangers.

Pasteurization

Plate heat exchanger:

- Consists of a series of thin vertical stainless steel plates, held tightly together in a metal frame.
- The plates form parallel channels.
- Liquid food and heating medium are pumped through alternate channels usually in a counter-current flow pattern.

Pasteurization



Pasteurization

Other heat exchanger:

- Concentric tube heat exchanger is suitable for more viscous foods and is used with dairy products, mayonnaise, tomato ketchup and baby foods.
- Consists of a number of concentric stainless steel coils, each made from double/triple-walled tube.
- Liquid food is passed from one coil to the next for heating and cooling.

Pasteurization

- Pasteurised food is immediately filled into cartons or bottles.
- Significant levels of spoilage and risks from pathogens can arise from post-pasteurisation contamination.
- Particularly when foods (for example milk) are not re-heated before consumption.

Pasteurization: Effect on foods

- Minor changes to the nutritional and sensory characteristics of most foods.
- The shelf life of pasteurised foods is usually only extended by a few days or weeks compared with many months with the more severe heat sterilisation.

Pasteurization: Effect on foods

Colour, flavour and aroma:

- In fruit juices enzymic browning by polyphenoloxidase.
- Other pigments in plant and animal products are also mostly unaffected.
- A small loss of volatile aroma compounds of juices causes a reduction in quality.

Pasteurization: Effect on foods

Vitamin Loss:

- In fruit juices losses of vitamin C and carotene are minimised by deaeration.
- Changes to milk are confined to a 5% loss of serum proteins and small changes to the vitamin content

Pasteurization: Effect on foods

Colour, flavour and aroma:

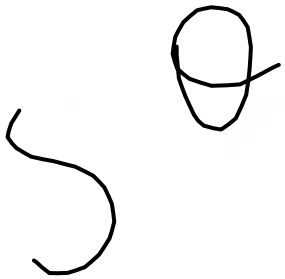
- In fruit juices enzymic browning by polyphenoloxidase.
- Other pigments in plant and animal products are also mostly unaffected by pasteurisation.
- A small loss of volatile aroma compounds during pasteurisation of juices causes a reduction in quality unmask other 'cooked'

Food Biotechnology

Module #67



Heat Sterilization



Heat sterilisation :

foods are heated at a sufficiently high temperature and for a sufficiently long time to destroy microbial and enzyme activity.

Sterilised foods have a shelf life in excess of six months at ambient temperatures.

Heat Sterilization

The length of time required to sterilise a food is influenced by:

- the heat resistance of micro-organisms or enzymes likely to be present in the food
- the heating conditions
- the pH of the food
- the size of the container
- the physical state of the food.

Heat Sterilization

Heat resistance of micro-organisms

- Most heat resistant spores have a *z* value of around 10°C. This temperature rise will result in a ten-fold reduction in processing time needed to achieve the same lethality.

Heat Sterilization

- In low-acid foods (pH 4.5), the heat resistant, spore forming micro-organism, *Clostridium botulinum* is the most dangerous pathogen likely to be present. It can cause lethality up to 65% cases by toxin botulin in anaerobic situations.

Heat Sterilization

- Thermal destruction of micro-organisms takes place logarithmically and a sterile product cannot therefore be produced with certainty no matter how long the process time.
- Commercial sterility means in practice that the heat processing inactivates substantially all micro-organisms and spores with potential to grow in storage conditions

Heat Sterilization

A 12*D* process is used when *C. botulinum* is likely to be present in low acid foods, but in foods that contain more heat-resistant spoilage microorganisms it would result in over processing so 2D to 8D process are used for an economical operation and quality

Prior care to avoid contamination is helpful.

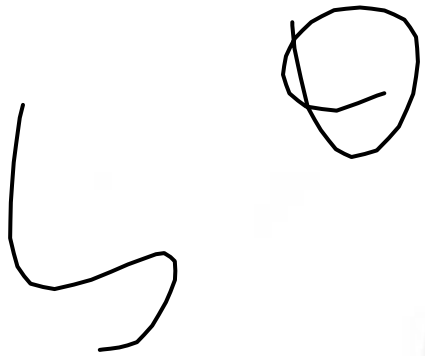
Heat Sterilization

Heating Methods:

- Plate heat exchangers.
- Concentric tube Consists of a number of concentric stainless steel coils, each made from double/triple-walled tube.

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Heat Sterilization



Rate of heat penetration:

- Type of product
- Size of container
- Agitation of the container
- Temperature of the retort
- Shape of the container
- Type of container

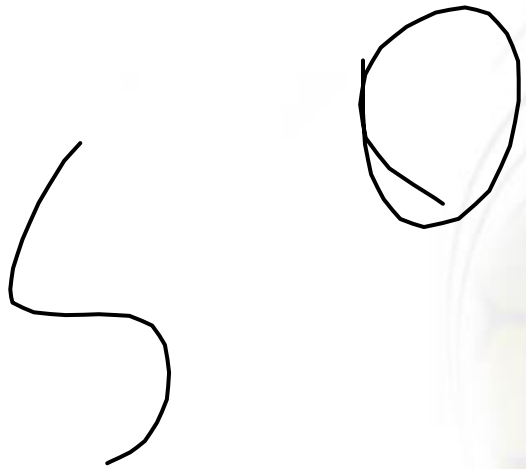
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Food Biotechnology

Module #68



UHT Process



- Also termed as *aseptic processing.*
- *Ultra-high temperature processing (UHT), ultra-heat treatment*, or *ultra-pasteurization* is a food processing technology that sterilizes liquid food, chiefly milk, by heating it above 135 °C (275 °F) – the temperature required to kill spores in milk – for 1 to 2 seconds.

UHT Process

- It is used to sterilise a wide range of liquid foods, including milk, fruit juices and concentrates, cream, yoghurt, wine, salad dressing, egg and ice cream mix
- Possible if the product is sterilised before it is filled into pre-sterilised containers in a sterile atmosphere.

UHT Process

Can also be used to process foods which contain small discrete particles, for example

- cottage cheese
- baby foods,
- tomato products,
- fruit and vegetables,
- soups and rice desserts.

UHT has an important additional advantage of a shelf life of at least six months at ambient temp.

UHT Process

- UHT processing is independent of container size
- Conventional retorting of A2 cans of vegetable soup requires 70 min at 121°C to achieve an F_0 value of 7min followed by 50 min cooling,
- aseptic processing in a scraped-surface heat exchanger at 140°C for 5 s gives an F_0 value of 9 min.

UHT Process

Increasing the can size to A10 increases the processing time to 218 min, whereas with aseptic processing the sterilisation time is the same. So very big containers can be used for sterilization unlike canning operations.

UHT Process

A theoretically ideal UHT process would heat the product instantly to the required temperature, hold it at that temperature to achieve sterility and cool it instantly to filling temperature.

In practice the degree to which this is achieved depends;

- part on the method used to heat the food
- on the properties of the food

UHT Process

Equipment used for UHT processing has the following characteristics:

- operation above 132°C
- exposure of a relatively small volume of product
- maintenance of turbulence
- use of pumps
- exchanger
- constant cleaning

UHT Process

Equipments are according to the method of heating into:

- direct systems (steam injection and steam infusion)
- indirect systems (plate heat exchangers, tubular heat exchangers (concentric tube or shell-and-tube) and scraped surface heat exchangers)
- other systems (microwave, dielectric, ohmic and induction heating

UHT Process

The main limitations of UHT processing are the

- Cost
- complexity of the plant,

which arises from the necessity to

- sterilise packaging materials,
- pipework/tanks,
- maintenance of sterile air and surfaces in filling machines,
- Higher skill levels required by operators and staff

End

Food Biotechnology

Module #69



Effects of heating on food

The purpose of heat sterilisation is

- to extend the shelf life of foods for storage at ambient temperatures,
- while minimising the changes in nutritional value and eating quality.

Effects of heating on food

Differences between the *D* and *z* values of micro-organisms, enzymes and sensory or nutritional components of foods are exploited to optimise processes for the retention of nutritional and sensory qualities

Effects of heating on food

COLOUR

- In meats the red oxymyoglobin pigment is converted to brown metmyoglobin, and purplish myoglobin is converted to red–brown myohaemichromogen
- Maillard browning and caramelisation also contribute to the colour of sterilised meats.

Effects of heating on food

COLOUR

- Chlorophyll is converted to pheophytin and anthocyanins are degraded to brown pigments.
- Iron or tin react with anthocyanins to form a purple pigment, or when colourless leucoanthocyanins form pink anthocyanin in pears and quinces.

Effects of heating on food

COLOUR

- In sterilised milk slight colour changes are due to caramelisation, Maillard browning and changes in the reflectivity of casein micelles.

Effects of heating on food

Flavor and Aroma

- In canned meats there are complex changes, like pyrolysis, deamination and decarboxylation of amino acids, degradation, Maillard reactions and caramelisation of carbohydrates to furfural and hydroxymethylfurfural, and oxidation and decarboxylation of lipids.

Effects of heating on food

Flavor and Aroma

- In fruits and vegetables, changes are due to complex reactions which involve the degradation, recombination and volatilisation of aldehydes, ketones, sugars, lactones, amino acids and organic acids
- Interactions between these components produce more than 600 flavour compounds in ten chemical classes.

Effects of heating on food

Flavour and Aroma

- In milk the development of a cooked flavour is due to denaturation of whey proteins to form hydrogen sulphide and the formation of lactones and methyl ketones from lipids.
- In aseptically sterilised foods the changes are better retained.

Effects of heating on food

Texture or Viscosity

- In canned meats, changes in texture are caused by coagulation and a loss of water holding capacity of proteins, which produces shrinkage and stiffening of muscle tissues.
- Softening is caused by hydrolysis of collagen, solubilisation of the resulting gelatin, and melting and dispersion of fats through the product.

Effects of heating on food

Texture or Viscosity

- Polyphosphates are added to increase tenderness.

Effects of heating on food

Texture or Viscosity

- In fruits and vegetables, softening is caused by hydrolysis of pectic materials, gelatinisation of starches and partial solubilisation of hemicelluloses.
- Calcium salts added increase the firmness of the canned products.
- Different salts are needed for different type of fruits.

Effects of heating on food

Nutritional Value

- Canning causes the hydrolysis of carbohydrates and lipid but the nutritional value is not affected.
- Proteins are coagulated and, in canned meats, losses of amino acids are 10–20%.
- Negligible vitamin losses in aseptically processed milk and lipids, carbohydrates and minerals are virtually unaffected..

Effects of heating on food

NUTRITIONAL VALUE

- Nutrient losses also occur during periods of prolonged storage.

Food Biotechnology

Module #70



Evaporation

- Evaporation, or concentration by boiling, is the partial removal of water from liquid foods by boiling off water vapour.
- It increases the solids content of a food and hence preserves it by a reduction in water activity.
- Evaporation is used to pre-concentrate foods (for example fruit juice, milk and coffee) to reduce their weight.

Evaporation

- During evaporation, sensible heat is transferred from steam to the food, to raise the temperature to its boiling point.
- Latent heat of vaporisation is then supplied by the steam to form bubbles of vapour, which leave the surface of the boiling liquid.

Evaporation

- Heat and mass balances are used to calculate the degree of concentration, energy use and processing times in an evaporator.

Evaporation

Factors influencing the rate of heat transfer:

- Temperature difference between the steam and boiling liquid
- Deposits on heat transfer surfaces.
- Boundary films

Evaporation

Factors influencing the economics of evaporation:

- Vapour recompression
- Preheating
- Multiple effect evaporation

Evaporation

Evaporators consist of:

- a heat exchanger (termed a *calandria*) which transfers heat from steam to the food
- a means of separating the vapours produced
- a mechanical or steam ejector vacuum pump. Mechanical pumps have lower operating costs but higher capital costs than steam ejector pumps.

Evaporation

Natural circulation evaporators

- *Open- or closed-pan evaporators* are hemispherical pans, heated directly by gas or electrical resistance.
- For vacuum operation, they are fitted with a lid. A stirrer or paddle is used to increase the rate of heat transfer and to prevent food from burning onto the pan.

Evaporation

Natural circulation evaporators

- *Long-tube evaporators* consist of a vertical bundle of tubes, each up to 5 cm in diameter, contained within a steam shell 3–15 m high.
- Liquor is heated almost to boiling point before entering the evaporator.
- It is then further heated inside the tubes and boiling commences.

Evaporation

Natural circulation evaporators

- The *short-tube evaporator* is an example of a tube-and-shell heat exchanger, also used in pasteurisation and heat sterilisation.
- It consists of a vessel (or shell) which contains a vertical, or less commonly horizontal bundle of tubes.

Evaporation

Forced circulation evaporators:

In forced-circulation evaporators a pump or scraper assembly moves the liquor, usually in thin layers, and thus maintains high heat transfer rates and short residence times.

- **Plate evaporators.**
- **The expanding flow evaporators.**

Evaporation

Effects on Foods

- Aroma compounds that are more volatile than water are thus lost during evaporation.
- This reduces the sensory characteristics of most concentrates;
- In fruit juices this results in a loss of flavour, although in some foods the loss of unpleasant volatiles improves the product quality.

Evaporation

Effects on Foods

- Some volatiles can be recovered and retained in the product:
- by volatile recovery by vapour condensation and fractional distillation
- by stripping volatiles from the feed liquor with inert gas and adding them back after evaporation

Food Biotechnology

Module #71



Distillation

- Distillation is separating the components or substances from a liquid mixture by using selective boiling and condensation.
- May be complete or partial separation that increases the concentration of selected components in the mixture.

Distillation

- Routine unit operation in chemical and food industry.
- Alcoholic spirits and
- Separation of volatile flavour and aroma compounds (for example, production of essential oils by steam distillation).

Distillation

- Some food contains components having different degrees of volatility.
- When heated, that have a higher vapour pressure (more volatile components) are separated first.
- These are termed the 'distillate' and components that have a lower volatility are termed 'bottoms' or residues.

Distillation

- Some industries perform batch distillation (in pots) for whiskey and other spirit distilleries.
- Most industrial distillation operations use more economical continuous distillation columns

Distillation

To enhance the separation of some components:

- a proportion of the distillate is added back to the top of the column called (reflux)
- a portion of the bottoms is vapourised in a reboiler and added to the bottom of the column

Distillation

Use of spinning cone column:

- To remove the volatile components from liquids
- It is used to recover flavours from beer, coffee, tomato products and fruit juices.
- To produce low-alcohol wines and beers
- To remove off-flavours.

Distillation

Equipment:

- A column containing a series of rotating inverted cones, which are intermeshed with stationary cones attached to the column wall.
- Steam or nitrogen is supplied to the base of the column and the feed liquor enters at the top.

Distillation

Equipment:

- Thin turbulent films are produced over the large surface area of the cones and rapid separation takes place.
- The gas passes out of the top of the column and volatile aroma compound are condensed and collected.

Distillation

Equipment:

- Separation is achieved by mechanical energy, rather than heat, there is less damage to flavours and lower energy consumption.

end

Food Biotechnology

Module #72



Dehydration

- ‘The application of heat under controlled conditions to remove the majority of the water by evaporation’ (or in the case of freeze drying by sublimation).
- Purpose of dehydration is to extend the shelf life of foods.

Dehydration

Large number of factors control the rate at which foods dry, which can be grouped into the following categories:

- those related to the processing conditions
- those related to the nature of the food
- those related to the drier design.

Dehydration

Drying using heated air:

Heat from drying air is absorbed by food and provides the latent heat needed to evaporate water from the surface.

There are three inter-related factors controling capacity of air to remove moisture:

- water vapour already carried by the air
- the air temperature
- the amount of air that passes over the food.

Dehydration

Drying using heated surfaces:

- Heat is conducted from the hot surface, through the food, and moisture is evaporated from the exposed surface.
- Slurries of food are deposited on a heated steel drum.

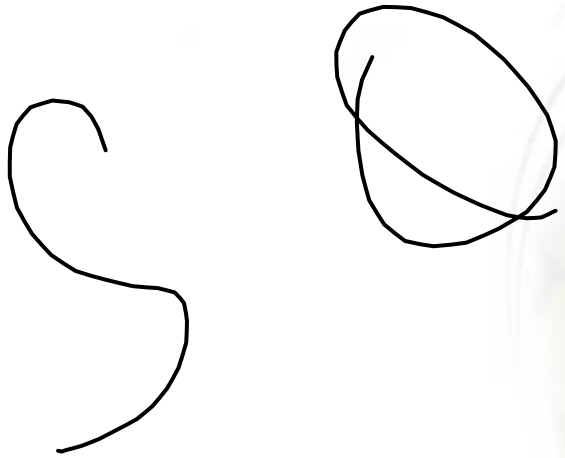
Dehydration



Equipment: Hot air driers:

- Bin driers
- Cabinet driers(tray driers)
- Tunnel driers
- Conveyor driers(belt driers)
- Fluidised-bed driers
- Klin driers
- Pneumatic driers
- Rotary driers
- Spray driers
- Sun and solar drying

Dehydration



Equipment: Heated surface driers:

- Drum driers(roller driers)
- Vacuum band and vacuum shelf driers

Dehydration

END

Effects on food:

aim of improved drying technologies is to minimise undesirable changes while maximising process efficiency.

- Texture
- Loss of flavour and aroma
- Changes in colour
- Nutritional value

Food Biotechnology

Module #73



Use of Low Temperature

Temperature of a food is reduced to between 1°C and 8°C.

- to reduce the rate of biochemical and microbiological changes, to extend the shelf life foods.
- It causes minimal changes to sensory characteristics and nutritional properties of foods

Use of Low Temperature

FRESH FOODS

Chilling reduces the rate of enzymic and microbiological change and retards respiration of fresh foods. The factors that control the shelf life of fresh crops in chill storage include:

- the type of food and variety or cultivar
- the part of the crop selected (the fastest growing parts have the highest metabolic rates

Use of Low Temperature

FRESH FOODS

- the condition of the food at harvest.
- the temperature of harvest, storage, distribution and retail display
- the relative humidity of the storage atmosphere, which influences dehydration losses.

Use of Low Temperature

PROCESSED FOODS

There are four broad categories of micro-organism, based on the temperature range for growth

1. thermophilic (minimum: 30–40 °C, optimum: 55–65 °C)
2. mesophilic (minimum: 5–10 °C, optimum: 30–40 °C)
3. psychrotrophic (minimum: <0–5°C, optimum: 20–30 °C)
4. psychrophilic (minimum:

Use of Low Temperature

COOK-CHILL SYSTEMS

Individual foods (for example sliced roast meats) or complete meals are produced by cook–chill or cook–pasteurise–chill processes.

The range of chilled foods can be characterised by the class of microbial risk that they pose to consumers as follows:

Use of Low Temperature

COOK-CHILL SYSTEMS

CLASS 1

Foods containing raw or uncooked ingredients, such as salad or cheese as ready-to-eat (RTE) foods (also includes chill-stable raw foods, such as meat, fish, etc.)

CLASS 2

Products made from a mixture of cooked and low risk raw ingredients

Use of Low Temperature

COOK-CHILL SYSTEMS

CLASS 3

Cooked products that are then packaged

CLASS 4

Products that are cooked after packaging, including ready-to-eat-products- for-extended-durability (REPFEDs) having a shelf life of 40+ days.

Use of Low Temperature

MECHANICAL REFRIGERATORS:

Mechanical refrigerators have four basic elements:

- an evaporator,
- a compressor,
- a condenser and
- an expansion valve

constructed from copper as the low thermal conductivity allows high rates of heat transfer and high thermal efficiencies.

Use of Low Temperature

Cryogenic Refrigerators

- A cryogen is a refrigerant that changes phase by absorbing latent heat to cool the food
- Cryogenic chillers use solid carbon dioxide, liquid carbon dioxide or liquid nitrogen.
- Solid carbon dioxide removes latent heat of sublimation and liquid cryogens remove latent heat of vaporization.

Use of Low Temperature

Control of Storage Conditions

- In all stores it is important to maintain an adequate circulation of air using fans, to control the temperature, relative humidity or atmospheric composition.
- Foods are therefore stacked in ways that enable air to circulate freely around all sides.

Use of Low Temperature

Effect on Foods

- Chilling foods to their correct storage temperature causes little or no reduction in the eating quality or nutritional properties of food.
- The most significant effect of chilling on the sensory characteristics of processed foods is hardening due to solidification of fats and oils.

Food Biotechnology

Module #74



Use of food preservatives

- compounds that help slow or prevent bacterial growth in a wide range of products, including foods, medicines, and personal care products.
- These compounds can be natural or synthetic.
- Preservatives play important roles in many products people use every.

Use of food preservatives

ANTIOXIDANTS :

Used to prevent, delay, or minimize the oxidation of the food to which they are added.

NUTRIENTS:

Losses of fractional amounts of nutrients during processing lead to the practice of adding a number of nutrients to the foods.

Flavorings :

Flavourings can be natural or synthetic, that are added to foods to produce flavors.

Flavor Enhancers :

intensify existing flavors. Monosodium glutamate (MSG) is one of the best known and most widely used flavor enhancers.

Use of food preservatives

Acidulants :

lower the pH of any food. Acidulants also enhance desirable flavors such as in pickled products.

Alkaline Compounds:

Compounds that raise the pH. Such as sodium hydroxide or potassium hydroxide.

Use of food preservatives

Sweeteners:

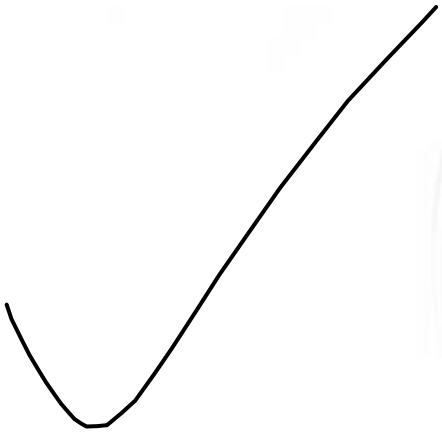
1. Fructose
2. Molasses
3. Honey
4. Maple sugar
5. Lactose
6. Maltose
7. Xylitol
8. Sorbitol
9. Mannitol
10. Aspartame
11. Saccharin
12. Starches

Use of food preservatives

ENZYMES:

1. Invertase
2. Pectinase
3. Cellulase
4. Proteases
5. Lipases
6. Glucose Oxidase
7. Calatase

Use of food preservatives



ENZYMES:

1. Invertase
2. Pectinase
3. Cellulase
4. Proteases
5. Lipases
6. Glucose Oxidase
7. Calatase

Use of food preservatives

Sequestrants:

Sequestrants combine with metals, forming complexes with them and making them unavailable for other reactions.

Polyhydric Alcohols

Used to improve texture and moisture retention because of their affinity for water. Many polyols are present in foods naturally, glycerine (glycerol) being the predominant one.

Use of food preservatives

Surface Active Agents:

affect the physical force at the interface of surfaces. Commonly called surfactants, they are present in all natural foods.

Leaving Agents :

Leavening agents are used to enhance the rising of dough in the manufacture of baked products.

Use of food preservatives

Ionizing Radiation :

Irradiation does not leave a residue in food and it does not make it radioactive. The levels of irradiation allowed in food processing do not induce measurable radioactivity.

Chemical Preservatives :

Added to create conditions unfavourable for food spoilage.

Use of food preservatives

COLORANTS :

Many colorants (compounds that add colors to foods) are natural, and these include the yellow from the annatto seed; green from chlorophyll; orange from carotene; brown from burnt sugar; and red from beets, tomatoes, and the cochineal insect.

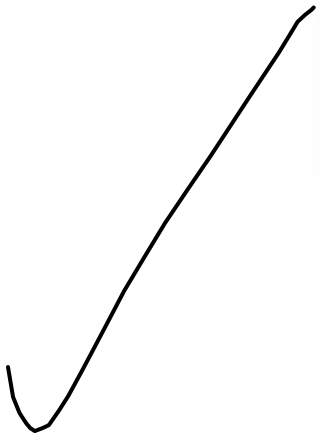
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Food Biotechnology

Module #75



Use of high pressure and ultrasound



ULTRASOUND:

It is used in food industry for many purposes including analysis methods and food processing such as:

- freezing,
- cutting,
- drying,
- Tempering,
- homogenization, degassing,
- antifoaming, filtration and extraction.

Use of high pressure and ultrasound

ULTRASOUND:

Ultrasound can be used as a promoter or alternative to food processing.

Use of high pressure and ultrasound

ULTRASOUND :

Ultrasound in food preservation causes:

a) Microbial inactivation
b) Enzyme inactivation

Use of high pressure and ultrasound

EFFECT ON FOOD

- Ultrasound is reported to have minimal effect on the quality of fruit juices.
- Such as oranges, guava, blackberry, and strawberry juices.
- Fruit juice colour is a primary factor in keeping the quality .

Use of high pressure and ultrasound

HIGH PRESSURE:

High pressure processing (HPP) has emerged as a commercially viable food manufacturing tool that satisfies consumers' demand for mildly processed, convenient, fresh-tasting foods with minimal to no preservatives.

Use of high pressure and ultrasound

HIGH PRESSURE:

Pressure treatment, with or without heat, inactivates pathogenic and spoilage bacteria, yeast, mould, viruses, and also spores and extends shelf life.

Use of high pressure and ultrasound

HIGH PRESSURE:

- applied at short periods of time (20minutes).
- Inactivation of Parasites,
- Plant cells.
- Vegetative micro-organisms. Some fungal spores.
- Many food borne viruses. Enzymes are selectively inactivated.
- Macro molecules can change conformation.

Use of high pressure and ultrasound

END

HIGH PRESSURE:

- Small molecules (eg: flavors) are generally unaffected
- High pressure is instantaneously and uniformly applied to the sample.
- Compression is fully reversible.

Food Biotechnology

Module #76



Electricity, radiation and microwaves

- Dielectric (microwave and radio frequency) energy and infrared (or radiant) energy are two forms of electromagnetic energy
- They are both transmitted as waves, which penetrate food and are then absorbed and converted to heat.

Electricity, radiation and microwaves

Direct methods:

- dielectric and
- ohmic heating

In which heat is generated within the product

Indirect method:

- Infrared

heat that is generated externally.

Electricity, radiation and microwaves

Microwave

consists of

- a microwave generator (*magnetron)*,
- aluminium tubes or w*ave guides*, and
- a metal chamber for batch operation, or a tunnel fitted with a conveyor belt for continuous operation.

Electricity, radiation and microwaves

Radio Frequency Heating:

similar principle to microwave heating, but at lower frequencies.

Food is passed between electrodes and a radio frequency voltage is applied across the electrodes.

Electricity, radiation and microwaves

Applications:

high rates of heating absence of surface changes.

The most important applications are

- Thawing and tempering
- Dehydration and baking.
- Other applications

(Pasteurisations and sterilization)

Electricity, radiation and microwaves

Effect On Foods

- no direct effect on micro-organisms, in contrast with ionising radiation.
- In pasteurisation and blanching applications result in reduced losses of heat-sensitive nutrients.

Electricity, radiation and microwaves

OHMIC HEATING:

'resistance heating' or 'electroheating', involves

- an alternating electric current is passed through a food, and
- the electrical resistance of the food causes the power to be translated directly into heat.

Electricity, radiation and microwaves

OHMIC HEATING:

can be used for UHT sterilisation of foods, and especially those that contain large particles (up to 2.5 cm) that are difficult to sterilise by other means.

Electricity, radiation and microwaves

OHMIC HEATING:

Commercially available:

- aseptic processing of high added-value ready meals, stored at ambient temperature
- pasteurisation of particulate foods for hot filling
- pre-heating for canning
- Value added prepared meals, distributed at chill temperatures

Electricity, radiation and microwaves

END

INFRARED HEATING:

The rate of heat transfer depends on:

- the surface temperatures of the heating and receiving materials
- the surface properties of the two materials
- the shapes of the emitting and receiving bodies.

Food Biotechnology

Module #77



Detection of yeast and moulds in dairy and bakery products

- Procedure is applicable to the enumeration of viable yeasts and moulds in foods and food ingredients
- It may also be used to material scraped from food plant equipment and the manufacturing environment.

Detection of yeast and moulds in dairy and bakery products

PRINCIPLE

A medium, containing

a. adequate nutrients for yeasts and moulds
b. Antibiotics is inoculated with a given quantity of the product or with scrapings from equipment or the manufacturing environment.

Detection of yeast and moulds in dairy and bakery products

PRINCIPLE:

incubated at 22-25°C for 3-5 days.

Colonies appearing on the medium

counted and/or examined.

Detection of yeast and moulds in dairy and bakery products

METHOD:

1. Preparation of medium
2. Preparation of dilutions
3. Plating
4. Incubation
5. Counting colonies and examining growth
6. Differentiation of colonies from interfering particles
7. Recording results

Detection of yeast and moulds in dairy and bakery products

Materials and Equipment:

Enumeration of xerophilic yeasts and moulds in grains, flours, nuts, and spices

- Dichloran-glycerol DG 18 agar (DG-18)

Enumeration of xerophilic yeasts and moulds in jams, jellies, and dried fruits

- 20% sucrose (diluent additive for osmophiles,
- Malt extract agar with50% (w/w) sucrose

Detection of yeast and moulds in dairy and bakery products

PLATING:

- surface spread-plate technique rather than with pour plates for maximal exposure of the cells to atmospheric oxygen and to avoids heat stress from molten agar.
- Agar spread plates should be dried overnight before being inoculated.

Detection of yeast and moulds in dairy and bakery products

Counting Colonies and Examining Growth:

- Colonies are counted according to their colonial morphology
- Microscopic examination with crystal violet stained smears may be used to distinguish yeast colonies from bacteria.

Detection of yeast and moulds in dairy and bakery products

END

Differentiation of Colonies from Interfering Particles:

- Food particles interfere with the enumeration of colonies.
- one extra plate of each dilution containing interfering particles, and holding it under refrigeration as a control for comparison during counting.

Food Biotechnology

Module #78



Detection of yeast and moulds in dairy and bakery products

Enumeration of Yeasts & Moulds in Foods:

These agars are suitable for foods where the a_w is above 0.95, such as fresh foods (fruit, vegetables, meat and dairy).

Detection of yeast and moulds in dairy and bakery products

1) Dichloran rose bengal chloramphenicol agar (DRBC)
2) Plate count agar with chloramphenicol (PCA-C)
3) Potato dextrose agar with chloramphenicol (PDA-C)
4) Potato dextrose salt agar with chloramphenicol (PDSA-C) (for analysis of 'spreader' moulds)

Detection of yeast and moulds in dairy and bakery products

OTHER:

- Peptone water (0.1%) (PW)
- 2% sodium citrate tempered to 45°C (diluent for high fat foods, such as cheese) (optional)
- 1N HCl and 1N NaOH
- Gram stain solutions
- Stomacher, blender or equivalent

Detection of yeast and moulds in dairy and bakery products

OTHER:

- pH meter
- Light microscope
- Colony counting device (optional)
- Incubator (darkened) capable of maintaining 22 to 25°C, 55°C
- water bath

Detection of yeast and moulds in dairy and bakery products

INCUBATION:

- Incubate plates in an upright position at 22 to 25°C for 3-5 days.
- Incubate plates in the dark.
- Count colonies on plates after 5 days.
- Or third day and if mould colonies are numerous,
- count them and then count again on the fifth day, if possible.

Detection of yeast and moulds in dairy and bakery products

INCUBATION:

- Handle the plates as little as possible when counting on day 3 so spores will not be dislodged, which may result in secondary growth.

Detection of yeast and moulds in dairy and bakery products

END

Recording Results:

- average count (arithmetic mean) is calculated
- To compute the yeast and mould count, use the formula:

N = A x D,

where N is the no. of colonies /g (mL) of product,

A is the average count/ plate, and

D is the dilution factor

Food Biotechnology

Module #79



Fermented foods

General Method of Production :

- exposing the raw or starting food materials to given conditions.
- Microorganisms grow and produce end-products with desirable acceptance qualities, many of which are attributed to the metabolic end products.

Fermented foods

Microorganisms Used:

- Variety of species and strains of bacteria, yeasts, and moulds are associated with fermentation of foods.
- Single or mixed strains or species

Fermented foods

Maximum growth of a desirable microorganism and optimum fermentation rate are dependent on environmental parameters such as:

- nutrients,
- temperature of incubation,
- oxidation-reduction potential, and
- pH.

Fermented foods

PROCESS:

Foods can be fermented in three different ways, based on the sources of the desirable microorganisms:

- natural fermentation,
- back slopping, and
- controlled fermentation.

Fermented foods

Fermented Dairy Products:

Fermented dairy products can be broadly divided into two groups:

- fermented milk products and
- cheeses.

Fermented foods

- In fermented milk products, all the constituents of the milk are retained in the final products, with the exception of those partially metabolized by the bacteria.
- In cheeses, a large portion of milk constituents is removed in whey to obtain the final products.

Fermented foods

Fermented Meat Products:

Fermented meat products are produced by first mixing

- meat,
- fat,
- salt,
- sugar,
- curing agents, and
- spices;

Filling the mixture in a casing; and fermenting it naturally or selected starter-culture bacteria.

Fermented foods

Fermented Vegetable Products:

- Almost all vegetables can be fermented through natural processes LA bacteria
- originated in the early years of human civilization and even now is widely used by many cultures.

Fermented foods

END

Fermented Vegetable Products:

sauerkraut (from cabbage), olives, cucumbers, carrots, celery, beans, peas, corn, okra, tomatoes, cauliflower, peppers, onions, citron, beets, turnips, radishes, chard, Brussels sprouts, and their blends.

Food Biotechnology

Module #80



Microbiology of fermented products

Cultured Buttermilk Fermentation:

- partially skim milk through controlled fermentation with starter cultures.
- It should have a pleasant acid taste and aroma (from diacetyl), with slight effervescence (from $C0_2$).
- It should have white colour with smooth, thick body and should pour easily.

Microbiology of fermented products

Microbiology

- *Lac. lactis* ssp. *lactis* or *cremoris* is used for acid and *Leu. mesenteroides* ssp. *cremoris* for diacetyl and $C0_2$.

Microbiology of fermented products

YOGHURT

- coagulation of milk (skim, or full fat) by starter-culture bacteria.
- sharp acid taste with a flavor similar to walnuts and a smooth mouth feel.
- The flavor is due to the combined effects of acetaldehyde, lactate, diacetyl and acetate, but 90% of the flavor is due to acetaldehyde.

Microbiology of fermented products

Cottage Cheese:

- from low-fat or skim milk and has a soft texture with Ca. 80% moisture.
- It is unripened and has a buttery aroma due to diacetyl (along with lactic acid and little acetaldehyde).
- Mixed strains of *Lac. lactis* ssp. *cremoris* and *lactis* are predominantly used for acid.

Microbiology of fermented products

- made from whole milk. contains less than 39% moisture, 48% fat
- Starters are selected mixed strains of *Lac. lactis* ssp. *cremoris* or *lactis. Leuconostoc,* which may be added for flavor.
- Starters can be used as frozen concentrates.

Microbiology of fermented products

Swiss Cheese:

- from partially skimmed milk (cow's milk) and coagulated with acid and rennin contains Ca. 41% moisture and 43% fat
- The cheese should have uniformly distributed medium-sized eyes (openings).It has a sweet taste, due to proline, and a nutty flavor.

Microbiology of fermented products

Microbiology of Swiss Cheese:

- Starters are *Str. thermophilus* and *Lab. helveticus* as primary for acid and *Propionibacterium* sp. as secondary for eye formation, taste, and flavor.

Microbiology of fermented products

Microbiology Of Semidry Sausages:

- The average composition is Ca. 30% fat, 20% protein. 3% minerals (salts), and 47% water.
- For high temperature and low pH, *Pediococcus acidilactici* strains are preferred;
- For low temperature and high pH, *Lab. plantarum* strains are preferred.

Microbiology of fermented products

- flavor imparted by the combined effect of lactate, acetate, and diacetyl, and some breakdown components from proteolysis and lipolysis.
- The use of spices also contributes to the flavor.
- Those containing nitrite have a pinkish color in contrast to the grayish color in products without it.

Microbiology of fermented products

Microbiology Of Sauerkraut:

- fermenting shredded cabbage.
- has a sour taste with a clean acid flavor.
- The raw material has a large number of undesirable organisms and a small population of lactic acid bacteria «1%).

Microbiology of fermented products

- Among the lactic acid bacteria, most are *Lactococcus* spp. and *Leuconostoc* spp and
- a small fraction is *Lactobacillus* spp. and *Pediococcus* spp.

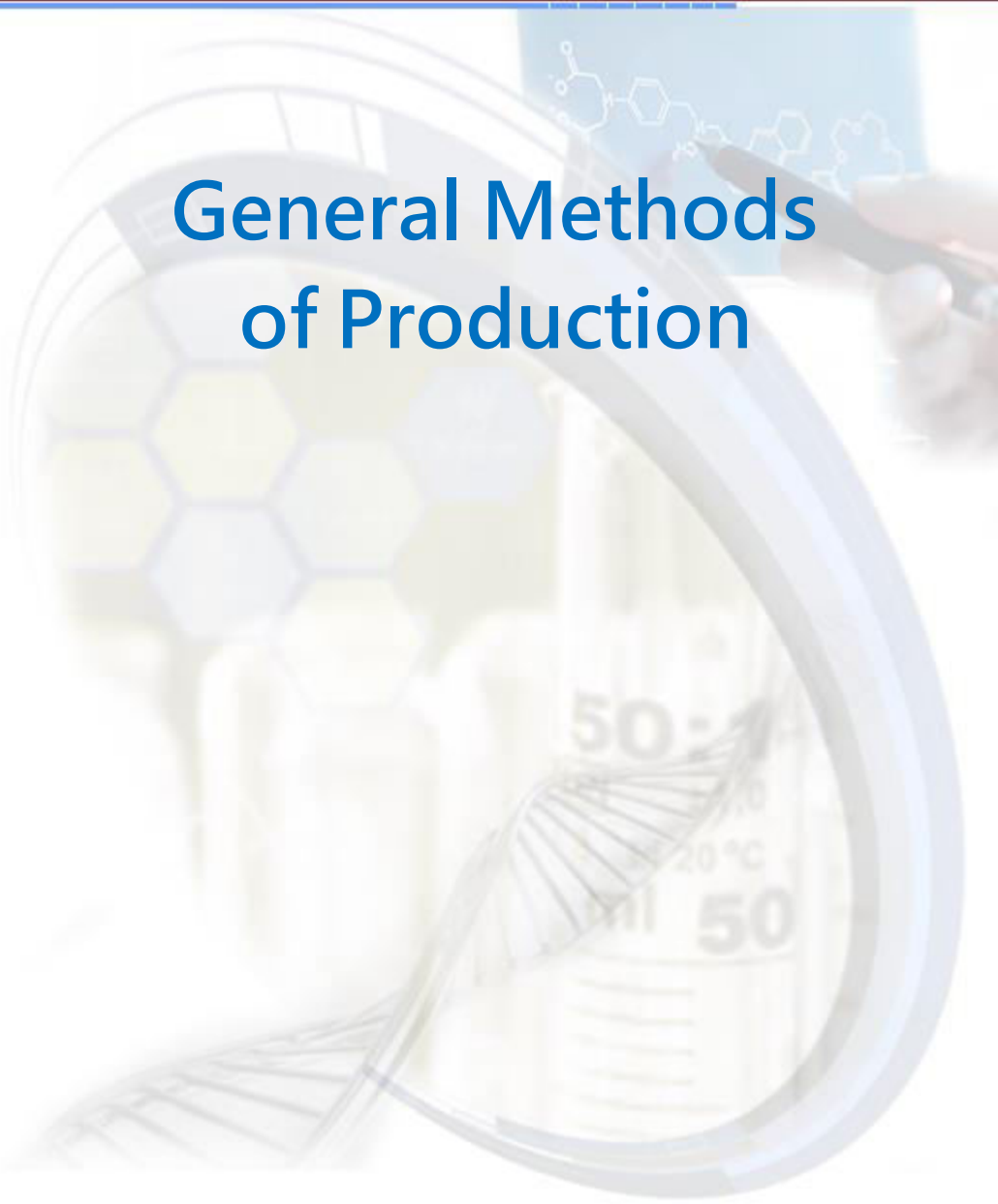
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Food Biotechnology

Module :- 81

Food Biotechnology

General Methods of Production



General Methods of Production

Fermentation :-

raw or starting food materials exposed to conditions that favor growth and metabolism of desirable microorganisms

→ Products that are made via fermentation are called fermented products

e.g. Pickles , Vinegar etc

General Methods of Production

→ Process of fermentation based on two conditions

1) Metabolic activities of micro-organisms

2) Storage condition

→ Type of micro-organisms selected for fermentation is crucial for fermentation process

General Methods of Production

Fermentation Process

Foods can be fermented in three different ways, based on the sources of the desirable microorganisms:

1) Natural Fermentation
2) Back Slopping
3) Controlled Fermentation

General Methods of Production

Natural Fermentation :-

both desired and non desired organisms are involved .

→ Rapid growth of desired one

→ Suppress the growth of undesired one

Benefits :-

Special Aroma

Drawback :-

Non-consistency in the organoleptic properties.

General Methods of Production

2) Back Slopping :-

conditions are set to facilitate the growth of the microorganisms coming from the previous product. This is still practiced in the production of many ethnic products in small volumes e.g. Yoghurt from Milk

General Methods of Production

END

3) Controlled Fermentation

starting materials (may be heat treated) are inoculated with a high population (10^6 cells/ml or more) of a pure culture of single or mixed strains or species of starter culture.

→ Larger volume can be fermented

→ Consistency

Food Biotechnology

Module :- 82

Food Biotechnology

Fermentation Process

Fermentation Process

As the desirable microorganisms grow in specific conditions on selected foods, they utilize some nutrients and produce some end products.

Fermentation Process

1) Alcoholic Fermentation :-
(Alcohol end product)

2) Acetic Acid Fermentation
(Acetic acid end product)

3) Lactic Acid Fermentation
(Lactic Acid end product)

Fermentation Process

General process

Substrate (Vegetables, Meat , Milk etc.) + Starter Culture (Bacteria, Yeast , mold) + optimum conditions (Time, Temperature etc.) -→ End products (yoghurt , Vinegar , Alcohol , acetic acid etc.)

Fermentation Process

Alcoholic Fermentation:

In this fermentation Alcohol is produced as a end product by the activity of micro-organisms (yeast)

→ Mostly used by Bakery industry for special aroma

Fermentation Process

Acetic Acid Fermentation :-

In this fermentation Acetic acid is produced as end product by the activity of Acetobacter bacteria on sugary / alcoholic food stuff

→ Used for the production of vinegar

Fermentation Process

END

Lactic Acid Fermentation

In this fermentation glucose and other six Carbon sugars are converted into cellular energy and metabolite lactate via Lactic acid producing bacteria as starter culture

→ Mostly used for the production of yoghurt from milk

Food Biotechnology

Module :- 83

Food Biotechnology

Fermented Dairy Products

Fermented Dairy Products

Huge variety of products produced globally

A few are produced by

→Controlled fermentation, and the microbial types and their respective contributions are known.

Fermented Dairy Products

→ In many others, fermented either naturally or by back slopping, the microbial profiles and their contribution are not exactly known.

→ Many types of lactic acid bacteria and some yeasts are found to predominate microbial flora in these products.

Fermented Dairy Products

1) Butter milk.:-

Made with Lactococcus species without or with Leuconostoc cremoris; . Some can have a ropy variant of Lactococcus species (landfill in Norway) or mold (Geotrichum candidum in villi in Finland)

Fermented Dairy Products

Yogurt:

Streptococcus thermophiles and Lactobacillus delbrueckii subsp. bulgaricus;

some with added Lab. acidophilus, casein, rhamnosus, and Bifidobacterium spp.;

Or

Lactococcus species and Lab. plantarum and lactose-fermentating yeasts (dahi)

Fermented Dairy Products

3)Acidophilus Milk.

Made with Lab. acidophilus.

4. Bifidus milk

Made with Bifidobacterium spp. **5.**

Yakult.

Made with Lab. casei; may contain Bifidobacterium spp.

Fermented Dairy Products

END

6. Kefir.

Made from Lab. kefir (several species of yeasts along with Leuconostoc, Lactobacillus, and Lactococcus spp.).

7. Kumiss.

Made from Lab. delbrueckii subsp. buigaricus and yeasts

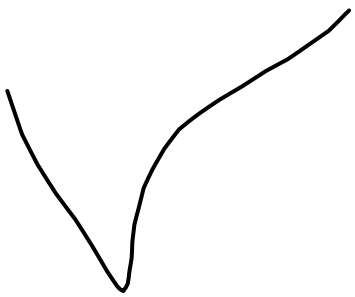
Food Biotechnology

Module :- 84

Food Biotechnology

Microbiology of Yogurt

Microbiology of Yogurt



Yogurt Fermentation Steps:-

1) Processed Milk
2) Starter Culture
3) Growth of Micro-organisms
4) Metabolites production
5) Flavor Production
6) Formate Production
7) Slime Formation (Glycan)
8) Proteolysis

Microbiology of Yogurt

1) Milk Treatment :-

Firstly milk is treated

→ Heat Treated (Pasteurize , Sterilize)

→ Homogenized

→ Standardized

Microbiology of Yogurt



2) Starters (Controlled Fermentation)

Frozen concentrates or direct vat set starters can be used.

Normally, *Lab delbrueckii ssp. bulgaricus and Str. thermophilus* are used.

Microbiology of Yogurt

other species may be combined, such as *Lab. acidophilus* and *Bifidobacterium spp., Lab. rhamnosus, or Lab. casei*.. For a good product, the two starter species should be added at a *Streptococcus Lactobacillus* cell ratio of 1:1; in the final product, the ratio should not exceed 3:2.

Microbiology of Yogurt

3)Growth of Micro-organisms :-

fermentation is conducted at ca. 110°F (43.3°C). both acid and flavor compounds are produced at the desired level If the temperature is raised above 110°F, the Lactobacillus sp. predominates, causing more acid and less flavor production

Microbiology of Yogurt

4)Metabolites Production :-

Lactic Acid is produced by the action of micro-organisms on lactose

Microbiology of Yogurt

5)Flavor Development :-

The major flavor compound in yogurt is acetaldehyde (25 ppm), with some diacetyl (0.5 ppm) and acetate. Acetaldehyde is produced by Lactobacillus sp.

Microbiology of Yogurt

6) Formate Production :-

Formate (necessary for Lactobacillus growth) is produced by Str. thermophilus from pyruvate by the action of formate lyase.

Microbiology of Yogurt

7) Slime Formation (Glycan)

B- galactosidase in some strains of *Str. thermopilus* polymerizes glucose to produce oligosaccharides and glycan, which may give a viscous texture to yogurt.

Microbiology of Yogurt

8) Proteolysis :-

Proteinase (LAB) reacts with milk protein and make curdles which leads to yoghurt

Microbiology of Yogurt

Microbial Problems :-

In plain yogurt. flavor problems can be associated with the concentrations of acetaldehyde. A low concentration gives a chalky and sour flavor, and too much acetaldehyde can give a green flavor.

Microbiology of Yogurt

END

Similarly, too much diacetyl gives a buttery aroma.

Too much acid production during storage causes a sour taste.

Proteolysis and accumulation of bitter peptides during storage are associated with bitter flavor

Food Biotechnology

Module 85 :-

Food Biotechnology

Fermented Product (Cheese)

Fermented Product (Cheese)

Introduction :-

Cheese are made by coagulating the casein in milk with lactic acid produced by lactic acid bacteria, without or with the enzyme rennin, followed by collecting the casein for further processing, which may include ripening.

Fermented Product (Cheese)

Types of cheese:-

Cheese varieties have been grouped in different ways. Examples of several varieties based on starter cultures used and some important secondary flora are listed here

1) Unripened cheese

→ Soft

2) Ripened cheese

→ Soft, Semi-hard,Hard

Fermented Product (Cheese)

1) Unripened Cheese :-

Soft

- Cottage cheese with starters *Lac. lactis ssp. lactis* and cremoris and *Leuconostoc mesenteroides ssp. cremoris*.
- Mozzarella cheese with starters *Stl: thermophilus* and *Lab. delbrueckii ssp. bulgaricus*.

Fermented Product (Cheese)

2) Ripened Cheese :-

→Soft

• Brie with *starter Lac. lactis ssp.; Penicillium sp*. and yeasts are secondary flora.

→Semi - hard

• Gouda with starters *Lac. lactis ssp. and Leuconostoc spp*.; dairy Propionibacterium may be secondary flora. • Blue cheese with starter *Lac. lactis ssp., Leuconostoc spp.; Penicillium roquefortii*, yeasts, micrococci are secondary flora

Fermented Product (Cheese)

Hard cheese :-

- Cheddar cheese with starters *Lac. lactis ssp.*; some *lactobacilli* and *pediococci* (and probably *enterococci*) are secondary (or associative) flora.

Fermented Product (Cheese)

Cottage cheese :-

Cottage cheese is made from low-fat or skim milk and has a soft texture with ca. 80% moisture. It is unripened and has a buttery aroma due to diacetyl (along with lactic acid and little

Fermented Product (Cheese)

END

Production of cottage cheese :-

1. Pasteurized, cooled to 70°F (22.2°C), starter added, and incubated for 12 h at pH 4.7.
2. Firm curd set, cut in cubes, and cooked at 125°F (51.7°C) for 50 min or more.
3. Whey drained off, stirred to remove more to get dry curd.
4. Salted, creamed, and preservative added.
5. Packaged and refrigerated

Food Biotechnology

Module 86



Food Biotechnology

Fermented Meat Products

Fermented Meat Products

Meat fermentation probably originated in the Mediterranean countries and later spread to European countries and North America. In the United States, semidry sausages are most popular, although some dry sausages are also produced.

Fermented Meat Products

Fermentation of Meat :-

1)Fermented meat products are produced by first mixing meat, fat, salt, sugar, curing agents, and spices.

2) Filling the mixture in a casing; and fermenting it either naturally or by adding (during mixing) selected starter-culture bacteria

Fermented Meat Products

3) Acids produced by the starters during fermentation and the curing agents used help control the growth of pathogenic and spoilage bacteria that might be present in the meat.

Fermented Meat Products

4) Depending on the type, the fermented products may be dried to reduce Aw or smoked or heated to ensure the safety and shelf life of the products

Fermented Meat Products

Types of Fermented Meat Products :-

1) Sausages

→ Semi-dry (Heated before consumption)

→Dry (Cooked before consumption to reduce Aw)

Fermented Meat Products

Types of semi-dry and dry sausages :-

→Semidry and dry sausages include pepperoni, Genoa salami, hard salami, summer sausage, beef sticks, beef logs, Italian salami etc.

→Most made with beef and pork, but recently meat from chicken and turkey is also used.

Fermented Meat Products

END

Microbiology of salami :-

→The average composition is ca. 30% fat, 20% protein. 3% minerals (salts), and 47% water.

→ They have a tangy taste with a desirable flavor imparted by the combined effect of lactate, acetate, and diacetyl, and some breakdown of protein by proteolysis enzyme

Food Biotechnology

Module 87

Food Biotechnology

Microbiology of Semi-dry Sausages

Microbiology of Semi-dry Sausage

Sausages :-

It's a cylindrical meat product usually made from ground meat ,often pork , beef along with salt, spices and other flavorings and breadcrumbs encased in a skin traditionally made from intestine

Microbiology of Semi-dry Sausage

Starters selection based on pH and Temperature for semi-dry sausages :-

→ For high temperature and low pH, *Pediococcus acidilactici* strains are used

→ For low temperature and high pH, *Lab. plantarum* strains are preferred

Microbiology of Semi-dry Sausage

raw meat used may contain pathogens and spoilage bacteria, it is extremely important that starter culture grows rapidly and produces acid in large amounts to reduce pH from the initial 5.7 to ca. 5.3 very quickly to retard their growth.

Microbiology of Semi-dry Sausage

Microbial Problems :-

→Slow acid production can be a serious problem if the starters used are not metabolically active or have lower numbers of viable cells or due to other factors such as low glucose and high salts in the mix.

Microbiology of Semi-dry Sausage

END

→Sour or no flavor can occur if the starter, especially *Ped. acidilactici*, grows very rapidly and reduces the pH to below 4.5.

→Gas formation can occur because of growth of *Leuconostoc spp*. during fermentation and during storage in vacuum packages

Food Biotechnology

Module 88 :-

Food Biotechnology

Fermented Vegetable Products

Fermented Vegetable Products

→Vegetable fermentation since early human civilization and even now is widely used now a days.

→ Almost all vegetables can be fermented through natural processes, because they harbor many types of lactic acid bacteria

e.g. sauerkraut , reddish, turnips , beets , beans, peas , corn etc.

Fermented Vegetable Products

Microbiology:

→ Fermented vegetable products have Sour taste with clean acid flavor

2) Processing :-

→ Wash, Clean and shred the vegetables

→ Put in a jar and air tight it and layered with salt (Anaerobic conditions for micro-organisms)

→ Ferment it for 2-3 months (18°C)

Fermented Vegetable Products

3) Starter (Natural) and growth :-

The raw material has a large number of undesirable organisms and a small population of lactic acid bacteria «1 %). Among the lactic acid bacteria, most are *Lactococcus spp.* and *Leuconostoc spp*.

FERMENTED VEGETABLE PRODUCTS

END

Microbial Problems :-

Off-flavor, soft texture, and discoloration of vegetables can occur by growth of molds and yeasts when air is not completely excluded.

Food Biotechnology

Module 86



Food Biotechnology

Fermented Meat Products

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Food Biotechnology

Module 87

Food Biotechnology

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Food Biotechnology

Module 88 :-

Food Biotechnology

Fermented Vegetable Products

Fermented Vegetable Products

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FERMENTED VEGETABLE PRODUCTS

END

Microbial Problems :-

Off-flavor, soft texture, and discoloration of vegetables can occur by growth of molds and yeasts when air is not completely excluded.

Food Biotechnology

Module 88 :-

Food Biotechnology

Fermented Vegetable Products

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FERMENTED VEGETABLE PRODUCTS

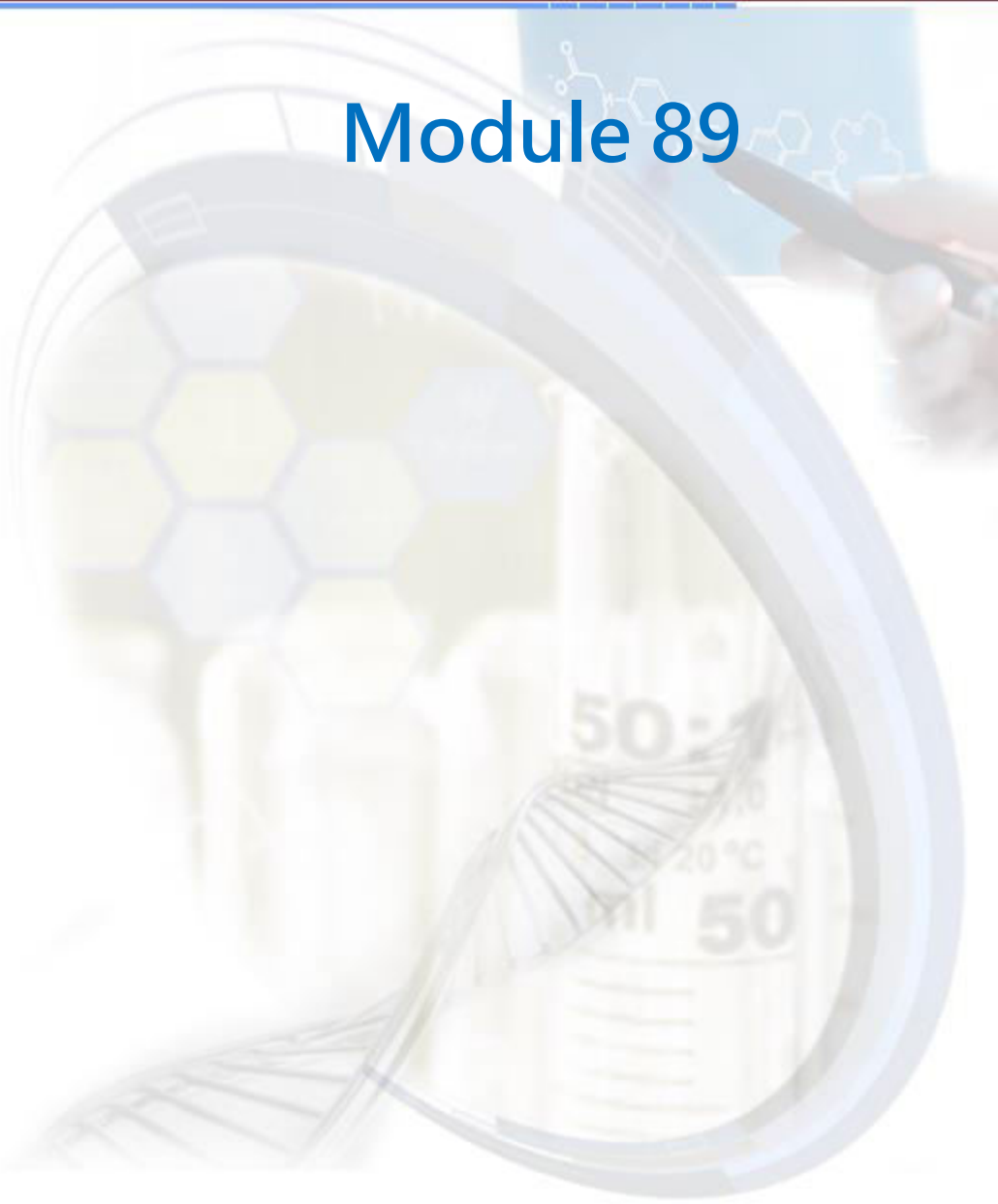
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Food Biotechnology

Module 89



Food Biotechnology

Sauerkraut

SAUERKRAUT

Sauerkraut is finely cut raw cabbage that has been fermented via various lactic acid bacteria which gives it a special sour flavor and aroma

SAUERKRAUT

Processing :-

1) Cabbage cleaned, trimmed, and shredded fine and uniform.

2) Packaged tight to exclude air in vat, and layered with salt (2.25%).

3) Top covered to exclude air, and fermented at 18°C (65°F) for 2 months.

SAUERKRAUT

3) Starter (Natural) and growth :-

The raw material has a large number of undesirable organisms and a small population of lactic acid bacteria «1 %). Among the lactic acid bacteria, most are *Lactococcus spp.* and *Leuconostoc spp.*

SAUERKRAUT

Biochemistry :-

Leuconostoc spp. metabolize sucrose, hexoses, and some pentose's in the raw material to lactate, acetate, ethanol, C02, and diacetyl

SAUERKRAUT

END

Microbial Problems :-

1)Off-flavor, soft texture, and discoloration of sauerkraut by growth of molds and yeasts when aerobic conditions..

2)A slimy texture occur due to overgrowth of *Leuconostoc spp*. in the presence of sucrose; they metabolize fructose but synthesize dextran's from glucose

Food Biotechnology

Module 90



Food Biotechnology

Vegetable Pickles

VEGETABLE PICKLES

Blends of different vegetables (Reddish, Carrot, Peas, Lemon etc.) which are fermented (Lactic Acid Fermentation) to have special aroma , flavor and increased shelf life are called vegetable pickles

VEGETABLE PICKLES

Types of Pickling :-

Pickling can be either done by

→ Direct immersion of vegetables in the salt and vinegar solution

→Anaerobic fermentation

VEGETABLE PICKLES

Process of Vegetable Pickling :-

1) Selection of Qualitatively good vegetables

2) Preparation of vegetables (Cutting, Slicing , Dicing , Shredding etc. and addition of salt)

3) Fermentation :- Anaerobic Fermentation mostly by LAB

VEGETABLE PICKLES

4) Filling :-

Fill in a air tight container and add oil .

→ Layer of oil over pickle act as a barrier for micro-organisms moreover oil has anti fungal property so preserve it efficiently

VEGETABLE PICKLES

END

Non-Fermented Pickles:

1) Selection of fruits
2) Preparation of Fruits (add salt /sugar)
3) Cooking
4) Containerization of fruits

Food Biotechnology

Module 91

Production of Cheddar and Related Hard Cheese

Production of Cheddar and Related Hard Cheese

→Cheddar cheese is classified as a hard cheese, ranging in color from nearly white (particularly if made from goat or sheep milk) to yellow to orange

→ Standards of identity for Cheddar include 39% moisture and 50% fat on a dry basis.

Production of Cheddar and Related Hard Cheese

Ingredients used for cheese making :-

) Milk

2) Calcium Chloride

3) Starter Culture

4) Adjunct Culture

5)Color

6) Enzyme

7) Salt

8) Optional ingredients

Production of Cheddar and Related Hard Cheese



Preparation for cheese making :-

1) Culture Selection / Propagation:-

→ Most important

1)Liquid (for propagation of mother culture; rarely used today);

2) Deep-frozen concentrated cultures (for propagation of bulk starter);

Production of Cheddar and Related Hard Cheese

3.)Freeze-dried concentrated cultures in powder form (for propagation of bulk starter

4. Deep-frozen or freeze-dried, super-concentrated cultures

Production of Cheddar and Related Hard Cheese

2) Cheese Milk Pretreatments

→ Centrifugation prior to standardization

→ Homogenization

→ Pasteurization

Production of Cheddar and Related Hard Cheese

Steps for cheese making :-

1) Addition of Culture , Calcium Chloride and color in pre-treated milk

2) Ripening for 15-30 minutes

3) Add coagulant and check after regular interval

4) After coagulation cut into proper sizes

Production of Cheddar and Related Hard Cheese

5) Cook it for 40-50 minutes at 102 Fahrenheit

6) Drain whey via proper filtration aid

7) Trenching

8) Cheddaring

9) Milling

10) Salting (TASTE)

Production of Cheddar and Related Hard Cheese

11) Hooping

12) Pressing (HARD CHEESE)

END

13) Packing

14) Ageing

Food Biotechnology

Module 92

Food Biotechnology

Ingredients used for Cheese Making

Ingredients used for Cheese Making

→Good cheese requires high-quality milk and carefully selected starter cultures.

→Additional ingredients are often utilized to enhance visual appeal (annatto), coagulation properties (calcium chloride or enzymes), and flavor development (adjunct cultures or enzymes) to make great cheeses

Ingredients used for Cheese Making

Ingredients for cheese production :-



1) Milk
2) Calcium Chloride
3) Starter Culture
4) Adjunct Culture
5)Color
6) Enzyme
7) Salt
8) Optional ingredients

Ingredients used for Cheese Making

1) Milk :-

→must start with high-quality milk.

→Cheddar and most Cheddar-like cheeses can be made from raw, heat-treated, heat-shocked, or pasteurized milk, nonfat milk or cream, alone or in combination

2) Calcium Chloride :-

→ Use as coagulant (0.02 %)

Ingredients used for Cheese Making

3) Starter Culture :-

→ Isolates that produce sufficient acid to reduce the pH of milk to 5.3 in 6 h at 30–37.8°C, and aid in curd digestion and flavor development

→ For cheddar cheese → *Lactococcus lactis Sub spp + lactococcus cremoris* is used.

Ingredients used for Cheese Making

4) Adjunct Culture :-

→An adjunct culture is a one that is added, along with starter culture, for the desirable characteristics it may impart upon the cheese other than acid.

→Adjunct cultures are select nonstarter lactic acid bacteria (NSLAB)

Ingredients used for Cheese Making

5) Color :-

→The recognizable yellow to orange color of Cheddar and Colby cheeses is derived from annatto, an extract from seeds of the “Lipstick tree,

6) Enzyme :-

1)Rennet

2) Chymosin

Other clotting enzymes

Ingredients used for Cheese Making

7) Salt :-

→Cheddar cheese typically contains 1.6–1.8% salt in the form of NaCl.

→Salt enhances flavor, encourages syneresis, and slows or stops growth of salt-sensitive bacteria

Ingredients used for Cheese Making

END

8) Optional ingredients :-

→ sodium benzoate ,
→sodium propionate,
→sorbic acid →Hydrogen peroxide is allowed, if followed by a quantity of catalase preparation sufficient to eliminate the hydrogen peroxide

Food Biotechnology

Module 93 :-

Selection of Cheese Culture

Selection of Cheese Culture

Introduction :-

→Culture quality is of the utmost

Culture manufacturers commonly work closely with processing facility operators to effectively meet specific needs.

→Culture manufacturers continuously develop unique culture combinations for a given product

Selection of Cheese Culture

Types of cultures :-

1. Liquid (for propagation of mother culture; rarely used today)
2. Deep-frozen concentrated cultures (for propagation of bulk starter)
3. Freeze-dried concentrated cultures in powder form (for propagation of bulk starter or DRI-VAC, for preparation of mother culture)

Selection of Cheese Culture

4) **Deep-frozen or freeze-dried,** super-concentrated cultures in readily soluble form for direct inoculation of the product (direct vat set, DVS).

Selection of Cheese Culture

→The availability of frozen or freeze-dried cultures eliminates the need for to make cultures

→The culture room is a separate room in the dairy plant because it limits opportunities for contamination by airborne yeast, mold, and bacteriophage

Selection of Cheese Culture

→ Larger plants are typically supplied with frozen or freeze-dried cultures for the manufacture of bulk starters in aseptic bulk culture rooms

→Cooling and storage conditions and shelf-lives of cultures vary.

Selection of Cheese Culture

Generally, deep-frozen and freeze-dried cultures can be stored for 9–12 months at -18 C and -45 C.

END

Maintenance of consistency across cheese lots requires constancy of culture handling

Food Biotechnology

Module:-94

Food Biotechnology

Pre-treatment of Milk for Cheese Making

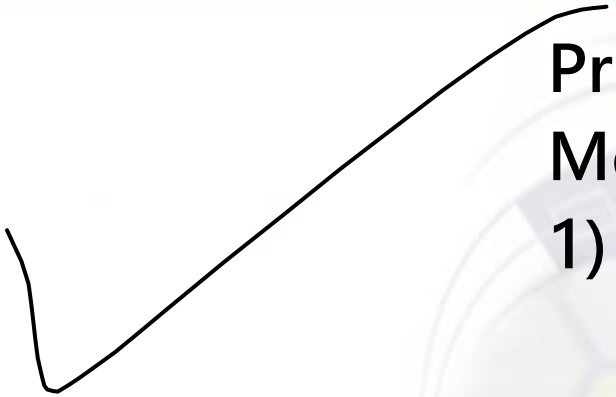
Pre-treatment of Milk for Cheese Making

Introduction :-

→ Pre-treatment of milk is essential for good cheese making

→ It ensures Quality and Safety of cheese

Pre-treatment of Milk for Cheese Making



Pre-Treatment Methods :-

1) Centrifugation :-

Centrifugal clarifier-separators are used to separate the cream and skim fractions, as well as remove solid impurities from milk

Pre-treatment of Milk for Cheese Making

Standardization :-

→Essential to maintain consistency in the quality of cheese

→ Fat addition with cream

→ Protein addition with Casein

Pre-treatment of Milk for Cheese Making

3) Homogenization :-

→Most efficient when fat globules are in the liquid state ,so milk is preheated in the plate heat exchanger in the regeneration section of the HTST pasteurizer, where the temperature is raised to at least 60C prior to homogenization

Pre-treatment of Milk for Cheese Making

→ In a two-stage homogenizer, pressures typically range from 10 to 25 MPa in the first stage and 5 MPa in the second stage

Pre-treatment of Milk for Cheese Making

END

4) Pasteurization :-

→The only step in the dairy processing system that guarantees the killing of pathogenic microorganisms is pasteurization.

→Thus, pasteurization may be considered the most critical segment of the cheese processing line

→ 72 C / few minutes

Food Biotechnology

Module 95 :-

Food Biotechnology

Cheese Making Process

Cheese Making Process

Ingredients used for cheese making :-

1) Milk
2) Calcium Chloride
3) Starter Culture
4) Adjunct Culture
5)Color
6) Enzyme
7) Salt
8) Optional ingredients

Cheese Making Process



Preparation for cheese making :-

1) Culture Selection / Propagation:-

→ Most important

1)Liquid (for propagation of mother culture; rarely used today);

2) Deep-frozen concentrated cultures (for propagation of bulk starter);

Cheese Making Process

3.)Freeze-dried concentrated cultures in powder form (for propagation of bulk starter

4. Deep-frozen or freeze-dried, super-concentrated cultures

Cheese Making Process

2) Cheese Milk Pretreatments

→ Centrifugation prior to standardization

→ Homogenization

→ Pasteurization

Cheese Making Process

Steps for cheese making :-

1) Addition of Culture , Calcium Chloride and color in pre-treated milk

2) Ripening for 15-30 minutes

3) Add coagulant and check after regular interval

Cheese Making Process

4) After coagulation cut into proper sizes Cutting of the curd is an extremely important step in the cheese making process because it influences whey drainage and cheese yield

Cheese Making Process

HEALING :-

Healing. Freshly cut curd is fragile and shatters easily, so curds are allowed to “heal” for 5–10 min prior to agitating and cooking

5) Cook it for 40-50 minutes at 102 Fahrenheit

6) Drain whey via proper filtration aid

7) Trenching

8) Milling

Cheese Making Process

9) Salting :-

1. Encourages further syneresis, 2. Inhibits further growth and metabolism of most microorganisms (thus arresting lactic acid production), and 3. Provides flavor.

10) Hooping

11) Pressing (HARD CHEESE)

12) Packing

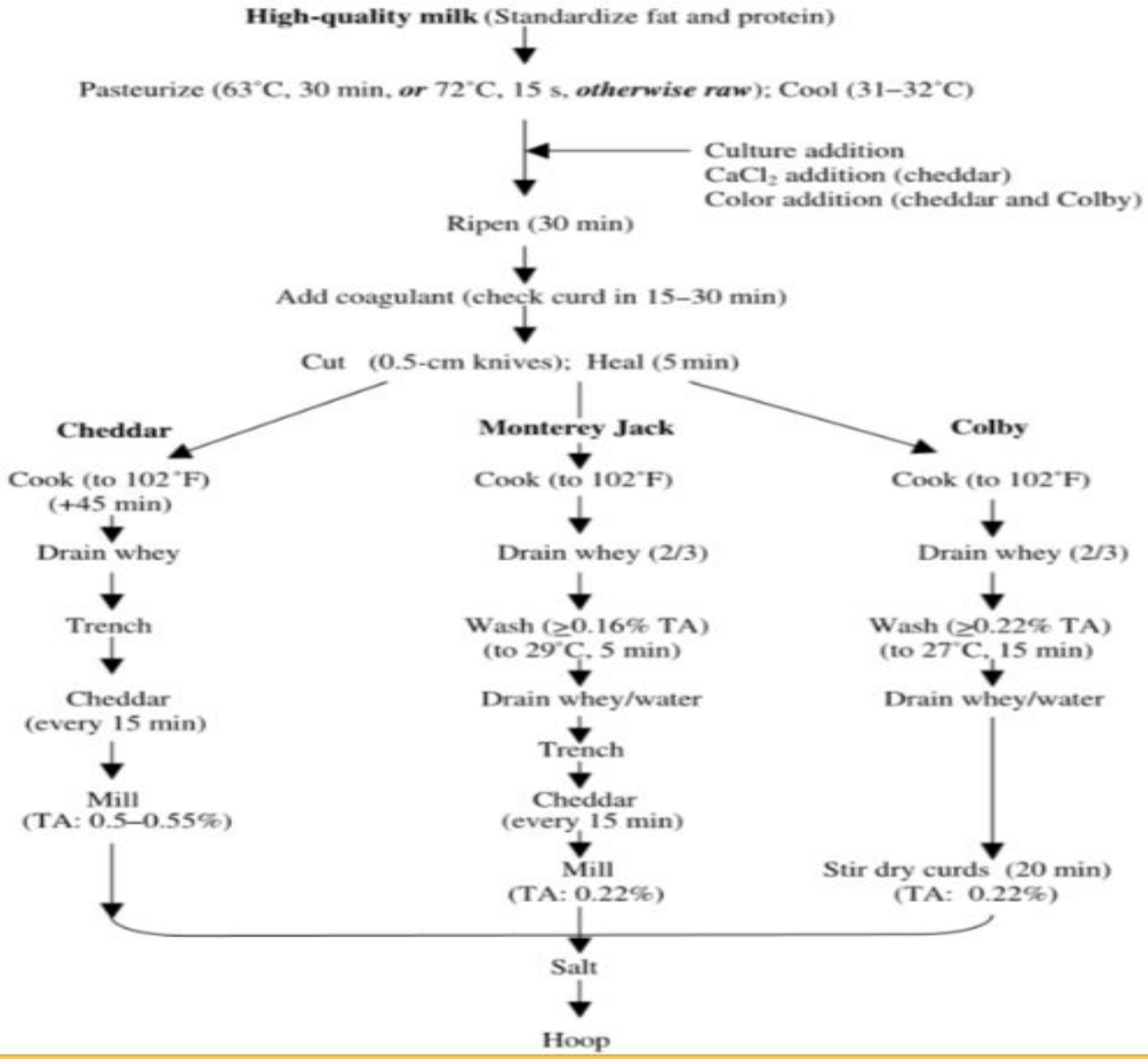
Cheese Making Process

Ageing :-

Aging enables flavor and texture development of hard cheeses

END

CHEESE MAKING PROCESS



Food Biotechnology

Module #96



Food Biotechnology

Quality Control of Cheese Making

Quality Control of Cheese Making

- The shelf-life of cheeses is limited by quality, not safety.
- The quality of good cheese improves with storage.
- ripend cheese may be removed from shelves due to flavor, body, or appearance defects.

Quality Control of Cheese Making

- Common flavor defects are high acid, bitter, unclean, and fermented /fruity.
- Common body defects are weak or crumbly body, gas holes, surface discoloration, and appearance of crystals on surfaces.

Quality Control of Cheese Making

Evaluation:

- High-quality Cheddar cheese has a full, balanced nutty, sharp, but not bitter flavor.
- The ideal texture is no gas holes and it should be firm, smooth and waxy.

Quality Control of Cheese Making

- There are as many as 200 different compounds that may contribute to cheese flavor however as few as 20 volatile compounds are pertinent to determinant of the eventual flavor of Cheddar cheese.
-

Quality Control of Cheese Making

- Cow diet, milk handling and sanitation practices, milk composition, and cheese manufacturing conditions all affect cheese chemistry.
- What appears to be critical is the relative proportions of the key chemical flavor compounds in providing peculiar flavor.

Quality Control of Cheese Making

- Ammonia-like and sulfur-like odors and bitter taste typically occur in aged cheeses, a consequence of amino acid breakdown.
- Although some attributes, such as sulfide, may be considered desirable in an aged cheese, mild cheeses are discredited for pronounced attributes.

Quality Control of Cheese Making

- Similarly certain functional properties in cheese (melting, grinding, slicing).
- Cheeses continually change during ripening, not only in flavor, but also in body/texture.
- Some of texture defects may be corky, crumbly, crystals, meaty, gassy, open, pasty, short and weak.

Quality Control of Cheese Making

Safety

- Cheddar cheese and other semi hard cheeses are generally considered safe and have rarely been associated with foodborne illness outbreaks.
- However, some pathogens can survive the cheese making process and during ripening.

Quality Control of Cheese Making

END

- Raw-milk Cheddar cheese was linked to major salmonellosis outbreaks in Canada in 1982, 1984, and 1998.
- Salmonella can survive in ripening Cheddar cheese for 7–10 months.
- *L. monocytogenes* can survive as long as 434 days in Cheddar cheese ripened at temperatures above 1.66 degree C.

Food Biotechnology

Module #97



Food Biotechnology

Food Enzymes

FOOD ENZYMES

- Enzymes, catalyze chemical reactions produced by living things and are critical for life.
- Enzyme reactions two steps:

In step 1:

E + S --- ES

In step 2,

ES + R --- P + E

FOOD ENZYMES

- Presence of necessary reactants plus energy input (activation energy) is required to start a particular reaction.

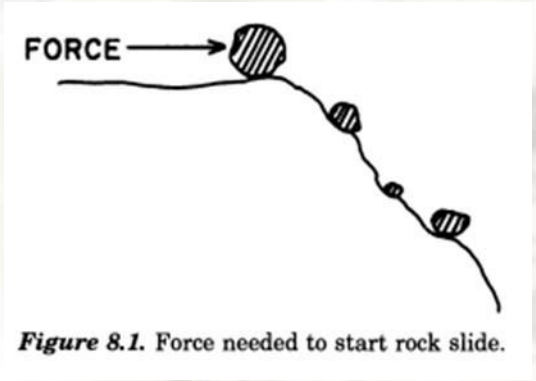


Figure 8.1. Force needed to start rock slide.

FOOD ENZYMES

- Enzyme reactions are dependent on temperature.
- reactions can be very quick at higher temperatures but living organisms have particular body temperatures at reaction have to take place so enzyme are needed.

FOOD ENZYMES

- Enzymes are needed by all levels of life, simple or complex, plant or animal kingdoms, including humans.
- All life depends on enzymes to convert foods or nutrients to a form in which they can be utilized, and to carry out cellular functions.

END

Food Biotechnology

Module #98



Food Biotechnology

The Nature of Enzymes

THE NATURE OF ENZYMES

- In composition, enzymes always contain a protein.
- They may also contain or require complex chemical compounds in order to become functional.
- These compounds are known as prosthetic groups (also called coenzymes) and are usually vitamins, especially those belonging to the B group

THE NATURE OF ENZYMES

- Some enzymes also require trace amounts of a metal, such as copper, to function.
- Enzymes therefore consist of either pure proteins, proteins with a prosthetic group, or proteins with a prosthetic group plus a metal cation.

THE NATURE OF ENZYMES

- Through their action, enzymes convert foods into less complex chemical substances that can be utilized for energy and for the building of cellular protoplasm.
- Proteins, fats, and carbohydrates are thus broken down to less complex compounds by enzymes in order that they may be utilized.

THE NATURE OF ENZYMES

- More complex carbohydrates are broken down to glucose, a source of ready energy eventually converted to carbon dioxide and water or built up into fats.
- When reserve energy is needed, fats are first hydrolyzed (water is inserted to split the molecule), breaking down into glycerine and fatty acids.

THE NATURE OF ENZYMES

- The fatty acids are then converted to acetates that can be utilized for energy, remaining may be reformed into fats and deposited as an energy reserve.
- Similarly, proteins break down to amino acids and incorporated into cellular protoplasm used for cell repair or growth.
- In some instances, oxidized to provide energy.

THE NATURE OF ENZYMES

- Enzymes always contain a protein, the protein itself having a particular shape or configuration.
- 20 to 22 known amino acids in a particular sequence produce proteins.
- Also, chains of amino acids are cross linked one to another and the different proteins form special configurations and shapes.

THE NATURE OF ENZYMES

- Proteins are also sometimes combined with carbohydrates, lipids, or phospholipids called as conjugated proteins.
- sequence of amino acids shape or relationship of these chains and conjugation with all affect the functional properties and determine the manner win which a protein will react to physical and chemical energy.

THE NATURE OF ENZYMES

- Ideal temperature in the range of 15.6 to 65.5°C,.
- However, some enzymes are able to react below freezing point or temperatures above 71°C.
- At temperatures between 71.1 and 93.3°C enzymes get inactivated due to structural changes.

THE NATURE OF ENZYMES

- An enzyme, which splits off fatty acids from fish phospholipids, is known to remain active even after steaming at 212°F (l00°C) for 20 min.
- Enzymes have an optimum pH at which they cause reactions to occur at the fastest rate.

END

Food Biotechnology

Module #99



PROTEOLYTIC ENZYMES

PROTEOLYTIC ENZYMES

- Breakdown or splitting of proteins are called proteolytic enzymes or, proteases.
- Proteinases split the protein peptones, then into polypeptides and peptides.
- Peptidases split polypeptides and peptides into amino acids thus liquefying proteins.

PROTEOLYTIC ENZYMES

- Fish enzymes are much more active than in meats even when fish is held in the eviscerated state in ice or under refrigeration.
- In fish held uneviscerated, a concentrated source of enzymes present in blind tubules (the pyloric ceca) attached to the intestines accelerate enzyme action leading to spoilage.

PROTEOLYTIC ENZYMES

- Members of the herring and mackerel families handled in the uneviscerated state are quite subject to this type of enzyme deterioration, especially if they have been feeding when caught.

PROTEOLYTIC ENZYMES

- Similarly, lobster especially at high temperatures, proteolysis is accelerated resulting in liquefied products and poor texture.
- While cooking the flesh will be soft and crumbly (short-meated) and part of the tail portion gets dissolved, leaving only apart of this section intact hence shall be used in fresh state.

PROTEOLYTIC ENZYMES

- Shrimp, proteolysis is not extensive if heads are removed shortly after the shrimp are caught.
- The relatively high activity of enzymes in marine species is attributed to the low-temperature conditions in the marine environment.
- That is, in order to make reactions proceed at low temperatures, the activation energy system must be more efficient.

PROTEOLYTIC ENZYMES

END

- Plants proteolytic enzymes usually contribute little to deterioration, especially as long as the tissues are intact.
- Bromelin is found in unpasteurized pineapple can result in bleeding or skin eroding in handlers.
- Papain from green papaya fruit.
- Ficin from fig trees.
- Meat tenderizers

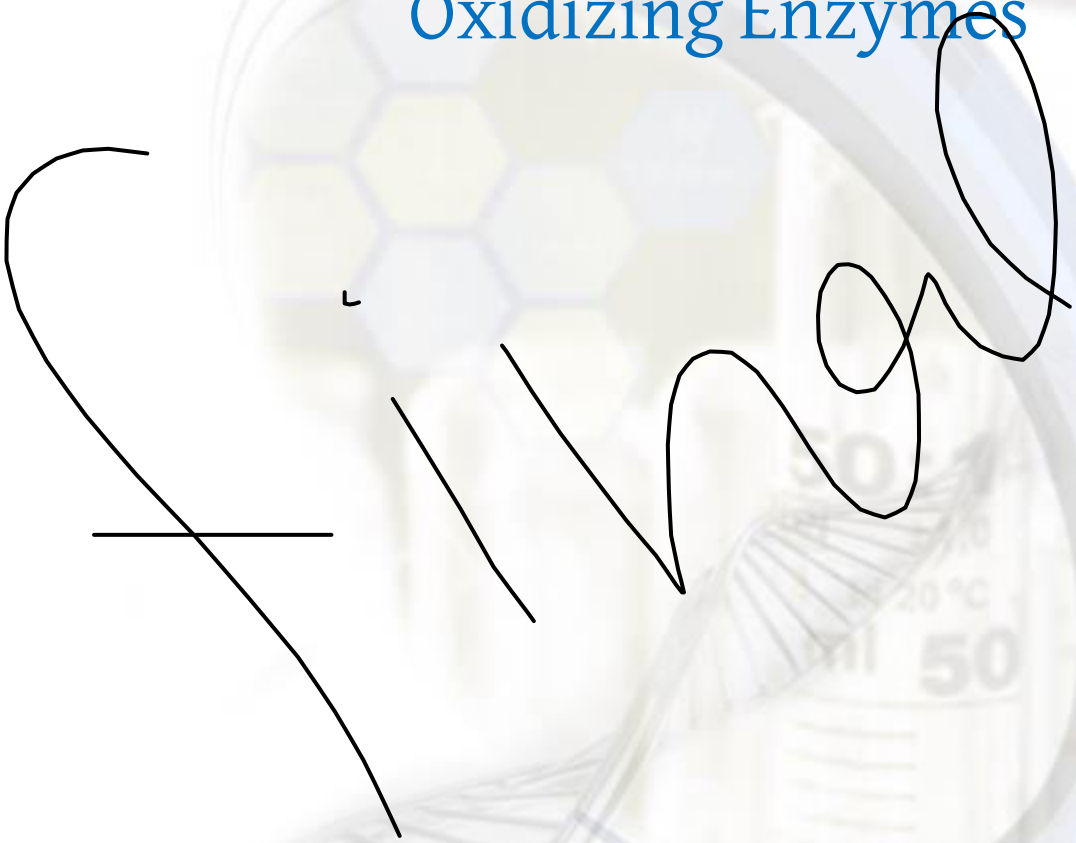
Food Biotechnology

Module #100



Food Biotechnology

Oxidizing Enzymes



Oxidizing Enzymes

- Loss of electrons in an atom or simply a chemical combination with oxygen.
- Such as wick of a candle burning.
- Oxidizing enzymes that bring about changes in foods that result in deterioration due to oxidation.

Oxidizing Enzymes

- In plants, peroxidases, ascorbic acid oxidase, tyrosinase, and polyphenolases may cause undesirable chemical reactions to occur.
- Peroxidases may oxidize certain phenol like compounds in root vegetables, such as horseradish, causing the prepared product to become darker in color when damaged.
-

Oxidizing Enzymes

- Ascorbic acid oxidase, in vegetables, oxidizes ascorbic acid cause loss of the vitamin C content of foods.
- Peroxidases also, indirectly, cause a loss of vitamin C by its by products.
- Phenolases are present in some fruits and vegetables cause brown or dark-colored compounds to be formed when the tissues are cut.

Oxidizing Enzymes

- Tyrosinase oxidizes tyrosine to form dark-colored compounds. The molecule rearranges and further oxidizes to form a red compound. Polymerization (combination of these compounds) results in the formation of dark-colored melanin compounds.

Oxidizing Enzymes

- The enzyme tyrosinase, cause discoloration of the cut tissue and will also oxidize compounds related to tyrosine.
- This enzyme is also present in shrimp and some spiny lobsters and may cause a discoloration called black spot as a black stripe on the flesh along the edges of the segments of the tail or as a pronounced band where the shell segments overlap.

Oxidizing Enzymes

END

- It is not generally recognized, but tyrosinases are also present in clams.
- Hence, shucked (deshelled) clams will darken at the surface if oxygen is present and if the enzymes have not been inactivated by heat.
- These reactions occur only after the shrimp or clams die.

Food Biotechnology

Module #101



Food Biotechnology

Fat Splitting Enzymes

Fat-splitting Enzymes

- Fats are composed of glycerine (glycerol) and fatty acids.
- Glycerine is a polyhydric alcohol (three alcohol groups), and fatty acids are short or long chains of carbon atoms to which hydrogen is attached, either to the fullest possible extent (saturated) or to a lesser extent (unsaturated), the latter resulting in reactive groups in the chain.

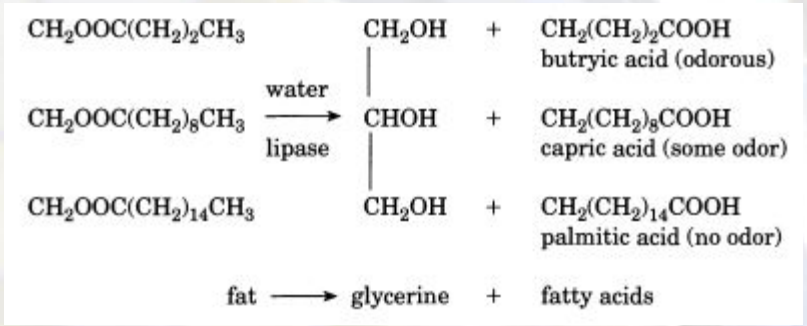
Fat-splitting Enzymes

- At one end of the fatty acid chain there is an acid group.
- In the formation of fats, each one of three fatty acids combines with one of the three alcohol groups of glycerine, splitting of water in each case.
- On the other hand, in the breakdown of fats, water and the enzyme lipase are present, and fats are split into their original component parts,

Fat-splitting Enzymes

- The fatty acids mostly consist of a chain of more than10 carbon atoms, and these fatty acids have no particular flavor or odor.
- Hence, when lipase acts on most natural fats, no bad odors are generated.
- However, when used for deep-fat frying, the oil may smoke during heating, which is undesirable.

Fat-splitting Enzymes



Fat-splitting Enzymes

- Short Chain fatty acids include butyric (four carbons), caproic (six carbons), caprylic (eight carbons), and capric (ten carbons) acids. All these fatty acids have an odor and flavor. Butyric acid, especially, is pungent and considered to be distasteful.

Fat-splitting Enzymes

- Thus butter releases butyric acid, which gives the butter a strong, undesirable rancid taste.
- Butter becomes rancid by the action of lipase produced by bacteria that grow in the water droplets.

Fat-splitting Enzymes

- Phospholipases, enzymes similar to lipase, are split phospholipids. Most phospholipids are similar to fats in that they contain glycerine, two alcohol groups of which are combined with fatty acids.
- The third alcohol group is combined with a molecule containing phosphoric acid.

Fat-splitting Enzymes

END

- Lecithin is a typical phospholipid. Phospholipase splits off fatty acids from phospholipids. Such action may cause deterioration in foods in resulting in destabilization of proteins causing a toughening of the tissues and a loss of succulence.

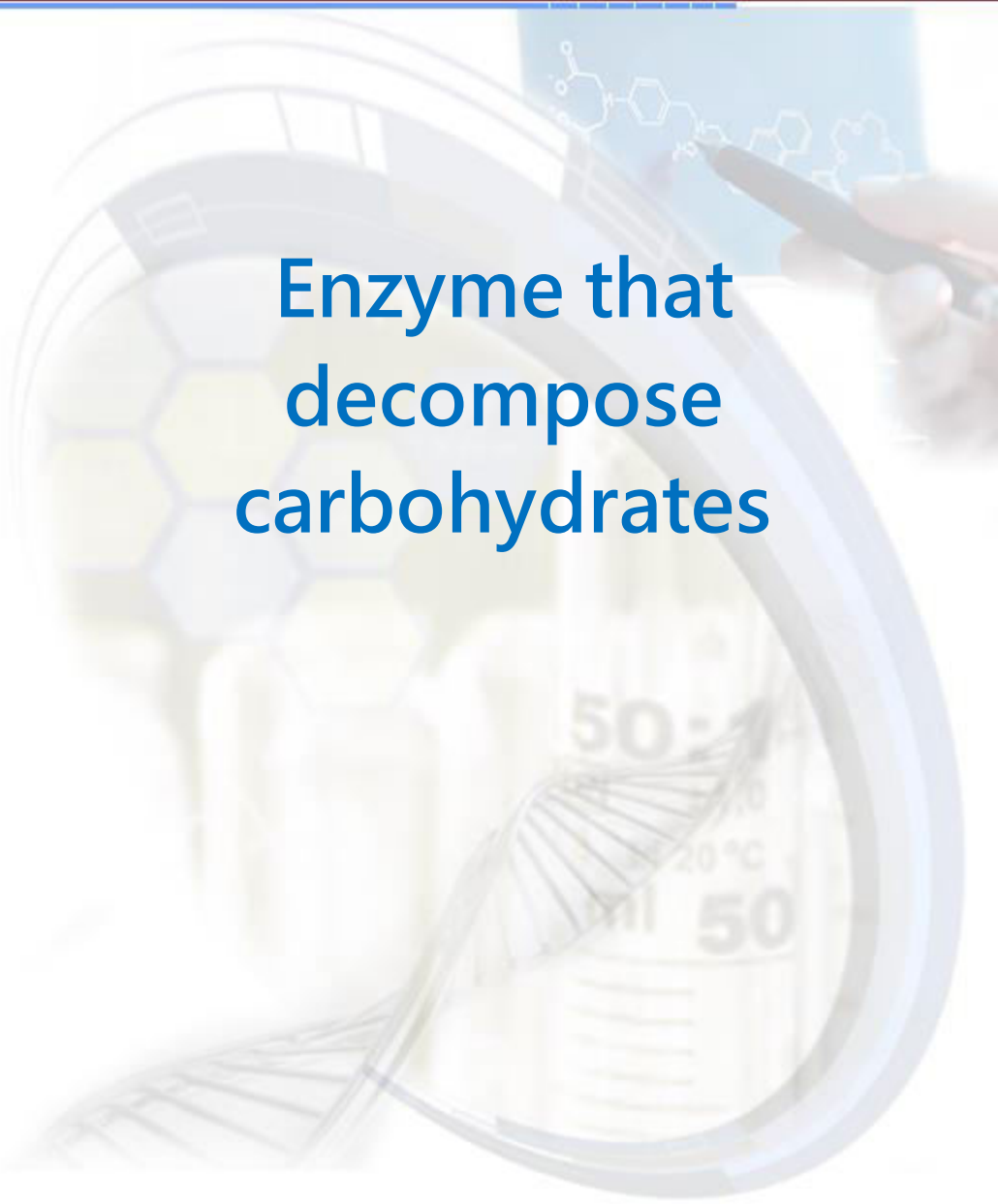
Food Biotechnology

Module #102



Food Biotechnology

Enzyme that decompose carbohydrates



Enzymes that Decompose Carbohydrates

- Pectin in fruits supports structure.
- In processed fruit juices broken pectin tend to settle to the bottom, leaving a clear serum on top.
- Pectin consists of a long chain of galacturonic acid molecules with the carboxyl groups partially esterified with methyl alcohol. It has a high water-holding capacity.

Enzymes that Decompose Carbohydrates

- There are pectic enzymes that will either break down the pectin molecule to smaller units or completely decompose the molecule to its primary unit, galacturonic acid.
- The emulsifying properties of pectin may be lost, causing settling in fruit juices and softening in fruit.

Enzymes that Decompose Carbohydrates

END

- Pectin in whole fruit breaks down resulting in deterioration of the fruit as a result of the action enzymes/microbes.
- In the sugar cane, invertase, breaks down sucrose, hence a part of sugarcane must be removed at harvesting.

Food Biotechnology

Module #103



Food Biotechnology

Glucose Oxidase

Glucose Oxidase

- Glucose oxidase is an enzyme that specifically catalyzes the oxidation of glucose to gluconic acid.
- Non enzymatic browning, is a undesirable browning reaction.
- The most important application of this enzyme is in the treatment of egg products, especially egg whites, prior to drying.

Glucose Oxidase

END

- In some cases, the enzyme is added to remove traces of oxygen to prevent oxidative degradation of quality. Examples of this type of application are mayonnaise and bottled and canned beverages (especially beer and citrus drinks).

Food Biotechnology

Module #104



Food Biotechnology

Catalases

Catalase

- Catalases are used to break down hydrogen peroxide (H_2O_2) to water and oxygen.
- Therefore, catalases are used when the presence of hydrogen peroxide is undesirable or when hydrogen peroxide is used for specific purposes, such as in bleaching, but then must be removed from the system.

Catalase

- Examples of the latter case are the uses of hydrogen peroxide for preserving milk in areas where heat pasteurization and refrigeration are unavailable and in the manufacture of cheese from unpasteurized milk.
- Hydrogen peroxide is produced during the spray-drying procss.
- Catalase is used to convert the unwanted H_2O_2 to water and

Catalase

- Catalases have health promoting activities in human by reducing the oxidative damage/inflammation etc.
- Damage proteins, cell membrane and even DNA, reduces with age and must come from food sources.

END

Food Biotechnology

Module #105



Food Biotechnology

Application of Enzymes

Application of Enzymes

- Whereas enzymes may cause a deterioration of foods, they may also be used in the processing of foods.
- Proteolytic enzymes obtained from plants may be used for tenderizing meat either by injecting animals with a solution of the enzyme prior to slaughter or by sprinkling the powdered enzyme on meat surfaces and allowing it to react, prior to cooking.

Application of Enzymes

- In the manufacture of certain kinds of milk powder (e.g., to be used in chocolate), the lipases may be allowed to act on the milk fat prior to drying to obtain a particular flavor in the finished product.
- Proteases may also be used to chill-proof beers, a technique used to remove proteins that would cause clouding during cool storage.

Application of Enzymes

- Protein hydrolysates can be produced from the use of proteases from both plant and animal sourees. They have many uses, for example, as flavorings, nutrients, and stabilizers.
- The characteristic flavor of certain cheeses is due to the action of lipases.
- In order to obtain the particular flavors of Roquefort or blue cheese, milk fat must first be broken.

Application of Enzymes

- fatty acids are produced by the molds allowed to grow in these cheeses. Enzymes produced by molds then oxidize the specific fatty acids that ultimately produce the unique characteristic flavors.
- Lipases also can be added to dried egg whites to improve the whipping quality.

Application of Enzymes

- Carbohydrate-splitting enzymes.
- In making malt, barley is germinated to obtain an enzyme that will convert starch to a sugar (maltose.

Application of Enzymes

- Carbohydrate-splitting enzymes are also used to modify starches used in foods and to modify starches used in sizing and laundering clothes.
- Corn syrup from corn starch and then isomerize glucose to fructose in the formation of high-fructose corn syrup which is used extensively in the manufacture of soft drinks.

Application of Enzymes

- Invertase (sucrase) is used in the production of chocolate covered cherries.
- During a short holding period, the enzyme splits the sucrose into a mixture of glucose and fructose, which has a higher water solubility and results in the sweet taste and creamy texture that is characteristic of the product.

Application of Enzymes

- The presence of pectins in fruit juices may cause clouding.
- Pectinases can be added to remove the pectin and clarify the juice.

Application of Enzymes

- One of the factors that will serve to widen the use of enzyme technology is the development of immobilized enzymes.
- It has been found that enzymes can be fixed chemically to the surface of inert substances, such as glass beads. In this form, they can be packed into a column through which a solution or suspension of the material to be acted on (called the substrate) is

Application of Enzymes

END

- Restoration and enrichment of flavor in some processed foods by addition of enzymes.
- e.g. conversion of all in in garlic into garlic oil by allinase and the addition of an enzyme preparation from mustard seeds to dehydrated cabbage which restores flavor by converting flavor precursors into the volatile sulfur-containing compounds responsible for the familiar flavors.

Food Biotechnology

Module #106



Food Biotechnology

Food Additives: Definition and Functions

Food Additives

- Food additives have been defined as chemical substances deliberately added to foods, directly or indirectly, in known and regulated quantities, for purposes of assisting in the processing of foods; preservation of foods; or in improving the flavor, texture, or appearance of foods.

Food Additives

- An additive may be reactive or inactive; it may be nutritive or nonnutritive; it should be neither toxic nor hazardous.
- Some substances, such as pesticides, are added to foods unintentionally, and these are, of course, undesirable, and may be hazardous to health.

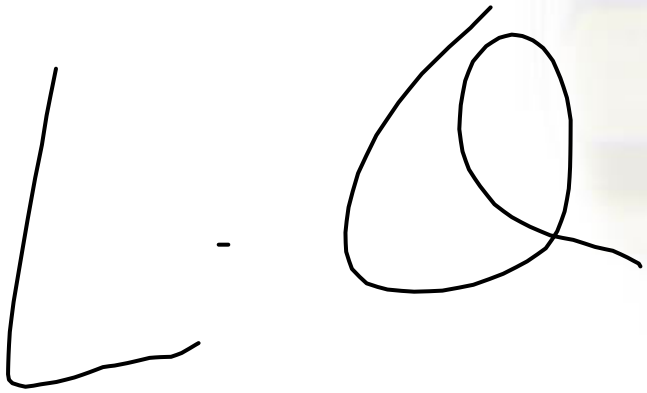
Food Additives

- Additives are classified as GRAS when they have been used without apparent harm for long periods of time, long before regulations had been put into effect.
- These include substances such as baking powder chemicals (e.g., sodium bicarbonate), fruit acids such as citric and malic, and gums such as agar-agar.

Food Additives

- Ionization radiation is considered an additive because the treatment may induce changes in food.
- In this case, the FDA tests foods that have been irradiated and approves irradiation sources and maximum dosages.

Food Additives



- Besides the requirements mentioned, food additives must also meet five general criteria:
  1. Intentional additives must perform their intended function.
  2. Additives must not deceive the consumer or conceal faulty ingredients or defects in manufacturing practices.

Food Additives

3. Additives cannot considerably reduce the food's nutritional value.
4. An additive cannot be used to achieve an effect that could be gained by good manufacturing practices.
5. A method of analysis must exist to monitor the use of the additive in foods or its incidental occurrence in foods (such as migration from a packaging material).

END

Food Biotechnology

Module #107



Philosophy of Food Additives

Philosophy of Food Additives

- Foods are made entirely of substances that, in the pure form, can be described as chemicals or chemical compounds.
- It is important to note that our knowledge of the composition of foods, because of its complexity, is by no means complete.
- For instance, it is reported that one of the most important of our natural foods, human milk, contains several hundred chemical compounds.

Philosophy of Food Additives

- Unfortunately, the interpretation of the word *chemical* is too often inaccurate.
- Refined sugar, vinegar, spices, and other substances that are routinely added to foods are also chemicals or mixtures of chemicals, and the use of these is not questioned.

Philosophy of Food Additives

- A large number of chemical additives are unfamiliar, and there is a need for regulatory agencies to question their use from the standpoint of safety.
- For example, indiscreet use of certain additives for coloring candy and popcorn was reported to have caused diarrhea in children, resulting in the removal of these dyes from the FDA approved list of additives.

Philosophy of Food Additives

- Additives can come from natural food sources.
- Any additive or chemical compound can be injurious to health on improper use.
- It is necessary to evaluate each additive for usefulness and toxicity in a sensible, scientific way.

Philosophy of Food Additives

- Given present capabilities, our most reasonable evaluation of an additive for safety can be made through conventional animal feeding studies.
- chemical materials added to foods must be approved for use unless their use.

Philosophy of Food Additives

- Most food additives are components of natural foods and that without these additives the quality of many foods would be greatly inferior to that to which we have become accustomed.
- The shelf-life or availability of many foods would also be greatly limited were all additives to be eliminated from foods.

Philosophy of Food Additives

- Food additives may be used for a number of reasons.
- At present, over 3000 intentional additives are allowed and they can be divided into several major groups.

END

Food Biotechnology

Module #108



Food Biotechnology

Antioxidants in Foods

Antioxidants

- Antioxidants are food additives used to stabilize foods against oxidation.
- Oxidative quality changes in foods include:

•the development of rancidity from the oxidation of unsaturated fats resulting in off-odors and off-flavors and

•discoloration from oxidation of pigments or other components of the food.

Antioxidants

- Antioxidants function in different ways, the purpose of each is to prevent, delay, or minimize the oxidation of the food to which they are added.
- antioxidants make their combination with oxygen.
- or prevent oxygen from reacting with components of the food, as in a hermetically sealed container.

Antioxidants

- Some antioxidants lose their effectiveness when they combine with oxygen;
- Precautions are also necessary to minimize oxidation (heat, light, and metals are pro-oxidants)
- Many of the antioxidants used commercially occur naturally in foods (e.g., vitamin C, vitamin E, citric acid, amines, and certain phenolic compounds).

Antioxidants

- amines and the phenolic compounds can be toxic to , potency of the naturally occurring antioxidants is less than synthetic antioxidants.
- butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), and propylgallate mostly used.
- These are used in combinations of two or all three of them, and often in combination with Citric acid.

Antioxidants

END

- Fats and shortenings, for bakery/fried foods, are subject to oxidative spoilage so chemical antioxidants in concentrations up to 0.02% of the weight of the fat component may be added.
- Their use is limited so that the maximum amount that can be added is generally 0.02% of the fat content of the food.

Food Biotechnology

Module #109



Food Biotechnology

Nutrient Additives

Nutrient Additives

- The need for a balanced and ample nutrient intake by the human body is well known.
- Although nutrients are available in foods, losses of fractional amounts of some of them through processing and other factors minimum daily requirements or sizable fractions of minimum daily requirements of a number of nutrients to popular foods.

Nutrient Additives

- Vitamin D is an example.
- The major source of vitamin D for humans is a precursor compound called 7-dehydrocholesterol which is produced in the liver.
- It circulates to an area just under the skin and is converted to pre vitamin D3 by the ultraviolet rays of sunlight.
- Previtamin D3 to active vitamin D.

Nutrient Additives

- However, in many cases, exposure to the sun is sporadic and insufficient, especially in areas where there is normally insufficient sunshine or in cases where sunlight exposure is of insufficient duration.

Nutrient Additives

- The addition of protein concentrate (produced from fish or soybeans) to components of the diet of inhabitants of underdeveloped countries has been used successfully to remedy the high incidence of protein malnutrition.
- It should be noted that soybean protein is incomplete and requires the addition of some amino acids in which it is deficient.

Nutrient Additives

- Children, especially, succumb in large numbers to the disease, kwashiorkor, that results from insufficient protein intake.
- Among minerals, iron has received major attention as a food additive, mainly because of its role in preventing certain anemias.

end

Food Biotechnology

Module #110



Flavorings and Flavor Enhancers

Flavorings & Flavor Enhancer

- Flavorings are compounds, natural or synthetic, that are added to foods to produce flavors or to modify existing flavors.
- In the early days of human existence, salt, sugar, vinegar, herbs, spices, smoke, honey, and berries were added to foods to improve their taste or to produce a special, desirable taste.
- The variety of natural and synthetic flavorings available to the modern food technologist is very large.
- Essential oils provide a major source of flavorings.
- Essential oils are odorous components of plants and plant materials that give the characteristic odors of the materials from which they are extracted.

Flavorings & Flavor Enhancer

- Because of the large production of orange juice, quantities of essential oil of orange are produced as byproducts.
- For this reason, there is little need for the production of synthetic orange flavoring.
- Fruit extracts have been used as flavorings, but these are relatively weak when compared to essential oils and oleoresins.
- An oleoresin is a solvent extract of spices from which the solvent, usually a hydrocarbon, has been removed by distillation.
- Because of their weak effects, fruit extracts may be intensified by combining them with other flavorings.

Flavorings & Flavor Enhancer

- Synthetic flavorings are usually less expensive and more plentiful than natural flavorings.
- On the other hand, natural flavorings are often more acceptable, but they are quite complex and difficult to reproduce synthetically.
- natural flavorings are variable due to season and other uncontrollable factors unlike synthetic flavorings.

Flavorings & Flavor Enhancer

- Many artificial flavors, such as amyl acetate (artificial banana flavor), benzaldehyde (artificial cherry flavor), and ethyl caproate (artificial pineapple flavor), are added to confectioneries, baked products, soft drinks, and ice cream.
- These flavorings are added in very small amounts, often 0.03% or less.

Flavorings & Flavor Enhancer

- Flavorings either impart a particular flavor to food or modify flavors already present.
- Flavor enhancers, on the other hand, intensify flavors already present, especially when the desirable flavors are relatively weak.
- Monosodium glutamate (MSG) is one of the best known and most widely used flavor enhancers.

Flavorings & Flavor Enhancer

- While it is effective at relatively low levels (parts per thousand), there are other compounds called flavor potentiators that also enhance flavors but are extremely powerful, effective in parts per million and even per billion.

Flavorings & Flavor Enhancer

- These compounds have been identified as nucleotides, and their effect is attributed to their synergistic properties (properties that intensify the effect of natural flavor components).
- Two in this group are disodium inosinate and disodium GMP.

Flavorings & Flavor Enhancer

- One theory is that they increase the sensitivity of the taste buds, thus increasing flavor.
- A second suggests that an increase in salivation as a result of the flavor enhancers.
- A third theory of intensified flavor perception is flavor enhancers produce certain physical sensations in the mouth such as coolness and heat.

end

Food Additives

Module # 111



Acidulants in Food

Acidulants in Food

- Acidulants are a class of compounds that tends to lower the pH of any food
- Enhance desirable flavor
- In some foods like pickle, they are a major taste component

Acidulants in Food

- Vinegar or acetic acid
- Added to relishes, chili sauce, ketchup and other condiments
- Acts as flavor component
- Also aids in preservation

Acidulants in Food

- By adding acidulants, it effectively decreases pH
- Lower pH or acidic conditions inhibit the growth of microbes

Acidulants in Food

- Many acidulants occur naturally in food

- Citric Acid - Citrus family
- Malic Acid - Apples
- Acetic Acid – Vinegar

- Figs contain all three acidulants

Acidulants in Food

- Tartaric acid is used commonly in food to provide tartness and enhance flavor
- Citric acid is used in carbonated drinks
- Phosphoric acid is an inorganic acidulant used in food

Acidulants in Food

- Can also be used as cleaners for dairy equipment
- Can also be used to improve gelling properties and texture

Acidulants in Food

- Used in processed cheese and cheese products: cheese spread
- Acidulants are used as emulsifying agents
- Also provide desirable tartness as to enhance flavor

Acidulants in Food

- Acid salts may be added to soft drinks

- Act as buffer to avoid excessive tartness
- Acid salts are also used to inhibit mold growth
- Calcium propionate added to bread

end

Food Biotechnology

Module # 112



Alkaline Compounds in Food

Alkaline Compounds in Food

- Alkaline compounds tend to raise the pH of foods

- Sodium Hydroxide
- Potassium Hydroxide

- Can be used to neutralize excess acid that can develop in natural or culture fermented foods

Alkaline Compounds in Food

- Manufacture of butter
- Acid in the cream may be partially neutralized
- Must be done prior to churning
- Failure to do so can result in undesirable flavor

Alkaline Compounds in Food

- Rendered fats can be refined by:
- Sodium carbonate
- Sodium bicarbonate

- Alkaline compounds added to chlorinated drinking water

- Adjusts pH high enough to control corrosion of pipes

Alkaline Compounds in Food

- Sodium carbonate along with other compounds used to reduce the amount of hardness in drinking water

- Sodium bicarbonate used as an ingredient of baking soda

Alkaline Compounds in Food

- Sodium Hydroxide used in production of caramel
- Also used to modify starches

- Alkaline compounds also used in production of chocolate
- Also used to adjust acidity levels in grape and other juices that are to be fermented for wine production

Alkaline Compounds in Food

- Sodium hydroxide
- Potassium hydroxide

- Powerful reagents
- Should be handled with care

- Sodium bicarbonate is relatively mild and safe to use

Alkaline Compounds in Food

END

- Sodium hydroxide
- Potassium hydroxide

- Powerful reagents
- Should be handled with care

- Sodium bicarbonate is relatively mild and safe to use

Food Biotechnology

Module # 113



Food Biotechnology

Sweeteners

Sweeteners

- Sweetening agents used largely in food and beverage industry
- Natural and synthetic agents
- Varying strengths and caloric value

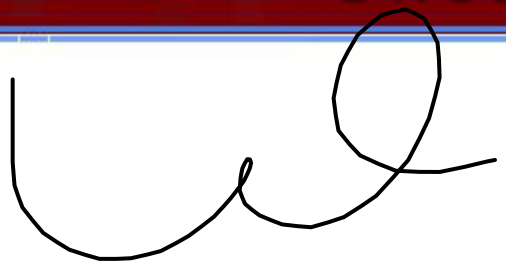
Sweeteners

- Most commonly used sweeteners:

- Table sugar (sucrose)
- Corn Syrup

- Many sweeteners are classified as nonnutritive

Sweeteners



- Fructose
- Molasses
- Honey
- Maple sugar
- Lactose
- Maltose
- Polyhydric alcohols
- Aspartame
- Saccharin

Sweeteners

- Fructose

- Natural
- Monosaccharide
- Twice as sweet as sucrose
- Most water soluble
- Hygroscopic

Sweeteners

- Greater flexibility of use over a range of temperatures
- Solution of fructose have low viscosity
- Does not crystallize out of solution
- Sometimes called fruit sugar due to its presence in fruits

Sweeteners

- Fructose
- Major component in honey, corn syrup, cane sugar, beet sugar
- Advised to diabetic individuals
- Accelerates the metabolism of alcohol

Sweeteners

- Fructose
- Rapid source of energy for athletes
- In combination with gluconate and saccharin; it is an effective, economic, safe and low-calorie sweetener for beverages

Sweeteners



- Molasses
- By-product of sugar production
- Use as sweetener mostly in baked goods
- Breads, cookies, cakes
- Adds flavor and acts as humectant

Sweeteners

- Molasses
- Greatest use of molasses is in animal feed
- Used in production of baked beans
- Also used in production of rum

Sweeteners

- Molasses
- Comprise of about 60% sucrose
- Can very depending upon raw material and grade of molasses

Sweeteners

- Honey

- Natural
- Viscous syrup

- Comprise mainly of invert sugar
- Relatively expsensive

Sweeteners

- Honey
- Produced from nectar of flowers
- Nectar is mainly sucrose
- Produced by action of invertase enzyme
- Enzyme secreted by honey bee

Sweeteners

- Honey
- Used as a direct sweetener
- Also used as an additive in many food products
- Included in some baked goods

Sweeteners

- Maple Sugar
- Sap of the sugar from maple tree
- Comprised mainly of sucrose and small amounts of other sugars like invert sugar

Sweeteners



- Maple Sugar
- Used in manufacture of candies, fudge, baked goods and toppings
- Among the most expensive sweeteners

Sweeteners



- Lactose
- Sugar component of mammalian milk
- Less sweet and lesser-water soluble than sucrose

Sweeteners

- Lactose
- Major source of lactose is whey, a cheese byproduct
- Because it is not as sweet as sucrose, it can be used in large amounts in those foods in which the texture benefits from a high solid content

Sweeteners

- Maltose

- Malt sugar
- Produced during enzymatic conversion of starch in the brewing process
- Converted to alcohol by action of yeasts
- Intermediate conversion to dextrose

Sweeteners

- Maltose
- Much less sweeter than sucrose
- Used in manufacture of baked goods and infant foods

Sweeteners

- Xylitol

- Polyhydric alcohol
- Non-carcinogenic property
- Used in chewing gums

- Occurs naturally
- Intermediary product of carbohydrate metabolism in humans and animals

Sweeteners

- Xylitol
- Produced commercially by hydrolysis of xylan to xylose
- Xylose hydrogenated to to produce xylitol
- Purified and crystallized

- Imparts a sweet taste and cooling effect
- Not metabolized by many microorganisms

Sweeteners

- Sorbitol

- Polyhydric alcohol
- Found in red seaweed and in fruits such as apples, cherries, peaches, pears

- First isolated from sorb berries

Sweeteners

- Sorbitol
- Used as an additive for its humectant property as well as its sweetening effect
- Used in
- Cough syrup
- Toothpaste
- Mouthwash

Sweeteners

- Sorbitol
- Produced industrially by electrochemical reduction or catalytic hydrogenation of glucose
- Not easily fermented by many microorganisms
- Largely transformed to fructose by liver enzymes

Sweeteners

- Mannitol

- Polyhydric alcohol
- Naturally occuring in many plants, algae and molds

- Can also be made by reduction of monosaccharides

Sweeteners

- Mannitol
- Occurs in sap on manna trees native to southern Italy
- Used in chewing gum and pharmaceuticals

Sweeteners

- Aspartame

- Common name for aspartyl-phenylalanine
- Combination of two amino acids

- 180 times sweeter than sucrose
- Leaves no bitter after taste

Sweeteners

- Aspartame

- Approved and later banned by FDA
- Reinstated as safe after exhausting evidence was presented

- About 200 times more expensive than sucrose
- Needed in much smaller quantities than sucrose

Sweeteners

- Saccharin

- Imide of *o*-benzosulfonic acid
- Used as sodium or calcium salt

- 300 times sweeter than sucrose

- Leaves bitter after taste

Food Biotechnology

Module # 114



Food Biotechnology

Starches

Starches

- Starch
- Although differs from each other somewhat
- Depending on the plant from which they are extracted

Starches

- Starch
- Usually sufficiently chemically similar
- Two basic starch polymers are
- Amylose
- Amylopectin

Starches

- Starch
- Used as a source of carbohydrate
- Relatively inexpensive
- Often used as an extender

Starches

- Starch
- Properties of starch make it useful as a thickening agent
- Major source is starch from corn
- Also produced from sorghum, potatoes and wheat

Food Biotechnology

Module # 115



Food Biotechnology

Gums

Gums

- Class of complex polysaccharides
- Defined as materials that are dispersible in water and make the water more viscous
- Many gums occur naturally

Gums

- Examples of gum are:

- Gum Arabic
- Agar

- Vinyl polymer gums are synthetic
- Cellulose derivate gums are modified or semi synthetic

Gums

- Used to:
- Stabilize ice-cream and desserts
- Thicken beverages and preserves
- Stabilize foam in beer
- Emulsify salad dressing
- Form protective coatings from meat and fish

Gums

- Gums add “body” and prevent settling of suspended particles in chocolate milk, ice cream and desserts
- May also prevent formation of large ice crystals in frozen desserts

Gums

- Use of gum also in production of certain low-calories food
- Oils in salad dressing can be replaced with gums
- Replacing oil with gum allows product to keep normal appearance, texture and taste while losing calorie count

Food Biotechnology

Module # 116



Polyhydric Alcohols

Polyhydric Alcohols

- Also called polyols
- Used to improve texture and moisture retention
- Present in foods naturally
- Glycerine (glycerol)

Polyhydric Alcohols

- Only four allowed polyols

- Glycerine
- Sorbitol
- Mannitol
- Propylene glycol

Polyhydric Alcohols

- Propylene glycol has an undesirable bitter after taste
- Acceptable in small amounts
- Sorbitol imparts a cool sensation
- Glycerine imparts a hot sensation

Polyhydric Alcohols

- Polyols are used in the production of dietetic products

- Beverages
- Candies
- Gum
- Ice cream

Polyhydric Alcohols

- These compounds have less adverse effect on teeth than sugar
- Not fermented as easily as sugar
- Usually washed away before they can be consumed

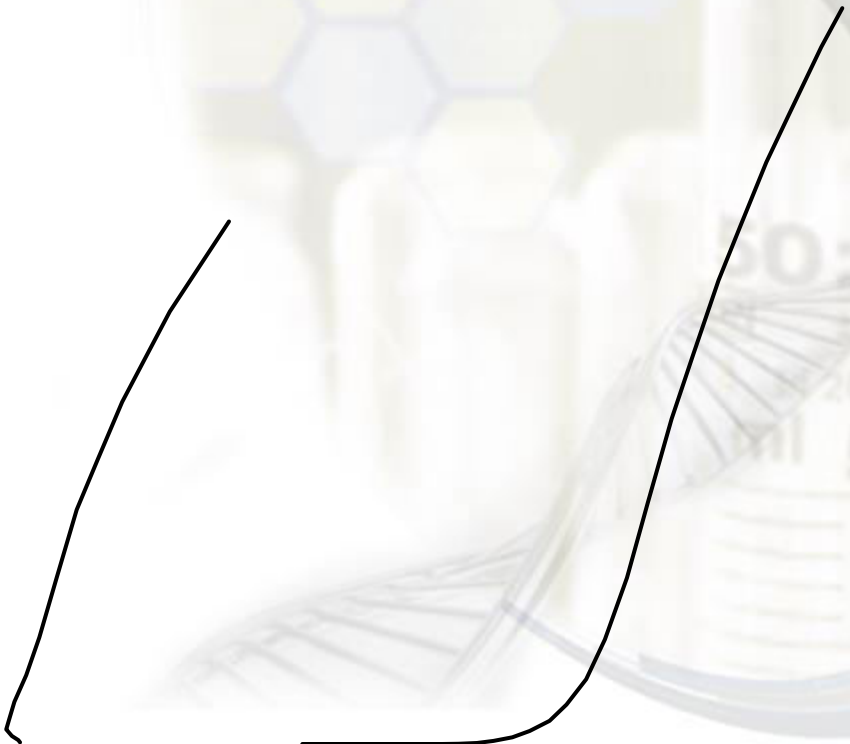
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Food Biotechnology

Module # 117



Surface Active Agents



Surface Active Agents

- Affect the physical force at the interface of surfaces
- Commonly called surfactants
- Present in all natural foods
- Play a role in growth process

Surface Active Agents

- Organic compounds that affect the surface activities

- Act as:

- Wetting agent
- Lubricant
- Dispersing agent
- Detergent
- Emulsifies

Surface Active Agents

- Wetting agents
- Reduce surface tension of materials to permit absorption of water by the material
- Powdered chocolate mixes

Surface Active Agents

- Used in processing of food to prevent sticking in untreated peanut butter
- Used in cleaning detergents used on food equipment
- Stabilize or break down foams

Surface Active Agents

- Emulsifiers such as lecithin
- May be added to bakery products
- Improve volume and texture of final product
- Also used in cake mixes, frozen desserts, ice cream

END

Food Biotechnology

Module # 118



Leavening Agents



Leavening Agents

- Used to enhance the rising of dough in the manufacture of baked products

- Chemical reagents that react to form carbon dioxide

Leavening Agents

- Inorganic salts mostly
- Ammonium salts
- Phosphate salts
- Favor growth of yeast that produces carbon dioxide
- Results in rise of dough

Leavening Agents

- Baking powder is a common household leavening agent
- Contains a mixture of chemical compounds
- React together to form carbon dioxide

Leavening Agents

- Baking soda can be of two types
- Single acting
- Double acting

Leavening Agents

- Single acting

- Sodium bicarbonate (base)
- Potassium acid tartrate (acid)

- Acid base reaction

- Carbon dioxide is liberated in one step

Leavening Agents

END

- Double acting

- Two acid reacting agents:
- Monocalcium phosphate monohydrate
- Sodium aluminium sulfate

- Portion of CO_2 released from sodium bicarbonate reaction with phosphate
- Remaining from sulfate

Food Biotechnology

Module # 119



Chemical Preservatives

Chemical Preservatives

- Foods can be preserved altering them to make them incapable of supporting microbial growth
- Usually used in concentrations of 0.1% or less

Chemical Preservatives



- Sodium diacetate and sodium or calcium propionate
- Used in breads
- Prevents mold growth and development of bacteria that may produce a slimy material known as rope

Chemical Preservatives

- Sorbic acid and its salts
- Used in bakery products, cheeses, syrups, and pie fillings to prevent mold growth

- Sulfur dioxide
- Used to prevent browning in certain dried fruits
- Prevent wild yeast growth in wines used to make vinegar

Chemical Preservatives

- Benzoic acid
- Sodium benzoate

- Used to inhibit mold and bacterial growth in some fruit juices, oleomargarines, pickles, and condiments

- Benzoic acid is a natural component of cranberries

Chemical Preservatives

- Salt is an excellent inhibitor
- Suppresses water activity
- Effectiveness enhanced in conjugation with drying and smoking

Chemical Preservatives

- Sodium Chloride
- Makes water unavailable to microorganisms
- Lowers the water activity
- Most bacteria require water activity of 0.95
- Chloride ion inhibits growth

Chemical Preservatives

- Sodium Nitrite

- Inhibit microbial growth
- Enhance color
- Prevents oxidation of myoglobin in meat
- Tends to prevent growth of spores of botulinum
- Not to be used in fresh foods

Chemical Preservatives

- Sodium Nitrite
- Nitrite cured products cooked at high temperatures may develop nitrosamine
- Known to be extremely carcinogenic

Chemical Preservatives

- Oxidizing Agents

- Not ordinarily used in foods
- Used to sanitize food equipment
- Even walls and floors of food processing area
- Small residuals can get into food

Chemical Preservatives

- Oxidizing agents

- Chlorine
- Iodine

eND

- Hydrogen peroxide may be used to destroy natural flora of milk prior to inoculation with culture

Food Biotechnology

Module # 120



Food Biotechnology

Food Colorants



Food Colorants

- Color additives used to make food more desirable
- Three major types:
- Natural
- Nature-identical
- Synthetic

Food Colorants

- Natural colors are pigments obtained from animal, vegetable, or mineral sources

- Yellow from the annatto seed
- Green from chlorophyll
- Orange from carotene
- Brown from burnt sugar
- Red from beets, tomatoes

Food Colorants

- Nature-identical
- Synthetic counterparts of color and pigment derived from natural sources
- Pure carotenoids:
- Canthaxanthin (red)
- Apocarotenal (orange-red)
- Beta-carotene (yellow-orange)

Food Colorants

- Some colorants are derived from synthetic dyes
- Dyes and lakes

- Dyes are water soluble and available in powder, granules, liquids, blends, pastes
- Lakes are water soluble dyes on a substratum of aluminum hydrate or aluminum oxide

Food Colorants

- Lakes are useful in foods that have very little water such as coloring oils

- Used in:
- Fondant coatings
- Icing
- Cake
- Doughnut mixes
- Hard candy
- Gum products

Food Colorants

- Some compounds are not color additives but are added to produce a white color

- Oxidizing agents including benzoyl peroxide, chlorine dioxide, nitrosyl chloride, and chlorine are used at the end of the production cycle to whiten wheat flour which is pale yellow in color if untreated

Food Colorants

ENd

- Titanium dioxide
- Considered a color additive
- May be added to some foods such as artificial cream or coffee whiteners to add a white color

Food Biotechnology

Module # 121



Food Biotechnology

Determination of titratable acidity of fruit juices

Determination of titratable acidity of fruit juices

- The titratable acidity of fruits is used as an indicator of maturity
- To determine the titratable acidity, a known volume of food sample is titrated with a standard base to an endpoint

Determination of titratable acidity of fruit juices

- The volume of titrant used along with the normality of the base and the volume of the sample are used to calculate titratable acidity

- Expressed in terms of predominant organic acid

Determination of titratable acidity of fruit juices

- pH is used to determine the end point of an acid-base titration

- An indicator dye is used

- Phenolphthalein
- Bromothymol blue

Determination of titratable acidity of fruit juices

- All food acids are weak acids
- Buffering occurs in food whenever a weak acid and its salts are present in the same medium
- Naturally occurring food acids do not buffer in the region of phenolphthalein endpoint

Determination of titratable acidity of fruit juices

- Phenolphthalein
- Changes color from clear to red in pH range 8.0 to 9.6
- Bromothymol blue
- Changes color from yellow to blue is pH range 6.0 to 7.6

Determination of titratable acidity of fruit juices

- Deeply colored samples present a problem for end point indicators
- A potentiometric method is used for such samples
- All indicators are either weak acid or weak base
- Influence titration in excessive amounts
- 2-3 drops used typically

Determination of titratable acidity of fruit juices

- Sodium Hydroxide is the most commonly used base
- NaOH titrating solutions must be standardized against a standard acid
- Potassium acid phthalate is used (KHP)

Determination of titratable acidity of fruit juices

- An aliquot sample often of 10ml is titrated with a standard alkali solution (often 0.1N NaOH) to a phenolphthalein endpoint
- Titrant is dispensed at a slow uniform rate until the endpoint is reached
- Then added drop-wise until the endpoint fades

Determination of titratable acidity of fruit juices

END

- Calculation of titratable acidity
- % acid (wt/wt) =
- [N x V x Eq wt / W x 1000] x 100
- N= normality of titrant
- V= volume of titrant
- Eq wt = equivalent weight of predominant acid
- W = mass of sample
- 1000 = factor (mg/g)

Food Biotechnology

Module # 122



Food Biotechnology

Introduction to Probiotics

Introduction to Probiotics

- Probiotics

- Best defined as live microorganisms which when reaching the intestines in sufficient numbers will exert positive effects
- Viable microorganisms that promote or support a beneficial balance of the microbial population of the GIT

Introduction to Probiotics

- Recent research has underlined the importance of a vital and "healthy" microbial population of the GIT

- The expression "probiotic" was probably first defined by Kollath in 1953

Introduction to Probiotics

- Although numerous definitions have been proposed
- None has been completely satisfactory
- Need for additional explanations, e.g., with regard to statements such as:
- “beneficial balance,” “normal population,” or “stabilization of the gut flora.”

Introduction to Probiotics

- A particular feature of probiotic cultures is that they regulate the balance of the gut bacterial population,

- By competition for epithelium contact sites and nutrients
- Also by modulation of the pH value

Introduction to Probiotics

END

- May not necessarily be constant inhabitants of the GIT
- Practically all organisms used in probiotic foods or food supplements are representatives of the genera Lactobacillus, Enterococcus or Bifidobacterium

Food Biotechnology

Module # 123



Administration and Consumption of Probiotics

Administration and Consumption of Probiotics

- Probiotics may be administered in different forms

- Traditional fermented foods
- Food products
- Pharmaceuticals

Administration and Consumption of Probiotics

- Traditional fermented food

- Lactic acid fermented food of plant or animal origin
- Consumed in raw or uncooked state

- Presence of LAB strains with ability to survive passage of GIT

Administration and Consumption of Probiotics

- Food products

- Novel-type fermented food containing strains selected for functional properties
- Probiotic “yoghurts”
- Nutritional supplements as capsules, tablets
- Cereal based food containing live microorganism and prebiotics

Administration and Consumption of Probiotics

- Pharmaceuticals
- Viable microorganisms selected for their functional properties
- Administered in capsules
- Sometimes as micro-encapsulated lyophilised powders

Administration and Consumption of Probiotics

- A continued increase is observed among the dairy-type probiotic foods
- Also in the range of nondairy probiotic food products such as:
- Fermented meats
- Vegetable
- Fruit juices

Administration and Consumption of Probiotics

END

- Wide range of potential (fermentable) substrates
- Different conditions under which LAB strains may be challenged for “functional performance
- Expected developments toward new food-based probiotics

Microbiota of GI Tract

Microbiota of GI Tract

- Increasing microbial populations are found throughout the GIT
- Many microbes isolated from the duodenum and jejunum are considered to be typical transients

Microbiota of GI Tract

- The estimated total population of 10^{14} viable bacteria in the adult human GIT represents about 10 times more than all body tissue cells

Microbiota of GI Tract

- Microbial population represents an immense metabolic potential
- Supports the digestion processes
- Interactive in detoxification and toxification processes
- Comprises the major part of the human immune system

Microbiota of GI Tract

- Bacteroides
- Eubacterium
- Bifidobacterium
- Predominate in the densely populate large intestine

- Role of the lactobacilli more important in small intestines where they comprise a higher proportion of the total population

Microbiota of GI Tract

- Considered as beneficial bacterial groups of major importance to the gut ecosystem, special attention was given to the Gram-positive genera Lactobacillus
- Bifidobacterium,
- And more recently, to Eubacterium

Microbiota of GI Tract

- LAB in the gut are mainly represented by the lactobacilli
- In contrast to their domination in the ileum, they only form a minor population in the colon

- Make up about 1% of the total bacterial population of human feces
- May be more numerous in the proximal colon

Microbiota of GI Tract

- In contrast to the oxygen-tolerant lactobacilli
- Study of anaerobic groups such as bifidobacteria and eubacteria was made possible by the development of anaerobic techniques developed in the early 1970s

Microbiota of GI Tract

- Compared to their domination in infants, bifidobacteria comprise only up to 3% of the total fecal bacteria of humans

- A reduced environment and special media are applied for the selective cultivation of bifidobacteria

Microbiota of GI Tract

- Functions of normal gut bacteria:
- Maintenance and restoration of barrier function
- Stimulation of the immune system
- Maintenance of mucosa nutrition and circulation

Microbiota of GI Tract

- Functions of normal gut bacteria:
- Improvement of bioavailability of nutrtients
- Stimulation of bowel motility and reduction of constipation

END

Food Biotechnology

Module # 125



Food Biotechnology

Probiotic Microorganisms

Probiotic Microorganisms

- Since at least 40 years in Japan and
- More than 30 years in Germany
- LAB cultures of human origin are applied in the manufacture of fermented milk products

Probiotic Microorganisms

- Viable probiotic strains with beneficial functional properties are at present found among a wide and diverse number of microbial species and genera

- Most strains currently in use as probiotics in food, nutrition, and in pharmaceutical preparations are members of the LAB

Probiotic Microorganisms

TABLE 1.5

Microorganisms Reported to Find Application as Probiotics Mainly for Humans

Lactobacillus Species	Bifidobacterium Species	Other LAB	"Non-lactics"[c]
L. acidophilus	*B. adolescentis*	*Ent. faecalis* [a]	*Bacillus cereus* ("toyoi")[a,c]
L. amylovorus	*B. animalis*	*Ent. faecium*	*Escherichia coli* (Nissle 1917)[c]
(L. casei)	*B. bifidum*	*Sporolactobacillus inulinus* [a]	*Propionibacterium freudenreichii*[a,c]
L. crispatus	*B. breve*		*Saccharomyces cerevisiae* ("boulardii")[c]
L. delbrueckii subsp. bulgaricus[c]	*B. infantis*		
L. gallinarum[a]	*B. lactis*[b]		
L. gasseri	*B. longum*		
L. johnsonii			
L. paracasei			
L. plantarum			
L. reuteri			
L. rhamnosus			

[a] Mainly for animals.
[b] Synonym of *B. animalis*.
[c] Mainly in pharmaceutical preparations.

Source: Modified from Holzapfel et al. (39).

Probiotic Microorganisms

- A number of “nonlactic” strains, e.g., *Bacillus cereus, Escherichia coli, and Saccharomyces cerevisiae*

- Available in the market mainly as pharmaceutic preparations and some also as animal feed supplements

Probiotic Microorganisms

- Three major requirements need to be met for approval of functional foods

1. Scientific evidence of the efficacy, including clinical testing
2. Safety for consumption
3. Analytical determination of the effective component

Probiotic Microorganisms

- Functional food products contain primarily strains of different bacteria

Probiotic Microorganisms

END

- Biotherapeutics for clinical applications are also based on selected probiotic strains
- Mainly LAB but may also include:
- *Escherichia coli* strains
- *Saccharomyces cerevisiae*
- *Enterococcus faecium*
- *E. fecalis*

Food Biotechnology

Module #126



Selection of appropriate strains

- The primary objective is to select microbial strains with functional properties.
- Requires efforts and resources for selection of an “ideal” strain.
- Desirable technical features and factors related to health promotion or sustaining health are important selection.

Selection of appropriate strains

Five key criteria for the selection of an appropriate functional strain:

1. General aspects:
2. Origin,
3. Identity, and
4. Resistance to mutations.
5. Technical aspects (growth properties *in vitro* and during processing, survival and viability during transport and storage)

Selection of appropriate strains

- **General aspects**
- resistance against environmental stress and to the antimicrobial factors prevailing in the upper GIT as encountered during the stomach-duodenum passage)

Selection of appropriate strains

- **Functional aspects and beneficial features:**
  - adhesion,
  - colonization potential of the mucosa,
  - competitiveness,
  - specific antimicrobial antagonism against pathogens,
  - stimulation of immune response,
  - selective stimulation of beneficial bacteria etc.

Selection of appropriate strains

- **Safety aspects:**
- No invasive potential
- No transferable resistance against therapeutic antibiotics
- No virulence factors.

Selection of appropriate strains

Major obstacles towards the large-scale production of functional foods containing probiotic strains like:

- Technical properties related to growth,
- Adaptation
- Persistence of some probiotic strains.
- The sensory properties of the resulting products.

Selection of appropriate strains

Use of *Lactobacillus* strains as probiotics.

- Association with traditional fermented foods, together with the high acceptability of lactic fermented foods.
- Their association with the human GIT, together with observations on their beneficial interactions in the gut ecosystem

Selection of appropriate strains

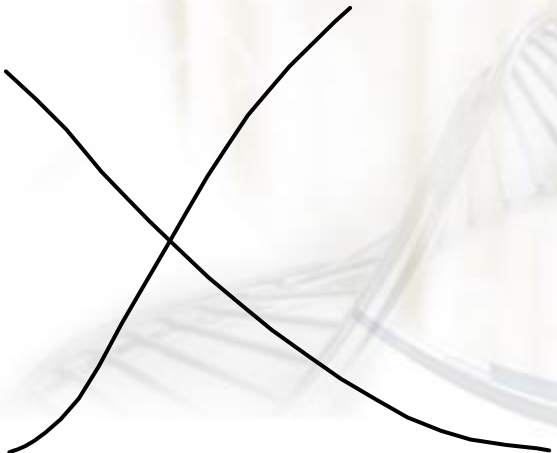
For the regulatory authorities,

- the safety and
- Non-pathogenicity of a new strain is considered of major importance.

END

Food Biotechnology

Module #126



Selection of appropriate strains

- The primary objective is to select microbial strains with functional properties.
- Requires efforts and resources for selection of an “ideal” strain.
- Desirable technical features and factors related to health promotion or sustaining health are important selection.

Selection of appropriate strains

Five key criteria for the selection of an appropriate functional strain:

1. General aspects:
2. Origin,
3. Identity, and
4. Resistance to mutations.
5. Technical aspects (growth properties *in vitro* and during processing, survival and viability during transport and storage)

Selection of appropriate strains

- **General aspects**
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Selection of appropriate strains

END

For the regulatory authorities,

- the safety and
- Non-pathogenicity of a new strain is considered of major importance.

Food Biotechnology

Module #127



Health benefits of Probiotics

NUTRIONAL BENEFITS:

- Vitamin production, availability of minerals and trace elements
- Production of important digestive enzymes (e.g., β-galactosidase) for alleviation of lactose in-tolerance.

Health benefits of Probiotics

Barrier, Restoration, Antagonistic Effects Against:

- Infectious diarrhoea (traveller's diarrhoea, children's acute viral diarrhoea)
- Antibiotic-associated diarrhoea, irradiation-associated diarrhoea.

Health benefits of Probiotics

Cholesterol-lowering Effects By:

- Cholesterol assimilation.
- Modification of bile salt hydrolase activities.
- Antioxidative effect.

Health benefits of Probiotics

Stimulation and improvement of the immune system:

- Strengthening of nonspecific defence against infection.
- Increasing phagocytic activity of white blood cells.
- Increasing IgA production.
- Regulating many immune functions.

Health benefits of Probiotics

Enhancement of bowel motility,

(relief from constipation & reduction of inflammatory or allergic reactions) **by:**

- Restoration of the homeostasis of the immune system.
- Regulation of cytokine synthesis.

Health benefits of Probiotics

END

Adherence and colonization resistance, Anti carcinogenic effects effects in the colon by:

- Mutagen binding
- Inactivation of carcinogens, or prevention of their formation.
- Modulation of metabolic activities of colonic microbes.
- Immune response.

Food Biotechnology

Module #128



Health Trials of Probiotics

- Human trials (clinical or dietary intervention studies) are essential for proving health benefits of probiotic strains.

Health Trials of Probiotics

Different designs have been applied such as

- pre- and postintervention designs and
- placebo-controlled designs,
- parallel and
- crossover designs, and
- case-control studies.

The best evidence is probably coming from

- double-blind,
- randomized,
- placebo- controlled trials.

Health Trials of Probiotics

- Epidemiologic evidence relating probiotics or probiotic containing foods and disease incidence would be valuable but is hardly available.
- The studies would be difficult to control as fundamental parameters such as:
- specific strain and dose would be unknown for most probiotic-containing food products.

Health Trials of Probiotics

- In human studies certain markers are usually applied in order to observe the effect of pre- and probiotics or functional foods in general on the human body.

Health Trials of Probiotics

END

Markers can be classified into three categories:

a) Markers that relate to the exposure to the food component under study.

b) Markers that relate to the target function or biological response.

c) Markers that relate to an appropriate intermediate endpoint.

Food Biotechnology

Module #129



Technological application of probiotics in food products

Application of probiotic cultures in food products.

- Probiotic bacteria are applied in many different products worldwide.
- In addition to food products, probiotic cultures are also used in pharmaceuticals and animal feed.

Technological application of probiotics in food products

Dairy products

- applied in fermented dairy products for many years.
- In some cases fermented milk products are monocultures of probiotic bacteria.
- Usually support cultures are applied to speed up the acidification process and provide the desired texture and flavor.

Technological application of probiotics in food products

Dairy products

- Many *lactobacilli* and *bifidobacteria* survive in fermented milk products for 4 to 8 weeks.

Technological application of probiotics in food products

Several parameters influence the growth and survival :

- the starter culture,
- fermentation temperature,
- pH,
- sugar content,
- presence of oxygen,
- packaging material,
- fruit preparations, and other ingredients.

Technological application of probiotics in food products

DAIRY PRODUCTS

Probiotics may also be applied to unfermented milk products such as milk-based sweet or acidified drinks and ice cream.

Technological application of probiotics in food products

The applicability of probiotics in food products depends

- water activity,
- processing and
- storage temperature,
- shelf life,
- oxygen content,
- pH,
- mechanical stress,
- salt content, and
- content of other harmful or essential ingredients.

Technological application of probiotics in food products

Other food products

- Products with an unfavourable water activity are, for example, cereals, chocolate, marmalade, honey, and toffees.
- These products are too “dry” for applying live bacteria and too “wet” for the application of freeze-dried bacteria.
- Freeze-dried bacteria could be applied in these products.

Technological application of probiotics in food products

Other food products

- Lactic acid bacteria have been used for the fermentation of meat products,today some strains are also utilized as protective cultures.
- Freeze-dried probiotic bacteria are applied to infant nutrition powders and powdered milk drinks.

Technological application of probiotics in food products

END

Food supplements and over-the-counter products

- Most probiotic food supplements and over-the-counter (OTC) products are available as powders, tablets, and capsules.

Food Biotechnology

Module #130



Concept of food safety

- Food safety is the assurance that food will not cause harm to the consumer when it is prepared and eaten according to its intended use.
- All requirements relating to the safety characteristics of a food must be met; there must be no unacceptable health risk associated with a food.

Concept of food safety

The assurance that a food will not cause harm, injury, or illness is determined by:

1) whether all harmful substances present in the food have been eliminated, reduced to an established acceptable level, or prevented from exceeding the acceptable level;

Concept of food safety

2) the food has been prepared, handled, and stored under controlled and sanitary conditions in conformance with practices prescribed by government regulations.

Concept of food safety

END

The harmful substances in foods are food safety hazards and the prescribed conditions and practices for preparing, handling, and storing food are considered GMPs

Food Biotechnology

Module #131



Concept of food quality

- Food quality, as distinct from food safety, is the extent to which all the established requirements relating to the characteristics of a food are met.
- Common examples of quality characteristics of food, excluding the food safety characteristics, are:
  - Identity of a food in relation to a standard (e.g., standardized food)

Concept of food quality

- Declared gross or net quantity (e.g., weight or volume) of a unit of the food or net fill of a food container
- Declared or claimed amount of one or more stated components of a food.
- Appearance (e.g., size, shape, color)
- Flavor
- Aroma
- Texture

Concept of food quality

- Viscosity
- Shelf-life stability
- Fitness for use as human food
- Wholesomeness
- Adulteration
- Packaging
- Labeling

Concept of food quality

- Some of these quality characteristics are covered in food laws and regulations.
- For instance, failure of a food to meet regulatory requirements relating to a standard of identity, the declared quantity, declared ingredients, or label claims, can be considered as misrepresentation, misbranding, or fraud.

Concept of food quality

- The spoilage, deterioration, or decomposition of foods with the absence of any resulting harmful substance that can lead to illness or injury, can be considered as failure to meet food quality requirements based on fitness for human use or wholesomeness criteria.

Concept of food quality

- In addition to the quality requirements established by government regulations, numerous requirements for food quality characteristics are also established by customers and consumers.

END

Food Biotechnology

Module #132



Responsibility of Food safety

The overall responsibility for food quality and food safety is shared by all segments of the food system, including

- the various food industry sectors,
- government regulatory agencies, and
- consumers in general.

Responsibility of Food safety

- The food industry has both the legal and moral responsibility for providing customers and consumers with foods that meet all established quality and safety requirements.
- Within a food company, overall responsibility for the implementation and effective use of these programs and systems rests with senior management.

Responsibility of Food safety

Governments worldwide have enacted food laws and regulations designed to ensure that foods are fit for human consumption. Such laws protect consumers from

- harm resulting from unsafe foods and
- from deception resulting from misrepresentation
- or fraud relating to certain established food quality characteristics

Responsibility of Food safety

Governments have also established various agencies that enforce these food laws and regulations; this legal framework is intended to provide consumers with confidence in the safety and quality of foods.

Responsibility of Food safety

Within the food supply chain, customers who purchase

- raw materials,
- ingredients and
- food contact packaging materials for manufacture of consumer foods,

must ensure that these materials are safe and fit for use.

Responsibility of Food safety

In particular, customers and consumers must pay attention to the instructions for handling, storage, preparation, and use of foods.

END

Food Biotechnology

Module #133



Factors of Food Quality and Systems

Food quality is the extent to which all the established requirements relating to the characteristics of a food are met.

Common examples of quality characteristics of food, excluding the food safety characteristics, are:

- Identity of a food in relation to a standard (e.g., standardized food)
- Declared gross or net quantity (e.g., weight or volume) of a unit of

Factors of Food Quality and Systems

- The food or net fill of a food container
- Declared or claimed amount of one or more stated components of a food
- Appearance (e.g., size, shape, color)
- Flavor
- Aroma
- Texture
- Viscosity
- Shelf-life stability
- Wholesomeness
- Fitness for use as human food

Factors of Food Quality and Systems

- Adulteration
- Packaging
- Labeling

Factors of Food Quality and Systems

Characteristics covered in food laws and regulations are:

- Failure of a food to meet regulatory requirements relating to a standard of identity,
- the declared quantity, declared ingredients,
- or label claims, can be considered as misrepresentation, misbranding, or fraud

Factors of Food Quality and Systems

Based on fitness for human use or wholesomeness criteria these can be considered as failure:

- The spoilage, deterioration, or decomposition of foods with the absence of any resulting harmful substance that can lead to illness or injury, can be considered as failure to meet food quality requirements based on fitness for human use or

Factors of Food Quality and Systems

Numerous requirements for food quality characteristics are also established by customers and consumers.

Factors of Food Quality and Systems

The food industry, used basic

- quality control programs,
- and more complex quality assurance programs and
- quality management systems,

in its efforts to achieve food quality; some food companies use the ISO 9000 Quality Management System Standard.

Factors of Food Quality and Systems

END

GMPs and the HACCP system can be integrated into a food industry,

- quality management system, or
- inspection and monitoring of materials, products, and
- processes for food safety hazards can be part of a quality control program.

Food Biotechnology

Module #134



Food Safety and Food Safety Systems

Systems and Programs for Food Safety

Food Safety and Food Safety Systems

- For decades, the food industry has depended on the use of quality programs based on inspection and testing of food products for hazards, and on GMPs for addressing food safety.
- Since the late 1980s, there has been widespread use of the HACCP system.
- HACCP system is designed to achieve food safety:
- Addresses food safety primarily on the basis of prevention or elimination of unacceptable hazard levels.

Food Safety and Food Safety Systems

- The GMPs, were used to address food safety requirements prior to the use of the HACCP system.
- GMP have also been incorporated into prerequisite programs for the HACCP system.
- A food company that does not operate with the HACCP system must continue to use the GMPs.

Food Safety and Food Safety Systems



Food Safety and Food Safety Systems

- **Food laws and regulations**
- The legal requirements protecting consumers and ensuring that foods are fit for human consumption
- scope varies from one country to another.
- The legal framework of food laws and regulations of a particular country depends on the overall government regulatory system of that country.

Food Safety and Food Safety Systems

- In the U.S. and Canada, the federal or national food laws are statements of government policies that cover both the general and specific aspects of adulteration and misbranding of foods, while the food regulations deal with the enforcement of government policies that are embodied in the food laws.

Food Safety and Food Safety Systems

Food laws and regulations apply to all foods produced

- domestically, as well as all
- foods imported into a country;
- foods cannot be imported if they do not conform to the food laws and regulations of the importing country.

Food Safety and Food Safety Systems

Examples of food laws are the:

- U.S. Federal Food, Drug, and Cosmetic Act (FDCA), which is the primary law governing the safety and quality of most foods in the U.S.
- Canada's Food and Drugs Act, which is the primary food law in Canada.

Food Safety and Food Safety Systems

The U.S. Code of Federal Regulations (CFR) Title 21 and Canada's Food and Drug Regulations are examples of food regulations that address food safety and food quality.

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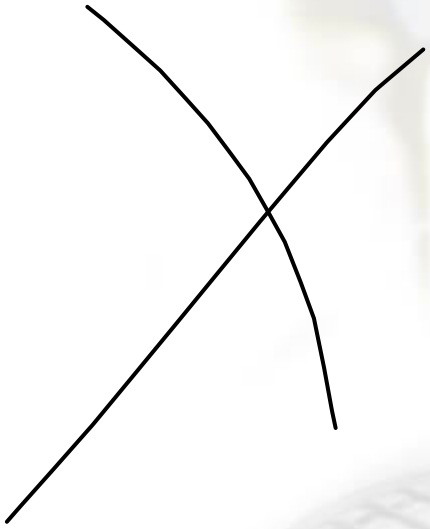
Food Safety and Food Safety Systems

END

- Food laws protect consumers from illnesses and injury by prohibiting the presence of any poisonous or harmful substance in foods that are intended for human consumption.
- For example, in the U.S., adulterated food is regulated primarily under the FDCA, which covers all aspects of food safety and certain aspects of food quality.

Food Biotechnology

Module #135



Food Laws and Regulations

The responsibility for enforcing is assigned to regulatory agencies.

Enforcement activities fall into two categories:

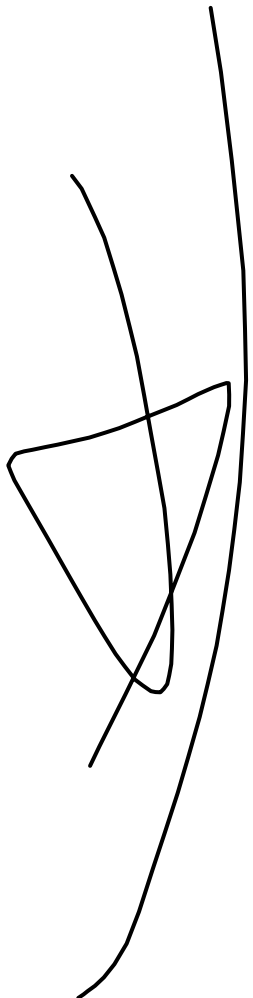
First, they include

- inspection and audit of establishments that process, handle, and store food to ensure that the required sanitary and controlled conditions are followed;

Food Laws and Regulations

- audits are used by some regulatory agencies that enforce HACCP-based regulations.
- Second, they include inspection and analysis of foods for harmful substances to ensure that there is conformance to established limits and tolerances.

Food Laws and Regulations



- When a misbranded/ harmful substance or agent is detected in a food, it is determined that there is a likelihood for a harmful substance or agent to be present in a food, or an actual food-borne illness or injury occurs,
- food companies and government regulatory agencies take the necessary action to protect consumers.

Food Laws and Regulations

END

When such thing happen it often result in

- the food being recalled from the marketplace.

In addition, if it is determined that adulterated or misbranded food has been produced as a result of negligence on the part of a food company, legal action can be taken against the company.

Food Biotechnology

Module #136



Food Standards

- Food standards in addition to food laws and regulations also establish requirements for the safety and quality of foods.
- A food standard is part of food regulations (e.g., standard of identity in the U.S. CFR Title 21), it is not a legal requirement.
- The Codex Standards are the best examples of food standards.

Food Standards

- The Codex Alimentarius Commission has the mandate to implement the joint Food and Agricultural Organization (FAO)/ World Health Organization (WHO) Foods Standards Program.
- This has resulted in the Codex Alimentarius, a collection of standards for food quality, food suitability, and food safety.

Food Standards

- The food standards have been adopted by countries worldwide and are intended primarily to protect consumers and to facilitate international food trade.

Food Standards

END

- They include codes of practice such as:
- The Codex General Principles Of Food Hygiene, standards for maximum residual levels (MRL) for pesticides and for veterinary drugs in foods, and
- standards for specifications for food additives.

Food Biotechnology

Module #137



Food Safety Hazards and Health Risk

Food safety hazards and health risks

Food Safety Hazards and Health Risk

The safety of a food can be related directly to certain harmful substances that are present in the food; these substances are food safety hazards.

Any substance that is reasonably likely to cause harm, injury or illness, when present above an established acceptable level, is a food safety hazard.

Food Safety Hazards and Health Risk

There are three recognized categories of food safety hazards:

- biological hazards,
- chemical hazards, and
- physical hazards

Food Safety Hazards and Health Risk

The origin of these hazards in foods can be from:

- naturally occurring substances or
- agents in foods, from
- deterioration or decomposition of foods,
- or from contamination of the foods with the hazard at various stages of their production, harvesting, storing, processing, distribution, preparation, and utilization.

Food Safety Hazards and Health Risk

For many hazards, government regulatory agencies have established an acceptable level of the hazard in a food;

- the Codex Alimentarius has also established acceptable levels of certain hazards as part of its Food Standards Programme.

Food Safety Hazards and Health Risk

- For a known food safety hazard, the extent of the harmful effects of the hazard on the health of the consumer is established by risk analysis and by hazard analysis.
- Risk analysis is usually conducted by a national food or health regulatory agency and addresses a public health concern regarding a particular food safety hazard associated

Food Safety Hazards and Health Risk

A risk analysis is comprised of:

- risk assessment,
- risk management, and
- risk communication.

- A primary objective of risk analysis is to establish a national food safety objective for a hazard in a food.

Food Safety Hazards and Health Risk

- The food safety objective for a hazard is the maximum frequency and concentration of a hazard in a food at the time of consumption that provides the appropriate level of protection from the hazard.
- The food safety objective can be considered as the maximum acceptable level for the hazard in a food.

Food Safety Hazards and Health Risk

- Hazard analysis is the first of the seven HACCP principles, and is performed to determine the health risk associated with a hazard present in a food when it is produced, processed, handled, or stored, according to an established sequence of steps at a particular location.

Food Safety Hazards and Health Risk

- Once a food safety objective for a hazard has been established by risk analysis, it must be considered during the hazard analysis step of HACCP plan development

END

Food Biotechnology

Module #138



Biological Safety Hazards

Pathogenic Bacteria

Biological Safety Hazards

More than forty different pathogenic bacteria are known; however, following are most commonly associated pathogenic bacteria:

- *Salmonella* spp.,
- *Eschericha coli 0157:H7,*
- *Lysteria monocytogenes,*
- *Clostridium perfringens,*
- *Clostridium botulinum,*
- *Staphylococcus aureus,* and
- *Campylobacter jejeuni.*

Biological Safety Hazards

Food poisoning from these organisms occur frequently, with symptoms that include:

- headache,
- muscle pain,
- nausea,
- fatigue,
- chills or fever,
- stomach or abdominal pain,
- vomiting, and diarrhea.

Biological Safety Hazards

Numerous severe and fatal illnesses occur as a result of food poisoning from pathogenic bacteria; infants and the elderly are particularly vulnerable.

Biological Safety Hazards

The foods that are commonly involved in these food poisoning incidents include:

- meat and poultry and their products,
- seafood and seafood products,
- egg and egg products,
- milk and dairy products,
- fruits and vegetables and their products,
- low-acid canned foods, and water.

Biological Safety Hazards

Viruses:

- Foods can be the medium for transmission of certain viruses.
- Examples of viruses that are known to be food safety hazards are :
  - the hepatitis A and E viruses,
  - the Norwalk group of viruses, and
  - rotavirus.

Biological Safety Hazards

END

Parasites:

- parasitic protozoan species (e.g.,
  - *Entamoeba histolytica,*
  - *Giardia lamblia,*
  - *Cryptosporidium parvum*), and
- parasitic worms (*Ascaris lumbricoides,*
  - *Taenia solium, Trichinella spiralis*).

Food Biotechnology

Module #139



Food Biotechnology

Chemical Safety Hazards

Chemical Safety Hazards

Permitted food additives

- These substances are intended to impart some improved nutritional effect
- (e.g., vitamin fortification) or some specific technical function
- (e.g., preservative action, sensory attribute, stabilizing effect, etc.).

Chemical Safety Hazards

Naturally occurring harmful compounds

- Many foods contain inherent components that can be harmful if they are present in excess of certain levels; examples are:
  - oxalate in rhubarb,
  - alkaloids in potatoes,
  - toxins in mushrooms and in shellfish.

Chemical Safety Hazards

Unavoidable contaminats

- harmful substances that are not normal or inherent components of the foods.
- cannot be removed through processing or manufacturing practices; examples are
- aflatoxins from molds in peanuts and in some cereals.

Chemical Safety Hazards

Agricultural residues

- They include residues of permitted pesticides,
- herbicides,
- fungicides,
- drugs,
- hormones, and antibiotics.

Chemical Safety Hazards

Industrial contaminats

- Several harmful chemicals that enter the environment as a result of industrial activity have been shown to be present in foods.
- These substances include heavy metals,
- organo-chlorinated compounds such as polychlorinated biphenyls (PCBs).

Chemical Safety Hazards

Chemical residues

- These substances include chemical compounds used for cleaning and sanitizing food contact surfaces of processing, handling, and
- storage equipment, and for lubricating certain parts of food processing equipment.

Chemical Safety Hazards

PROHIBITED CHEMICALS

- No chemical substance is permitted for use in a food unless it meets all of the requirements that are covered in the applicable food laws and regulations.

Chemical Safety Hazards

END

Food allergens

- Most common food allergens:
  - peanuts,
  - soybeans,
  - milk, eggs,
  - fish,
  - crustacea,
  - tree nuts, and wheat.

Food Biotechnology

Module #140



Physical Safety Hazards

Broken glass

- In a food plant, the common potential sources of broken glass include light bulbs, glass containers, and gauges with glass covers.
- In addition, many foods are packaged, distributed and sold in glass containers. For these foods, the glass packaging itself can be a source of broken glass.

Physical Safety Hazards

Plastics:

- Both hard and soft plastic foreign objects are sometimes found in foods.
- In some food plants, some utensils and tools used for cleaning of equipment are made from hard plastic material.

Physical Safety Hazards

Metal pieces

- The most common sources of metal pieces in a food plant are food processing equipment, metallic cleaning tools, and equipment maintenance activities.
- In many food plants, magnets are used to eliminate some metals from foods, and metal detectors are used to detect the presence of metals in foods.

Physical Safety Hazards

Wood pieces

- The most common sources of wood pieces in a food plant are wood structures and wood pallets.
- The presence of these sources should be avoided whenever possible in food processing and production.

Physical Safety Hazards

Stones:

- Many plant foods and particularly field crops such as peas and beans can contain small stones that become incorporated with the foods during harvesting.
- In addition, in food processing plants, a common source of stones is concrete structures, particularly concrete floors.

Physical Safety Hazards

Personal Articles:

- Personal articles that have been found in foods include jewellery, pens or pencils or their parts, Band-Aids, and ear plugs.

Physical Safety Hazards

END

Other food safety concerns

- These include concerns relating to the safety of foods from biotechnology and particularly from genetically modified organisms, the safety of irradiated foods, and the safety of some herbal supplements and botanical products.

Food Biotechnology

Module 141



Good Manufacturing Practices (GMP)



Good Manufacturing Practices (GMP)

(GMPs) that have been mandated by government agencies on the basis of scientific knowledge relating to known health hazards in foods, and the need to prevent unacceptable levels of these hazards or to eliminate them from food

Good Manufacturing Practices (GMP)

→ Some GMPs Addresses Quality assurance

→Consequently, the traditional GMPs, along with some additional requirements, are now universally regarded as Prerequisite Programs for the implementation of an HACCP system

Good Manufacturing Practices (GMP)

GMP programs or they should cover the basic controls required for raw materials, ingredients, packaging materials and products, and for the plant's facilities, employees, equipment, operations, and environment that influence the safety of a food.

Good Manufacturing Practices (GMP)

Components of GMP

1) GHP (Good Hygiene Practices)

2) GPP (Good Personnel Practices)

3) GVP (Good Vetrinary Practices)

→ Healthy, Fit , Clean animal

4) Good Sanitary Practices

Good Manufacturing Practices (GMP)

END

5) Good raw material practices

→ No cross contamination
→ Proper storage
→ Air, humidity ventilation

6) Good environment practices

Food Biotechnology

Module 142



HACCP Systems

HACCP Systems

HACCP is a systematic preventive approach to food safety from biological, chemical and physical hazards in production processes that can cause finished product to be unsafe and designs measures to reduce these risks to a safe level

HACCP Systems

Pre-requisite of HACCP :-

→ Different for different countries and foods

In Canada, Agriculture and Agri-Foods Canada Food Safety Enhancement Program (FSEP), HACCP prerequisite programs are covered by the following: Premises, Transportation and Storage, Sanitation, Equipment, Personnel Training, Recall Program

HACCP Systems

USDA have Food Safety and Inspection Services (FSIS) mandatory Pathogen Reduction: HACCP inspection program for the meat and poultry sector.

HACCP Systems

HACCP prerequisite programs are covered in sections on Establishment's grounds and facilities, Equipment and utensils, Sanitary operations, Employee hygiene, Tagging unsanitary equipment, utensils, rooms or compartments and mandatory Sanitation standard operating procedures (SSOPs).

HACCP Systems

Seven Principles of HACCP:-

1) **Conduct a hazard Analysis**

2) **Identify Critical Control Points :-**

Ccps are identified

For Milk Processing Ccps are

→ **Pasteurization**

→ **Cooling Temperature**

HACCP Systems

3) Establish Critical Limits :-

→ After the identification of Ccps Critical limits are established in order to control Ccps

e.g. For pasteurization Critical limits are 72ºC / 15 secs

HACCP Systems

4) Monitor Critical Limits :-

When Ccps are identified and Critical limits are set then Critical limits are monitored in order prevent safety hazards

HACCP Systems

5) Establish Corrective Actions :-

When Critical limits crossed then Corrective actions are taken in

→ Corrective actions are costly, time consuming so better monitoring is better than corrective actions

HACCP Systems

6) Verification :-

verification is done in order to see whether the corrective actions overcomes the problem or not

7) Record Keeping :-

→ Record of each and every step is compulsory in order to have better future plan

END

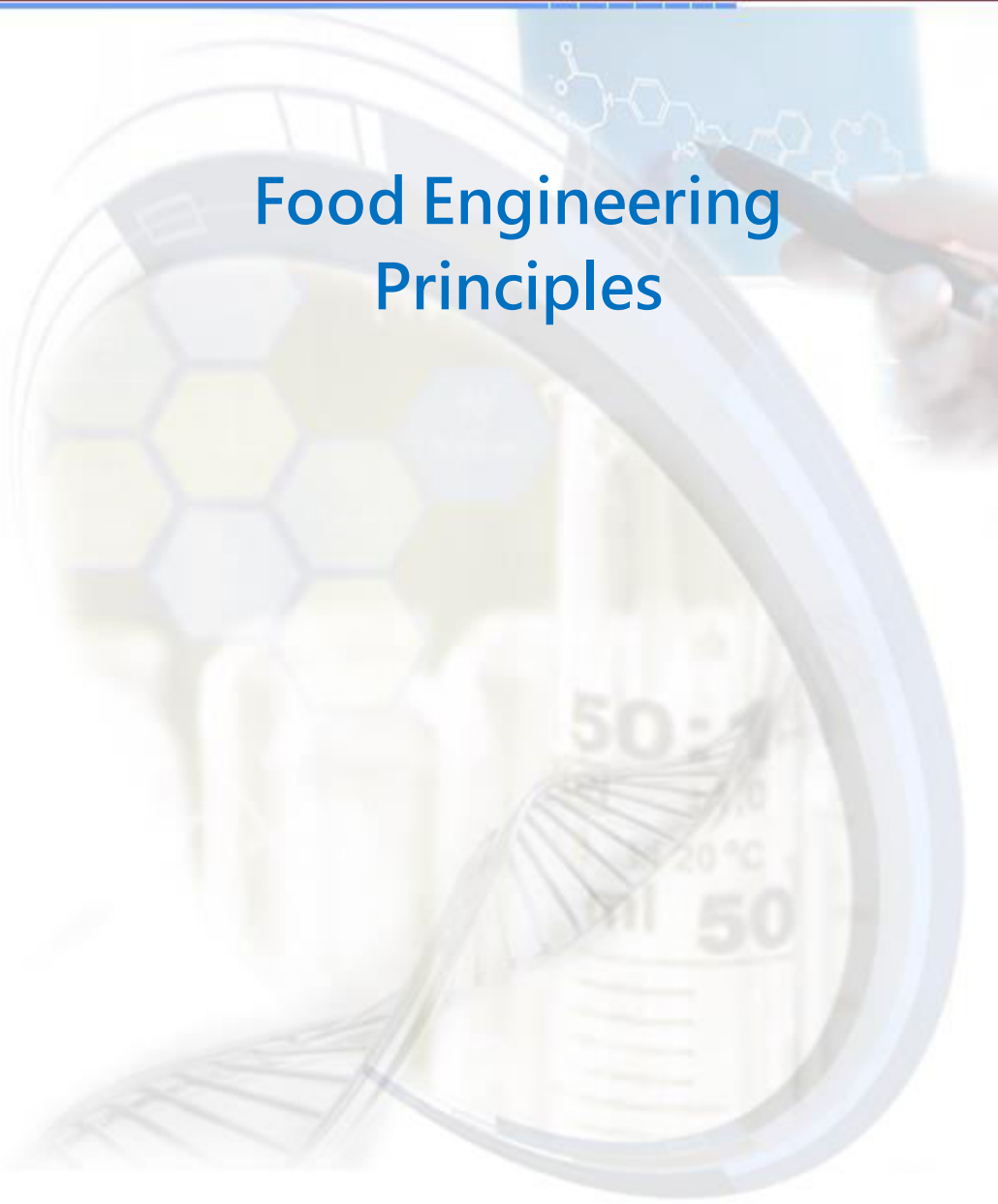
Food Biotechnology

Module 143



Food Biotechnology

Food Engineering Principles



Food Engineering Principles

Converting raw food into desired products involves equipment & processes. To design equipment, an engineer must consider numerous criteria that are inherent to a process. For example, in designing a heat exchanger, one must consider the heat transfer, fluid flow, and various physical, chemical, and biological aspects of food.

Food Engineering Principles

Overall engineering process involve energy balance and material balance.

Overall process can be broken down into series of operations. These are called unit operations.

Efforts are directed towards maximizing efficiency and reducing losses in every unit operations.

Food Engineering Principles

The essential concept is therefore to divide physical food processes into basic unit operations, each of which stands alone and depends on coherent physical principles. e.g. heat transfer is a unit operation and the fundamental physical principle underlying heat transfer principles.

Food Engineering Principles

Because of the dependence of the unit operation on a physical principle, or a small group of associated principles, quantitative relationships in the form of mathematical equations can be built to describe them. The equations can be used to follow what is happening in the process, and to control and modify the process if required.

Food Engineering Principles

Important unit operations include fluid flow, heat transfer, drying, evaporation, contact equilibrium processes (which include distillation, extraction, gas absorption, crystallization, and membrane processes), mechanical separations (which include filtration, centrifugation, sedimentation and sieving), size reduction and mixing.

Food Engineering Principles

The law of conservation of mass and energy is applied. Hence losses are calculated and discouraged in both materials and energy for best optimization of process.

END

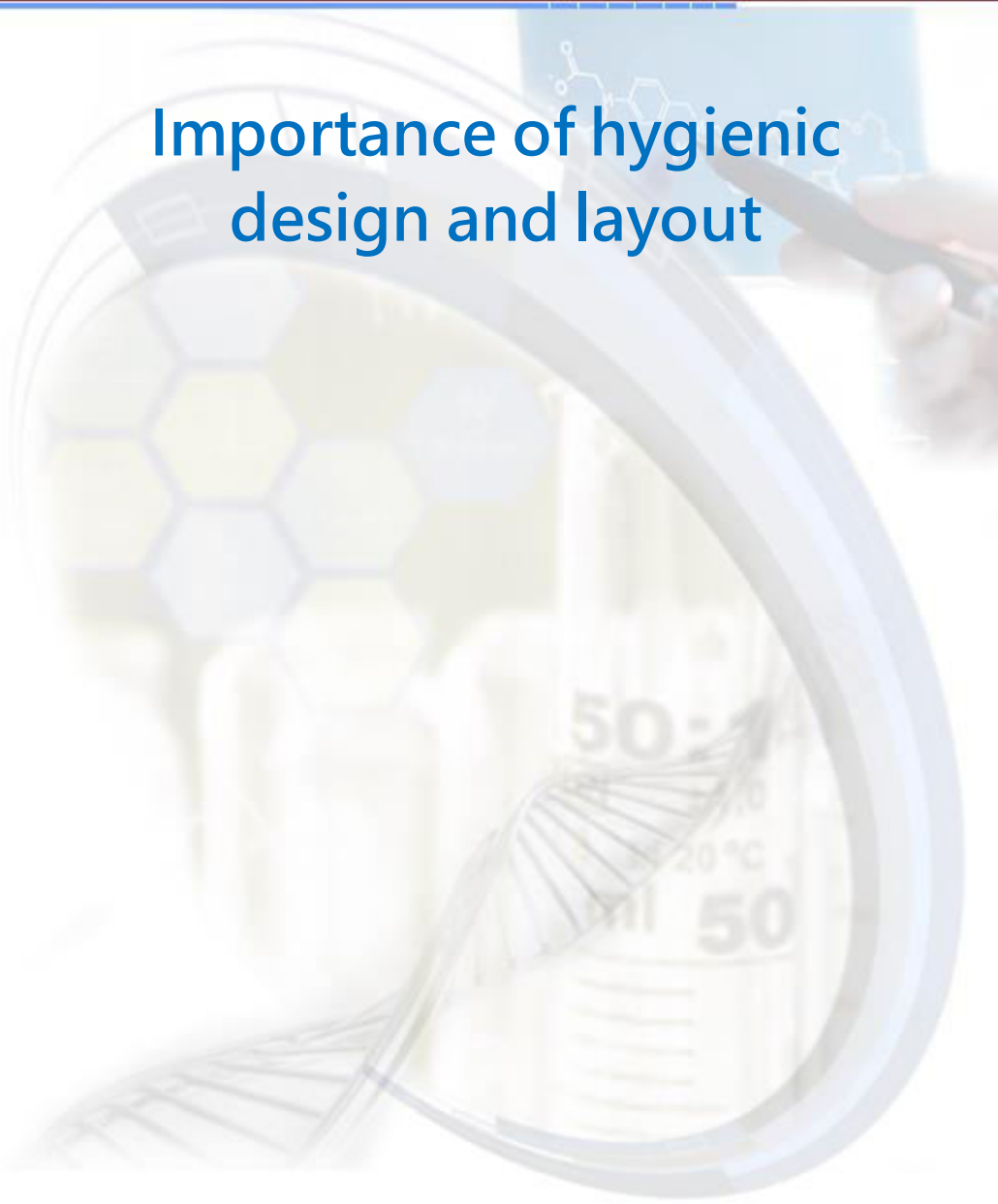
Food Biotechnology

Module 144



Food Biotechnology

Importance of hygienic design and layout



Importance of hygienic design and layout

Hygiene is vital for the **production of a quality of a food product.**

The product quality depends not only on the **capability and safety of the process**

Preventive measures like **GMP, HACCP, maintenance**, etc.

But also on **the quality of the equipment** used.

Importance of hygienic design and layout

The efficiency of cleaning procedures is also **highly dependant** on **the design and installation** of equipment.

Therefore, when considering food safety, **hygienic design** and **appropriate cleaning** are, together with GMP, hygiene, HACCP, maintenance, etc., **essential preventive measures**.

Importance of hygienic design and layout

A prerequisite for high quality and safe products is **correct hygienic design** and **maintenance of the production systems.**

A correct hygienic design of process equipment has a tremendous impact on **diminishing the risks of contamination** of food during production.

Importance of hygienic design and layout

This also means that the shelf life of products is **lengthened**.

If the process equipment is of **poor hygienic design** residues (soil) may be retained in crevices and dead areas,

Which would allow micro-organisms **to persist, multiply** and then **cross-contaminate** subsequent batches of product.

Importance of hygienic design and layout

Poor hygienic design can affect the **equipment manufacturer**, the **food processing client**, **government legislative bodies** and the **final consumer**.

Many cases of food poisonings are caused by **incorrect equipment design**.

Importance of hygienic design and layout

For example the presence of salmonella in milk powder with the evidence of 76 cases of illness and 1 event of death were caused in 1985 in UK by contaminated water.

The source of contamination was traced to the fact that **birds had access to water** as a result of an **inadequate closure** of a washing water tank.

Importance of hygienic design and layout

Good hygienic building design **should minimize** the possibility of infestation by insects,

- birds,
- animals,
- micro-organisms,
- any accumulation of dust, surface water or condensed water or product.

These events are all **risk of contamination**.

END

Food Biotechnology

Module 145



Food Biotechnology

Food process equipment design



Food process equipment design

Risks due **to poor hygienic design** are mainly caused by **incorrect positioning of equipment and utility installations** by **horizontal surfaces**, **hollow bodies**, **dead spaces**, **bad drainage**, **insufficient cleanability / accessibility**, **non-resistant materials** etc.

Food process equipment design

The main issues of hygienic design are:

1. to **avoid contamination by foreign organisms** and materials like animals, insects, birds, micro-organism, migrations, construction/ equipment parts, lubricants

Food process equipment design

2. residues of other products and cleaning agents, etc.

3. to **avoid conditions which enhance the growth of micro-organism**, like accumulation of dust, water, moisture (condensation) and product

4. to **improve cleanability**

Food process equipment design

Product contact surfaces must be **inert** to the product as well as to

- detergents,
- disinfectants
- or sanitizers under the intended conditions of use.

Food process equipment design

They must **not migrate to, or be absorbed by, the food** and have to be

- corrosion resistant,
- non-toxic,
- mechanically stable,
- and their surface finish must not be adversely affected under conditions of use.

Food process equipment design

To ensure clean food processing equipment, the product contact surface has an important role. If the surface is rough and/or porous it will most likely allow food particles to build up and it will be more difficult to clean when compared with a smooth, polished surface.

Food process equipment design

The contact surface must not chemically interact with the food and its ingredients. It should be free of corrosion and it must be inert to the chemicals used during cleaning

Food process equipment design

→If the contact surface is hidden during visual inspection, it would be impossible to know whether it is properly cleaned.Therefore,all contact surfaces must be accessible during inspection and cleaning.

Food process equipment design

3)Incorporate design features that prevent harboring microbial accumulation and their growth

→ Porous surface → Microbial accumulation

→ Non accessible surfaces can also accumulate micro organisms and their growth

Food process equipment design

→ Equipment's that accumulate dust , water , oil etc. should be kept away from food to avoid cross contamination

Food process equipment design

Control measures in order to have control over cross contamination form

→ Equipment to material
→ Minimize microbial hazards
→ For proper inspection
→ To fulfill GHP
→ Better inspection
→ Minimize hazards

Food process equipment design

→ Motors used to power equipment should be placed where any lubricant used in the motor does not contaminate the product. Direct drive systems are generally preferable

Food process equipment design

→ Another source of contamination in drive systems is bearings. Use of food grade material such as nylon, sealed or self-lubricating bearings are preferable

→ Seals should be nontoxic and nonabsorbent. It should be easy to remove seals for inspection and sanitation purposes

Food process equipment design

→ For processing liquid foods, kettles should be of the self-draining type. For kettles equipped with mixers, the mixer lubricant should not enter the product.

END

→If steam is to be directly injected, then any additives used in the boiling water must be food grade

Food Biotechnology

Module 146



Cleaning and sanitizing in food industry

Cleaning and sanitizing in food premises

Introduction :-

Cleaning and sanitizing in a food processing plant involves a number of steps.

1) Removal of gross soil
2) Chemical agent used for removal of visual soil
3) Rinse of cleaning agent
4) Sanitizer (For micro-organism removal, killing)
5) Final rinse

Cleaning and sanitizing in food premises

Cleaning is influenced by

→Temperature

→Time of cleaning,

→Concentration of chemicals,

→ Mechanical action used in cleaning

Cleaning and sanitizing in food premises

→Use of higher temperatures during cleaning of fat and grease is beneficial; however, it should not be excessively high to cause protein adhesion to a surface.

Cleaning and sanitizing in food premises

→ Many properties of soil are important in determining the ease or difficulty in removing it, for example, particle size, viscosity, surface tension, wettability, solubility of a liquid soil in a solid soil, the chemical reactivity with the substrate, the attachment of soil to a surface or entrapped in voids and attachment.

Cleaning and sanitizing in food premises

The selection of a cleaning compound depends upon:
→the type of soil on the surface; →the type of surface to be cleaned; -
→the amount of soil on the surface; →the method of cleaning (such as soaking, use of a foam, or clean-in-place);
→the type of cleaning agent – liquid or powder;
→ the quality of the water;

Cleaning and sanitizing in food premises

Cleaning involves first the separation of the soil from the surface, followed by dispersion of the soil in the detergent medium. It is important that the soil must not redeposit on the surface. The effectiveness of a detergent is evaluated based on its:

Cleaning and sanitizing in food premises

→penetration and wetting ability; →control of water hardness; →efficient removal of soil

→ease of rinsing;

→non corrosiveness to the surface.

END

→The desired properties of detergents are obtained by proper mixing of selected chemicals

Food Biotechnology

Module 147



Food Biotechnology

Process Control



Process control

Introduction :-

→Process control may be defined as manipulation of process variables so that the desired product attributes are obtained

→ The variables employed in a process have a marked influence on the final attributes of the product

Process control

→ By controlling process variables, the desired consistency of operation is achieved along with reduced costs of production and improvements in safety

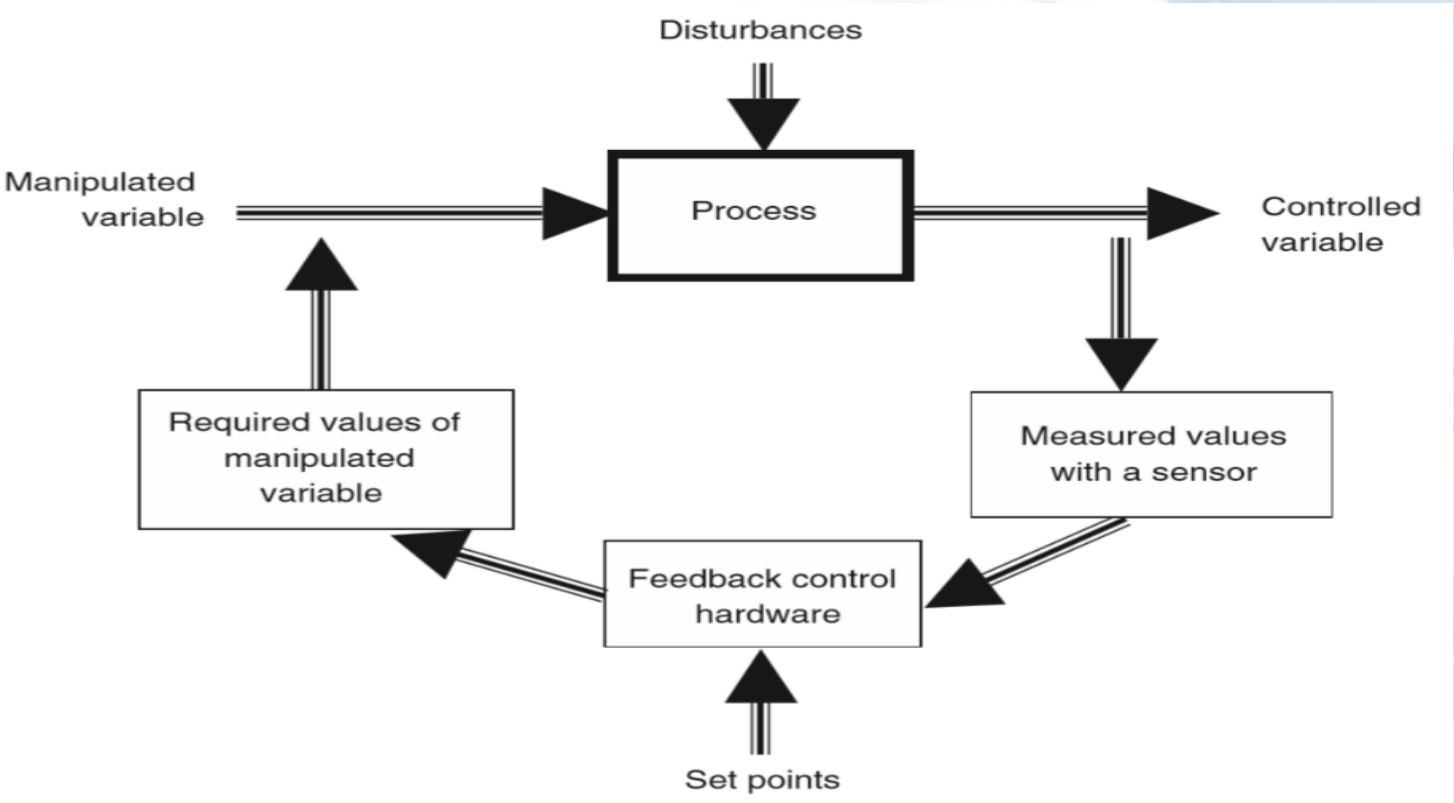
→ When equipment's deviate from what it is designed for, automatic controls provide a higher level of consistency than if human intervention is allowed that often leads to higher levels of variability.

Process control

→ The production productivity is enhanced as less out-of-specification product is produced

→ Automatic controls provide a higher level of screening of unsafe conditions to improve overall safety of the equipment

Process control



In this feed back process control diagram when a disturbance/change in a process or product specifications occurs then a control variable measure deflection or change in process with different sensors and sensor sends feedback to operator then operator make changes and adjustments to have constant and desired production

Process control

→Different sensors are employed in a process to detects undesirable changes during a process

→ Different valves are employed in order to have adjustments

END

Food Biotechnology

Module 148



Food Biotechnology

Handling of solid foods in processing plant

Handling of solid foods in processing plant

→The design and layout of the plant and how materials are handled between various processing equipment have a major impact on the production efficiency

→ In the case of transporting solid foods in a food processing plant, it is typical to consider the movement of product in any direction, horizontal or vertical

Handling of solid foods in processing plant

A variety of conveyors and elevators are used for this purpose. These include

→ belt,
→ chain,
→ screw,
→gravity,
→ pneumatic conveyors,
→ bucket elevators

In some cases fork-lifts and cranes are employed

Handling of solid foods in processing plant

1) Belt Conveyers :-

→ An endless belt operating between two or more pulleys is one of the most ubiquitous conveyors

→ used in transporting solid foods in a processing plant

Handling of solid foods in processing plant

2) Screw Conveyors :-

Screw conveyors are suited for handling

→ powders,
→ sticky
→ viscous products such as peanut butter, and granular materials

Handling of solid foods in processing plant

3) Bucket Elevators :-

→The bucket elevator consists of an endless belt with buckets attached to it

→ Bucket elevators are highly efficient as there is absence of frictional loss between the product and the housing material

Handling of solid foods in processing plant

4) Pneumatic Conveyors :-

In a pneumatic conveyor, particulate food is conveyed in a closed duct by a high-velocity stream of air

Handling of solid foods in processing plant

END

Belt , screw , pneumatic Conveyors and bucket elevators

increase the efficiency of →material handling
→reduce cost ,
→Minimizes losses and
→ most importantly it is time effective

Food Biotechnology

Module 149



Food Biotechnology

Food transport (Screw conveyors)



Food transport (Screw conveyors)

Screw conveyors consists of a helix turning inside a circular or U-shaped trough

Food transport (Screw conveyors)

→ Screw conveyors are suited for handling powders, sticky and viscous products such as peanut butter, and granular materials

→ Screw conveyors are useful to empty silos of flour and powder materials.

→ Often they are used as metering devices

Food transport (Screw conveyors)

→The flights of screw conveyor are made of a variety of materials including stainless steel.

→Their operating power requirements are high,

→They are used for distances less than 25 m

Food transport (Screw conveyors)

→Screw conveyors are suitable for horizontal as well as incline up to20 degrees.

→For horizontal screw conveyors an oval trough is used

→For a steep incline a cylindrical trough is necessary

Food transport (Screw conveyors)

The power requirement of a screw conveyor depends upon a number of factors including:

→the length of the conveyor

→elevation;

→pitch;

→speed;

→type of flights

→weight and properties of the material being conveyed;

→the coefficient of friction between the product and the material of the flights and housing.

Food transport (Screw conveyors)

The startup power requirement of a screw conveyor is generally higher than for continuous operation.

END

Food Biotechnology

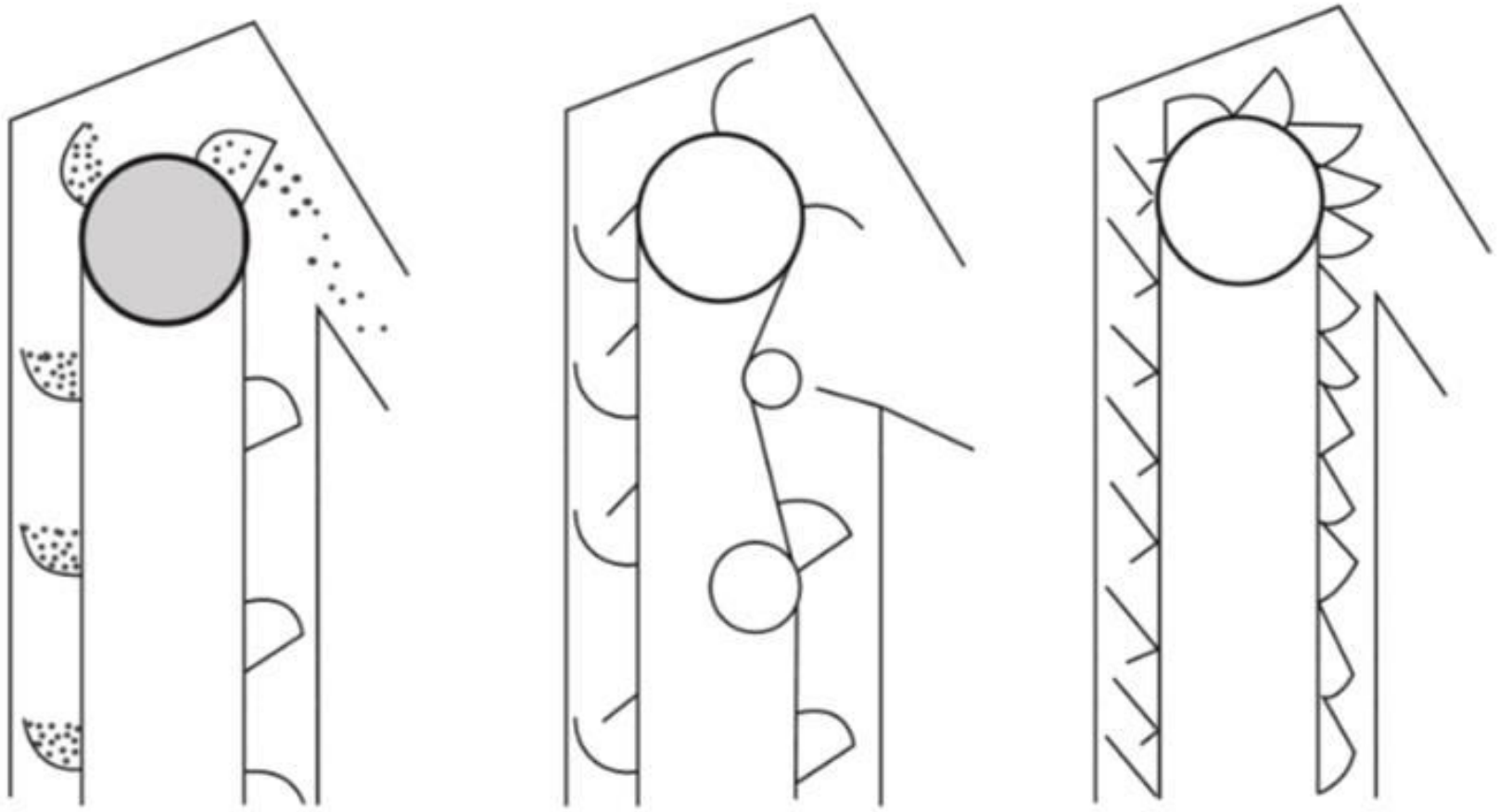
Module 150



Food transport (Bucket elevators)

The bucket elevator consists of an endless belt with buckets attached to it. The belt operates on two wheels; the top wheel is referred to as the head and the bottom is called the foot

Food transport (Bucket elevators)



Food transport (Bucket elevators)

→Bucket elevators are highly efficient as there is absence of frictional loss between the product and the housing materials

→The bucket elevators are enclosed in a single housing referred to as a leg; in certain cases, there turn is housed in a second leg

Food transport (Bucket elevators)

→ A chain or a belt is used to carry the buckets that are shaped with either rounded or sharp bottoms.

→ The belt or chain operates between two wheels – the head and foot. For longer lengths, idlers are installed to prevent belt whip

Food transport (Bucket elevators)

→The product carried by the buckets is discharged at the top when the bucket turns around the head wheel,

→Product is thrown out by the centrifugal force

Food transport (Bucket elevators)

→ The speed of the bucket as it goes around the head wheel must be maintained within limits to ensure that the product is discharged in a desired region

Food transport (Bucket elevators)

The conveying capacity of a bucket elevator depends on the

1) Product density,

2) Belt speed

3) Bucket size

4) Spacing of buckets on the belt

Food transport (Bucket elevators)

→Typical applications of bucket elevators are for Handling cereal grain, animal feed, and meal.

→The energy requirements of bucket conveyors for cereal grains range from 0.1 to 0.2 kWh/m3.

END

Food Biotechnology

Module 151



Storage of Fruits and Vegetables

Storage of fruits and vegetables

→post harvest management, proper techniques of handling and storage of fruits and vegetables are essential to minimize losses.

→The range of post harvest losses varies anywhere from 5 to 50% or even higher

Storage of fruits and vegetables

In industrialized countries, major progress has been made in developing proper systems for the handling of fruits and vegetables.

Storage of fruits and vegetables

1) Respiration process and storage of fruits and vegetables :-

→Fruits and vegetables continue to undergo physiological changes after harvest. These changes are largely the result of the respiration process

→ Oxygen plays an important role in the respiration process

Storage of fruits and vegetables

→Aerobic Respiration (with oxygen)

→ Anaerobic Respiration (Without Oxygen)

Anaerobic respiration results in the production of ketones, aldehydes, and alcohol. These products are often toxic to the plant tissue and hasten its death and decay

Storage of fruits and vegetables

→ So if we limited the amount of O_2 then we can increase the shelf life of fruits and vegetables

2)Temperature Control

Heat is generated as a end product of aerobic respiration so by lowering temperature we can increase the shelf life of product

Storage of fruits and vegetables

3) Ethylene control:

→Climacteric (Produce ethylene and CO_2)

→ Non Climacteric (does not produce ethylene)

Ethylene control also prevent fruits decay

Storage of fruits and vegetables

Climacteric fruits		Nonclimacteric fruits	
Apple	Muskmelon	Blackberry	Olive
Apricot	Nectarine	Cacao	Orange
Avocado	Papaya	Cashew apple	Pepper
Banana	Passion fruit	Cherry	Pineapple
Blueberry	Peach	Cucumber	Pomegranate
Breadfruit	Pear	Eggplant	Raspberry
Cherimoya	Persimmon	Grape	Satsuma mandarin
Feijoa	Plantain	Grapefruit	Strawberry
Fig	Plum	Jujube	Summer squash
Guava	Sapote	Lemon	Tamarillo
Jackfruit	Soursop	Lime	Tangerine
Kiwifruit	Tomato	Loquat	
Mango	Watermelon	Lychee	

Storage of fruits and vegetables

Proper use of low temperature prevents

→ Chilling injuries
→Freezing injuries

Storage of fruits and vegetables

Fruits and vegetables must be stored under

→ Low oxygen
→ High CO_2
→ Proper storage temperature

END

Storage of fruits and vegetables

Fruits and vegetables must be stored under

→ Low oxygen
→ High CO_2
→ Proper storage temperature

END

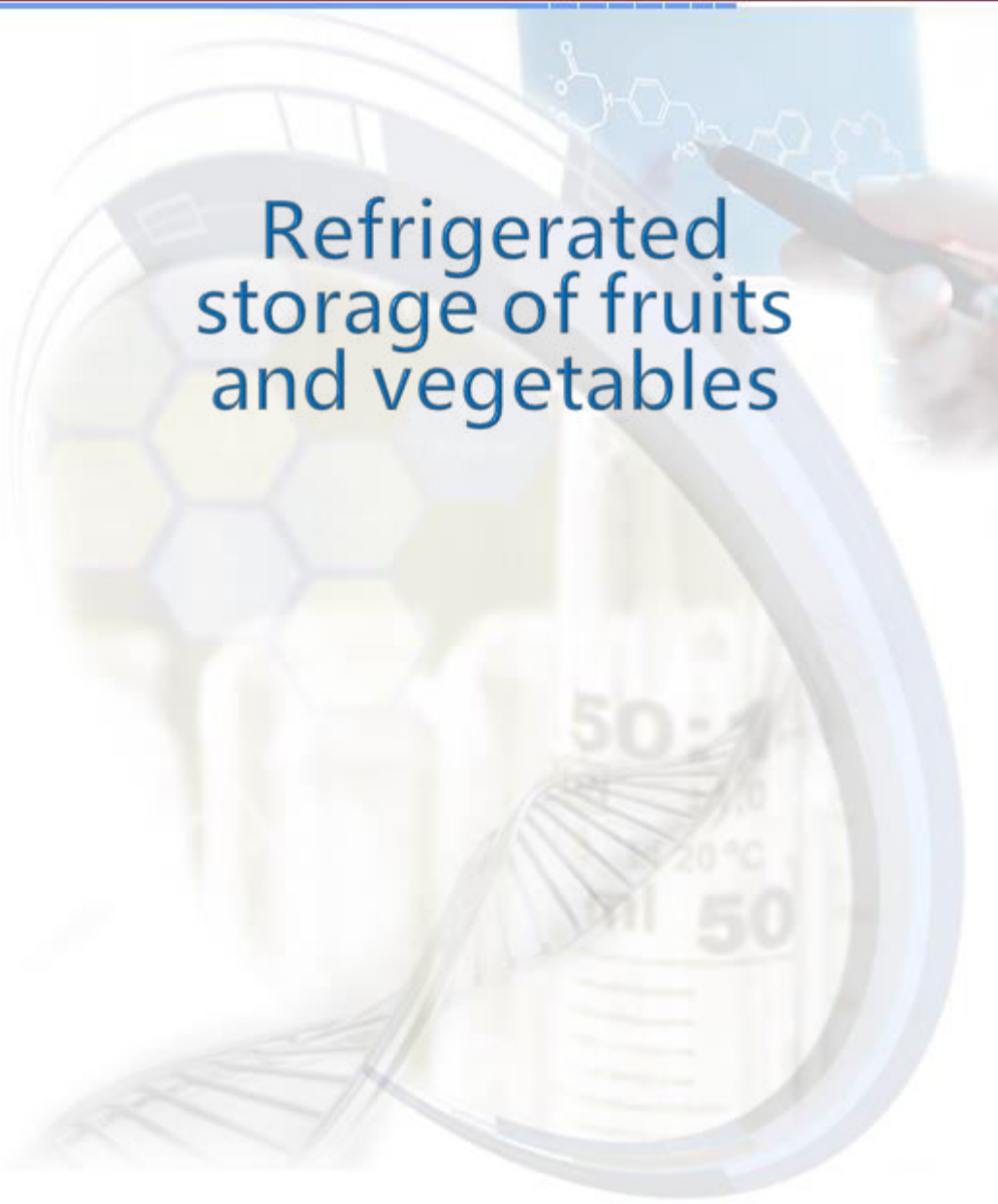
Food Biotechnology

Module 152



Food Biotechnology

Refrigerated storage of fruits and vegetables

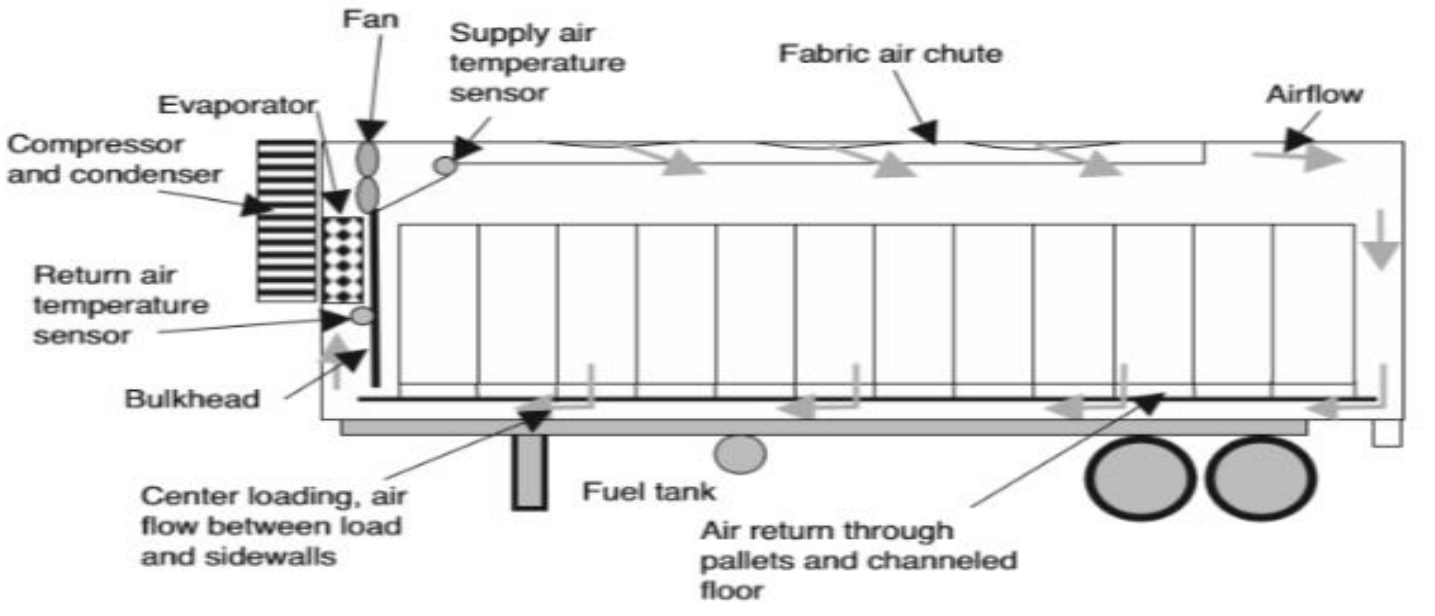


Refrigerated transport of fruits and vegetables

→One of the most common ways to transport perishable foods such as fruits and vegetables is by using a refrigerated trailer that may be either pulled via a truck on a highway or placed in a ship for trans-ocean shipment

→ Any perishable product stored in the trailer must be kept refrigerated for the entire duration

Refrigerated transport of fruits and vegetables



A typical refrigerated trailer, shown includes a refrigeration system to cool air and an air handling system to distribute air within the trailer. It is vital that there is uniform air distribution within the trailer; otherwise regions with no air circulation can lead to product heating and spoilage

Refrigerated transport of fruits and vegetables

→Temperature control of the air circulating inside the trailer is vital to the shipment of perishable foods

→ Modern trailers are equipped with temperature sensors and controllers that automatically control the refrigeration unit based on the temperature of air exiting the refrigeration unit

Refrigerated transport of fruits and vegetables

→ The temperature of the thermostat in these systems is set within 0.5◦C of the long-term storage temperature

→ The trailer used for frozen products should be set at −18◦C or colder.

→ Temperature of frozen product during loading should be less than -12 C

Refrigerated transport of fruits and vegetables

→ When a trailer load contains more than one type of produce, care must be taken that these are compatible in terms of their storage temperature and ethylene sensitivity

→ Ethylene-sensitive vegetables should not be mixed with ethylene-producing fruits

Refrigerated transport of fruits and vegetables

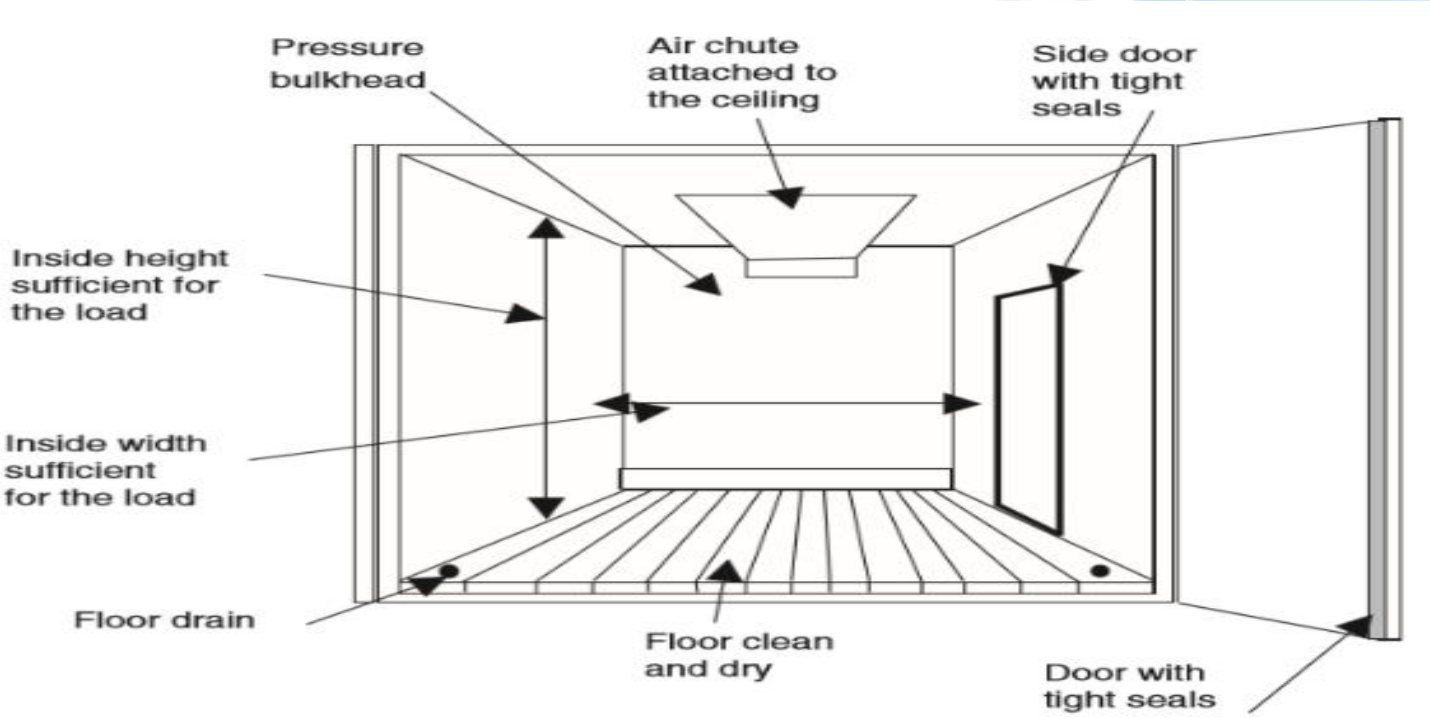


Figure shows a check list for trailer conditions before loading food

Refrigerated transport of fruits and vegetables

→ In long-distance shipping, vibration damage to the produce can be a significant factor.

→ Vibration damage to product is more severe for trailers using steel spring-suspended axles.

Refrigerated transport of fruits and vegetables

→ To prevent vibration loading of material should be done in such a way that there would be no or minimum inter material spaces

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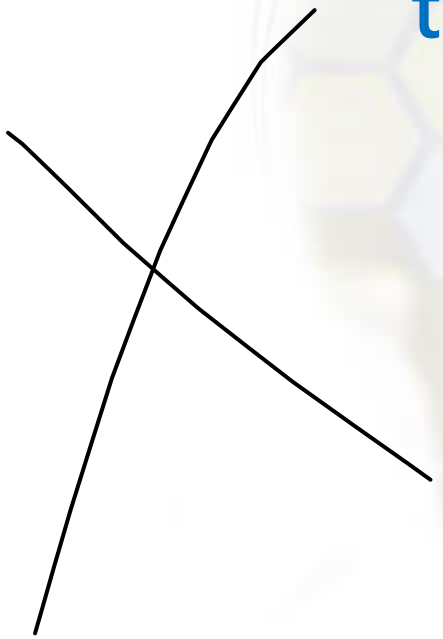
Food Biotechnology

Module 153



Food Biotechnology

Water quality and waste water treatment in food processing



Water quality & waste water treatment in food processing

→Water use in food processing has increased with the widespread mechanization of harvesting operations; raw agricultural products arriving at a food processing plant require large quantities of water for cleaning

Water quality & waste water treatment in food processing

→Inside the plant, water is used in a variety of processing and handling operations such as product conveying, peeling, blanching, cooling, generating steam, and washing equipment and floors

Water quality & waste water treatment in food processing

→ The quality of water used in food processing depends on the function of water in the manufacturing process. For example, water quality for initial cleaning of raw produce has a different requirement than water required for the formulation of carbonated beverages, beer, and bottled drinking water.

Water quality & waste water treatment in food processing

1)Treatment of hard water :-

If the water has high level of hardness due to the presence of dissolved solids, then processes such as

→ precipitation,

→ion exchange,

→distillation,

→reverse osmosis may be used.

Water quality & waste water treatment in food processing

2)Turbid water Treatment:

Turbid water often pretreated using

→coagulation

→flocculation

→sedimentation

→filtration

Water quality & waste water treatment in food processing

The presence of dissolved organics in water causes

Off-flavor
0ff-odor
Off-color

That is often removed by

→activated carbon adsorption.

Water quality & waste water treatment in food processing

Waste Water Treatment :-

→Primary treatment is often a physico-chemical process involving sedimentation;

→secondary treatment comprises biological treatment with sedimentation

→tertiary treatment involves removal of residual and non biodegradable materials.

Water quality & waste water treatment in food processing

→Dissolved organic matter in the wastewater is removed using biological treatment and adsorption

→Dissolved inorganic matter requires the use of ion exchange, reverse osmosis, evaporation, and/or distillation.

→Any suspended organic matter is typically removed using physico-chemical and biological treatment methods.

Water quality & waste water treatment in food processing

→→Other suspended inorganic or organic content is removed by screening, sedimentation, filtration, and coagulation

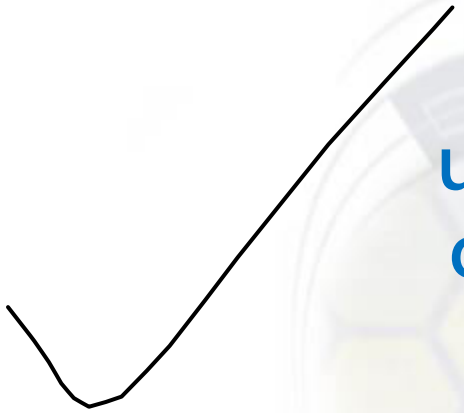
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Food Biotechnology

Module 154



Food Biotechnology



Main techniques used for production of food ingredients by microbes

Main techniques used for production of food ingredients by microbes

→Microorganisms play an important role in the generation of natural compounds, particularly in the field of food ingredients

→ Microbial processes seem to be the most promising methods.

Main techniques used for production of food ingredients by microbes

→ Many microorganisms are capable of synthesizing useful compounds when growing on a culture media

→ They are also an economical source of enzymes, which can be utilized to enhance or alter the flavors of many food products

Main techniques used for production of food ingredients by microbes

→Biotechnological processes involved in the production of flavor compounds can be divided into two groups:

1)Microbiological Methods

→Biosynthesis
→ Biotransformation

2) Enzymatic Methods

Main techniques used for production of food ingredients by microbes

1)Microbial Methods (Biosynthesis and Biotransformation)

→The first is the production of chemical compounds by cells (fermentation or secondary metabolism).

→The second refers to the use of microbial cells in the specific modification of chemical structures

Main techniques used for production of food ingredients by microbes

→In fermentation, the production of flavors starts from cheap and simple sources such as sugars and amino acids.

→ The product is generated by the complex metabolism of the microorganism. When microorganisms are used in order to catalyze specific conversions of precursors and intermediates, the process is called biotransformation

Main techniques used for production of food ingredients by microbes

→Fermentation requires C and N sources, a specific substrate is necessary for microbial transformation. The enzymatic catalysis precedes a simple and specific transformation of the substrate molecule

Main techniques used for production of food ingredients by microbes

END

Flavors obtained from Fermentation :-

→ Yoghurt Flavor
→ Alcohol flavor
→ Wine
→ Beer
→ Pickles etc.

Acidic and alcoholic flavor due to fermentation

Food Biotechnology

Module :- 155



Selection of microorganisms and development



Selection of microorganisms and development

Micro-organisms are

Prokaryotes
Eukaryotes
Autotrophs
Heterotrophs
Gram +ve
Gram –ve
Mesophiles
Thermophiles
Physchrophiles

Selection of microorganisms and development

Selective Enrichment:

Microorganisms of public health significance associated with foods are usually present as a small minority of the total microbial population. For both detection and quantitation of such microorganisms, selective enrichment cultivation is usually undertaken.

Selection of microorganisms and development

Selective enrichment involves conditions of cultivation that favor the development of the target organism over that of the usual majority of extraneous microorganism

Selection of microorganisms and development

For example most culture media for selective enrichment of *E. coli* make use of the fact that the organism is a common inhabitant of the intestinal tract of mammals and hence is normally in contact with surface-active agents such as bile salts, which are frequently used as selective agents for the organism.

Selection of microorganisms and development

Temperature Control for Selective Enrichment :-

→Incubation temperature has also been used successfully with selective chemical agents for the selective cultivation and enumeration of certain bacteria.

Selection of microorganisms and development

→The use of 45.5°C for the most probable number (MPN) enumeration of *E. coli* using the culture broth known as EC functions selectively in conjunction with the content of 0.15% bile salts .

Selection of microorganisms and development

Use of Selective incubation temperature for isolation :-

→ 55-65 ºC (Thermophiles)

→ 20 ºC (Physchrophiles)

Selection of microorganisms and development

(The Use of Mineral Salts Media for the Isolation of Unique Carbon Sources Utilizing Microorganisms)

The isolation of microorganisms capable of utilizing unique carbon sources is greatly facilitated if the organism being sought is capable of growth in a strictly glucose–mineral salts medium.

Selection of microorganisms and development

Unique carbon and energy sources such as methanol, cholesterol, and naphthalene, can then be added as sole carbon sources with the assurance that only organisms capable of attacking these unique substrates will grow.

Selection of microorganisms and development

→ Selective is for selective organisms by providing them suitable conditions

→ By knowing the temperature , pH , Salt etc. requirements of micro-organisms we can selectively grow them

END

Food Biotechnology

Module #156



Food Biotechnology

Culture Media and Upstream compounds



Culture media and upstream compounds

- The ideal culture medium will use inexpensive nutrient requirements.
- These ingredients are crop, animal, marine or yeast based components.
- The culture medium alone can represent 30 to 70% of the fermentation production costs.

Continue

- Slight changes in medium micronutrients can have a major impact on the fermentation.
- Thus, the food industry demands a consistent product from suppliers of these complex components.
- Failure to provide a consistent product will eliminate the commercial use.
- What decides the culture medium makeup?

Culture media and upstream compounds

- Essentially, it is the nutritional requirements of the microorganism of choice and its ability to biosynthesize essential elements such as amino acids, vitamins, lipids, and carbohydrates.
- For example, bacteria and yeast are high in protein (40–50%), whereas molds are not (10–25%).
- Yeast can generally grow in a minimal medium, whereas lactic acid bacteria require a host of micronutrients to grow.

Culture media and upstream compounds

- Each microorganism has different micronutrient requirements.
- Typically the exact micronutrient which benefits the fermentation that is being supplied by the complex nutrient employed is unknown.
- Generally it involves specific amino acids, vitamins, trace elements, and lipids.
- Furthermore, the concentration ratio of carbohydrate to nitrogen and phosphorous has a dramatic impact on microbial growth.

Culture media and upstream compounds

- Carbon–nitrogen imbalances can result in the production of other byproducts such as extracellular polysaccharide, and fermentation end products such as ethanol.
- In the case of ethanol production in yeast, excess glucose in the presence of oxygen will direct yeast to produce ethanol.
- This is called the Crabtree effect.
- Yeast cells typically consist of 48% carbon and 8% nitrogen on a dry weight basis.

Culture media and upstream compounds

- When the C:N ratio is 10:1, yeasts grow aerobically consuming little substrate while producing maximum cell mass, CO_2 and H_2O, but when the C:N ratio is 50:1 yeasts grow anaerobically, consuming much substrate while producing little cell mass, and much CO_2 and ethanol.
- This difference in yeast growth is also called the Pasteur effect.

Culture media and upstream compounds

- Why the increase in industrial microbiology fermentation processes over the past 20 years? It is the result of the corn syrup sweetener industry and computer technology.
- Corn is 70% starch, and when dried to 19%, it can be stored for more than two years.
- glucose syrups from corn is consistent, low cost supply of substrate.

Culture media and upstream compounds

- For example, corn syrups containing 19% dextrose, 14% maltose, 12% maltotriose, and 55% higher saccharides are used to control microbial growth rates and biological heat production in many fermentations.
- Thus, glucose is the platform chemical used for the microbial production of organic acids, amino acids, vitamins, and more.
- In fact some food grade fermentation facilities have located adjacent to a corn sweetener facility to permit glucose syrup to be piped directly to their fermentors.
- Glucose corn syrup is also shipped via truck or rail cars as liquid or dried product.

Culture media and upstream compounds

END

- Furthermore, the advent of computer process controls of industrial scale fermenters has removed many of the fears associated with commercial scale fermentation by providing reliable and easy to operate dissolved oxygen, pH, foam, temperature, and sterilization controls of the process.

Module #157



Bioreactor monitoring systems and design

Bioreactor monitoring systems and design

- The stirred tank bioreactor design is the most common fermentor and consists of agitator, baffles, aeration sparger for aerobic culture growth, sterilizable monitoring probes for pH, dissolved oxygen, temperature, and antifoam, filling and draining ports, and often culture medium sterilization capabilities in the reactor tank.
- Reactor agitation is essential for temperature control, pH adjustments, oxygen absorption into the liquid medium, overall culture health, and mixing of any required additions of substrate, and nutrients.

Bioreactor monitoring systems and design

- Fermentation health requires real time monitoring system.
- Microbial growth can be monitored via several methods.
- The most common method is indirect measurement of biomass by absorbance of the fermentation broth at 620 nm by using a spectrophotometer.

Bioreactor monitoring systems and design

- The measured absorbance values can then be used to estimate biomass concentrations by using a standard curve.
- Standard curves are developed by collecting fresh log phase cells via centrifugation, washing cells with water or 0.1 M ammonium acetate pH 7.0 buffers, then serially diluting the pellet.
- Absorbance for each dilution is then determined spectrophotometrically at 620 nm.

Bioreactor monitoring systems and design

- The actual dry weight biomass (g/l) is determined for each dilution via direct biomass measurement after oven drying of each dilution in pre weighed boats at 70°C for 24–36 hr.
- This needs to be performed in at least replicates of three. By washing biomass with water, any influence of culture medium on dry weights can be eliminated or minimized.
- Finally, a standard curve can be constructed by plotting absorbance versus actual dry biomass weight (g/l).
- This method allows for quick, reliable, and easy conversion of absorbance to dry biomass (g/L).

Bioreactor monitoring systems and design

- For determining microbial health, viable cell counts can be rapidly performed by flow cytometer.
- A two color histogram is collected with gating on the bacteria only population from the two parameter light scatter distribution and is used for the analysis of green only (live bacteria), red only (dead bacteria), and both colors (stressed bacteria).

Bioreactor monitoring systems and design

- Substrate consumption and product formation rates can be monitored by high pressure liquid chromatography (HPLC) or by membrane bound enzymes biosensors, which requires 20 and 1 min to run, respectively.

Bioreactor monitoring systems and design

- Some compounds currently measurable are glucose, ethanol, maltose, lactic acid, and lysine. Sample preparation is simply filtration (0.45 :m) and dilution with water if the value falls outside the instrument window.
- Organic acid production can be continuously monitored via alkali addition rates for pH control.

Bioreactor monitoring systems and design

- Alkali consumption can be easily monitored by feeding alkali solution from a sterile burette.
- Microbial respiration for aerobic cultures can be continuously monitored via dissolved oxygen probe or the concentration of CO2 in the exit gas, which can be monitored via off gas analyzers or simply via alkali (4 N NaOH) traps followed by pH titration.
- However, more technologies are needed to acquire real time measurements of microbial growth, ensuring optimal fermentation time, and product formation in the shortest time possible.

Bioreactor monitoring systems and design

- For rapid analysis of biomass, substrate and product concentrations without any sample preparation, Fourier transform mid infrared (FT-MIR) spectroscopy has been successfully utilized for lactic acid and ethanol fermentation.
- This method can be used for online, real time analysis for monitoring and process control purposes.

ENd

Module #158



Fermentation Types Employed:
The Actual Production Process

Fermentation types employed: The actual production process

- batch and fed-batch fermentation are most common.
- sequence starts with medium sterilization, reactor inoculation, incubation for complete microbial growth cycle with lag, log, and stationary phases, fermentation termination, draining the reactor for product recovery down stream, cleanup of the reactor, and starting over.

Fermentation types employed: The actual production process

- Percentage yield is calculated by taking the concentration of the product formed (g/l) divided by the concentration of substrate consumed (g/l) times 100, whereas productivity is a measure of the product formation rates. It is calculated by dividing the product concentration (g/l) by the fermentation time (hours); thus, it is presented in g/l/hr.

Fermentation types employed: The actual production process

- In fed-batch fermentation, additional carbohydrate is supplied to the batch fermentation during the run.
- High carbohydrate concentrations in the initial culture medium are toxic to many microorganisms.
- Thus, an optimal carbohydrate concentration is employed initially, which permits maximum culture growth to late log phase.

Fermentation types employed: The actual production process

- When the carbohydrate concentration is reduced to almost zero, additional sterile carbohydrate is injected into the bioreactor to bring the carbohydrate concentration back to the starting level.
- When this is consumed, the process is repeated until end product inhibition forces the whole bioconversion to stop. Ideally, at the end of the fermentation you will have a product concentration which is three to four times

Fermentation types employed: The actual production process

- This will generate the highest yield possible. Also, due to increased end product accumulation with each carbohydrate addition, culture production rates will decrease.
- Thus, the decision as to how many fed-batch fermentations to perform before harvest is based on the desired final product concentration and the optimal fermentor use time.

Fermentation types employed: The actual production process

- Continuous fermentations are open fermentations, whereby fresh medium is continually added to the bioreactor, while spent culture medium, cells, and product are continually leaving.
- It is desirable because it is 10 to 100 times smaller than batch fermentations, a steady stream of fermentation product is produced which will optimize downstream processes, bioconversion rates are always at maximum, operation costs are less, and the system can be fully automated and computer controlled to the point where only two operators are needed to manage the fermentation each day.

Fermentation types employed: The actual production process

- However, it requires a continuous supply of sterilized or pasteurized culture medium, dilution rates are linked to microbial growth rates and the operational speed of downstream recovery process.
- Startup is slow, so any facility shut downs have an impact on production, and you are constantly fighting contamination. Thus, not all fermentations can be operated this way.

Fermentation types employed: The actual production process

- Specific growth rate equals dilution rate. Finally a holding tank at the end is used to ensure complete bioconversion of any residual sugars to ethanol.
- The CO2 is collected and concentrated, then sold as another valuable byproduct. Fermented beverages (i.e., wine and beer) are still performed in batch.

Fermentation types employed: The actual production process

END

- For some continuous fermentations, an increased concentration of biomass in the reactor is required.
- This can be achieved by cell recycling, immobilized cell, and biofilm fermentations. Cell recycling reactors employ a filtration unit that allows for the constant bleeding of culture supernatant while retaining biomass.
- However, filtration unit

Module #159



Bioreactors design



Bioreactor design

- Entrapment type is very common bioreactor, where high concentrations of cells are trapped in a polymer matrix.
- high cell density is continuously retained in the fermentor continuously converting substrate to product.
- This results in higher productivities and yields.

Bioreactor design

- Migration of substrate through the matrix to the cell and the migration of product out is disadvantage, similarly end product inhibition, cell leakage from the polymer matrix due to cell growth, and bead swelling and disintegration over time causing the whole fermentation to be stopped, cleaned, and restarted.

Bioreactor design

- Biofilms type bioreactors are a natural form of cell immobilization in which microorganisms are attached to a solid surface.
- In this bioreactor, cells are continually growing, and sloughing off. Thus, the reactor is a mixture of immobilized and suspended cells.
- Continuous biofilm fermentations are truly open immobilized cell bioreactors.

Bioreactor design

- This is similar to a suspended cell continuous fermentation with the added advantage of increase biomass concentrations in the bioreactor.
- Biofilms are typically resistant to harsh conditions.

Bioreactor design

- Filamentous microorganisms such as fungi and actinomycetes are natural biofilm formers.
- For nonfilamentous bacteria to form a biofilm, an extracellular polysaccharide needs to be generated by the bacterium.

Bioreactor design

- Some bacteria will form biofilms on any surface such as metal, plastic, and glass.
- However lactobacilli, require something to stimulate this biofilm development. Plastic composite support (PCS) stimulate biofilm development of *Lactobacillus casei*, *Zymomonas mobilis*, *Saccharomyces cerevisiae*, and *Actinobacillus succinogenes*.

Bioreactor design

- PCS is a high temperature extruded material consisting of at least 50% polypropylene, plus ground soybean hulls, bovine albumin and various culture micronutrients.
- Soybean hulls keep the extruded product porous due to the release of steam as the PCS leaves the extruder die.

Bioreactor design

- Bovine albumin performs as a natural plastizer which protects the temperature sensitive micronutrients.
- Micronutrients are selected based on the specific cultural requirements for amino acids, vitamins, and lipids.
- Monosaccharides are avoided due to poor PCS production.

Bioreactor design

- For example, the PCS blend for lactobacilli contains 50% (w/w) polypropylene, 35% (w/w) ground dried soybean hulls, 5% (w/w) bovine albumin, 5% (w/w) yeast extract, 5% (w/w) soybean flour, and mineral salts.
- PCS have shown higher percentage yields and productivity rates than suspended cell lactic acid fermentations.

Bioreactor design

- It will continue to grow as a biofilm and have good longevity.
- biofilm reactor washed with concentrated ammonium hydroxide, rinsed with mineral salts solution, and then reinoculated with a fresh culture and medium will reestablish itself overnight.
- quick vinegar process is the most common biofilm application.

Bioreactor design

- Solid substrate fermentation is when a substrate such as soybeans is ground, inoculated with *Aspergillus oryzae*, then incubated for three days for the production of soy sauce.
- It is a simple fermentation process and commonly used for aerobic fermentation due to its large surface area.

Bioreactor design

- Thus, oxygen concentration is high without using any mechanical forced air systems.
- Solid substrate fermentations require large areas or incubation space.
- A temperature controlled environment, intermittent monitoring for contamination and quality of starting material is essential for success.

END

Food Biotechnology

Module #160



Food Biotechnology

Future research in food fermentation



Future research in food fermentation

- Research is still needed for isolation of new microbial strains with improved production efficiencies and higher yields.
- More real time measurements are needed for culture conditions and metabolite formation.
- Recovery will continue to be the key factor associated with final product purity and cost.

Future research in food fermentation

- As an industry we cannot rely solely on genetic engineering as our method of improving current fermentations.
- As we have illustrated, there are many other techniques which can be employed to improve productivity and yield, including new inexpensive medium ingredients, more continuous fermentation processes, and new exotic microbial reservoirs in nature and in the food industry.

Module #161



Genetic modification of plant starches for food applications

Genetic modification of plant starches for food applications

- Most of the maize starch mutants affecting starch structure were discovered before biotechnology understanding.
- names were based upon the phenotype or on the properties of the starch, rather than that of the gene or enzyme affected by the mutation.

Genetic modification of plant starches for food applications

- For example, waxy, amylose extender, dull, shrunken, sugary-2, and sugary) have been described extensively with regard to their effects on carbohydrate composition and response to genetic background, allelic dosage, or interaction with other mutations.

Genetic modification of plant starches for food applications

- Examination of maize kernels with differing starch phenotypes has been instrumental in determining which enzymes are required for starch synthesis in this storage organ.
- Mutations that are responsible for most of the abnormal starch phenotypes have been located in genes encoding starch synthetic enzymes.
- Related isoforms, for which there are as yet no mutants available, have also been identified and characterized.

Genetic modification of plant starches for food applications

- It is possible to modify or engineer starch synthesis enzymes using modern biotechnology.
- This in turn opens up oppurtunities to consider using tools of breeding, genetics and biotechnology to genetically control starch deposition in plants so as to enhance starch yield and starch quality.
- instead of crudely controlling enzyme activity, it is possible to more precisely reduce the expression or activity of certain enzymes (such as by using antisense or RNAi technology).

Genetic modification of plant starches for food applications

- Unlike breeding, enzymes may be overexpressed:
- Additionally, novel enzyme isoforms can be selected in order to act synergistically with the existing starch pathway enzymes.
- With full sequencing of many homologous genes from different plants, we can identify domains within enzymes, which impart particular catalytic properties.
- This is an important new opportunity to gain new understandings of the origins, mode of action and potential for genetic modification of different enzymes.

Genetic modification of plant starches for food applications

- The starch created from such work has the potential to be extremely novel and valuable.
- Despite limitless opportunities, constraints remain for understanding links between starch structural changes and functionality.
- Additionally limitations are regarding our understanding of what determines the specific catalytic properties of the enzymes.

Genetic modification of plant starches for food applications

Enzyme and Specific Isoform	Mutant (Maize)	Effect on Starch
Granule Bound Starch Synthase		
GBSSI	Waxy (*wx*)	Low amylose content.
Soluble Starch Synthase		
SSI	None known	Not known
SSII	Sugary2 (*su2*)	Lacks intermediate glucan chains in amylopectin
SSIII	Dull1 (*du*)	Lacks longer glucan chains in amylopectin
SSIV	None known	Not known
Branching Enzyme		
BEI	BEI (*be1*)	None/Minimal
BEIIb	Amylose extender (*ae*)	High amylose content.
Debranching Enzyme		
ISAI	Sugary1 (*su1*)	Forms compound granules and phytoglycogen
ISAII	None known	Not known
ISAIII	None known	Not known
PUI	Pullulanase1 (*pu1*)	None/Minimal

Food Biotechnology

Module #162



Modification of starches

Modification of starches

- Modifications in GBSS (granule bound starch synthase) activity result in changes in amylose content or amylose structure.
- In maize, the GBSSI mutant phenotype was named waxy because of waxy phenotypic appearance.
- waxy mutants of cereals is the only enzyme determining the amylose contents in cereals, the amf mutants of potato, the lam mutants of pea, and GBSSI antisense lines of potato show either a reduction or elimination of GBSS activity and a specific reduction of amylose in starch from tubers.

Modification of starches

- Some studies have attempted to restore the production of amylose in amylose free potato plants by transforming the plants with genes for GBSSI enzymes produced by other plants.

Modification of starches

- 3.5%-13% amylose was restored to amylose free mutants of potato by transformation with the cassava GBSSI enzyme.
- Similarly, amylose free potatoes transformed with pea GBSSI isoforms resulted in potatoes with amylose contents 0.8% -1%.
- Like, the other low amylose potatoes and pea starch, heterogeneity in amylose and amylopectin content was observed within the granules: the granules stained with iodine revealed amylose in concentric rings or having blue staining granule cores.

Modification of starches

- Japanese scientists has identified wheat mutants lacking GBSSI activity.
- substitution of genes producing functional GBSSI enzyme with the null alleles resulted in starches with a 22 to 23% amylose content rather than the 25.5% normal control.
- Similarly, elimination of an active GBSSI enzyme at 2 of the 3 loci in wheat endosperm results in a wheat starch that has an amylose content of 16% to 21%.

Modification of starches

- Thus, the presence of a functional GBSSI enzyme from a single locus pair is sufficient to produce a starch with an amylose content of at least 16%.
- between 14.1 and 16.7% amylose content wheat can be created through ethyl methanesulphonate (EMS) mutagenesis of the seeds.
- Amylose free wheat starches were created using triple null combinations of GBSSI mutants and using mutagenesis of a double null wheat known as Ike to generate a non null wheat.

Modification of starches

- Two functional Wx alleles of rice exist: Wxa, which produces a large amount of amylose, and Wxb, which produces a smaller amount of amylose.
- Studies of the effects of the two alleles on the gene expression at the waxy locus in rice showed that the Wxb allele resulted in an ineffective production of GBSSI enzyme and amylose in japonica rice, while the Wxa allele produced larger quantities of GBSSI enzyme and amylose in indica rice.

Modification of starches

- Wxa allele is shown less effective in the production of amylose than the Wxb allele based on analysis of 40 rice varieties.
- Similarly, two wild-type rice alleles, Wxa and Wxb, Wxb had a tenfold lower GBSS expression than Wxa at the protein and mRNA levels.
- The reason for this effect was result of a point mutation within the genetic sequence for the normal rice enzyme (Wxa allele).

Modification of starches

- Modifications in starch synthase activity result in changes in amylopectin content or amylopectin structure.
- Starch synthase may have either a subtle or profound impact on the starch, Although SSI is a relatively minor isoform in potato, it is the predominant isoform of SS in the cereals.

Modification of starches

- Of the starch synthases, the effects of SSIIa and SSIII on starch structure and composition are the best elucidated, especially in maize.
- The first and only reported SSI mutant was in rice having only a minor effect on starch content and quality.
- In maize, SSIIa maps to a locus known as Sugary-2 which when mutated produces the sugary-2 mutant.

Modification of starches

- The structural gene for SSIII in maize is known to be the Dull1 locus which when mutated gives the dull1 mutant.
- Starch from the dull1 mutant is relatively lacking in longer amylopectin chains.
- In potato the simultaneous antisense inhibition of SSII and SSIII resulted in a grossly modified amylopectin, with yet further changes in structure if GBSSI as well as SSII and SSIII are inhibited.

Modification of starches

- The first naturally high amylose starch was reported 50 years ago a maize endosperm mutant termed *amylose extender* (*ae*).
- We now know that the structural gene for SBEIIb is the *Ae* locus, which when mutated produces *ae* mutant starch having an amylose content of up to 50%.

Modification of starches

- relatively clear demarcation between amylose and amylopectin existing for normal starch is blurred for *ae* starch, giving rise to a third type of starch termed intermediate starch.
- Additional evidence for this comes from material collected as amylopectin from *wxae* intermediate starch which has no GBSSI activity and hence produces no true amylose, and has reduced amylopectin branching.

END

Module #163



Functionality of Starches

Functionality of starches

- Crystalline packing of double helices of starch chains from amylose fiber shows diffraction pattern through wide angle x-ray scattering.
- When starch granules are heated sufficiently in excess water, this pattern is lost and an endothermic transition is observed.
- Double helical order is additionally lost during this endothermic event, and it is this loss and not the loss of crystalline order that is believed to be responsible for the endothermic event

Functionality of starches

- This loss of granular, crystalline, and double helical order is called gelatinization, and it is irreversible with respect to the three dimensional granule organization.
- Subsequent to gelatinization, starch chains of both amylose and amylopectin may organize into new double helical and crystalline structures in a process called retrogradation.

Functionality of starches

- In its native granular form mixed with water, starch makes a high solids mixture, useful in making batters and doughs.
- During heating, starch granules swells 30-fold its dry weight.
- This process is reversible provided temperatures and pressures are lower than those that lead to disruption of the organized intramolecular glucan chain associations within the granule.

Functionality of starches

- Above these temperatures and pressures, the granules will undergo irreversible changes and glucan chains will dissociate, permitting further swelling.
- additional heating and shear forces, swollen hydrated granules will eventually collapse to form an unorganized paste of starch molecules.
- results in a thickening effect, imparting texture and structure.
- In functional terms, this process of starch granule swelling and dissociation is known as gelatinization.

Functionality of starches

- The glucan polymers of a gelatinized starch can associate with each other or other components of food to impart stickiness, tackiness or a rubbery texture.
- amylose also readily leaches out of the granule during gelatinization and interacts with the food components.
- In mild cooking conditions, amylopectin does not so readily leach and hence is enriched inside the gelatinized granule.
- extreme cooking conditions, the amylopectin is much more dispersed, resulting in products with a tacky or sticky character

Functionality of starches

- Once a cooked starch is allowed to cool, the glucan chains begin to reassociate in a process resulting in a change in the functional characteristics, including a decrease in paste clarity and gelation of the paste. In functional terms, this process of reorganization is known as retrogradation.
- Normal starches are generally recognized for their ability to retrograde within hours and usually form opaque gels.
- Retrogradation of amylopectin gels occurs during days or weeks of storage.

Functionality of starches

- As a consequence of their different molecular weights and chain length profiles, the rates of retrogradation of amylose and amylopectin are not the same.
- The rapid setting of the structure of breads is believed to be due to rapid (within seconds or minutes) amylose retrogradation to form a network structure.
- Starch functionality is therefore a consequence of the degree of gelatinization and is influenced by retrogradation, time, temperature, concentration, etc.

END

Food Biotechnology

Module #164



Application of Starches

Application of starches

- Role of starch is in improving processing, shelf life, consistency and appearance by providing texture and thickening capability for suspending solids.
- Extensive applications of starches in foods and these uses have been extended due to application work by food technologists using physical and chemical modifications.
- These include oxidation, monostarch phosphorylation , distarch phosphorylation , acetylation, hydroxypropylation, and octenylsuccination

Application of starches

- **Amylose free starch:**
- Obtained from waxy maize and rice and in all cases are considered non-GM natural variants of normal maize and rice.
- Staining starch with iodine readily identifies amylose free types: normal starch will stain blue or purple whereas the starch produced from amylose free plants will stain red or brown or brownish red in color.
- Amylose free maize starches are generally considered to contain zero or almost zero amylose.

Application of starches

- Amylose free rice starches have been shown to contain between 0 and 3% amylose, though collectively these starches are referred to as amylose free or waxy rice starches.
- With these amylose free rice starches, it has been assumed that the differing cooking and paste properties are due to differences in the structure of the amylopectin of the starch rather than variations in the low levels of amylose of the starch.

Application of starches

- Amylose free starches are considered useful as water binders, viscosity builders, and texturizers in food as well as in industrial applications.
- However, these starches are less resistant to shear, acid, and high temperatures than are normal starches, and extended cooking results in stringy, cohesive pastes.
- Amylose free starches are generally recognized for their improved transparency and have better freeze and thaw stability compared to normal starches once cooked.

Application of starches

- They are also recognized for their improved long term storage capability as they require weeks to gel if they could be considered to gel at all.
- To correct for some of the negative paste attributes of amylose free starch, such as poor stability to temperature, shear and acid and undesirable paste quality, most amylose free starches are chemically modified by substitution, cross linking, or both.

Application of starches

- Developing amylose free starches in other crops so as to take further advantage of any species specific qualities of the starch produced from that species is being sought.
- However, although mutants are readily found in some plant species (such as maize, rice, and barley) this is more difficult in other plant species where there are multiple copies of each gene (such as the polyploid species like potato, oats, and wheat).

Application of starches

- In the case of wheat, the advances have been made by screening for mutants (non-GM), while in potato the waxy types were made using biotechnology (GM).
- Amylose free wheats are finding applications in foods such as noodles and baked goods including breads .

Application of starches

- Commercialized resistant starches include unprocessed starches (green banana, legumes, potato) or processed forms of high amylose starches.
- High amylose starches from other sources are also being advocated for their resistant starch properties. The most widely available high amylose starches originate from maize using the *amylose extender* mutation that results in a starch having 40–50% amylose.

Application of starches

- As a result of extensive breeding and selection work higher amylose contents (up to 90%) have been achieved.
- High amylose starches are also available in barley and rice.

END

Food Biotechnology

Module #165



Novel Starches

Novel starches

- Starches obtained from different crops having varied functionalities due to differences in composition may now theoretically be readily transferred into crops from one or another different plant species.
- In general we can consider making changes in any aspect of amylose/amylopectin, phosphate, protein, phospholipids, crystallinity, gelation, and pasting characteristics, flavor and starch granule morphology.

Novel starches

- One interesting example of a cross species difference that may be exploited is phosphate content in potato.
- starch phosphate content may be attributed to a novel glucan water dikinase enzyme. This discovery opens up the possibility of creating novel genetically modified starches with variable phosphate contents.
- Application in food include changing starch digestibility or starch viscosity after cooking.

Novel starches

- Another avenue associated with the genes responsible for endosperm texture in some cereal crops.
- In this case the puroindoline genes, described as PinA and PinB are interesting candidates. Transforming the wheat genes into rice successfully created the softness trait in the seed.

Novel starches

- Although starch granule morphology varies extensively across species and starch from different species is highly valued for certain applications, there are few specific reports that granule morphology is the attribute desired for valuable food applications.
- Furthermore, although there have been some reports of progress in understanding the mechanisms that control and influence starch granule size, for granule morphology there is little that can be defined as clear enough to arouse interest in a genetic approach.

Novel starches

- Similarly, flavor can be considered to vary extensively across species and a bland flavor of (e.g., tapioca starch) is valued over certain cereal starches that impart a mealy character. However, as with granule morphology, our understanding of the genetic components determining this character is rather poor, making flavor a difficult target for genetic modification at the present time.

Novel starches

END

- Another novel concept has come from bifunctional domains involving cellulose and starch, which may enable developments with novel biomaterials.
- Another possibility is to explore incorporation of inulins with starch.
- Even more novel possibilities to consider include adding new functionality to starch using gene constructs containing fusion proteins.
- The fusion proteins can be proteins, which might affect starch properties or enzymes, or other bioactive proteins.

Food Biotechnology

Module #166



Biotechnology of food flavors production

Biotechnology of food flavors production

- Plants were the sole sources of flavor compounds historically.
- However, active compounds are present in low concentrations , which makes their isolation difficult.
- flavors are dependent on factors such as the weather or plant diseases.
- The production of flavor by biotechnological methods is an alternative due to consumers' prefernatural ingredients. Microbial processes seem to be the most promising methods

Biotechnology of food flavors production

Main Technologies:

- micro organisms growing on a culture media.
- They have the ability to perform conversions that would require multiple chemical s steps.
- They catalyze specific steps.
- They are also an economical source of enzymes , which can be utilized to enhance or alter the flavors of many food products.
- production of flavor compounds can be divided into two groups : microbiological and enzymatic.

Biotechnology of food flavors production

- Microbial methods are of two types.
- **Biosynthesis** is production of chemical compounds by cells (fermentation or secondary metabolism).
  Biotransformation is use of microbial cells in the specific modification of chemical structures.
- In fermentation, the production of flavors starts from cheap and simple sources such as sugars and amino acids. When micro organisms are used in order to catalyze specific conversions of precursors and intermediates, the process is called biotransformation.

Biotechnology of food flavors production

- Although fermentation requires C and N sources, a specific substrate is necessary for microbial transformation.
- The enzymatic catalysis precedes a simple and specific transformation of the substrate molecule.
- “non-natural” substrates can also be biotransformed.
- It is important to distinguish research with the purpose of obtaining complex products with natural characteristics from those that try to obtain isolated molecules.

Biotechnology of food flavors production

- The first consists in the experience of nature imitation and in developing a process with one or more microorganisms and enzymes .
- The second tries to obtain a higher yield of the characteristic components.
- The choice between them determines the methodology, which will be employed in vivo or in vitro, through biosynthesis or bioconversions .

Biotechnology of food flavors production

How to obtain flavors

- **Flavor from Fermented Foods:**
- The organoleptic properties of fermented foods are superior to unfermented substrate and are depend ant upon the biochemical activities of the associated microorganisms.
- Fermentation is technology for the production of organic acids, flavor compounds , and other biologically important chemicals.
- New aroma and flavors includes acids,alcohols, carbonyl com pounds, esters, and pyrazines.

Biotechnology of food flavors production

- Fermented products are preserved, modified, and flavored using micro organ isms.
- Lactic acid and alcoholic fermentations are the two important processes responsible for fermented food flavors .
- The creation of new fermented products can result in the development of novel flavors and textures.

Biotechnology of food flavors production

How to obtain flavors- Dairy Products:

- Cheese , yogurt and buttermilk flavors find application in snacks, sauces, baked goods, and several other products.
- The cheese flavor results from the action of microorganisms and enzymes on milk's proteins, fats, and carbohydrates.
- Numerous breakdown products are formed, among them, short-chain fatty acids, acetic and lactic acids, alcohols, aldehydes, ketones, esters, sulfur and nitrogen compounds .

Biotechnology of food flavors production

- The starter cultures used in dairy technology are mainly prokaryotes like *Lactococci, Lactobacilli, Leuconostocs, Bifidobacteria, Propionibacteria, Streptococci, and Brevibacterium linens.*
- Yeasts, such as *Kluyveromyces, Debaromyces, Candida* or *Trichosporon* are present in many manufactured milk products.
- These microorganisms modify the sensory characteristics of the products by synthesizing or assimilating volatile nitrogen and sulfur compounds

Biotechnology of food flavors production

- Several chemical reactions take place in the surface of ripened cheeses such as Camembert and Brie during maturation due to fungi growth.
- The Fungal mycelium of Penicillium roqueforti grows rapidly and the resulting products are used directly for flavoring foods with a blue cheese-type flavor.

Biotechnology of food flavors production

How to obtain flavors-Alcoholic beverages:

- Flavor compounds are produced as byproducts of yeast metabolism during alcoholic fermentation. Many flavor compounds have been identified in alcoholic beverages.
- The main compounds are: During alcoholic fermentation, yeasts transform sugars (glucose, fructose, and sucrose) into ethanol and carbon dioxide by the Embden–Meyrhof–Parmas pathway.

Biotechnology of food flavors production

- This is the main bio reaction, but several secondary byproducts are formed.
- Higher alcohols, organic acids, and esters are the main flavor compounds.
- Higher alcohols are major portion of the secondary products of yeast metabolism. They include n-propanol, isobutyl alcohol, 2-methyl butanol, amyl alcohol, isoamyl alcohol , and 2-phenyl ethanol.
- Isoamyl alcohol accounts for more than 50% of the total concentration of higher alcohols.

Biotechnology of food flavors production

- Esters at appropriate concentrations impart flowery and fruity flavors . They are formed by esterification of fatty acyl -CoA or of organic acid by alcohols.
- Diacetyl and pentane-2,3-dione (vicinal diketones) have a characteristic flavor described as “buttery”, “honey” or “toffee-like”. They have a very high off-flavor potential, dependent on the fermentation temperature.
- The threshold for diacetyl in lager-type beers is 0.10–0.14 mg /L. At levels above 1 mg /L, it becomes increasingly “cheese-like” and sharp.

Biotechnology of food flavors production

- Volatile acids are usually present in beer at concentrations of 20–150 mg /L. Butyric and iso-butyric acids in concentrations of 6 mg /L cause a “butyric” or “rancid” flavor.
- Valeric and iso-valeric acids cause an “old-hop” and “cheesy” flavor. Fatty acids with 6 to 12 carbon fatty acids give the characteristic flavor of “cheesy”, “goaty”, or “sweaty” (Smogrovicova´ and Do¨me´ny 1999).

Biotechnology of food flavors production

END

How to obtain flavors- Bakery products:

- Although Candida yeast has occasionally been used for baking and some *Saccharomyces carlsbergensis* strains have been patented for use as baker's yeast, pure strains of *Saccharomyces cerevisae* are almost universally employed.

Food Biotechnology

Module #167



Food Biotechnology

Production of esters and aldehydes

Production of esters and aldehydes

Ester Production:

- Esters are flavor compounds of fresh fruits and fermented foods, which are found in concentrations 1-100 ppm
- The production of the ester ethyl acetate by the yeast *Candida utilis* from glucose is observed when the yeast grows on a medium containing a specific initial ethanol concentration.

Production of esters and aldehydes

- Esters of low molecular weight are responsible for fruity odors and consist of acids and their derived compounds such as acetates, propionates, and butyrates.
- Some examples are ethyl butyrate and isoamyl acetate, which are found in strawberry and banana .

Production of esters and aldehydes

- The presence of esters such as ethyl acetate and butyric acetate in the culture medium can eventually describe a detoxification mechanism by which the microorganism avoids the accumulation of toxic compounds.
- The production of acetates occurs in order to detoxify the medium by converting acetic acid and high alcohols .
- Two metabolic pathways can be followed in the formation of esters : alcoholysis of acyl-CoA compounds and the direct esterification of an organic acid.

Production of esters and aldehydes

- Yeasts follow predominantly the first path way, and filamentous fungi and bacteria prefer the second .
- Some ramified amino acids are important precursors of flavor compounds and are related to fruit maturation.
- The initial reaction is called the enzymatic Strecker degradation.
- Several micro organ isms including yeasts and bacteria such as *Streptococcus lactis* can modify the majority of the amino acid structures.

Production of esters and aldehydes

- Ethyl acetate comprises, with other compounds, banana flavor.
- 2-Methyl-ethyl-butyrate has a great impact on characteristic apple flavor .
- fruity banana flavor produced by the yeast *Hansenula mrakii* and by the fungus *Geotrichum penicilliatum* in submerged fermentation using a synthetic medium.
- Seventeen compounds were identified in concentrations greater than 50 mL /L, including ethanol, ethyl acetate, isobutanol, ethyl propionate, isobutyl acetate, and isopenthyl acetate.

Production of esters and aldehydes

- Alcohols were formed in the exponential growth phase, but the esters were formed in the stationary phase.
- Ethyl acetate was the main product. In both studies, some precursors of fruity esters were added, such as vanillin, leucine, isoleucine, and phenylalanine.

  Thirty-three compounds were identified in concentrations greater than 50 mL /L: ethanol, ethyl acetate, ethyl propionate, and others.
- Ethyl acetate was produced in the highest concentrations (9924.1 mL /L).

Production of esters and aldehydes

- Strains of Ceratocystis were also identified as ester producers. Lanza and others (1976) studied the production of acetates with different carbon (glucose, galactose, and glycerol) and nitrogen (urea and leucine) sources.
- They concluded that the type of flavor compounds produced depended on both sources (carbon and nitrogen), which is different for other microorganisms such as Trichoderma viride.
- Collins and Morgan (1961) identified esters synthesized by different species of Ceratocystis (C. moliniformis, C. major, C. coerulescens,

Production of esters and aldehydes

- A strong banana flavor was detected when using dextrose and urea due to the presence of isoamyl acetate, which was also identified in leucine- or isoleucinebased media.
- For the combination galactose–urea, the main flavor was citric due to the formation of terpenes.
- The yeast Kluyveromyces marxianus produced some compounds with characteristic fruity flavor, with ethyl acetate found in higher concentrations in solid-state fermentation of different agro-industrial residue s such as cassava bagasse (Medeiros and others 2000, 2001).

Production of esters and aldehydes

- Banana flavor has also been identified in some plants when micro organisms (Erwinia caratovora ssp atroseptica) infected them.
- Spinnler and Dijian (1991) identified the volatiles formed in a synthetic medium, similar to the ones in the infected plant.
- From 13 isolated microorganisms with capacity to produce esters and alcohols, several compounds were identified; including ethyl acetate, 2-methyl-1-propanol, and propyl acetate.
- In this study, different media were tested with different nitrogen and carbon sources

Production of esters and aldehydes

- The microbiological population was identified and quantified by Park and others (1982), with counts between 6 *105 and 1.9 *106 CFU, as being predominantly Aspergillus niger and Pecylomyces sp.
- Yoshizawa and others (1988) identified some volatiles produced by a strain of Neurospora, isolated by Park and others (1982), in submerged fermentation.
- These include ethanol, ethyl acetate, isoamyl acetate, ethyl hexanoate, and acetoine.

Production of esters and aldehydes

Aldehyde production:

- Aliphatic, aromatic, and terpenoid aldehydes are important contributors to the flavor of fermented dairy products.
- They are synthesized by microorganisms as intermediates in the formation of alcohols from keto acids.
- An example is the bioconversion of ethanol to acetaldehyde by Candida utilis.

Production of esters and aldehydes

END

- Flavor production using immobilized lipase from the yeast Candida cylindracea in a nonaqueous system has been studied for producing a broad range of esters including ethyl butyrate, isoamyl acetate, and isobutyl acetate.
- Ethyl butyrate has a pineapple–banana flavor, which has a large market demand, and sells at a price of US$150 /kg upwards.
- This process has shown a great stability of the enzyme (more than a month) if kept hydrated intermittently (Singhal and Kulkarni 1999).

Food Biotechnology

Module #168



Production of alcohol and terpenes

Production of alcohol and terpenes

Alcohol production:

- Alcohols do not contribute as a flavor component unless present in high concentration.
- They are formed as a primary metabolite from microorganisms' activity or due to the reduction of a carbonyl.
- Fusel alcohols can be formed from either carbohydrate or amino-acid metabolism and are the predominant volatiles of all fermented beverages, in addition to ethanol.

Production of alcohol and terpenes

- Different alcohols can be found in the culture of yeasts such as ethanol, propanol, isobutanol, and phenyl ethyl alcohol.
- In filamentous fungi it is possible to find methyl-3-buthanol, butanol, isobutanol, pentanol, hexanol, octanol-3, and phenyl ethanol from the metabolism of amino acids such as leucine, valine, isoleucine, and phenylalanine (Welsh and others 1989).

Production of alcohol and terpenes

Terpens production:

- Terpenes are the most important natural components of essential oils to be used as flavors.
- Microorganisms are able not only to synthesize but also to degrade or transform terpenes.
- Fungi are the main microorganisms responsible for terpene production, but bacteria are capable of synthesizing a few volatile terpenoids, such as geosmin and cadin-4-ene-1-ol (Berger 1995).

Production of alcohol and terpenes

- The synthesis of monoterpenes by *Ceratocystis variospora* is reported.
- Numerous other microorganisms are able to synthesize monoterpenes, among them *Ceratocystis moniliformis, Kluyveromyces lactis, Sporobolomyces odorus, Trametes odorata* and *Trichoderma viridae*.
- Microbial bioconversion of terpenes has been extensively studied.

Production of alcohol and terpenes

- Monoterpenoid compounds like citronellal, citral, limonene, and menthol (acetates) can be biotransformed in citronellol, geranic acid, carveol and 1-menthol, respectively.
- Valencene is a sesquiterpenoid available from orange oil and has little commercial use.Some bacteria are capable of transforming valencene to nootkatone, a main flavoring component of grapefruit.

Production of alcohol and terpenes

- From the economic point of view, the development of biotechnological processes for the production of terpenes is not viable due to the low yields obtained and the abundance of vegetable sources available.
- The real meaning of these studies is the understanding of the steps of the catabolism of terpenes.

END

Food Biotechnology

Module #169



Determination of specific gravity of sugar

Determination of specific gravity of sugar

- Density, the mass per unit volume at a specified temperature, can be used to identify foods.
- A related parameter, specific gravity, is often used.
- Specific gravity is the ratio of the density of a substance at a specified temperature to that of water at 4°C. The specific gravity of water at 4°C is 1.0.
- The density of water at 20°C is 0.99823 g/cm3, which is close enough to unity that the temperature of the water will not need to be at 4°C for most uses.

Determination of specific gravity of sugar

- Specific gravity helps determine adulteration, quality, and composition of food such as the water or butterfat content of milk, the syrup concentration of canned fruits or juices, and the alcohol content of beverages.
- Specific gravity of liquids can be measured with a hydrometer.
- The function of the hydrometer is based on Archimedes' principle that a solid suspended in a liquid will be buoyed up by a force equal to the weight of the liquid displaced.

Determination of specific gravity of sugar

- The lower the density of the sample, the lower the hydrometer will sink.
- Saccharometers are hydrometers graduated to indicate percentage of sucrose by weight (degrees Brix).
- The refractometer provides an alternative means of measuring the soluble solids of sugar syrups and fruit products and is based on the index of refraction of the solution.

Determination of specific gravity of sugar

- Specific gravity of solids:
- Measures:
- Specific gravity by weight of known volume.
- Uses:
- Measure amount of air incorporated in products such as whipped cream, egg white foams, creamed shortening, and cake batters.

Determination of specific gravity of sugar

Proceedure:

- Weigh a dry container to the nearest gram.
- Fill container with cooled, boiled deionized water at room temperature. Complete fill on balance; judge at eye level. Weigh to nearest gram.
- Fill dry cup with test material. Do not pack. Remove excess with spatula. Wipe outside of container. Weigh to nearest gram.

Determination of specific gravity of sugar

- Calculate the specific gravity as follows:

Specific G= wt. of filled container- wt. of container/vol. of container.

- where volume container = (weight container + water) – weight container.
- Since specific gravity is the density of a substance relative to water it has no units.

END

Food Biotechnology

Module # 170



Food Biotechnology

Biotechnological approaches to improve nutritional quality and shelf life of fruits and vegetables

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Fruits and vegetables consumed as fresh or processed
- Important ingredients of a healthy diet
- They are valuable sources of :
- Vitamins
- Minerals
- Antioxidants
- Fiber

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Important quality factors of fruits and vegetables are:
- Color
- Flavor
- Texture
- Nutritive value

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Quality is a concept involving measurable attributes :
- Degree of purity
- Firmness
- Flavor
- Color and size
- Maturity condition,
- Any other distinctive attributes or characteristics of the product

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Qualities of the produce are influenced by :
- cultivar
- environmental conditions affecting growth
- Cultural practices
- pests and diseases
- Time of harvest
- Post harvest and storage conditions

Biotech approaches for nutrition and shelf life of fruits and vegetables

- With advancement of technology in several areas, the only factor on which the grower has no control of is the environment of the field conditions

- Heredity plays a major role in determining the quality of fruits and vegetables

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Traditional crop breeding is still being used as one means of crop improvement
- Continuing advances in knowledge and technology have dramatically expanded the biotechnological tools available for genetic improvement and production of vegetables and fruits

Biotech approaches for nutrition and shelf life of fruits and vegetables

- The term biotechnology is broad, encompassing a wide range of disciplines in science focus will be on plant transformation:

- Genes are modified or transferred by molecular means, and the resulting improvements in fruit and vegetable quality

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Parameters and mechanisms that affect nutritional quality and shelf life
- Other than pests and diseases
- Which can be improved or modified by genetic engineering

END

Food Biotechnology

Module # 171



Factors effecting reducing sugars in potatoes

Factors effecting reducing sugars in potatoes

- Potato is an important agri commodity.
- Non enzymatic browning of the product occurs at high temperature, when reducing sugars combine with the alpha-amino groups of amino acids Resulting in dark and more bitter products that are unacceptable to the consumer
- Known as Maillard reaction

Factors effecting reducing sugars in potatoes

reducing sugars accumulate in potatoes by:

1. Low temperature sweetening (LTS) when stored below 10° C in storage.

2. Metabolism of starch in tuber.

- To avoid browning potatoes are generally stored at around 10° C but it leads to sprouting.
- chemical sprout inhibitors are used

Factors effecting reducing sugars in potatoes

The only solution to avoid this problem is using cultivars that are resistant to LTS. Benefits include

- Control of sprout growth and senescent sweetening
- Reduction of physiological weight loss due to decreased respiration and losses associated with bacterial and fungal pathogens
- Extended marketability

Factors effecting reducing sugars in potatoes

Disadvantages:

- Associated with it is the accumulation of reducing sugars and the resulting browning of processed products such as chips and fries
- The mechanism is complex and may involve the interaction of several pathways of carbohydrate metabolism and the genes that regulate these pathways

Factors effecting reducing sugars in potatoes

Metabolism of starch in tuber

- Potato tubers contain 60–80% starch Of which sugars up to 3% on a dry weight basis
- The principal event in LTS is the cold induced synthesis of sucrose
- The carbon needed for the synthesis of sucrose and reducing sugars for LTS is generally, but not always, provided by a net breakdown of starch

Factors effecting reducing sugars in potatoes

- Increase in potato tuber sugar content occurs early during cold storage;
- Over two to three months at storage temperatures of 1–3° C tubers can convert as much as 30% of their starch content.
- sweetening response to low temperatures is fairly consistent
- Is influenced by cultivar, locality, and conditions prior to cold storage

Factors effecting reducing sugars in potatoes

- After cold storage, potatoes are reconditioned at 18° C where sugar content decreases and starch content increases
- Response to reconditioning is neither consistent nor completely restorative
- Tends to be cultivar dependent

Factors effecting reducing sugars in potatoes

- Preconditioning treatment has been used to lessen chilling injury in chilling sensitive plants

- Storage at 10° C prior to cold storage can acclimatize tubers and lessen the LTS effect

END

Food Biotechnology

Module # 172



Starch Synthesis

Starch Synthesis

- Starch is synthesized in plastids (amyloplasts) upon tuber initiation
- number of starch grains and grain size increase during tuber growth.
- Potato starch is comprised of 21–25% amylose and 75–79% amylopectin

Starch Synthesis

- Starch is synthesized from ADPglucose by the concerted action of:
- ADPglucose pyrophosphorylase (ADPGPase),
- Starch synthase
- Starch branching enzymes

Starch Synthesis

- It starts with conversion of sucrose into hexose phosphates in the cytosol:

- Then Glucose-6-phosphate is transported into the amyloplast where it is converted into Glucose-1-phosphate.
- Subsequently converted to ADP glucose by the action of ADPGPase.

Starch Synthesis

Starch synthases:

- Catalyze the polymerization of the glucose monomers into alpha-1,4-glucans
- Using ADP glucose as a substrate

Starch branching enzyme catalyzes the formation of the alpha-1,6-linkages of amylopectin

Starch Synthesis

- ADPGPase is a rate-limiting step in starch synthesis subjected to allosteric activation by 3-P-glycerate and inhibition by inorganic phosphate
- Strategies to alter the starch metabolism in tubers by genetic manipulation of ADPGPase may be helpful in reducing the accumulation of reducing sugars during LTS

END

Food Biotechnology

Module # 173



Starch Degradation

Starch Degradation

- The differential response of potato cultivars to LTS is result of starch properties that affects enzymes ability to degrade it.
- Chemical modifications of glucose units by attachment of covalently bound phosphate esters attached to C3 or C6 glucosyl residues of amylopectin in inter branch chains but absent around branching points. This affects the cleavage sites and degradation pattern.

Starch Degradation

- Similarly negative surface charge from surface phosphate, lipid, or protein can affect the properties of enzymes and other soluble compounds.

Starch Degradation

- The degradation of starch is believed to occur in the amyloplast
- The widespread distribution of alpha-glucan phosphorylase, alpha-amylase, beta-amylase, and maltase suggests that starch breakdown could be phosphoryltic, hydrolytic, or both.

Starch Degradation

upon transfer of tubers to chilling temperatures sucrose is the first sugar to accumulate

Amylase is quite inactive at such cold temperature to catalyze the required rate of starch degradation

Starch breakdown in cold stored potato tubers is mainly conducted by starch phosphorylase

Starch Degradation

Two types of potato phosphorylases are recognized based on:

- Glucan specificity
- Monomer size
- Intracellular location.

They are non-interconvertible proteins with different primary structures and different immunological properties

Starch Degradation

- Type 1 isozyme also known as type H localized in the cytoplasm has low affinity for maltodextrins and high affinity for branched polyglucans,
- Cross reacts with type H phosphorylase from potato leaves

Starch Degradation

- Type 2 isozyme also known as type L located in the amyloplast has a high affinity for maltodextrins, low affinity for branched polyglucans
- Cross reacts with type L enzyme from the leaf

Starch Degradation

Effect of Inorganic Phosphates:

- Intracellular compartmentalization of Pi influence carbon partitioning in non-photosynthetic potato tubers
- Increased inorganic phosphate in the amyloplast shifted metabolic activities toward starch degradation rather than accumulation

Starch Degradation

- Increased Pi concentration inhibits ADPGPase and enhances starch breakdown by alpha-glucan phosphorylase

- Another source of Pi in plant cells is the vacuole. major portion of Pi is stored in potato tuber vacuoles where it is compartmentalized away from the cytoplasm

END

Food Biotechnology

Module # 174



Food Biotechnology

Starch-Sugar Balance



Starch-Sugar Balance

Relationship between starch and sugar levels when potatoes are cooled from 10 to 2° C and then re-warmed from 2 to 10° C after an interval, has given rise to a misleading concept of a starch to sugar balance in which the overall change between the two compounds is seen as being reversible

Starch-Sugar Balance

- Sucrose is formed from starch by an irreversible pathway

- Starch is formed from sucrose by separate, but likewise irreversible, routes

Starch-Sugar Balance

The very close relationship between starch and sucrose in stored potatoes may be due to the fact that starch is the only possible source of carbon for sugar synthesis in the cold

Strong evidence to indicate that G-6-P is transported into the amyloplast of potato tubers to support starch synthesis

Starch-Sugar Balance

Pathways of starch and sugar biosynthesis compete for the same pool of precursors

The coexistence of the pathways of sucrose synthesis and starch breakdown in stored tubers may be regulated by fine control mechanisms

Starch-Sugar Balance

Genotypic variation in the capacity to maintain an active starch synthesizing system may help in alleviating the rate of sucrose accumulation

END

Food Biotechnology

Module # 175



Food Biotechnology

Concept of Edible Vaccines

Concept of Edible Vaccines

Involves the process of incorporating the selected desired genes into plants and then enabling these altered plants to produce the encoded proteins

The process is known as transformation

Altered plants are known as transgenic plants

Concept of Edible Vaccines

Edible vaccines like traditional subunit vaccines consist of antigenic proteins and are devoid of pathogenic genes.

Concept of Edible Vaccines

Traditional subunit vaccines are:

- Unaffordable
- Technology-intensive
- Require purification, refrigeration
- Produce poor mucosal response

Concept of Edible Vaccines

Edible vaccines

- Posses good genetics
- Heat stability
- Do not need cold-chain maintenance, can be stored at the site.
- Syringes and needles are also not required, thus reduces the incidence of various infections

Concept of Edible Vaccines

- Act by stimulating mucosal and systemic immunity
- Dual action of edible vaccines provide first-line defense against pathogens attacking via mucosa
- Mycobacterium tuberculosis and carriers causing diarrhea, pneumonia, STDs, HIV etc.

Concept of Edible Vaccines

At present edible vaccines are produced for various human and animal diseases such as measles, Cholera, FMD, Hep B, C and E.

Also be used to prevent exceptional diseases like:

- Dengue, Hookworm, Rabies by combining with other vaccination programs enabling multiple antigen delivery.

Concept of Edible Vaccines

Methods:

Selected gene obtained from the microbes encoding specific antigen can be handled in two different ways

1. Suitable plant virus is engineered to produce desired peptides/proteins is incorporated into the plant, which enables it to produce a huge number of new plants from which chimeric virions are isolated and purified and applied.

Concept of Edible Vaccines

END

2. In another method, the desirable gene is incorporated with plant vector by transformation

   Many other approaches have been utilized which can be categorized into following groups:

- Agrobacterium mediated gene transfer
- Biolistic method
- Electroporation

Food Biotechnology

Module # 176



Examples of Edible Vaccines

Examples of Edible Vaccines

Transgenic potatoes for diarrhea:

- First successful human trial was conducted in 1997
- Transgenic potatoes, which possessed the b-subunit of the *E. coli* heat-labile toxin, responsible for diarrhea
- 4-fold increase in serum antibodies was manifested in ten out of the 11 volunteers

Examples of Edible Vaccines

Next clinical trial took place at the Boyce Thompson Institute at Cornell University, USA, in which 20 volunteers ate the potatoes containing the Norwalk virus (responsible for vomiting and diarrhea), out of which19 showed an immune response.

Major drawback is cooking potato denatures protein and makes it ineffective

Examples of Edible Vaccines

Transgenic tomatoes against diarrhea:

- Transgenic tomatoes were produced against the Norwalk virus
- Responsible agent for severe diarrhea.
- The transgenic tomatoes are capable to produce surface protein specific to the virus.
- mice fed with transgenic tomatoes showed immune response towards the virus

Examples of Edible Vaccines

Other transgenic plants:

Presently, banana is being exploited as a good source for edible vaccine production

- It does not require cooking
- Locally grown plant

Examples of Edible Vaccines

- The protein expression in transgenic banana is tissue specific promoter dependent.
- Several other examples involve rabies glycoprotein expressed by viral vectors in spinach
- Hepatitis B surface antigen in case of lettuce and potato

Examples of Edible Vaccines

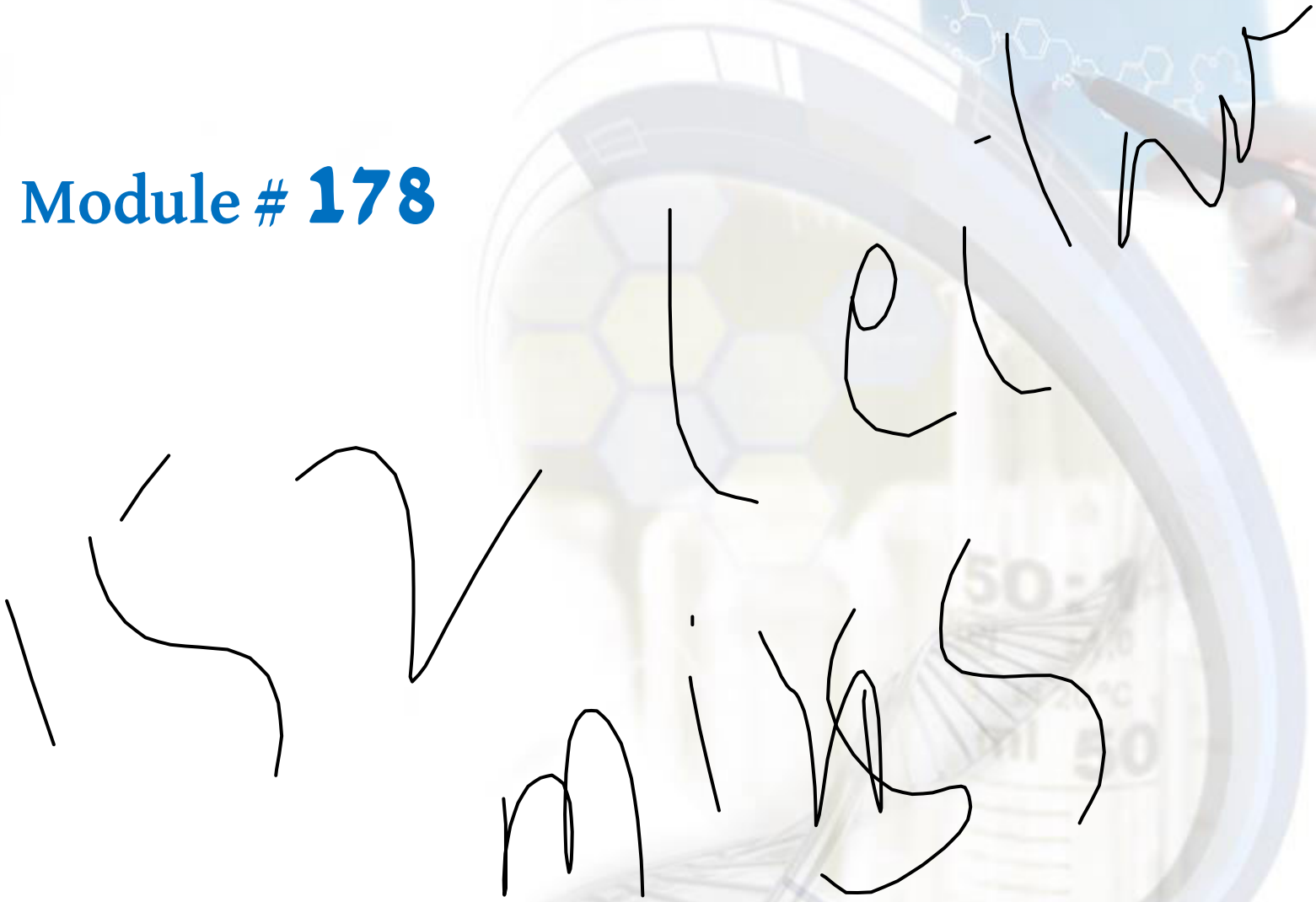
- Applications of edible vaccines include:

- Cancer therapy
- Birth control
- Chloroplast transformation
- Role in autoimmune disease
- Recombinant drugs/proteins

END

Food Biotechnology

Module # 178



Food Marketing Principles

Food Marketing Principles

Marketing activities are generally carried out in all types of industries

However, the nature of the industry does lead to some differences in the activities

Food Marketing Principles

Food companies contribute to the development of marketing theory as objects of case study

Nestle, Unilever, Coca-Cola, McDonald's

Popular cases in recent years

Food Marketing Principles

Methodology for the entire food industry are confronted with one difficulty

'Food' is a general term for edible products that human beings ingest in daily life

Different companies produce different kinds of food

Food Marketing Principles

Nestle is famous for dairy products, chocolate, coffee

Unilever make margarine

Coca-Cola is a soft drink company

Mc-Donald's is a hamburger chain

Food Marketing Principles

The food industry comprises plural sub-industries

There is a need to consider many types of marketing applications that correspond to the different products in the food industry

Food Marketing Principles

Marketing activities implemented by companies and organizations in the food industry are described as ‘food marketing’

END

Food Marketing Principles

- The marketing concept
- The marketing process
- The marketing research
- The formulation of a marketing strategy
- The devising of a marketing plan

Food Biotechnology

Module # 179



Marketing Principles



Marketing Principles

Marketing is different than selling and promotion only, now a days it is consider much broader, such as:

"the total of activities involved in the transfer of goods from the producer or seller to the consumer or buyer, including advertising, shipping, storing, and selling”

Marketing Principles

The basic principles of marketing include the 4 P's

- Product
- Price
- Place
- Promotion

Marketing Principles

Two parts

1. Create value for customers
2. To build strong customer relationships

Marketing Principles

The 4 P's have been updated to now 7 P's

- Product
- Price
- Place
- Promotion
- People
- Positioning
- Packaging

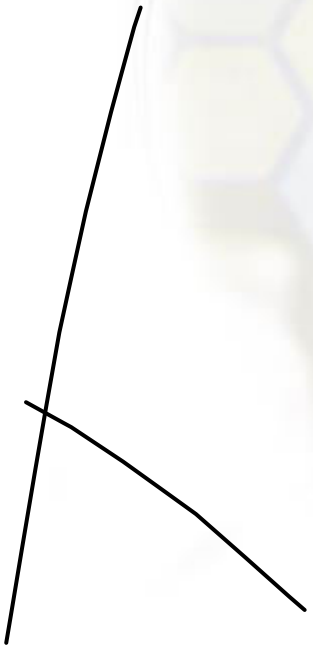
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Food Biotechnology

Module # 180



The Marketing Concept



The Marketing Concept

Marketing is defined as:

- ‘Marketing is the process by which companies create value for customers and build strong customer relationships in order to capture value from customers in return’

- Advertisements are a part of marketing

The Marketing Concept

- Sometimes, you are requested to sample new products such as a new flavor of cheese or a new taste of beverage, in a supermarket.
- Sales people explain to you how the new sample differs from existing products.
- This selling activity is also a part of marketing

The Marketing Concept

- Advertising and selling constitute only a part of marketing
- Marketing begins with understanding the customer's needs and wants
- Finding potential needs and wants that the customer is still unaware of also plays a key role

The Marketing Concept

- In order to create value for customers, it is necessary to realize customers' needs and wants

The Marketing Concept

- Knorr's soup stock
- Accepted by customers since it realized one such customer need:
- It not only reduced the soup to powder but also met the demand for quick and easy soup preparation

The Marketing Concept

- Building strong customer relationships is also integral to marketing
- Goal of marketing is to capture value from the customer
- Advertising, sales and promotions are marketing activities carried out to deliver the value created for customers

The Marketing Concept

- Once a customer buys a product, some measures must be taken to ensure that he/she makes repeat purchases

- Retaining customers is an important part of marketing

The Marketing Concept

- It is important to establish a brand name for the customer to recognize a product
- Branding helps in customer retention

END

Food Biotechnology

Module # 181



Marketing Process

Marketing Process

As mentioned previously, the aim of marketing is to capture value from the customer.

To arrive at this value, marketers go through five steps

Collectively referred to as the ‘marketing process’

Marketing Process

1. Understand the marketplace and customer needs and wants

2. Design a customer-driven marketing strategy

3. Construct a marketing program that delivers superior value

Marketing Process

4. Build profitable relationships and create customer delight

5. Capture value from customers to create profits and customer equity

Marketing Process

understanding marketplace and customers

- Marketing research is one method
- Understand main influences on marketing activities
- Collect, analyze, and evaluate information and data

Marketing Process

Design of customer-driven marketing strategy

- Marketing strategy formulation depends on the targeted customers
- Forced to decide who customers are
- Carryout market segmentation
- Deliver goods to target customers

Marketing Process

Preparing a practical marketing program.

- Begin with planning the details of the marketing mix
- Marketing mix is a combination of the firm's marketing tools – product, price, promotion and place
- Draft a marketing program that will meet the wants of the market

Marketing Process

Customer and Partner relationship management:

- The former pertains to the manner in which firms build relationships with their customers
- The latter pertains to how firms build relationships with their trade partners.

Marketing Process

Capturing value from the customer

- Attempt to build customer loyalty by enhancing customer satisfaction
- Focus on increasing their market share
- Also on increasing customers' mind share

Marketing Process

- By building customer loyalty, firms are able to increase their profits substantially

END

Food Biotechnology

Module # 182



Food Biotechnology

Application of social, legal, ethical and environmental principles to marketing situations



Application of social, legal, ethical & environmental principles

Corporate Social Responsibility (CSR)

The idea of social responsibility supposes that the corporation has not only economic and legal obligations, but also certain responsibilities to society which extend beyond these obligations

Application of social, legal, ethical & environmental principles

Economic responsibilities demand that the corporation should produce goods and services that society wants and sell them at fair prices



Application of social, legal, ethical & environmental principles

Fair prices do not naturally imply low prices

A fair price is one that includes dividends to the investors and sufficient profits for the continuance of the business

Application of social, legal, ethical & environmental principles

In recent years, legal responsibilities have come to the force due to the keen market competition

Compliance with laws is an important aspect of a business responsibilities

Application of social, legal, ethical & environmental principles

A company's ethical responsibilities toward society go beyond its economic and legal obligations

Ethical responsibilities embrace those activities and practices that are expected or prohibited by societal members even though they are not codified into law

Application of social, legal, ethical & environmental principles

Application of CSR to marketing situations is linked to producing a positive image that the company contributes to society and that its existence is valuable to society

Positive image is not a tangible asset of the company but is stored in the minds of customers

Application of social, legal, ethical & environmental principles

- Neglecting CSR can lead to major problems
- Food scandals and reputation loss

Application of social, legal, ethical & environmental principles

CSR includes the important topic of environment protection

The significance of this topic came to light with the environmental pollution of the 1970s

In the 980s, there emerged the environmental concerns, such as global warming, depletion of the ozone layer, reckless deforestation and sea pollution.

Green marketing

Application of social, legal, ethical & environmental principles

- For companies to build their reputation in the market, it is important that they carry out green marketing

- Customers who are very sensitive to global environmental problems constitute an important market that firms cannot afford to ignore

END

Food Biotechnology

Module # 183



Marketing Research



Marketing Research

For marketing research, we must first:

Understand the main influences on the marketing process within the macro and micro environments

Marketing Research

- Analyze and respond to the issues associated with buying behavior in both consumer and organizational markets
- Evaluate the need for information in marketing and understand the different methods of collecting and analyzing data
- Apply the marketing research process in selected food markets.

Marketing Research

- For effective marketing, marketers need to understand all relationships that surround the company
- These relationships are called the:marketing environment'. It is imperative that companies adapt to the changing environment

Marketing Research

- The marketing environment comprises a macro environment and a micro environment

- Macro environment is formed of some societal forces that influence its marketing activities

Demographic, Economic, Natural, Technological, Political, Cultural

Marketing Research

Demographic : study of the human population. Includes the study of the size, structure and distribution of populations and how populations change over time

- Baby boomers (born between 1946-1964),
- Most powerful market
- Developing products for this market has gained prominence

Marketing Research

Economic: level and distribution of the income effect and consumer purchasing power and purchasing patterns

In the case of a company that approaches a foreign market, the price of the imported food is higher than local food

Important to know the size of the market for the high-priced product

Marketing Research

Natural: concern for the global environment has been increasing steadily

Companies need to develop products that do not lead to global warming and air and water pollution

The production processes should be modified to suit this need

Marketing Research

Technological: new technologies create new markets and opportunities

New technology replaces older technology

New technology also creates new regulations.

For example, it is mandatory to check the safety of genetically modified organism (GMO) foods before they are delivered to the market.

Marketing Research

Political: laws, government agencies and lobby groups form the political environment

These factors influence businesses in various ways

The number of laws and regulations is increasing year by year

Marketing Research

Cultural: people in a society have common basic beliefs and values, namely, culture.

These core beliefs and values affect their perceptions, preferences and behaviors.

Eating habits are especially influenced by the food culture

Marketing Research

The micro environment consists of the actors close to the company that affect its ability to serve its customers

Following are the forces considered to constitute micro environment

Company, Suppliers

Marketing intermediaries

Customer markets, Competitors, Public.

END

Marketing Research

- Company: marketing plan that requires the coordination of many intra-firm functions – top management, finance, research and development (R&D), purchasing, operations and accounting – is formulated.
- These interrelated functions constitute the internal environment

Marketing Research

- Suppliers: companies need many kinds of resources to produce goods and services
- These resources are provided by suppliers. If there are shortages and late deliveries of these resources, marketing activities are affected.
- The prices of resources influence the prices of products and services.

Marketing Research

- Marketing intermediaries: in making products accessible to the final customer, marketing intermediaries, for example, resellers, physical distribution firms, marketing services agencies and financial intermediaries, play an important role
- Companies need to build good relationships with their marketing intermediaries

Marketing Research

- Customer markets: the three well-known types of customer markets are personal consumption markets, business markets and government markets
- Personal consumption markets are subdivided into individuals and households.
- Marketers today need to recognize both the international market and the domestic market

Marketing Research

- Competitors: companies produce goods in order to satisfy the needs of their target consumers

- These goods must realize greater customer satisfaction than those of their competitors

Marketing Research

- Public: public relations influence marketing activities
- A company's relationships with investors, the media, government, consumer and citizen groups and local communities play an extremely important role
- Workers, managers and the board of directors constitute the internal public

Marketing Research

- It is important to make employees feel good about their company

Food Biotechnology

Module # 184



Food Biotechnology

Consumer Behavior

Consumer Behavior

As markets are classified; Buying behavior is also classified

- Consumer buyer behavior in the consumer market
- Business buyer behavior in the business market

Consumer Behavior

- Marketers have to consider two things:
- Factors influencing consumers
- The buyer decision process

Consumer Behavior

Factors influencing consumers, the following four are well known:

- Cultural
- Social
- Personal
- Psychological

These four factors are presented in a kind of progression, from factors that have a broad influence to those that have a personal influence

Consumer Behavior

Marketers need to understand the three components of the buyer decision process

Need to understand:

- The buying decision maker
- Types of buyer decisions
- The steps in the buyer decision process

Consumer Behavior

The buying decision maker has a role in the buying decision process

A man who purchases daily food items from a supermarket by referring to a list is the purchaser

His partner, who made the list, is the decision maker of these purchases

Marketers have to identify the buying decision maker and approach him/her

Consumer Behavior

The decision-making process of a purchaser depends on what he/she wants to buy

The types of buyer decisions are classified along two axes:

- The commitment level to the buying process
- The differences between brands

Consumer Behavior

This model depicts the five steps in the buyer decision process. The buyer:

1 recognizes a need

2 seeks information

3 evaluates alternatives

4 decides the purchase

5 makes a post-purchase assessment

Consumer Behavior

The buying process begins long before the actual purchase

Continues long after it is completed

Therefore, marketers need to focus on the entire buying process rather than on merely the purchase decision

Consumer Behavior

Business buyer behavior

- Consists of organizations that purchase other companies' products and services in order to produce their goods
- Characteristics of a business market differ from those of a consumer market

Consumer Behavior

- The business market has fewer buyers than a consumer market
- The buyers in a business market are bigger than those in a consumer market
- There exists a close relationship between the supplier and the customer

Consumer Behavior

Business buyer behavior is influenced by various factors:

- Environmental
- Organizational
- Interpersonal
- Individual factors

These factors include many factors that differ from those that influence the general public and those that are related to individual transactions.

Consumer Behavior

Marketers also need to understand the three components of the buyer decision process:

- Buying decision maker
- Types of buyer decisions
- Steps in the buyer decision process

Consumer Behavior

- First, the decision is made in the buying centre.
- The buying centre includes organization members who play five roles in the purchase decision process:
- ‘users’ use the product or service
- ‘influencers’ help to prepare the specifications

Consumer Behavior

- ‘buyers’ select the supplier for routine buying
- ‘deciders’ have the final authority to select suppliers
- ‘gatekeepers’ control the information flow into the buying centre

Consumer Behavior

- Second, there are three orientations in the case of the buyer decision
- ‘purchasing orientation’ is a market transaction; the buyer aims to buy goods that are cheaper
- ‘sourcing orientation’ builds more cooperative relationships with suppliers

Consumer Behavior

- By building close relationships, the buyer aims to simultaneously achieve quality improvement and lower costs.
- ‘supply management orientation’, the buying centre plays a broader role than in other orientations; the company aims to enhance value through the entire value chain that it formulates

Consumer Behavior

END

Following eight steps are suggested to constitute the buyer decision process:

1 recognize problems

2 describe the general needs

3 specify the products

4 seek suppliers

5 solicit proposals

6 select the supplier

7 order the desired specification

8 review the performance

Food Biotechnology

Module # 185



Market Research Process in Food Market



Market Research Process in Food Market

Making decisions needs exact information about market. This information comes through market research.

Market research has 4 steps.

1. Define problem & objective.

2. Develop research plan.

3. Implement research plan.

4. Interpret and report findings

Market Research Process in Food Market

Defining Problem & Objective:

Most difficult step to assign cause to some effect/market observation. e.g. decrease in demand for a product. Then we make objectives through 3 approaches:

1. Exploratory research
2. Descriptive research
3. Cause & effect research

Market Research Process in Food Market

2. Developing Research Plan:

It outlines the sources of existing data and spells out the specific research approaches, contact methods, sampling plans and instruments to gather new data. It comes in the form of research proposal.

Secondary and primary data collection.

Market Research Process in Food Market

3. implementing research plan.

collection, processing and analysis of information.

Collection through research methods

For processing and analysis of information, we need to isolate the important information and findings.

Market Research Process in Food Market

4. Interpret and report findings

Interpretation is vital and must be correct to understand new perspectives.

Sharing must be reported to internal and external experts and be ready to embrace results.

Market Research Process in Food Market

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